ULTRAMAX®

THE BLUE BOOK;
Maximizing Profits through Production

The ULTRAMAX Method:
Concepts and Procedures

Carlos W. Moreno

Advanced Process Management™

Edition: November 2004
Revision: November 20, 2006

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Carlos W. Moreno
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Editions: May 1995, January 2000, February 2004, November 2004
Latest revision: November 20, 2006

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Patent pending US60/591,045.

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The rights to use the concepts and procedures described herein which characterize the ULTRAMAX® Method is given only to those organizations which have a valid user agreement with the Ultramax Corporation.

Relevant quotes for the challenges that businesses face in today's increasingly demanding production environments:

W. Edwards Deming: **"Learning is not compulsory ... and neither is survival."**

Peter Drucker: **"The computer … does it much faster and demands logical clarity that forces us to be rigorous."**

W. Edwards Deming: **"No customer asked for electric lights. No customer asked for photography. No customer asked for telephones."** … in other words, creativity is largely the thrust of a single person or a group of people with vision (and perseverance).

Thomas Jefferson: **"I'm a great believer in luck, and I find the harder I work the more I have of it."** (Let's change "harder" for "smarter"!)

Robert E. Peary: "***Inveniam viam aut facium***" **"I shall find a way or make one."**

Western Union, 1876, internal memo: "**This 'telephone' has too many shortcomings to be seriously considered as a means of communication**."

Robert Byrne: **"Everything is in a state of flux, including the status quo."**

Murphy's Lesser Known Laws:

Those who live by the sword get shot by those who don't.

The things that come to those who wait will be the things left by those who got there first.

Søren A. Kierkegaard: **"Life can only be understood (looking) backwards; but it must be lived (looking) forwards."** (Similar to ULTRAMAX – learn from the past to do better in the future)

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Preface

Most production assets are significantly underutilized... not in terms of time-usage... but in terms of operations contributing to comprehensive performance metrics selected by the user, including overall profit contribution. Optimizing regular operations captures most of the unused potential in the existing production assets.

Many high-volume production / manufacturing operations have untapped potential to increase contributions to company margins by a few hundred to a few thousand dollars per shift.

You already have these latent process capabilities -- all paid for. You just need to uncover this source of financial gains by applying the right multivariable learning "intelligence" to regular operations. The toolset to use is ULTRAMAX®, with advanced quantitative analyses for adjusting process settings (setpoints, biases, etc.), detecting problems, and leading plant personnel to understand problems and barriers and imagine better solutions... all without requiring changes to existing equipment or controls. It is safe, effective and simple. It is a high pay-back proposition.

We expect that very soon all executives responsible for production / manufacturing will no longer be willing to tolerate current production assets generating anything less that the maximum possible value for the firm. Neither will CFOs tolerate their investments in the existing production equipment yielding any less profit contribution than that of which they are capable.

This book provides proven and practical answers to these important questions for achieving excellence in daily operations of industrial production / manufacturing:

- ***How to adjust production / manufacturing equipment – or any other repetitive, measurable system -- while engaged in operations to generate the most value for the firm?***
- ***How to define performance metrics to evaluate (overall, total) improvements in value generation or achievement by operations?***

This book describes how to materialize these gains by applying the ULTRAMAX® Method, also known as *Advanced Process Management*™ (APM™) or *Optimizing Performance Operations*™ (OPO™). This solution can be used for:

- ... short-term optimization studies [e.g., to define optimal Standard Operating Procedures (SOP)], or more generally, when doing experiments for optimization
- ... all the way up to on-line continual applications to effectively make true optimization part of daily SOP in production / manufacturing when there are important changes in conditions, mostly applicable to high-volume industrial processes.

The application of this toolset achieves operational excellence in industrial production with the existing process (equipment and controls); a part of 'best business practice'. It yields significant payback:

- increasing quality and throughput
- reducing costs, losses, undesirable byproducts and emissions;

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- while satisfying safety, regulatory and other requirements

all with performance metrics as defined by the user.

The self-learning "intelligence" technology used in ULTRAMAX includes the fastest and most effective empirical optimization available today. Implementation or deployment is, as well, the simplest in the market. It does not require up-front the time consuming creation of process models or collection of risky experimental data – it learns from on-going operating data. In fact, it is so universal, thorough and simple that suppliers are not required to have expertise about the processes being optimized. Thus the slogan:

"If it can be measured, it can be optimized. ™"

Further – and perhaps even more important for the business – having an optimization culture in place requires and provides an environment for plant personnel to understand the process better in terms of quantitative business impact, and to discover breakthrough ideas for further improvements.

ULTRAMAX is meant to expand the capabilities of enabled personnel, not to be a substitute for them – even in closed-loop optimization where much of the routine labor is removed. As complete and automated as it is, it is meant to support an environment where people ultimately make things happen. ULTRAMAX is fully compatible and complementary with Maintenance, and with Engineering (equipment and controls), Six-Sigma and Lean Manufacturing approaches for making improvements to the equipment and processes.

The ULTRAMAX Method and the associated software have been well proven over many years, and have been readily deployed in hundreds of widely differing processes. The software is the same for all applications – with significant upgrades every two-to-four years.

ULTRAMAX has been applied successfully to (see specific examples in www.ultramax.com/Applications.htm):

- chemicals (a great variety),
- food processing (cooking, smoking), coffee roasting,
- soap / detergents,
- fiber manufacturing, nonwovens,
- paper making (including energy conservation), converting,
- machining and metal removal,
- boilers for steam and electric power generation (many, with emphasis on emissions control and fuel efficiency),
- pharmaceuticals: fermentation (antibiotics), encapsulation,
- manufacturing pigments and dyes,
- painting and coating,
- plastic manufacturing (including injection molding, extrusion, blow-molding and composite materials),
- welding and soldering,
- semiconductor fabrication (photolithography, dry etching, film deposition, thermal processing and ion implantation).

and several other processes.

A significant number of these applications have yielded bottom-line gains exceeding a few $1,000,000/year.

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Representative customers have included: Procter & Gamble, IBM, BASF, PPG, Sara Lee, Colbond Nonwoven, over a dozen power generating utilities with close to 100 boilers, Rockwell, Merck, Alpharma, General Electric, Pierre Foods, Oxy Vinyls, Golden Aluminum, Johnson Controls, Collins & Eickman (Textron, Evart, Rantoul, etc.); in several countries.

These are the three primary ways by which ULTRAMAX contributes to improving the company bottom-line or profits:

1. **Making Adjustments** to existing equipment / production process settings (manual, setpoints, biases, control parameters) for tangible improvements in operating performance.

   "Performance" is as defined **by the end-user**, and measured / calculated with data collected **by the end-user**. "Improved performance" is a better combination of all important outputs[1]; such as the right balance of higher throughput, better quality (averages and consistency), lower losses and rework, lower costs incurred, lower contamination; all that while process parameters are maintained within safety, reliability, quality, government regulations and legal constraints. The end-user represents this balance by a Production Operations **Performance Index** (a single metric to be maximized or minimized, depending on its nature) as long as **Constraints** are satisfied; sufficiently complete or holistic to avoid the possibility of improving something as the sacrifice of a higher loss elsewhere – what is often called "systems approach".

   ULTRAMAX has superior tools to define the single Performance Index metric that is the balance of various objectives (Goal Setting, page 101) including full economic analysis impact on the P&L Statement and effective quasi-economic analysis otherwise. Daily use of a representative Performance Index (and constraint satisfaction) is of great value even if not doing optimization, to enhance people's understanding of issues related to production and the business.

   What is adjusted to get better performance are, e.g.: speeds, flows rates, cycle times, pressures, temperatures, positions, angles, amount of ingredients, voltages, amps, levels, feeds, depth-of-cut, gain coefficients, etc. The adjustment decisions being optimized are often setpoints for the regulatory process control, or biases if already being adjusted by control logic.

   Opportunities for these gains exist when there are several adjustable inputs where performance is hurt if more than one is too large or too small.

   The adjustments are the result of the SEO technology learning and analyzing the characteristics of the process **to maximize performance** while:

   Compensating[2] for the known values of the important physical conditions (those that one has to accept in daily operations, such as raw material and fuel characteristics, ambient conditions, process conditions, as long as measured beforehand);

   Adapting to factors which represent current economic conditions and customer/business requirements (which are readily updated by management);

---

[1] Including Key Performance Indicators (KPI)

[2] Without introducing knowledge about conditions – such as when defining SOP adjustments-- the optimum is such for the family of conditions experienced, obviously with lower average performance.

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2. **Recognizing process upsets** -- often long before alarms sound -- and providing information (and rationale) to resolve the problem to avoid the same upset in the future – an example of ongoing Quality Control procedures; and
3. **Enhancing the understanding** of what contributes to the bottom line or profits, what are the capabilities of the process, and what are barriers to higher performance. This provides a basis to discover changes to the variables and objectives and/or improvements to the process to generate more value. An example of changes to the process is to vary the constraint values (e.g., a larger pump) – more generally, constraint management. Physical changes to the process is the first outcome from ULTRAMAX that might actually involve capital expenses; an activity likely to be done by Engineering using standard solution methods, and perhaps Lean Manufacturing and Six-Sigma approaches.

An advantage of optimization is not only to achieve improvements in terms of the current metrics of performance. Once one realizes that there are tools to inexorably improve metrics, the opportunity becomes obvious of having sharper metrics, more aligned with ultimate corporate objectives – as compared to vague indicators such as people's productivity (production / manpower) or time percent asset utilization. A good optimization solution does provide handy solutions for this refinement of the definition of performance metrics, as ULTRAMAX does.

ULTRAMAX is a product of industrial engineering; and it incorporates concepts and technology from management sciences, operations research, economics, quality assurance, statistics, decision sciences, and control theory.

Basic Requirements for Success

- Within the end user's organization there must be a culture of being **driven by improving results in measurable terms**, rather than <u>just</u> following procedures.
- The production / manufacturing process to be optimized must…
  - Produce data on a timely basis (ideally on-line or within a few hours of the completion of a "run". This requirement applies mostly while extrapolating to an optimum.
  - Have the inherent capability of satisfying any narrow constraints -- even if some constraints are not satisfied to start with. A barrier to satisfy constraints is high noise. ULTRAMAX usually gets as close to the optimum as the level of the noise.
- The user's role in assuring success includes:
  - Define your criteria for improvements and business success in measurable terms; usually some combination of production rate, quality, cost savings and/or emissions reduction; plus safety, capacity and regulatory requirements.
  - Provide basic knowledge about your product and process, such as adjustable inputs, uncontrolled inputs, and outputs.
  - Be well acquainted with the concept of "standard deviation" and recognize that a deviation exceeding 3-sigmas is an indication of a possible assignable or special cause that is better to identify and manage.

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Objectives of the Book, and Who Will Profit

This book provides a complete description of the ULTRAMAX Method. There are separate guides for using the software and for integration.

The book serves two primary audiences:

1. ***Production, Manufacturing and Finance Executives; Operations and Plant Managers***. Personnel with the overall responsibility for production and cost control. Ideally all important process operations are continually optimized for maximum contribution to the company bottom line or profits, and having a uniform operating culture across crews and production lines.

   Managers should become sufficiently acquainted with ULTRAMAX so that they participate in the definition of the objectives subject to changing economics and requirements. Also, they need to be able to assign / authorize execution of optimization to the appropriate personnel and to regularly recognize successes and barriers.

2. ***Plant Personnel, shift managers, operators and engineers*** with the responsibility for production / manufacturing.

   The most difficult issues in achieving near-optimal performance is related to resolving occasional process problems, such as dealing with an upset (the process going out of control). This book is particularly devoted to providing clear procedures to recognize these conditions and to focus the team's attention on the most effective approaches to resolving such situations in order to maintain the optimization culture.

This book is dedicated to the customers and personnel of the Ultramax Corporation, who through their feedback and dedication have contributed tremendously to the maturing of its concepts, procedures, and technology. We hope that our customers will continue to share their successes with us as the ULTRAMAX Method moves along its traditional path of ongoing improvement, addressing the new challenges of today and tomorrow.

Carlos W. Moreno, February 1995, January 2000

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Perspectives on Optimization

Applicability of ULTRAMAX optimization

Optimization is a decision process to aim at the best possible outcome in overall (holistic, multivariable) quantitative terms within a particular scope.

The "scope" includes the domain of decisions considered with an existing system plus the definition of objectives: a Performance Index or Objective Function, and Constraints to be satisfied. (True) optimization requires serious multivariable mathematical technology.

The art, or engineering, of optimization includes the definitions of a practical scope, usually a balance between quick returns by solving a problem of smaller scope, and not to sub-optimize too much to miss greater improvement opportunities. A practical optimization technology allows for easy changes to the scope.

Anything less results in "sub-optimization", even when using optimization technology for the reduced scope. Aspects of sub-optimization are:

- The selection of decisions being a subset of all practical possibilities
- The definition of objectives being less comprehensive, or holistic, in terms of organizational objectives

Managing regular production / manufacturing operations is, fortunately, a problem with a well-structured mathematical solution. This problem is well understood and there are several companies offering comprehensive solutions, many that can be used on-line.

ULTRAMAX is a decision support system with these characteristics:

- It has multivariable quantitative technology to propose better operating decisions (among a very large number of alternatives). This is an area where computer power excels.
- The criterion for improvements is any metric defined by the user, as holistic as practical to avoid improving something at the sacrifice of a higher loss elsewhere.
- In addition, it helps analyze barriers to greater performance to guide future re-engineering projects (including constraint management).

Related to ULTRAMAX there are three pivotal, slightly over-simplified, concepts:

1. **Regulatory Control** is to maintain controlled variables close to setpoints – where Advanced Process Control (APC) has many levels of advanced technology. APC also includes **Control Logic** to adjust setpoints as a function of conditions (similar to the main function of ULTRAMAX but unlikely to include full optimization).

   Regulatory control, as a specific scope, can be optimized (for which there are several suppliers) such as optimal PID tuning – but this is very different than optimizing directly performance or economic value generation, as explained in the next paragraph.

2. **Operating Performance Optimization, Operations Optimization** or **Process Optimization** (or just Optimization in this book) is to maximize the value generated by production operations for the firm by the **adjustment** of settings such as of: *setpoints* for regulatory control; *biases* to improve the adjustments made by built-in control logic, and *manually* set elements (such as a damper) and other production operations

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decisions, given the existing equipment; and concurrent with important varying conditions. This optimization also does regulatory control (balanced with other performance metrics) for those outputs out of direct reach of direct single loop or multi-variable control schemes.

3. ULTRAMAX's Optimization (for dynamic systems) can get **as close to the optimum** as the order of magnitude of the "**noise**" (unexplained output variations). Better regulatory control reduces the noise in the output data, which in turn enables empirical optimization being closer to ideal optimum (without noise).

The key technology used in ULTRAMAX to optimize the performance of operations is Sequential Empirical Optimization (SEO). The decision domain consists mostly of Adjustments described in #2 above for repetitive "normal" process operations (usually excluding startups, shutdowns, changeover, etc. since these operations do not happen as often, it takes a long time to optimize). It balances impacts on other processes, products and business units outside the scope through the internal marginal values of the resources it uses. ULTRAMAX is not applied to decisions related to scheduling, production planning and maintenance, but there could be exceptions.

More generally, SEO is applicable to any operations whose performance depends on quantitative adjustment decisions, where the system runs frequently, and performance can be measured. Imagine, for instance, these examples: skippering a sailboat, carburation in a truck engine, formulation of an industrial food recipe, and treating a chronic disease.

As cultural impacts ULTRAMAX also includes inducing the user to discover possible improvements (innovation) related to system being managed. The **Macro-Cycle of improvements** it is (a) optimize, (b) discover barriers and innovative opportunities to justify improvements in the production assets and procedures; and with a new process or system, do Optimization again (a). A good optimization technology allows for easy re-start with a revised scope; one of the breakthroughs achieved in ULTRAMAX with the version 7.0.

In production / manufacturing the main path for significant improvements is innovation on products and the production system itself. In turn, ULTRAMAX is an innovation in decision support systems. It is a practical answer to the business consideration: *why get any less than current production assets are capable of?*

Some distinguishing characteristics of ULTRAMAX

To distinguish ULTRAMAX's optimization solution from other approaches that use the term "optimization" for various kinds of improvements, let us clarify that:

1. Some solutions provide useful information or visibility for the end-user to appreciate reality better, which may lead to making better and quicker decisions. The ULTRAMAX solution has technology that actually makes / advises better decisions.

   Thus, displaying Key Performance Indicators (KPI) -- which is also done by ULTRAMAX -- is good to see improvements and especially to detect problems, but that alone is way short of optimization.

2. Some suppliers offer improvements of indicators, such as running faster, increasing machine utilization (in terms of the amount of hours it runs, not its net contribution to profits) or getting closer to constraints – which is a sub-optimization that can easily result in significantly different decisions and outcomes than full optimization. ULTRAMAX offers optimization of overall, comprehensive, performance metrics (including economic,

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holistic, bottom-line metrics if so chosen); all together representing the balance of any number of multiple and potentially conflicting objectives.

3. The Process Control System (PCS) is what governs the operation of production equipment. Production Personnel adjust settings to the PCS to manage production performance, and ULTRAMAX is a tool to do it more effectively. PCS is necessary for ULTRAMAX to operate, and as part of the equipment itself, it is used as it is.

   The more basic PCS installations include regulatory control of several controlled variables (or parameters) with single-loop feedback mechanisms (typically PID[3] loops or subsets). Advanced Process Control (APC) have solutions for being closer to setpoint values by e.g., applying multivariable dynamic matrix control (DMC) and model predictive control (MPC). APC may also have control logic to adjust setpoint values as a function of one or more physical conditions, often represented as "curves"; and the most advanced APC have true optimization logic (such as ULTRAMAX, First Principle Models, or Neural Networks).

   There is synergism between Process Control and ULTRAMAX:

   - The closer Regulatory Control schemes can keep controlled variables to their setpoints the better it is for optimization solutions. Better regulatory control results in less "noise[4]", with two benefits: (1) there is less need to provide protection against individual run data violating output constraints, which means that operations can approach performance closer to the ideal optimum (without noise); and (2) it enables an empirical solution to learn more accurately about process performance characteristics, and thus better identify the optimal adjustments.
   - Optimization can also include the objective of reducing the deviation of outputs from targets. Thus, optimization is a practical solution for those outputs beyond the reach of regulatory control (as it tends to be true for most KPI) (see An Illustrative Example, page 9).

4. Built-in control logic could in principle be optimal but it virtually never is because:

   - The built-in logic is likely to be generic, to represent basic, fundamental relationship to make process operations "acceptable" in general; and it tends to deal with a small subset of physical conditions and adjustable inputs. It is unlikely that before commissioning all business considerations were known, quantified and used in creating the optimal control logic.
   - Built-in logic is likely to remain fixed while changes take place, such as: (1) specific business needs, which may change with time (such as quality, emissions and economics); (2) upgrades to the equipment; and (3) degradation in equipment performance. Thus, static built-in logic, if ever optimal at one time, becomes "out of tune" or obsolete.

   That built-in logic is usually sub-optimal is proven by the very many processes whose performance has been improved significantly while the built-in logic continued operating during (true) optimization.

---

[3] Proportional / Integral / Derivative – sometimes with some gains set to zero

[4] Noise is the unexplained variations in outputs for constant known inputs.

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ULTRAMAX is a great complement on top of static built-in control logic by completing the picture:

- Refining the setpoints adjusted by the built-in logic by adjusting biases
- Adjusting all other desired adjustable inputs
- Taking into account all other desired known conditions
- Taking into account current business needs

all the above with a dynamic flexibility managed by the user.

As a further area of refinement, with ULTRAMAX optimization, the coefficients or parameters of the built-in logic can also be optimally adjusted (tuned) if they are accessible to be modified through the process digital control system.

5. Some optimization solutions are "model-centric" based on prediction models created up-front (e.g., those based on first-principles, design of experiments, or neural networks). These are also "true" optimization solutions, their usefulness depending upon the accuracy of the models.

   These models need to be maintained as the process changes with time or as the scope of optimization evolves with time. ULTRAMAX creates and maintains the prediction models automatically in full (and not by a few constants, e.g., to correct for drift) – the burden being that it requires on-going collection of operating data (which in turns allows for detecting upsets or aberrant behavior).

   With these static model-centric solutions the "optimum" they can identify is in the area where their models are valid. By contrast, ULTRAMAX sequential method will converge to the optimum regardless of whether or not there has been prior experience or data around it[5].

About the Empirical Optimization Technology used

- ULTRAMAX's Sequential Optimization starts with the current adjustments, and sequentially extrapolates effectively and safely towards improvements until it operates around an optimum, wherever it is (or optima as known uncontrolled inputs change).

  The single most important reason why this sequential technology is the fastest optimizer is that it extracts and uses information from operating data immediately after each operating run to learn and decide what to do next[6]. This alone makes ULTRAMAX very unique; it is certainly not "just another optimization solution".

  This technology bypasses the requirement to have prediction process models up front – it creates and updates automatically its own prediction models with the stored operating data. Thus it saves the time and costs required to create sufficiently accurate

---

[5] However ULTRAMAX may converge to a distinct local optimum if it is on the way to the global optimum. For the model-centric solutions to identify various local optima the experimental or validation runs need to have operated there. Of course, if this data were provided to ULTRAMAX it could also find that optimum.

In the spirit of ULTRAMAX optimizing while engaged in production, the experimental activity of searching for other "mountains" is really a research responsibility of Product / Process Development.

[6] As compared to collecting a lot of data and analyzing all of it at the end of the study.

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First Principles[7] models (if possible at all); or the time and disruption (costs) created by having to collect massive amounts of experimental data for Neural Networks and Design of Experiments[8] models.

The model-centric alternative solutions need to be validated around all likely optima before one can use them; which means having to operate the process around the optima in order to find the optima – a fairly demanding requirement, indeed! This issue is aggravated by: (1) there being an optimum for each combination of uncontrolled inputs or influences; and (2) empirical solutions[9] needing a lot of data around each optimum to pick up the shape of the output behaviors, and thus a relative accurate location of the optimum. Also, the Optimization Plans of these model-centric solutions are static, and need to be correct and complete up front.

As a distinct improvement over these traditional optimization technologies, ULTRAMAX is dynamic -- constantly adapting to the changing behavior of the process and changing conditions and objectives, not limited by prior assumptions or by old run data. New developments in the traditional technology seem to be moving in this direction as well.

First Principle models has an advantages over ULTRAMAX when they are more accurate than the noise in the process outputs, even as known uncontrolled inputs change.

- Empirical optimization works best when the process is sufficiently consistent or reproducible -- i.e., when there is good control, when output variability (for given known inputs) is not important in business terms. Contributors to consistency (low noise) are: good regulatory process control, sufficiently accurate sensors, and including all the inputs which have significant effect on the outputs. High noise is particularly detrimental when it is so large that narrow constraints cannot be satisfied consistently.

  Historically, in the last 20+ years, some 1/3rd of the processes did not meet the above requirement. In such cases applying optimization has to be delayed until, and if, the problems are fixed. Since 2004 there is a separate early Capability / Consistency test (page 72) to identify this situation, which gives customers specific information when and where problems exist.

---

7 Sufficiently accurate models based on first principles are desirable because: (1) creating them demands thorough understanding of the process, and (2) they maintain viability when sensors fail. However they are very costly and time consuming to create and validate, and difficult to maintain as the process behavior changes with time, maintenance and various conditions.

8 Models built with DOE, due to orthogonality, do not lead to optimization as well or as accurately as ULTRAMAX, but provide better insights into the effects of the inputs, which is good for R&D and design.

9 Including ULTRAMAX

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Introduction to ULTRAMAX

ULTRAMAX in a Nutshell

Some basic concepts:

A. If you can measure something, it can be optimized. With ULTRAMAX you can get as close to the optimum as the amount of noise in the process outputs. Noise is the variation in outputs for constant values of the inputs in the Optimization Plan. Gains continue to improve with more data in a static process -- however few processes are static; they keep on changing due to unknown reasons.

B. You, the end-user, select the inputs: the adjustment decisions that you want to make better, and the uncontrolled inputs or influences as required for predicting output values and determining the corresponding best adjustments. [10]

   Decisions are often the adjustments of setpoints for automatic process control logic. Those adjustments which you already know to make best are implemented as "ruled" inputs. Other adjustments that have only minor effects on the outputs are kept fixed.

C. To get improvements the ULTRAMAX technology:

   a. Learns about the process behavior characteristics by "listening" to the data generated by process operations. To enable this learning you need to enable changes of the input adjustments to observe the changes in the outputs – and further, you will need to change to optimize outputs. The changes in each cycle do not need to be larger than such that would create a change on (some) outputs as large as their noise – thus, the lower the noise the lower the necessary cycle changes in adjustments.

   b. Based on such knowledge, the software gives Advice to re-adjust all inputs[11]. Advices are either accepted or modified by plant personnel to satisfy and take advantage of current expertise -- thus there are virtually no risks of failures. Then go back to (a) with the consequent new data, to repeat a cycle.

      When you build confidence on the suitability and reliability of the implementation, you enable closed-loop, autonomous optimization simply by pressing a button.

Implementation: Steps to put into practice true optimization as part of standard operating procedures:

0. **Select the team** to define optimization strategy and manage it. In the first three implementations use an optimization expert facilitator (internal or provided by an ULTRAMAX Supplier) to train the team.

   Often, do a preliminary Capability Analysis (step #3 below) to understand the state of process operations.

---

[10] If you feel that you do not sure about which are the important inputs to include in the Optimization Plan, we recommend against running some sort of screening experiment or DOE to find out. That effort is better invested into doing optimization with what is known, and incuding more variables in the course of time. Recall, in SEO the Optimization Plan is fluid; YOU are in control of optimization. .

[11] The first few readjustment advices are mostly random, within limits defined by the end-user.

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1. The Team **defines** the **Optimization Plan**: i.e., selects the process, defines inputs (adjusted and uncontrolled) and outputs or consequences, and defines objectives (resulting in the Performance Index and Constraints). There is a powerful, holistic approach to define performance indices for production operations to balance multiple objectives when full economic analysis is impractical as explained in Goal Setting, the Approach 2: Quasi-Economic Analysis in page 112. This Performance Index and Constraint Satisfaction are an excellent management tool to evaluate overall operations performance, ***even if not attempting technical optimization***.
2. **Integrate** ULTRAMAX software (in a dedicated computer) with the digital process control system for automatic data exchange with OPC (or DDE) protocols. Integration is optional; else, enter data and make adjustments manually, as done for projects and studies (including R&D and Product and Process design).
3. Do **Baseline Capability Analysis** with operating data generated at Baseline adjustments (i.e., with adjustments at Baseline values), to prevent the surprise of discovering process incapabilities while engaged in production with optimization
4. **Start sequential optimization** in production. Sequentially implement this cycle: adjust process inputs, get consequent output values, get new uncontrolled inputs values, enter all data into software, get advice for new adjustment, modify if deemed necessary by the operator; repeat cycle. Eventually this can be done fully closed-loop, when you are ready to automatically accept advices without modifications (will revert to advisory when a problem is detected).

   Note the Alerts to detect special situations. If process problems are detected, fix problem to prevent it in the future and delete invalid data.

   Significant improvements are often obtained in 50-200 cycles.

   Technical note: ULTRAMAX generates the Advices with a Sequential Empirical Optimization (**SEO**) technology with these characteristics:

   - **Learning**: Creates prediction models for each output as a function of all inputs, based on stored run data. Models are short-lived and dynamic; updated with each new run data – thus converting the information into useful knowledge immediately. Models are created with Bayesian statistics to extract useful knowledge with very little data; and become locally accurate around the optimum (using weighted regression) with abundant run data. No prior models or prior data is required; SEO learns on its own.
   - **Synthesis** (optimization): The Advice for next adjustments is produced by a search engine on all the prediction models simultaneously to best fulfill optimization objectives with safety. The advice will extrapolate towards improvements until there is data around the optimum -- when it interpolates as done by other model-based solutions. The optimum changes as the uncontrolled inputs change.
5. Make **weekly reviews**; and make quarterly reviews with thorough **progress analysis**. Eventually understand barriers to further gains, for possible process improvements projects.

   The team regularly refines the Optimization Plan as awareness of the business and the process develops – which production with optimization encourages. Upper management maintains objectives consistent with on-going business economics and requirements, as often as required for responsive, agile production. **It is imperative that the objectives**

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in ULTRAMAX match the objectives by which people's performance are evaluated. Senior Management needs to enable and encourage a continuous optimization attitude – which makes macro-optimization a journey without limits. This is probably nothing more than Leadership.

Successful industrial applications include: Chemicals (a great variety), food processing (cooking, smoking), coffee roasting, soap / detergent, fibers, nonwovens, paper making (including energy conservation), encapsulation, machining and metal removal, boilers for steam and electric power generation (almost 100 application with emphasis on emissions control and fuel efficiency), fermentation (antibiotics), pigments and dyes, painting and coating, plastics (including injection molding, extrusion, blow-molding and composite materials), welding and soldering, semiconductor fabrication (photolithography, dry etching, film deposition, thermal processing and ion implantation).

An Illustrative Example

This section presents highlights of the CHEM process, including the impact both on results and on the attitude of the personnel involved. CHEM is simulated by the DOS program CHEM.exe, and it is based on an actual application.

The Optimization Plan

Situation

A batch process produces roof tar, a seasonal product, which cannot meet demand because of rework due to *viscosity* frequently falling way outside of specifications. Profitability has suffered. If the problem is not resolved, the business unit is to be closed after this summer.

Business Objectives

Improve the quality of the finished product (reduce deviation from targets) while operating at a reasonable production rate and maintaining safety and reliability constraints. In particular resolve the problem of viscosity falling regularly outside of specifications, creating some 35% rework.

With current adjustments of input settings, the batch on average produces these results:

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Results Table			
	Going in: **Baseline** Avg. 10 Baseline Runs		
Crude Feed	26		
Reactant	14		
Temperature	170		
Viscosity	915.5		
Solids	45.15		
Pressure	309.8		
Rework	35 %		

The Process and Individual Objectives

Key variables (parameters) from running the batch process are established as: production rate, which should have at least a minimum value; two quality characteristics -- viscosity and solids -- that should not only be within specification but close to targets for better customer satisfaction; and one safety result -- pressure -- for which there is an upper constraint. These results are affected by three critical control (adjusted) inputs: crude feed, reactant, and temperature. Other control inputs, such as batch cycle time, remain fixed for this optimization plan.

More specifically, following are the five individual objectives identified in running the process, together with their associated measurable variables, as displayed in Fig. 1. Sketch of Decision Input/Output Diagram.

1. *Crude feed* must be larger than 25 K Gal/batch in order to meet minimum production volume required.
2. Reaction *temperature* must be less that 190 Deg. F, because at higher temperatures undesirable reactions take place.
3. *Pressure* must be less than 320 PSI so as not to open the safety valve of the reaction vessel, which results in a mess and the loss of one day's production. The constraint is set at a conservative 310 PSI to make sure that it does not exceed 320 PSI.
4. *Viscosity* has an ideal target of 850 K CPS (K centipoises), with specifications represented by a tolerance deviation of ± 50 K CPS. While there will be a Performance Index that includes this objective, it is possible also to set the upper spec as a constraint. In fact, this is recommended since this is an pre-eminent quality issue at this time, and in reports and plots we will wish to see how the violation of this constraint is resolved.
5. *Solids* have an ideal target of 45% wt, with specifications represented by a tolerance deviation of ± 2.5% wt.

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Although the *reactant* was rather expensive, management did not worry (at that time) about this issue in comparison to the higher-priority quality/production rate considerations.

Output performance for each set of inputs was measured by making one batch of roof tar.

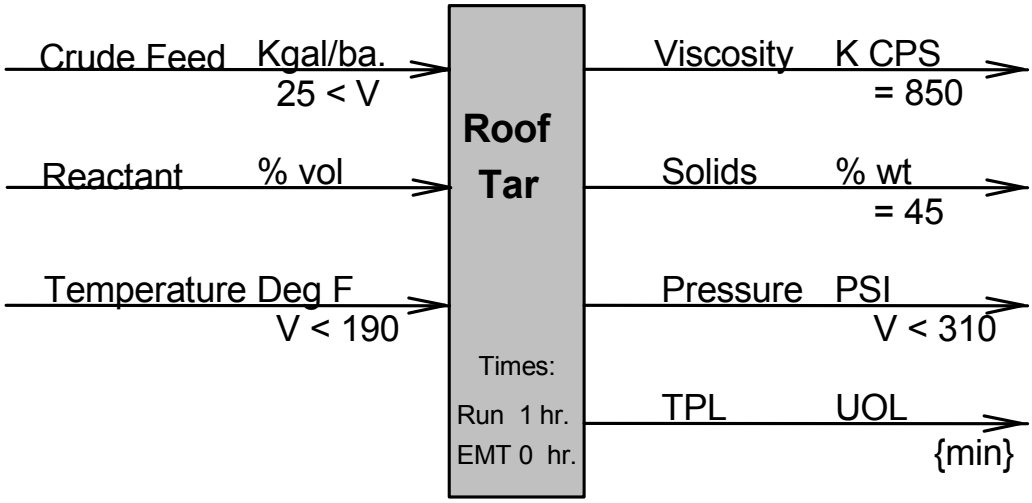


Fig. 1. Sketch of Decision Input/Output Diagram

EMT is the Extra Measurement Time after the start of the next run

The Performance Index is TPL (Total Performance Loss) as defined below in Goal Setting, page 112. The TPL is defined in additive abstract Units of Loss (UOL), where UOL=1 is the threshold into intolerable values.

Goal Setting

Goal setting is the process of balancing multiple individual objectives. Four individual objectives were already represented through constraints: *crude feed, temperature, pressure* and the upper constraint for *Viscosity*. The two other objectives, related to a balanced reaching of targets more closely for *viscosity* and *solids* (actually, an objective of regulatory process control), are represented within a Performance Index to be maximized or minimized; either an economic analysis or, in this case, a quasi-economic analysis using the simpler "performance loss function" approximation, with the factors summarized in the table below related to each of the variables (these factors are associated with the calculations, not the variables themselves).

Performance Loss Functions are additive, standardized losses from ideal or reference conditions based on tolerances. The tolerances are maximum tolerable deviation from target, the threshold into intolerable deviations. The addition of the losses for each objective gives the Performance Index = Total Performance Loss (TPL), which as the concept of "loss" indicates, it should be minimized. Each loss function is defined by the factors in the table below (with the actual calculations explained through examples in page 121 and in the Optimization Plan below). Thus,

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for instance, the calculation of the parabolic performance loss for *Viscosity* is:

$$PL(Vis\cos ity) = \left(\frac{Vis\cos ity - 850}{50}\right)^2$$ [=0 when *Viscosity*=850)

Performance Index: Table of **Global Factors** to calculate Performance Loss			
for Variable	**Target or Reference**	**In-tolerance**	**Form**
Viscosity	850 K CPS	± 50 K CPS	Parabola
Solids	45 % wt	± 2.5 % wt	Parabola
Crude Feed[12]	26 Kgal/batch	-10 Kgal/batch	Linear

The data in this table to calculate the Performance Loss is entered into the software through the "Global Factors". Global factors, as well as certain constraints, are understood by management and changed to reflect evolving company/customer needs and business conditions.

The Optimization Plan, with variable definitions and TPL equation (for the second Round explained below) is:

```
                        PROCESS OPTIMIZATION PLAN
                                                          16:41 09 JUN 2005
APPLICATION: C:\Devel...ent\chem.umax Simulation of chemical process
ULTRAMAX® 7.0.02 ©1982-2004, Ultramax Corporation. All rights reserved.
================================================================================
_____________________________________________________________________________
VAR|    NAME    | UNITS  | DEPENDENCY |RO|MO|  MID   |   CONSTRAINTS   |
  #|            |        |            |LE|DE|        |   LO   |   HI   |
---|------------|--------|------------|--|--|--------|--------|--------|
  1 CRUDE_FEED   K GAL/BA               1 H       0.2      25.       --
  2 REACTANT     % VOL                  1 H       0.2       --       --
  3 TEMPERATURE  DEG.F.                 1 H        5.       --     190.
 11 VISCOSITY    K CPS                  5 H       10.       --     900.
 12 SOLIDS       % WT                   5 H       0.5       --       --
 13 PRESSURE     PSI                    5 H        5.       --     310.
 14 TPL          UOL                    6 C       0.1       --       --
---|------------|--------|------------|--|--|--------|--------|--------|
  MINimizing objective function, variable # 14, "TPL" (Type 6)

 List of GLOBAL FACTORS
         #     Value      Units       Name
  GLOBAL( 1) = 850.      K CPS       VISCtarget
  GLOBAL( 2) = 50.       K CPS       VISCtol
  GLOBAL( 3) = 45.       % WT        SOLtarget
  GLOBAL( 4) = 2.5       % WT        SOLtol
  GLOBAL( 5) = 26.       KGal/Ba     CFref
  GLOBAL( 6) = -10.      KGal/Ba     CFtol

 TPL          = (( VISCOSITY - VISCtarget )/ VISCtol )^2
                +(( SOLIDS - SOLtarget )/ SOLtol )^2
                +( CRUDE_FEED - CFref )/ CFtol

Definitions:        Role = 1   Input - Control (Adjusted)
           Role = 2       Input - Uncontrolled
           Role = 5       Output - important (has constraints or is used in
                                  calculations)
           Role = 6       Performance Index
```

The third table below shows values of these TPLs.

[12] The Crude Feed role in defining objectives will be seen later on, when illustrating the dynamic aspect of optimization in a more refined round of optimization starting in run #51.

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Improving Performance

Performance improvements are achieved by readjustment of the control inputs through Sequential Empirical Optimization (SEO).

The following chart shows the sequence of runs following these three stages:

1. 10 runs with adjustments at Baseline to study process capability / consistency (normally many more)
2. 40 runs (11-50): first round of optimization, with adjustments following SEO advice; and engineering / management analysis at the end
3. 30 runs (51-80): second round of optimization with revised Optimization Plan, with adjustments following SEO advice

Thereafter production continued managed with SEO, possibly with a more refined Optimization Plan.

More details follow Fig. 2.

Baseline Runs

The first 10 runs[13] were the Baseline Runs at current normal adjustments to check that the process and procedures ran smoothly, and to perform the Baseline Capability Analysis to observe the level of control in the process (or lack thereof, the noise). The variations in the outputs are the level of existing noise (unexplained variations). For a potentially dynamic process ULTRAMAX can approach the optimum, on the average, approximately as close as the noise level.

First Round of SEO

The essence of SEO is to provide advice on how to adjust the setting of the control inputs for each run (here a batch). This advice is based on the regularly updated database of input and output values which are obtained from past process runs conducted with previously provided advice. The database starts empty. Thus, the SEO software creates knowledge from past data in order to safely move the adjustments of the control inputs towards and around optimal values. The flow of data and advice is displayed generically in Fig. 3. Bird's-eye view of the ULTRAMAX Method, page 21.

Through this sequence of adjust-and-run optimization cycles, ULTRAMAX arrives at the near-optimal setting of the control inputs, and quickly reacts to known changes in situations and objectives. The software also recognizes symptoms indicating when there are conditions that would prevent or slow the achievements of these results.

[13] Current procedures call for having some 45 Baseline Runs for this size application.

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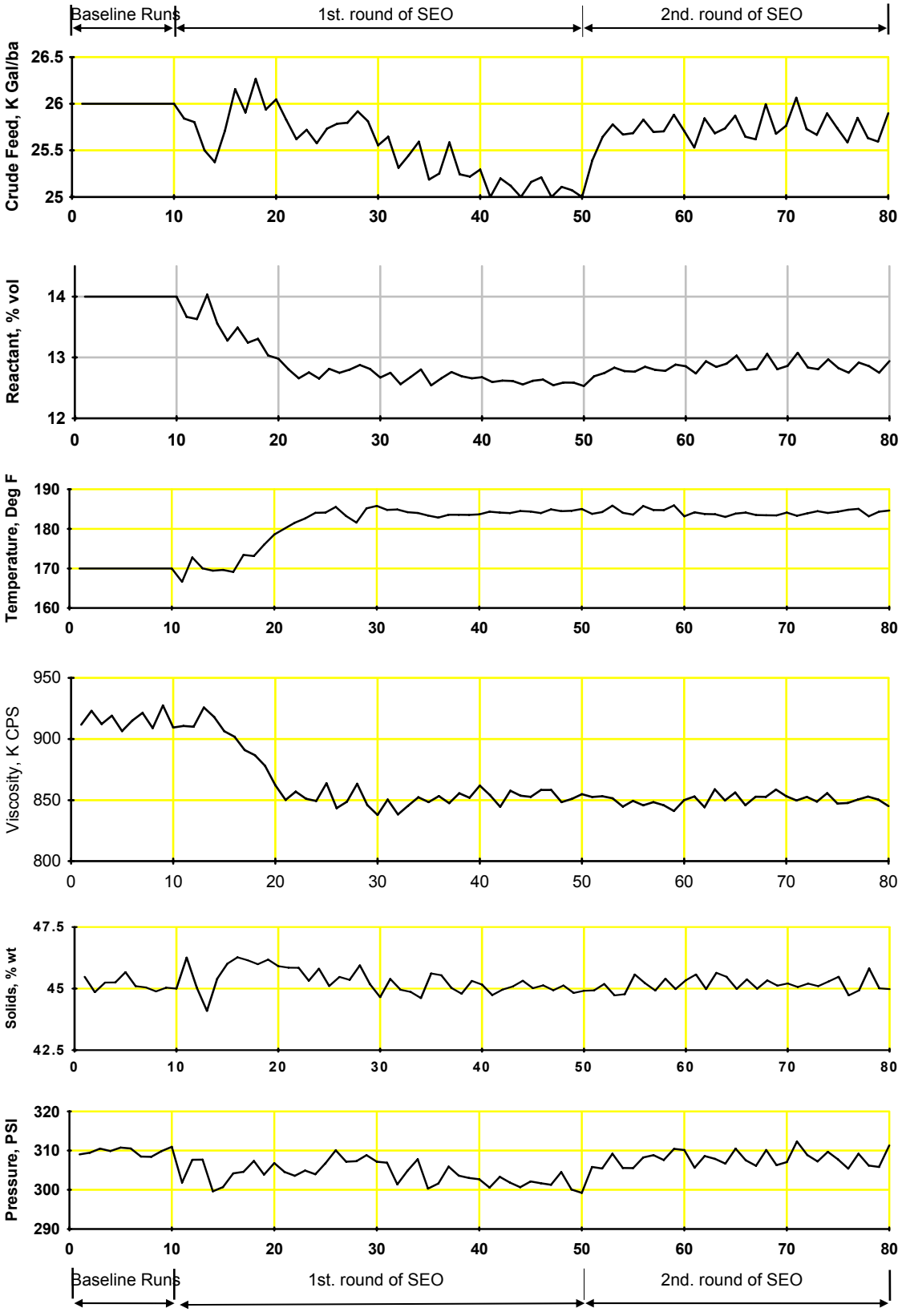


Fig. 2. Improvement Profile for the Example

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The first round of SEO consisted of 40 production runs (runs 11 through 50). As shown in Fig. 2. Improvement Profile for the Example *Viscosity* (the original problem area) was brought within specifications after seven runs, and was close to targets in some 12 runs. The result was total avoidance of rework. Net throughput increased considerably because of the elimination of rework, even though individual batch volume decreased by almost 4%. In particular:

- *Crude feed* settled close to the lower limits of 25 Kgal/batch.
- *Reactant*, surprisingly, was decreased by a tenth, and not for economic reasons.
- *Temperature* increased, but was always comfortably below the upper limit of 190 Deg. F.
- *Viscosity* was brought within specifications in about seven runs, and regularly close to target in about 12 runs.
- *Solids* deviated further from targets, although always remaining within specifications, but after some 20 runs it was close to target again.
- *Pressure*, remained comfortably away from the constraint.

Fortunately, this application of ULTRAMAX did not encounter any conditions that would have required extra steps. The advice was always deemed acceptable. The inputs and outputs were always under control. The spread of the output data (as compared to the noise) was such that the models were suitable for optimization. The noise level was sufficiently low to achieve significant improvements.

Results Table			
	Going in: Baseline Avg. 10 Baseline Runs	**After 1st. round of SEO** Avg. last 10	
Crude Feed	26	25.09	
Reactant	14	12.59	
Temperature	170	184.4	
Viscosity	915.5	853.4	
Solids	45.15	45.01	
Pressure	309.8	301.5	
Rework	35 %	0 %	

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First Appraisal

After relative optimization was achieved to meet the individual objectives in the first optimization plan (basically, quality within constraints), an Appraisal was conducted in order to explore further potentials for further profitability. The evolved main business objective at this stage was to increase revenues by increasing production rate.

The solution was simply to add to the previous Performance Index a performance loss component to reflect the desire to increase batch size, i.e., *Crude Feed.* This loss function reflects the short-term (marginal) economic impact of losing production -- while the constraint was a simple rule to reflect the shipment contractual requirements (with complicated long-term economic effects).

Note in the table of Performance Loss Factors above, showing tolerances, that the (in)tolerance deviation for *crude feed* is negative, and being linear, that positive deviations are more valuable than the reference. With this new Performance Index applied to the old data, optimization indicated that there was a potential to increase batch size and still be close to quality targets, while obeying all constraints. Thus a new round of SEO was started, including the old data, but with a new Performance Index in the Optimization Plan.

Second Round of SEO and Appraisal

Production continued under the second round of SEO with the Performance Index calculated with a more complete TPL evaluation (and still using all the physical data obtained previously). The next 30 runs confirmed the engineering/management analysis predictions, producing these results (summarized in the expanded Results table below):

Results Table			
	Going in: Baseline Avg. 10 Baseline Runs	**After 1st. round of SEO** Avg. last 10	**After 2nd. round of SEO** Avg. last 10
Crude Feed	26	25.09	25.77
Reactant	14	12.59	12.87
Temperature	170	184.4	184.2
Viscosity	915.5	853.4	850.1
Solids	45.15	45.01	45.16
Pressure	309.8	301.5	308.4
Rework	35 %	0 %	0 %
TPL	1.720	0.096	0.027

- Production rate (*crude feed*) increased so as to become close to original levels.
- *Reactant* increased somewhat in percentage volume.

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- *Pressure* increased, so that it was again borderline with its constraint (but still far from the 320 PSI limit).

Note in the plots that the output variables that should be close to a target, viscosity and solids, displayed only a little more variability under SEO, when inputs kept changing, than when inputs were kept constant during the Baseline Runs.

It is worth noticing how quickly the process was brought up to optimization during the second round of SEO. This was probably due to the availability of historical data somewhat close to the optimum for the new Performance Index.

The table above also displays the TPL with the calculations for the second round.

The net result of the project was that production was then unhampered by the delays of rework and much closer to targets than ever before. This result satisfied customer needs, and the business unit was made profitable again. Corporate decided not to close it.

Future Plans

Now the team is ready to take a fresh look during the next season at what the meaning of this process really is for the total business. As a result, they will expand their conceptual understanding and control over the operation. The team intends to manage the process with ULTRAMAX, taking a more thorough approach to maximize performance in terms of customer needs with a more complete and explicit economic analysis. In addition, they may expand the scope of their Optimization Plan to adjust the cycle time, as well as perhaps other inputs.

Management is encouraging this evolution and actively participating in the definition of objectives and goal setting. They intend to keep abreast of improvements and remain alert for signs of any barriers that might arise to slow this productivity improvement.

The rest of the book is divided into several parts. In the first one there is a more comprehensive introduction and a refresher very useful for people with already some exposure.

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More Specifics about ULTRAMAX

Fundamental Plan to apply ULTRAMAX in Production Operations

After enabling the process to operate with ULTRAMAX optimization:

⇒ Manage operations, specifically, the adjustments of production equipment, with SEO cycles

Detect and solve process problems as they become evident -- basically, Statistical Quality Control (SPC) – which in the long run increases consistency (decreases noise)

⇒ Proactively refine Optimization Plan and/or Reduce Noise[14] -- especially after reaching Relative Optimum (for current Optimization Plan and noise levels).

⇒ Repeat the two above until reaching a Practical Optimum, and then continue operations with routine SEO to "Maintain and Refine the Optimum".

⇒ At any time, react to changes in objectives, costs and conditions; and incorporate Discoveries for sharper, focused SEO results.

The Four Critical Success Factors

ULTRAMAX's SEO is most likely to deliver gains if the process/product largely satisfies the following four requirements. There are many cases of failing some requirements and still getting improvements. Also, there exists the possibility that the requirements are fully satisfied and that gains are not forthcoming: when the process is already regularly run near the optimum.

The level of fulfillment of the first three requirements below is readily revealed while formulating the Optimization Plan. The last one, "consistency", requires collecting and processing operating data:

- **Measurability:** the variables must be measurable – ULTRAMAX is a quantitative solution; and performance level is determined by these measurements.

  This is meant to exclude processes where achievements are determined subjectively according to the "feel" of the operator, or even artistically. The problem is not to have numbers, or not to have fairly repetitive numbers for the same state of production. Do notice, however, that there has been fully satisfactory success using scales assigning values to a sequence of physical samples or photos (they need to have more than six levels, of which two or three should be for "acceptable" production).

  This requirement also includes that levels of achievements are measurable. Thus, it excludes processes where achievement is assured only by making adjustments following fixed predetermined rules – namely, where production procedures preclude any flexibility. In the latter case, use ULTRAMAX only for optimization studies to optimize the fixed procedures.

[14] For instance, implement Engineering changes or Six-Sigma projects.

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- **Timeliness: Data must be available on a timely basis after each run, especially in the early stages when extrapolating towards the optimum.**

  The critical advantage of SEO is the immediate use of information in the new data to refine its learning (prediction models). Thus it is a great advantage that the data for each sequential run be available typically within a few hours after the next re-adjustment of the control inputs. The problem is ameliorated with Staggered SEO (Glossary, page 191). See Appendix "Timeliness", page 195.

  After converging to an optimum, timeliness of data is not critical to maintain the optimum.

- **Graduality: Small changes in the inputs must result in small changes of the mean output values.**

  This requirement does not apply to the uncontrolled "component inputs".

  Similar concepts: continuity, smooth response; and the opposite of "categorical" (such as on-off).

  Adjusted inputs need to be able to change in small steps.

  ULTRAMAX can handle isolated output discontinuities away from the optimum. The issue of noise and disturbances from run to run is part of the "consistency" issue below.

- **Capability / Consistency: Capability is how well the process can satisfy constraints and how close to ideal Performance Index values an empirical optimization solution can get. This is most often affected by consistency, having sufficiently good control of outputs, i.e., get relatively repetitive output data for the same input values[15].**

  The end-user needs to accept: (a) (any) losses in effective output operating area in order to regularly satisfy constraints – and sometimes the process data is incapable of satisfying upper and lower constraints on the same output, in which case the process it just **incapable**; and (b) how far from the ideal optimum empirical optimization will take operations. Else, the end-user needs to fix the requirements or fix the problems with the process – as has been happening in about 1/3rd of the process which tried to use ULTRAMAX in the last 15 years.

  The Ultramax Method offers some solutions to increase consistency (reduce unexplained variations) through better adjustments, when unexplained variation is one of the measures included in the definition of the Performance Index (see Noise Sources and Reduction, page 85).

Implementation Plan

The implementation of ULTRAMAX consists of these steps, performed by a team including, ideally, lead operators, trained by an optimization expert either supplied by an ULTRAMAX Agent or by the company;

1. **Planning,** to represent management objectives and preferences; criterion of performance (Performance Index and Constraints); and the variables of interest in the process. This results in the Optimization Plan.

[15] This does not mean that the inputs need to remain constant.

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2. **Integration**, to exchange data automatically from the process control system (DCS, SCADA, or intermediaries such as historians) and the optimization computer with the ULTRAMAX software (otherwise use ULTRAMAX Stand-Alone, with manual data entry and process adjustments)
3. **Baseline Capability Analysis,** to evaluate that the process is capable to satisfy the constraint requirements (and if not, whether the process can be proven to be incapable to satisfy certain constraints), and to see how close to the optimum one can get. The main barrier is usually “noise”, the lack of consistency, the variability of outputs for constant known inputs (control or uncontrolled), that is, the lack of perfect process control.
4. **Start Sequential Optimization**: This is the last training (on-the-job) of the team to manage production with optimization and to cope with the frequent process/team start-up problems. The optimization expert issues a report about team/process conformance with applying the ULTRAMAX Method (or perceived barriers). Thereafter there is remote support by the Agent, as contracted for.
5. The above leads to the realization of improved value added; and periodic evaluations of gains and of barriers to further gains, for engineering and management assessments. The first progress report is written by the Agent (if part of the agreement). Thereafter there are quarterly reviews if part of the agreement -- highly recommended for at least one year for long-term success. The biggest barriers are: process problems not solved in a timely basis; lack of expectations from operators by higher management. Maintaining training for new personnel is important.

ULTRAMAX is applicable to batch and continuous processes in the process industries, and for batch processing and continuous production in discrete manufacturing / fabrication.

Steps 1 through 4 can be completed in two months from project start-up (two weeks less if Stand-Alone), and thereafter near-optimal results are approached asymptotically typically in a few weeks to a few months. Optimization is refined with more production data, and also as the optimization plan is improved.

The following figure is an abbreviated presentation of applying ULTRAMAX as a flow diagram.

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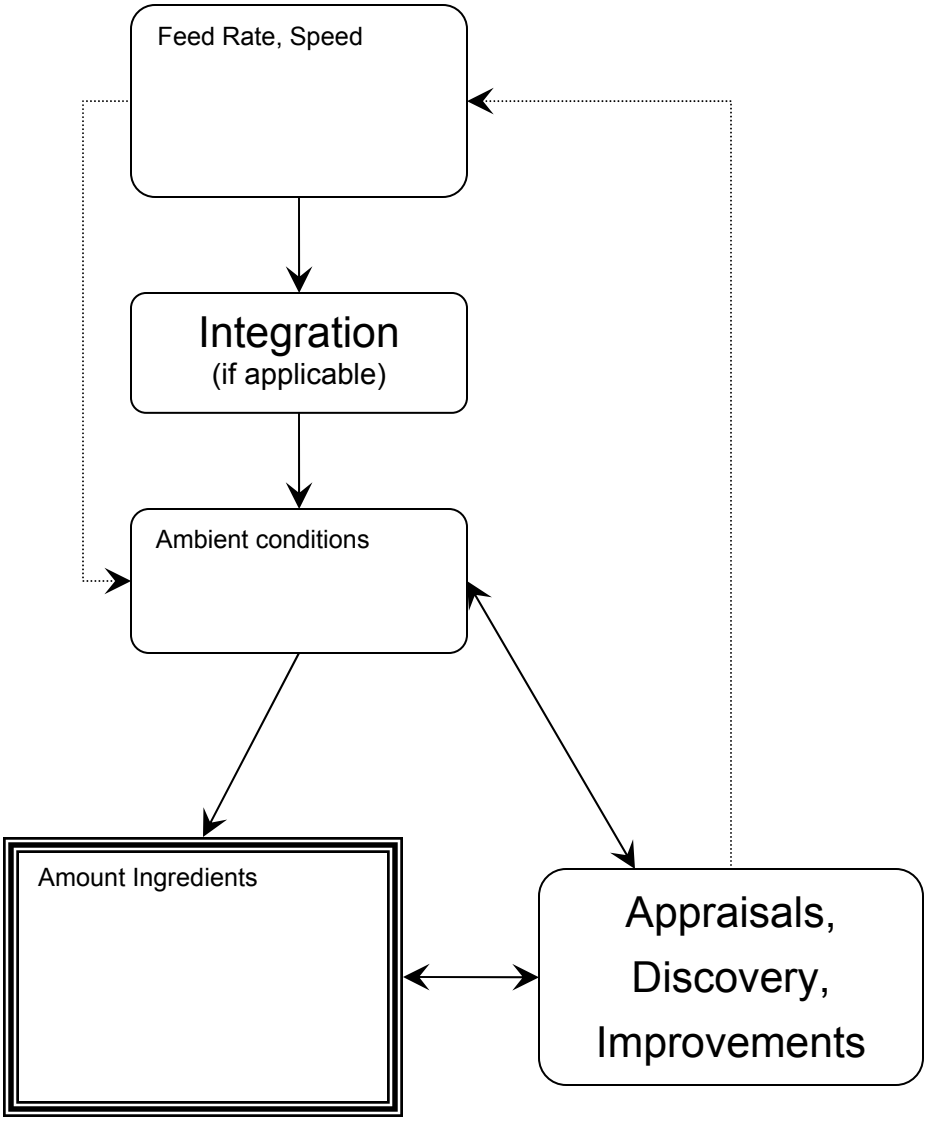


Fig. 3. Bird's-eye view of the ULTRAMAX Method

Detailed ULTRAMAX Summary

The operating concepts central to the ULTRAMAX Method and APM are summarized below. This summary is a significant expansion of the ULTRAMAX in a Nutshell, page 1; it is also a reminder for people already exposed to ULTRAMAX.

Definitions of specialized terms can be found in the Glossary (page 180).

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Configuration (Formulation)

a. ULTRAMAX production management is typically applied by a **team**, although it has been used profitably by individuals as well, especially for projects or studies.

   The optimization team leader is frequently an operations manager or a "champion" from some functional area. The core of the team is made up of plant personnel who are responsible for production and/or can contribute practical theories and empirical knowledge about the process and product, such as process and quality engineers, operators, and technicians.

   The team also includes the timely participation of top manufacturing management to provide an up-to-date review of business objectives and to help prevent entrenched, unsupported, or obsolete traditions from being a barrier to profitability improvements.

   The comptroller has a role to play as well when defining individual objectives in economic terms – however, this by itself may not sufficient to quantify the economic consequences of quality deviations from ideal – as done with the Performance Loss Functions approach.

b. The team -- often with the direct participation of the operations manager -- describes the general business objectives, the driving considerations for performance improvements, in terms familiar to top management.

   In order to understand performance in terms of the general objectives, the team also describes the **individual objectives** related to the general business objectives in **quantitative and measurable** terms, namely, the **outputs** (or inputs) that should be measured or calculated to evaluate process performance. They often include Key Performance Indicators (KPI).

   Calculated outputs could include the following: Performance Index economic or quasi-economic metrics to reflect impact on the P&L statement through marginal costs and revenue calculations, averages or extremes of several variables, values from a mass and energy balance equation; prediction models created by other means; statistical measures of satisfaction of specifications (such as C_{PK}[16]), profile modulation; etc. Economic and quasi-economic calculations usually include **global factors** to reflect unit costs, sales prices, targets, specifications, etc. These factors can be readily changed to bring up to date the representation of business objectives and economic reality.

c. The team identifies which inputs (factors, parameters) affect the important outputs. There are two types of inputs:

   – The adjustable **control inputs**, the decisions made to manage & control production. A very important consideration is that the inputs be **independent**, i.e., adjusting one does not create an automatic adjustment of another[17]. One characteristic of "independence" is that the values of any control input will not change when re-adjusting any other one. The team decides which control inputs to keep fixed at the desired adjustments; and which to re-adjust periodically to achieve improvements.

---

[16] C_{PK}, a measure of how well individual raw data meets specifications.

[17] Independence does not preclude interactions -- where the effect of one input depends on the value of other inputs.

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- Usually there are also **uncontrolled** inputs which one cannot or chose not to control, but which impact the outputs. The uncontrolled inputs need to be independent of the control inputs, but can be interdependent among themselves.

All consequences of the inputs, including **intermediate** variables, are considered "outcomes", "consequences", "dependents" or "outputs."

d. The ideal scenario is to manage the process from a holistic or comprehensive point of view, keeping in mind simultaneously all the important consequences and results from running the process.

   This is the opposite of focusing periodically and rather exclusively on different key performance factors (e.g., quality, then yield, then safety, then cycle time, then cost containment, then back to quality, etc.), where you can always obtain improvements in today's topic at the expense of other results.

   With the selected process variables the team defines the improvement **Goals** -- the balance among the various individual objectives. This is accomplished by defining:

   - A Performance Index (or Objective Function), typically a calculated output, to be maximized or minimized. ULTRAMAX offers a very powerful method to define Performance Index for production operations when purely economic analysis is not practical. Among other features, it includes "costs" from deviation from targets, to implement regulatory process control out of the reach of process control technologies.
   - Constraints on inputs and outputs within which the process must be operated to obtain satisfactory operations. The constraints might include quality limits, safety and reliability protection, process capacity, agreements with internal and external customers, or legal requirements, typically concerns more important than those included in the Performance Index. Satisfying Constraints is beyond economic analysis and is more important (a higher priority) than improving the Performance Index.

   Operators, managers and engineers -- and ULTRAMAX -- will all work towards the very same goals on a daily basis. This is a very important consideration for success; clearly, to the degree that this fails, the "life" of optimization will be limited. The goals are not static, they are kept up to date by management to reflect business dynamics.

e. The team determines the adjustments at Baseline, that is, the normal operating adjustment before SEO. SEO starts with a run at Baseline adjustments, whose Baseline results will be used to evaluate improvements.

f. All of the above considerations are reflected in the optimization plan (which could also have been called "optimization strategy"). The optimization plan normally evolves as the team acquires sharper awareness of the business objectives and the physical realities of the process/product.

   Changes to the optimization plan, or any thing else about the process and optimization, are recorded in an ULTRAMAX **operations log**.

   Entering the variable definitions in the optimization plan and an application file is the main "configuration" that must be made to the software for each application (the other is the mapping of variables with DCS/SCADA tags when doing

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integration). The software itself remains the same for all applications, as is the case with a spreadsheet or accounting software.

g. Integration is the automatic data exchange with computer process control systems such as Distributed Control Systems (DCS) or Supervisory Control And Data Acquisition (SCADA) systems. ULTRAMAX has OPC and DDE "clients" for communications protocol. The customer is responsible for having a "server" with access to the necessary data.

   Ultramax can also be used "**Stand Alone**" – with data entered by hand. This is the usual mode of use for projects and studies, both with production equipment and for R&D and Process / Product Development.

Operations

a. A **run data** (also run data set, multivariable data point) consists of the values of all the inputs and consequent outputs for one process run -- often about 30-60 minutes of continuous production, or one batch of production.

   Run data must be valid. That is, it must be representative of how the process operates now and will operate in the future. It is important to assure that new sequential data is valid, and ULTRAMAX has good detection of unusual data.

   Outputs need to be the **consequence** of the values of the inputs (not, for example, of values of inputs of previous runs). The new "Transient ULTRAMAX" removes this requirement, but is more complex to use. Ideally, measurement accuracy should be sufficient to positively detect a **minimum important difference (MID)** change in each output.

   ULTRAMAX output data often consists of **summaries**, such as averages and standard deviations of several raw measurements within the run. Summaries over longer periods of time help reduce the noise of random variations in the data.

   Prior data is valid data available before the application of ULTRAMAX. Prior data (if any) plus the data obtained through running the process with ULTRAMAX, constitutes the **historical data**. Data does not need to be entered through the sequential SEO cycles for ULTRAMAX to learn from it (e.g., it can be imported in blocks of runs at one time).

   The graphic portrayal of the process performance as a function of time is called the **improvement profile**, such as in page Fig. 2

   Sequential Empirical Optimization (SEO) is the key technology inside ULTRAMAX (it is formally described in The SEO Technology, page 138). SEO gives periodic, iterative advice for adjusting the control inputs. Adjustments are made on ALL control inputs at (almost) the same time – which when using the right technology makes optimization much faster than changing, e.g., one variable at a time. SEO aids and automates this aspect of supervisory control for regular operations – i.e., not for startup, shut down, cleaning, etc.

   As run data becomes available from running the process and is accumulated in the database, knowledge and the sequential advice will be smarter, moving from the Baseline or starting adjustments, extrapolating towards improvements up to optimality.

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The SEO technology creates advice by performing these two functions:

- **Learning:** it is to create **models** based on historical data to predict outputs as a function of the inputs. The quality of a model for conducting SEO is indicated by its **prediction error** and eventually its **signal**. These models are highly dynamic, adapting to circumstances. What is permanent with ULTRAMAX is the database of run data, which constantly grows with new production.
- **Synthesis:** a search engine probes all the prediction models to determine an **advice** on how to adjust the inputs for the next run. The advice includes compensation for the current known values of the uncontrolled inputs (if any). The advice also represents a balance between increasing performance now and generating useful data for faster future improvements.

You are not required to follow the advice. You can modify it or select from among several advices. The advice should make sense, according to your experience, before you implement it. ***You are in control*** *(and responsible for any outcomes).*

SEO can be used **advisory only**, where an operator must review and possibly modify each new set of adjustments, or **automatic closed-loop** that is started when you certify that the application of ULTRAMAX is appropriate and reliable. Closed-loop will return automatically to advisory if an Alarm is detected.

SEO is almost always effective improving performance when the four The Four Critical Success Factors in the next section are true. The biggest exception is when the end-user is already running operations as close to the optimum as the noise level. The benefits of detecting upsets and giving plant personnel a better quantitative understanding of the process and inducing valuable discoveries always remain.

b. Noise, the unexplained variation of outputs for constant inputs, is a barrier to approaching ideal optimization in empirical solutions. The noise level is estimated by the expected error. Unlike other solutions, variations in outputs created by known changes in uncontrolled inputs are not noise, as the prediction models explain these changes.

   Because ULTRAMAX learns from data, it will not be as effective when using non-representative data such as created by upsets, aberrant behavior, breakdowns, changeovers, cleaning, etc. Thus, ULTRAMAX has powerful diagnostics to detect potentially invalid data to avoid the GIGO syndrome (garbage-in, garbage-out).

   There are several ways to reduce noise. The most obvious solutions are to eliminate sources of noise or to block their effects. When this is impractical, and if the cause is sufficiently constant within a run, a variable representing the noise source (e.g., ambient humidity or raw material viscosity) can be placed in the optimization plan as an uncontrolled input.

   Similar to noise is the issue of process upsets, whether affecting the variables in the Optimization Plan or not. A benefit of the ULTRAMAX Method is that it focuses so sharply on results that people become quickly aware of barriers and forces them to resolve the issues. The main approach is to proactively do SEO and respond reactively to the upsets. However, when failures are so frequent that they demand too much attention, it is less stressful for the team to pause temporarily applying SEO and to concentrate on solving the upsets; including implementing

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methods – such as better maintenance – to prevent these failures from occurring so frequently.

c. SEO will approach optimization roughly speaking as close as the noise level. The smaller the noise, the faster and closer SEO converges to the ideal (no noise) optimum.

   Being close to the optimum is called the **relative optimum**. Further improvements with the same optimization plan would require the reduction of noise effects. **Practical optimum** is achieved when at the relative optimum and the noise effects represent unimportant changes in outputs.

   Further improvements require a new focus for the optimization plan or improvements to the components of the process itself. Indications of what must be done may become evident at any time by **discovery**, e.g., when doing **engineering / management analysis** (some conducted with the software).

d. If management wants individuals and teams to perform as described above, and to achieve the consequent unusually high profitability improvements, they must understand the method, recognize and reward success, and help resolve barriers on an ongoing basis.

e. Once ULTRAMAX is properly established for operations management, the global factors and constraints (as explained earlier in this section) become the main mechanism by which top management guides the evaluation of performance, and SEO maintains the corresponding optimum.

   The ULTRAMAX Method was designed to be adopted as a procedure for continuously managing process operations, and enabling personnel to be focused, more aware, and more effective.

   However, several customers have used ULTRAMAX as an optimization tool for a project or study as well. This alternative is particularly useful when each run or test has to be done sequentially and it is expensive or time consuming, due to the effectiveness of Sequential Optimization in approaching the optimum in the fewest tests. In the latter case there is a “final” report, rather than periodic “progress” reports.

   The flow diagram in Fig. 3 portrays a bird's-eye view of the major stages in the application of the ULTRAMAX Method.

   In particular, for the SEO stage, each step is described in a separate page. Each step starts with a description of the reasons for performing it (the driving force, the motivation). If you know that the reasons do not apply, or you can manage the risks the step is trying to prevent, then in principle you can bypass the step. However, in practice, when some people take more shortcuts than they should, they depend on luck to achieve improvements with ULTRAMAX. Thoroughness leads to more reliability, better understanding and tighter control.

For an in-depth summary of the technical issues related to Sequential Empirical Optimization see SEO Concepts and Metrics, page 159.

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Part 1: SETUP PROCEDURES

Planning and the Optimization Plan

The Optimization Plan (OP) defines the domain, scope, objectives and methods in optimizing production. It is the "conceptual model" of the key aspects of the production process[18]. It represents the decision input/output situation for production operations management.

Once reliable implementation is achieved, the OP is to be approved by the organization as all Standard Operating Procedures (SOP) are approved. In fact, production operations with optimization guided by the OP become part of the new SOP.

The Optimization Plan is described in the "Optimization Plan Document". The document is known and easily accessible to all members of the optimization team. The document is rather complete, even including the aspects that are ignored – for the time being -- in doing optimization.

In addition, the active part of the Optimization Plan is entered into the software; also called "configuring" the application. The software OP matches exactly the OP document.

The document is dynamic.

- As the business changes, such as economic factors and product requirements, they are represented by readily modified "handles" (the Global Factors). Higher management keeps up to date the Global Factors in order for production to truly respond to business dynamics.
- As experience is gathered, the OP is periodically refined, improving the definition of objectives and the selection of variables and their definitions (including, e.g., constraints).

In addition, the Optimization Plan achieves these results:

1. Guides the team to a definition of objectives from a business point of view and to determine a framework within which to achieve improvements.

   It is typical for individual team members (process managers, operators, quality and process engineers, etc.) to start with very diverse goals. The steps in the formulation of the optimization plan get everyone on the same page with unity of purpose and scope, including understanding of basic issues, and concurrence of higher management. This stage in itself is extremely valuable to the organization, and to make several crew operate more uniformly.

2. Determines early on whether the process satisfies the Critical Success Factors (described below) for the team to improve its performance through better adjustments and resolution of barriers with SEO

[18] The "conceptual model" is not to be confused with the mathematical "prediction models" that emulate the way the input values are converted into the output values by the process.

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Steps to Formulate the Optimization Plan

The procedure for formulating the Optimization plan is for the team, with occasional participation by various experts and managers, to perform the following steps.

There are two basic plant configurations:

Configuration	The "process"
There is one primary process or operating unit, such as a boiler for steam production, a paper mill, a continuous oven, an injection molding machine, etc	The primary process
Production is highly linked between several processes, as frequently seen for multiple-step discrete production or a sequence of chemical processes	"Squeaky wheel" approach... A recognized bottleneck in terms of production rate, costs incurred, quality shortcomings, and/or undesirable losses or byproducts (including emissions and production). After this process is optimized another one is likely to become the bottleneck. Sometimes a set of bottleneck "processes" are defined where all should be optimized with consistent production rate for increased line profitability.

The general "process" to be optimized is decided by plant personnel before starting the detailed "Formulation of the Optimization Plan".

1. Describe Business Objectives

⇒ Write a 100-word or less statement which would make sense to higher management. It is a key consideration that the team and higher management buy into this statement.

2. Draw Engineering Sketch, and select the "Process"

⇒ Graphically represent the Process that needs to be optimized in order to satisfy the Business Objectives.

This can be a block diagram where each block is a well understood step or unit operation of the production line (flow) (Fig. 4), or a more graphic representation (Fig. 5), as required for communications between team members.

When the process consists of several steps – such as various unit operations feeding each other – then it is very helpful to draw a boundary around the units that will delimit the portion of the production line that constitutes the "process" to be optimized, as done in Fig. 4. With this it is much more straightforward to determine which are the inputs (control and uncontrolled), the intermediate variables, and the output variables.

Notes:

The input for an adjustable variable is the **decision** made for its adjustment, its setting. In such case the "controlled" variable (that may be the physical input to the process) in a consequence or outcome of the decision – and possibly also of other inputs.

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Inputs from recycle loops are neither “control inputs”, nor “uncontrolled inputs” because they are affected by the control inputs of past runs. There are no simple statistical methods to deal with recycle inputs; thus ideally they should be internal intermediate variables included within the process boundary. If this cannot be done, then treat the recycle inputs as uncontrolled inputs, but then be mindful that short-term optimization does not lead you to creating recycle flow with detrimental effects.

Intermediate variables usually are not included in the Optimization Plan. If you wish to know how they behave, they have to be included as “outputs”, that is, as consequence or outcome of the inputs.

For a new user it is a good idea to start with a smaller process with one or very few units where measurements can be obtained quickly. As your experience develops, then you can proceed with more involved processes, including cascading optimization.

Including the existing Process Control System (PCS)

As with the production equipment itself, the Process Control System (PCS) – hardware and the control scheme -- is usually used as is.

Managing or controlling process performance is done through plant personnel making **adjustments** to setpoints (and biases, parameters, etc.) that the PCS implements, including those with “Advanced Process Control” (APC) or manual physical adjustments. The main impact of ULTRAMAX® is to provide the PCS with **adjustments which optimize process performance**. Hence the motto “*Advanced Process Management™*”.

A PCS is usually configured in Distributed Control Systems (DCS) or Supervisory Control and Data Acquisition (SCADA) systems. Their control scheme includes:

- **Regulatory Control** (to keep controlled variables close to setpoint adjustments), almost always.

  Normally the regulatory control scheme is used as is (if working properly).

  As a complement to direct regulatory control ULTRAMAX can be used to effectively implement balanced regulatory control to downstream process variables beyond the reach of single loop or multi-variable control schemes -- see the CHEM example for *Viscosity* and *Solids* in the Blue Book.

- **Control Logic** (to determine setpoint values as a function of process conditions), sometimes.

  One needs to decide whether to *supplement* or *replace* any Control Logic with ULTRAMAX, as they do similar jobs. This decision has a large impact on the scope of optimization, including objectives and variables.

  If supplementing it – as done usually -- add a “bias” to each setpoint value determined by the Control Logic. For each, the PCS implements the addition of the value determined by the Control Logic plus the bias optimized by ULTRAMAX (which also depends on conditions). If the control logic is optimal – which is seldom seen -- then the optimal bias will be near-zero for all conditions.

In addition, some users have reported success using ULTRAMAX to optimize adjustable **parameters** (constants, factors, gains, limits, etc.) in the equations or

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algorithms of PID Process Control and/or Control Logic.

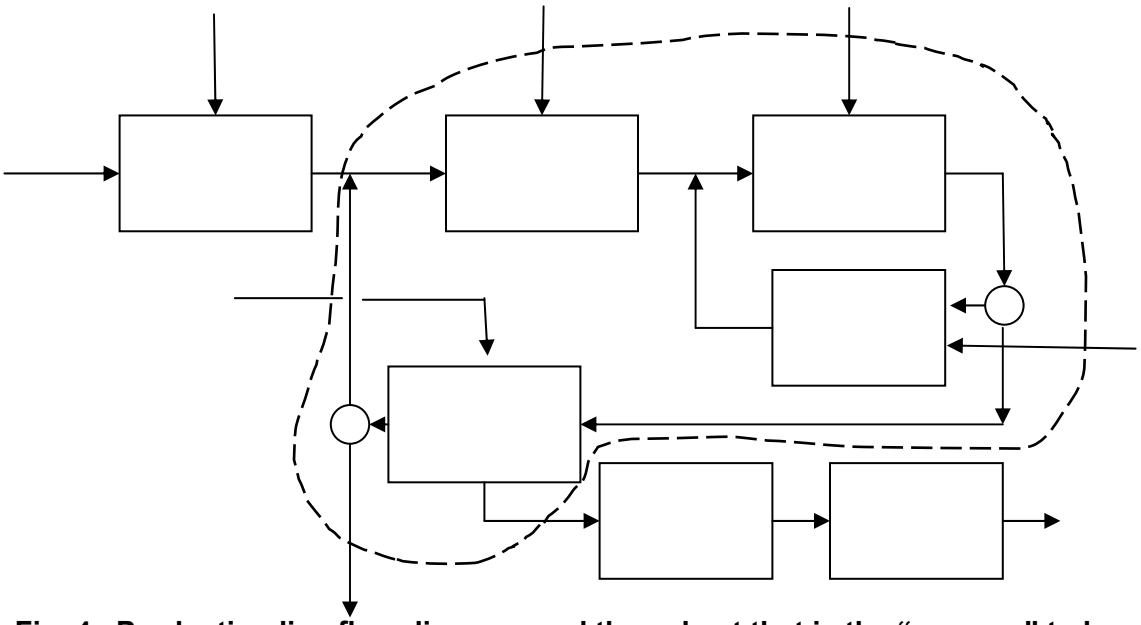


Fig. 4. Production line flow diagram, and the subset that is the "process" to be optimized.

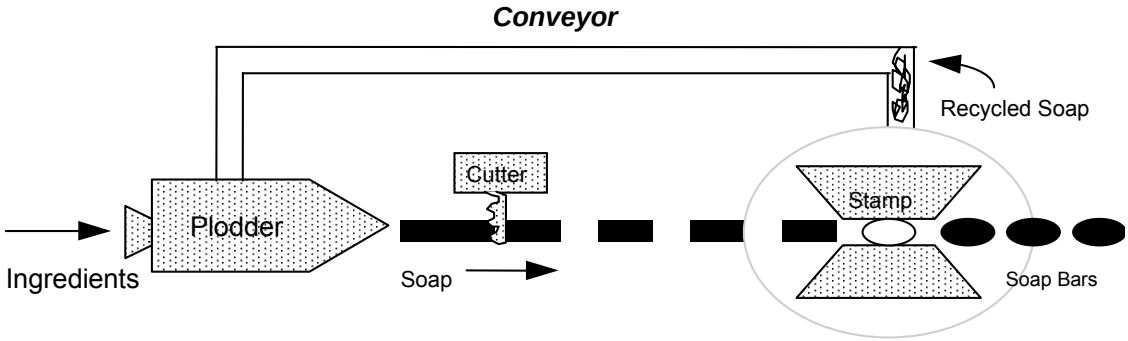


Fig. 5. A more graphic flow diagram, for a soap manufacturing process

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3. Start Decision Input/Output Diagram

⇒ For the defined Product / Process to be optimized, write in process box as in Fig. 6 for the CHEM illustrative example in page 9.

⇒ Determine Timeliness issue: the likely Extra Measurement Time vs. Run Time (see Appendix "Timeliness", page 195).

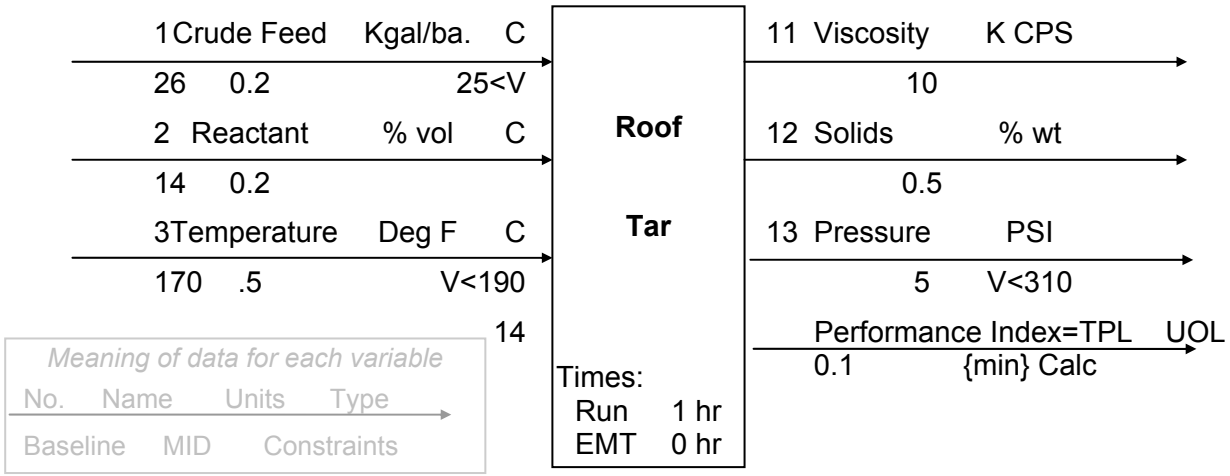


Fig. 6. Decision Input/Output Diagram (after completion of the Optimization Plan)

4. Specify Individual Objectives, and the associated process/product variables to be measured or calculated

Describe individual objectives to represent the general business objectives plus all the other aspects of production which should be minded for satisfactory business (e.g., quality, safety, regulations, etc.). Then determine the associated variables (frequently outputs, but also inputs) that need to be measured or calculated. These include, of course, the Key Performance Indicators (KPI).

Typical families of objectives:

a) Production Rate, Yield, Throughput, etc.;

b) Quality (product characteristics and consistency), as many as necessary;

c) Costs, Losses, Revenues, Byproducts;

d) Safety, Reliability (protect process/product);

e) Emission regulations.

Enter variable in the Input/Output Decision Diagram as in Fig. 6.

⇒ For each Output / Consequence / Outcome, include the units in which it is measured, and if they are summaries of raw data such as averages or standard deviations, include sample size (see Appendix "ULTRAMAX Data", page 195). Some outputs may be calculated such as Mass-balance equations, CPKs, Performance Loss Functions, Economic Analysis, or prediction models based on more easily available measurements (as thoroughly as you need).

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⇒ Define a "Run". For batch processes it is usually a batch. For continuous processes determine "how long", or "how many parts", and the time it takes to reach practical steady state due to minor changes in the adjustments of the control inputs.

5. Identify all Inputs that affect Important Results

Inputs are “control” or “adjustable” (decisions), and “uncontrolled” (determined elsewhere).

Control inputs can be “adjusted”, “fixed” or “ruled”. (Due to historical reasons sometime “control” is also used for “adjusted”; namely, actually adjusted control inputs.)

Potential **adjusted inputs** are inputs that are or can be adjusted (set) relatively easy by a plant personnel. It can be a setpoint, a directly adjusted device, a bias (to refine adjustments made by control logic) or process control parameters (such as gains, limits, profile parameters, etc.).

Adjusting the control inputs are the decisions made to apply supervisory control.

Control or Adjustable inputs need to be independent of each other and of the uncontrolled inputs). This means that new adjustments or changes in the uncontrolled inputs would not automatically result – by some internal mechanism in the process – in a change to another adjustment[19].

Uncontrolled inputs should not be affected directly or indirectly by the control inputs (but can be interdependent among themselves).

Adjusted inputs are those selected to be adjusted in the optimization cycles. Do not select as adjusted inputs those which you are quite certain of how to adjust, that is, you have little to learn about their adjustment. For instance, do not include:

- Control inputs which are regularly adjusted to the extreme safely allowed by the process, such as maximum speed, pressure or temperature. These become “fixed” at the known extremes.
- Control inputs adjusted with a special rule (useful if applied consistently). These become “ruled” inputs.

If either are included in the Optimization Plan and consequent data entry, by turning them to control inputs you can empirically optimize them, and thus confirm or refine your rules.

[19] This requirement is to assure having cause-and-effect models (as compared to just correlation models); that is, the models will remain predictive even when one intervenes by adjusting various inputs to get better outputs

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⇒ List all of the likely inputs (later on there will be a mechanism to ignore some inputs). This is equivalent to creating a Fishbone Diagram.

⇒ For Control inputs (even those that will remain fixed), include the units in which the operators adjust them; for Uncontrolled inputs, include the units in which they are measured.

When there are several inputs of the same type you can sometimes significantly simplify the problem by defining a rule (a profile) to determine their values as a function of parameters, which in turn are the control (decision) inputs in the Optimization plan. There is a new software option to implement such approach.

E.g., if there are ten temperature heaters along a nozzle or oven, you could impose as a reasonable beginning simplification that the adjustments should change linearly. Then their values can be determined by two decision inputs, the first and last temperature, while the setpoints in between are calculated by interpolation.

Another example is to start with a particular temperature profile and increase or decrease all of them by the same delta, the delta being a control input (decision variable).

6. Select Goals (Performance Index and Constraints): the balance of all individual objectives:

Unless you have a lot of experience executing these steps in formulating the optimization plan, be sure that step 5 is completed before doing this one.

⇒ To balance the several individual objectives (or KPIs), decide on an output to be maximized or minimized (the Performance Index or Objective Function, that frequently is a calculated output); and the Constraints on inputs and outputs within which production is acceptable.

Note: However, do not place constraints to rule out output physical impossibilities, such as defining Standard Deviation ≥ 0, Fraction Defective ≥ 0, or Yield $\leq 100\%$. ULTRAMAX treats output constraints as something from which to stay away, and you do not want to impede, e.g., having 0% defectives. However these types of constraints need to be placed on the control input decisions (e.g., valve flow ≥ 0).

There are three major approaches to goal setting (explained in full in Chapter "Goal Setting", page 101)

1. The Performance Index or Objective Function is a **full economic** calculation (ideally $/time, likely including the KPIs); plus constraints for considerations which have "hard limits", such as capacities, contractual agreements, safety/reliability limits, regulatory requirements, etc.
2. Same as #1, but a **quasi-economic** analysis such as:
   - addition of Performance Loss Functions
   - indicators of satisfaction, such as linearly weighted variables
4. Maximize or minimize a **single output**, and all other requirements are constraints on other variables.

The calculation in approaches 1 and 2 usually include **Global Factors** to represent variable business factors to determine profitability or other metrics of performance, such as cost of materials, revenues from the product, specifications, etc.

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By the maintenance of the Global Factors, upper management guides production operations to contribute dynamically the most to profits and thus achieve highly agile and responsive operations.

Any of these three approaches ends up with the end-user defined *Performance Index and Constraints that provide real-time performance measures*, updated as often as the summary measurements are taken, such as every half-hour.

There is an important line of management thinking that makes having this metrics alone a significant contributor to gains in operations performance. This helps plant personnel resolve often conflicting demands placed on them, and thus they can concentrate in resolving logical potential contradictions and be more productive.

⇒ Create a "Watch-for" list of outputs that you want to observe to check that they remain acceptable, but which you do not wish to measure carefully because they are rarely a problem.

7. Select final structure and quantitative details:

⇒ Select Baseline Adjustments, the current adjustments or Standard Operating Conditions (SOP). The Baseline constitutes both the starting values of the sequential optimization cycles, and the reference values for measuring performance improvements.

⇒ Select Minimum Importance Differences for outputs and for inputs. This is described in full detail in page 197.

Basically, for

- **Performance Index**: it is the minimum change in value that management cares about
- **Outputs with constraints**: the minimum violation by **data** that is considered important by management
- **Inputs**: it is **both** the minimum difference that is necessary to create changes in important outputs that are noticeable (in comparison to the noise); **and** the maximum change that the customer determines they can make from run to run. Generally, the larger the value the faster the learning and optimization.

⇒ Select any Dependencies; that it, variables which have values only when certain process components are operating.

⇒ Select the fewest control and uncontrolled inputs that the team considers critical to include to achieve some significant improvements. This is based on team knowledge and experience. It is practical to think in terms of:

which adjustments do I need to learn to adjust better than the Baseline?

which uncontrolled inputs really affect outputs?

The other inputs can be made active later on just by changing their role in the future. Thus we start simple and make things more thorough and sophisticated later on.

The Optimization Plan document will include the list of all the inputs, and especially the Baseline to be kept constant for those adjustable inputs not included in the

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software. It is a practical to include most uncontrolled inputs in the software configuration even if marked unused, as one may discover later on that their effects is indeed important.

8. Plan Procedures

⇒ Identify operating procedures which might not be followed regularly and record them in the Optimization Plan Report, to assure more consistency in production in the constant search for avoiding sources of noise.

⇒ Make sure that major maintenance projects are not planned for the first 30 or 50 optimization cycles, i.e., during the first round of SEO. On the other hand, maintenance should be ready to help in the course of SEO cycles when trouble spots are detected (e.g., a worn belt, improper feedback control operations, dirty filter, etc.)

⇒ Start Event Log Sheet so as to have on record events that may be useful to interpret data for engineering / management analysis, such as reaction to Alarms and important changes to the process.

⇒ Start collecting data for the 40+ Baseline Runs (adjusted at Baseline inputs) for the Capability / Consistency Analysis as soon as possible to:

- make sure that there are no implementation barriers and the procedures are sound,
- assure that integration is working reliably and operators are comfortable with it
- evaluate that either: constraints can be satisfied, or exclude that they cannot be satisfied.
- evaluate effective output operating area to satisfy constraints, and that any loss is acceptable; and evaluate how close to the optimal Performance Index within the effective operating area sequential optimization can take us, and that this is acceptable. Both items are determined by the noise.
- evaluate Baseline performance.

9. Assure Success

Make sure, both informally and formally, that if one satisfies Constraints and improves the Performance Index as defined, that all people who are affected truly appreciate it as a success. This will be certified after you pass the Optimization Plan Document to the team members and other stakeholders, thus achieving Unity of Purpose.

⇒ Write Optimization Plan Document (see next section)

⇒ Get Organizational Acceptance

Optimization Plan Document

The Optimization Plan Document helps the team arrive at a consensus and at approval by all stakeholders. Below is a proposed outline. The potential for getting recognition and to avoid organizational problems can be enhanced significantly

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by drafting the report and getting sign-off on the plan. Obtaining understanding and agreement among all people working on the plan contributes to the success of the plan by ensuring unity of purpose and scope. Don't forget the operators!

Ultramax Corporation provides a sample Optimization Plan Document, with some explanatory text (in dark green) which remains the same for all applications.

Outline for the Optimization Plan Document

Process Description

Approach *(basic principles of ULTRAMAX optimization)*

Production Operations Objectives

Business Objectives

Individual Objectives, including constraints

Structure, scope of Sequential Optimization

Decision Input/Output Diagram

Variables, Definition Table (see sample below), usually with an outline such as

Performance Index to minimize. MID = min change important to management
Important outputs to measure or calculate. MID = min constraint violation deemed important. MID ignored for outputs without constraints – such as for those used in calculations (then the MID of the final calculated variable rules).
Outputs to be monitored and entered, that do not affect the evaluation of performance
Outputs monitored outside the ULTRAMAX application file, some used to externally calculate outputs entered into the file
Component Inputs – components or conditions of the process that may or may not be operating. They enable corresponding variables to require or not values – else, they are just Uncontrolled inputs.
Control inputs to ADJUST. MID = max desirable change in adjustments and min necessary to create detectable output change on some output
Control inputs adjusted with rules
Control inputs to keep FIXED (usually not entered, and all other Adjustments with potential impact @ SOP)
Uncontrolled inputs entered and active
Uncontrolled inputs entered but unused (noise source)
Miscellaneous variables entered but unused
Uncontrolled inputs only in Data Log Sheet or Historian (not entered, noise source)
Uncontrolled inputs not measured (noise source)

Optimization Plan in the Software

Variables, further explanations

Calculated variables

Reasons for Constraints

Procedures

Description of sequential optimization runs

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Measurement Procedures

Watch-for list

The Optimization Plan is not supposed to remain static in a proactive, continuous-improvements environment; the Optimization Plan Report is kept up to date.

Variable Description	Variable Name in software (max 12 characters)	Units (max 8 char.)	Dependency (max 12 characters)	Role[20]	Entry Mode[21]	Base Point (control inputs)	MID	Constraints Low	Constraints High
Performance Index to minimize; MID = min change important to management									
Boiler Efficiency	Total_eff	eff		6 (max)	C		0.1	--	--
Important outputs to measure; MID = min constraint violation deemed important									
LP turbine reheat temperature B Side	Reheat_T_B	Deg C		5	H		0.5	528.	545.
Final steam exit pressure	Steam_Press	Mpa		5	H		0.2	--	16.1
Flue gas exit temperature	Gas_Exit_Tem	Deg C		5	H		0.5	100	130
Flue gas oxygen content A side	FlueGas_O2_A	%dry vo		5	H		0.2	2.5	4
Flue gas carbon monoxide content	FlueGas_CO	Ppm		5	H		10	--	150
Flue gas dust content A side	DustBurden_A	G/m^3		5	H		0.05	--	0.4
Flue gas dust content B side	DustBurden_B	G/m3		5	H		0.05	--	0.4
Rate of coal delivery	Unit_Fuel	Kg/s		5	H		0.2	--	--
Carbon in fly ash B duct	C_in_Dust_B	%wt		5	H		--	--	--
Carbon in fly ash C duct	C_in_Dust_C	%wt		5	H		--	--	--
Monitored outputs to measure									
Boiler Efficiency Direct	B_Eff_Dir	%		5	H				
Boiler Efficiency Loss	B_Eff_Loss	%		5	H				
Component Inputs (MID will be updated automatically)									
Mill running: A B C D	Mill_Run_A B C D	Yes-no		2	A		1		
Mill feed adj. by UMAX (manual): A B C D	Mill_S_U_A B C D	Yes-no	Mill_Run_A B C D	2	A		1		

[20] C=1=control input; U=2=uncontrolled inp.; R=8=ruled inp.; F=0=fixed input; I=0=ignored; M=5=motinored output; O=5=important output; OF=6=objective function

[21] H=manual or by hand; C=calculated; A=automatic (from integration)

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Control inputs to adjust; MID=max change in adjustments									
Set point of residual oxygen in the flue gas	O2_Setpoint	% dry vo		1	H	3.3	0.2	2.8	--
Mill Feeder Speed: A B C D	Mill_Speed_A B C D	??	Mill_S_U_A B C D	1	A	??	??		
Mill Air Bias: A B C D	Mill_Air_B_A B C D	??	Mill_Run_A B C D	1	A	??	??		
LP turbine reheater steam	RHTR_A_T_SP	Deg C		1	H	530	1	530	545
Mill bias adds more or less air to the air fuel ratio	Mill_A_bias	%		1	H	100	5	--	--
Mill outlet temperature set point can be set for individual mills or set up the same for all mills	Mill_Temp_SP	Deg C		1	H	80	1	--	90
Induced Draft fan bias, positive bias makes A side fan work harder	ID_Bias	% bias		1	H	0	2	10	--
Control inputs adjusted with rules (ruled inputs)									
				8	H				
Uncontrolled inputs to include									
Unit load	Load	MW		2	H				
Time since sootblowing	Sootblow	H		2	H				
Forced draft inlet temperature	FD_InL_Temp	Deg C		2	H				
Composite sample of coal specific energy for the day	Coal_SE	MJ/kg		2	H				
Composite sample of coal moisture content for the day	Coal_Moist	%ar		2	H				
Feed water temperature	Fwater_Temp	Deg C		2	H				
Control inputs to keep fixed (not entered, and all other control inputs with potential impact, adjusted at SOP)									
Main Steam Temperature SP						538			
Miscellaneous variables entered but unused									
Uncontrolled inputs entered but unused (noise source)									
Feed water flow rate	Fwater_Flow	Kg/s		0	H				
Mill bias adds more or less air to the air fuel ratio	Mill D_bias	%		0	H	100	5	--	
				0	H				
Uncontrolled inputs only in Data Log Sheet or Historian (not entered, noise source)									
Uncontrolled inputs not measured (noise source)									

Enter the Optimization Plan into an Application File

The Optimization Plan Document includes a table as illustrated above with the variables names, units, etc. The active variables are entered into a new application file, variable by variable, using the stand-alone version of ULTRAMAX. In turn, the software generates an Optimization Plan Report, which is included in the Optimization Plan document.

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A reasonable next step is, when possible, to import prior data to make preliminary analyses which may enable you to discover some interesting facts – and sometimes provide a head-start for SEO -- as described below.

Import and Analyze Prior Data (optional)

Prior Data is run data obtained before starting the project of applying ULTRAMAX.

Prior data is valid if it represents the process behavior now and in the future. Even if the run data is not of the quality that will be generated and collected with the ULTRAMAX procedures, it can be worthwhile to import the prior data into a dedicated application file to make various analyses that might give insights about the process and its operations.

Basically, this is equivalent to doing some aspect of Engineering / Management analysis as described in full detail in Maintenance, Discovery and Improvements, page 65.

Analyses with Prior Data

1. Do a Capability & Consistency Analysis™ (CCA) as described in page 167. In particular check that the noise levels are as expected, in the order of magnitude of the MIDs.
2. Reset all Parameters PAR(n) to default, and recreate the models. Find the optimum estimate. The optimum estimate has meaning if the control inputs are truly independent, and if they have changed sufficiently to learn about the process operating characteristics (the Standard Deviation of the input values in the Capability Report are larger than their MID).

   For those inputs that did change, and if the optimum estimate has a Standard Distance Factor (SDF) < 0.6, this is a fairly good indication that you already have data close to the optimum (for the fixed values of the unchanging control inputs). Then what is left may be to confirm or refine the optimum, and to maintain it.

   If in addition you get a Signal > 0.65 for all active outputs (see Signal report), it might be beneficial to start SEO with the prior data[22].

   All this very seldom happens.

   Reminder:

   - it is critical that the presumed control inputs be independent of each other
   - it is misleading to delete runs that have outliers unless it is known separately that something was not representative with that data
3. Displays plots to see any unusual behavior:
   - Plot all variables vs. Run#

[22] Recall that is it possible to leave the prior data and not use it, by setting PAR(1) to the run number of the first run to be used.

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Integration / Automation

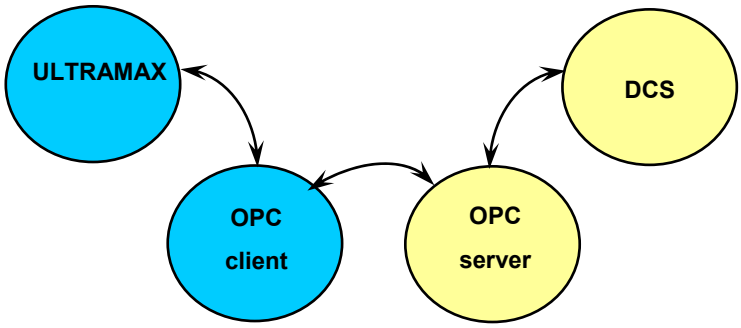
"**Integration**" is the automatic data exchange (communication) between digital control systems[23] (one way or bi-directional) and the ULTRAMAX software.

This step is not done in Stand Alone implementations, which are typical of optimization projects rather than operations management. Also, Integration is sometimes delayed until a later step, especially after passing the Process Capability analysis (which can be performed as a prior low-cost project).

The benefit of integration is greater success in optimizing production operation performance by:

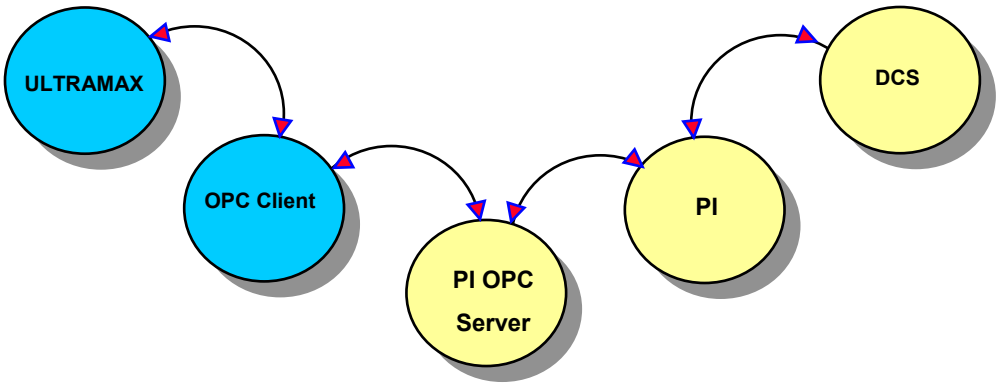
1. Making the application of SEO easier for the operators and reducing the probability of human errors with manually data entry.
2. Increasing process consistency from operator to operator, shift to shift.
3. Enabling closed-loop optimization, maintaining optimal operations even when operators are distracted with other responsibilities.
4. Reducing the time it takes for the process to respond to changes in conditions and objectives.
5. Imbedding the concept of process optimization as an integral part of process management/control – as compared to applying optimization on a 'time available basis', as a 'project', or as a 'problem solver', which results in static knowledge.
6. Integration typically involves:
    - Establishing communications with the DCS through an existing **OPC** (Object linking and embedding – OLE -- for Process Control) or **DDE** (Dynamic Data Exchange) **Server**, sometimes through a historian such as PI.
    - Configuring the "ULTRAMAX Event Driver" (UED) module and its OPC / DDE Client to manage communications
    - Maintaining the above as the process is changed

Following are two typical diagrams for integration. The blue modules are provided with ULTRAMAX; the yellow modules are part of the end-user system.



[23] E.g.: Distributed Control System (DCS); Supervisory Control and Data Acquisition (SCADA) system.

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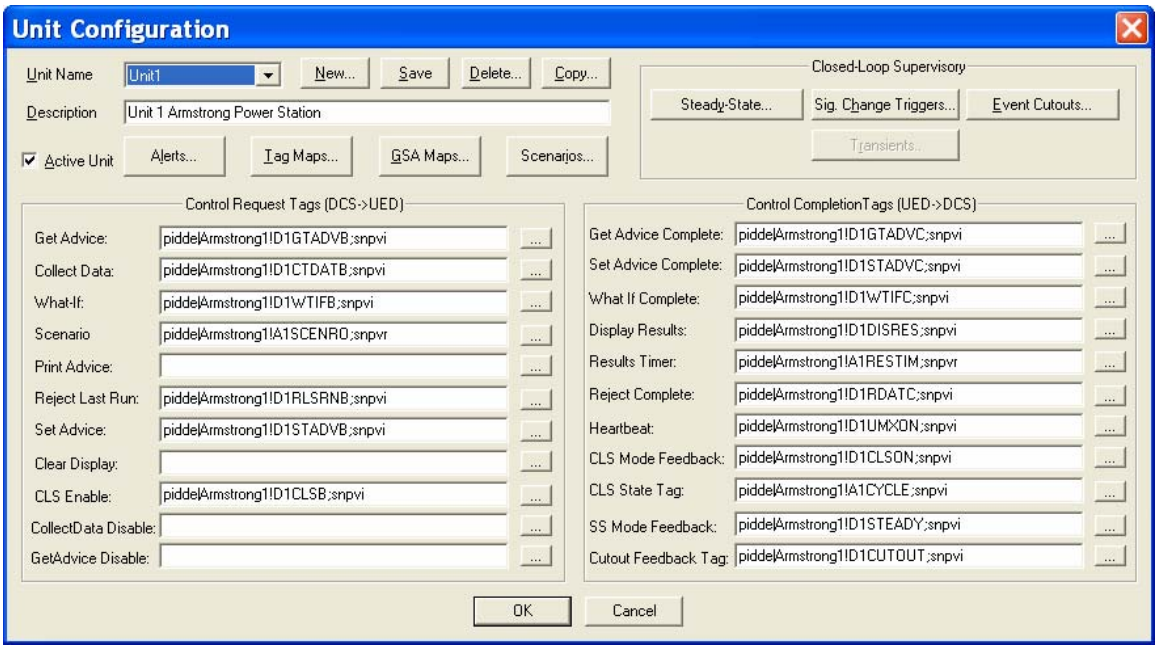
The ULTRAMAX Event Driver (UED)

The key module of integration is the ULTRAMAX Event Driver (UED), which:

- has a Client end that supports OPC and DDE communications with the Server. The Server is provided by the customer
- manages the communications with the rest of ULTRAMAX

The UED assumes automatically the variables in the Optimization Plan (in the application file). In addition the UED has a GUI to configure the tags associated with each variable, and other parameters associations with integration. There is separate documentation to install and maintain the UED.

Below is a sample the first screen of the UED GUI.

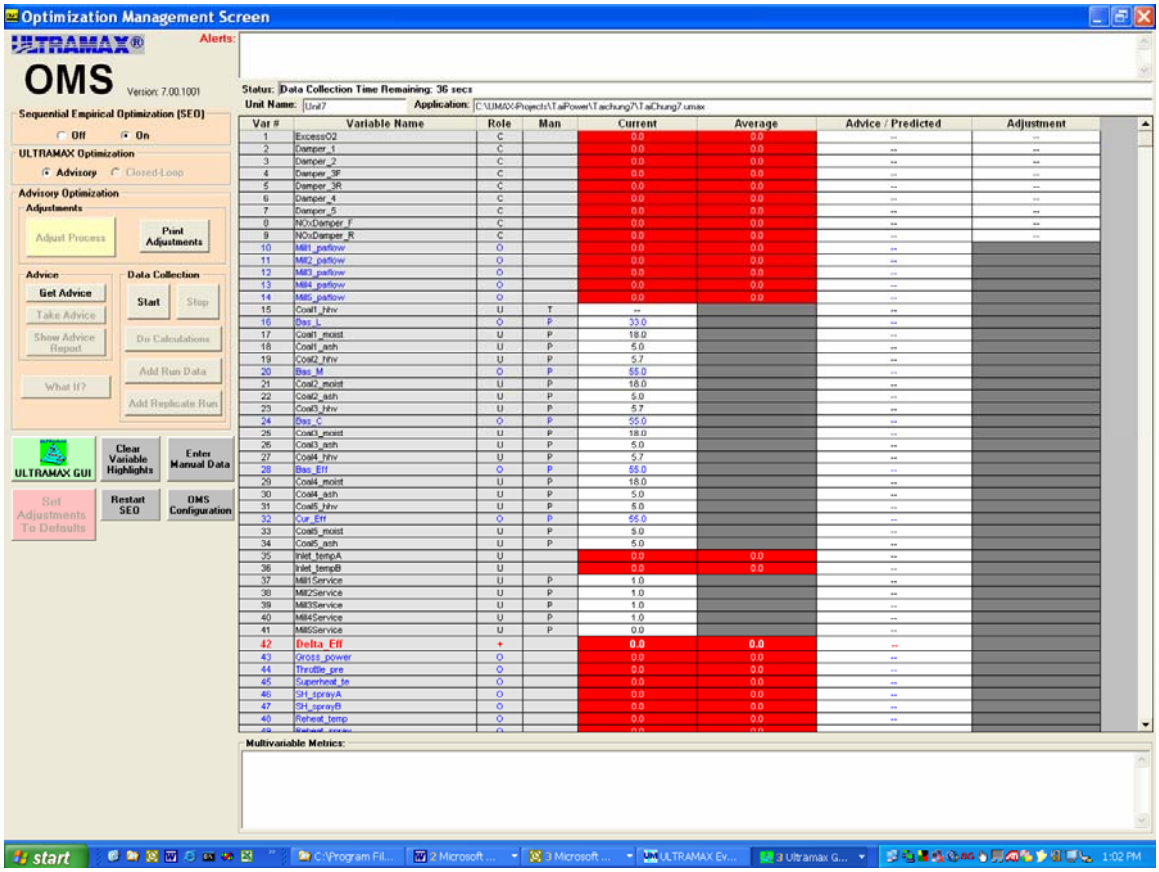


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The Optimization Management Screen (OMS)

Integrated ULTRAMAX is managed from the Human-Machine Interface (HMI) Optimization Management Screen (OMS)[24] GUI. There is a separate documentation provided with the product about how to use the OMS.

ULTRAMAX supports an automatic OMS at the optimization workstation monitor. Below is a sample of one screen (without data for privacy purposes):



The UED also supports an OMS at the control room console, which is designed and implemented (usually) by the customer. The console OMS is driven by the DCS tag values which are controlled by the UED module.

Among various functions, the automated OMS at the optimization workstation displays current data, run-average data, the adjustment advice provided by the SEO technology, the operator desired next adjustments, and the Alerts statements. There are buttons, for instance, to:

- Make the adjustments (the whole set of control inputs in a predefined time profile)

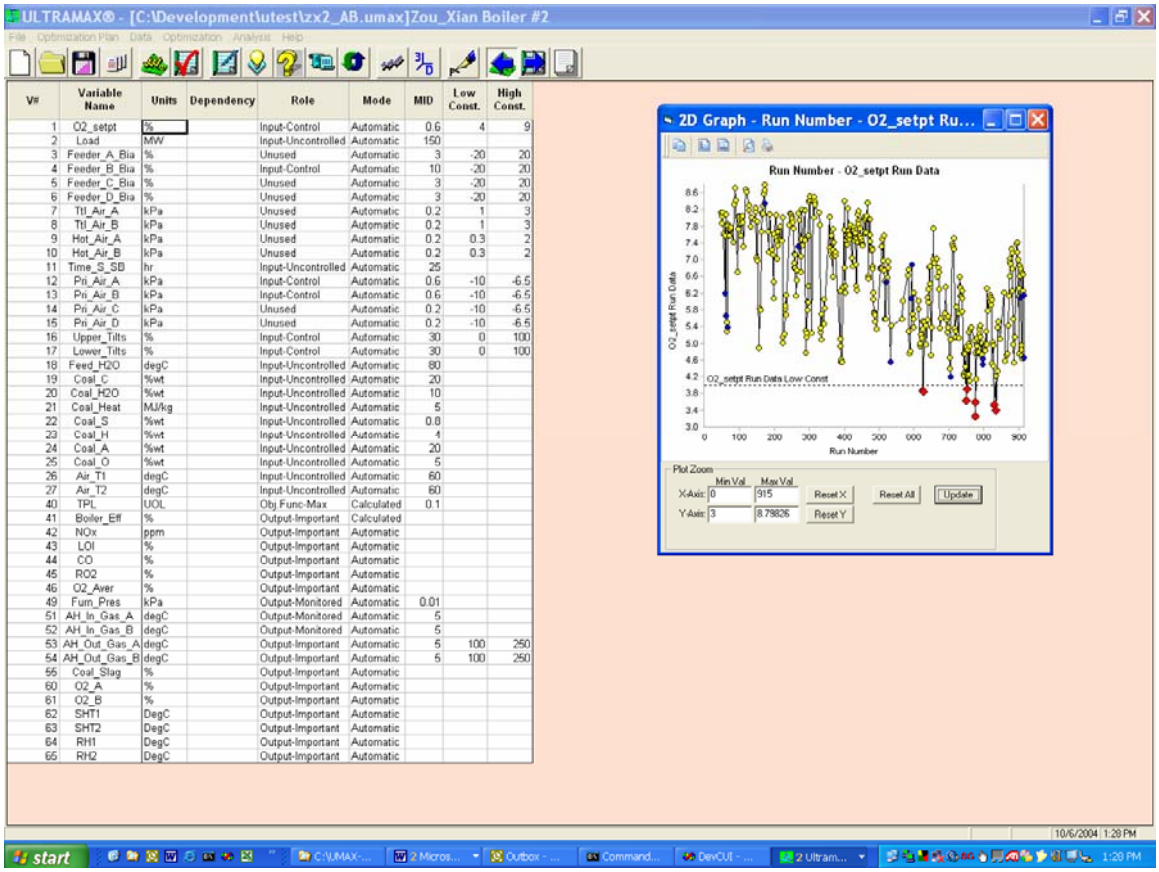
[24] Previously called the Operator Ultramax Interface (OUI), and only at the control consoles.

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- Choose among three operating modes:
  - Manual Supervisory Control: where the operator adjusts setpoints, biases, etc. as he/she would in the console but now in a dedicated central location, without the aid of SEO. This is very useful to develop practice and confidence that the integration installation and data collection and adjustment procedures are correct; and also to collect data for the Baseline Analysis.
  - Advisory Optimization: where the operator has to approve each new adjustment advised by SEO – or modify it -- and presses the "Take Advice" button to implement the new Advices.
  - Closed-Loop Optimization: where the Advice is implemented automatically. Closed-loop reverts to Advisory automatically when the UED or the SEO detects an error or upset, or when the operator turns it off. Automatic closed-loop is turned on by the end-user when they verify that the application of ULTRAMAX is reliable (when the sequence of Advices are accepted regularly without need for revisions).

The OMS supports an icon to bring up the stand-alone GUI for virtually all its features except changing the Optimization Plan. Changing the Optimization Plan is done off-line with the Stand-Alone ULTRAMAX.

Below is a sample of the main screen of the Stand-Alone GUI and of a selected plot. There is a separate documentation about how to use the Stand-Alone GUI.



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Baseline Capability Analysis

Driving issues:

There are processes with Capability problems due to low Consistency (high noise) than cannot be resolved with better adjustments. For instance, they are incapable to satisfy defined constraints. Some of these problems need to be resolved just to meet production requirements; others to enable running SEO.

Thus, it is practical to perform an up-front Baseline Capability Analysis before starting SEO. The analysis is done with data from normal process operations, such as adjusted at Baseline values.

The analysis performed is the Capability & Consistency Analysis™ (CCA) *described in Part 4, page 167. Note that this capability analysis is done for all performance metrics, not only for quality (as commonly done elsewhere).*

Just doing this analysis is worth doing even if not interested in optimization.

This Baseline Capability Analysis multivariable analysis is of significant value for the customer to know…

- Whether the process has capability problems that present barriers for normal process operations
- Whether the process is sufficiently capable for SEO
- The noise levels, how close ULTRAMAX can get to the optimum.
- Since the data is generated with adjustments at Baseline, the analysis of the outputs provides a reference for evaluating future SEO performance improvements.

While performing this analysis is part of the ULTRAMAX normal procedures for a new process, note that the up-front Baseline Capability Analysis is not technically required to start and proceed with SEO[25]. One could start right away if:

- There is already sufficient confidence that the process is capable, or
- One would not think that process incapabilities discovered during SEO is a problem generated by SEO.

Passing the Baseline Capability Analysis is necessary before a guarantee of performance improvements is issued. The guarantee also requires that the process maintains its capability.

Principles

The Section in Engineering and Management Activities, Capability Analysis explains the principles being used.

The Capability Analysis indicates which specific variables have any historical or capability problems. In particular it analyzes:

1. how well the selected actual data has satisfied constraints

[25] By comparison, it is necessary to collect large amounts of experimental data upfront to "train" Neural Network models.

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2. the capability of future runs (with same mean outputs) to satisfy constraints
3. the Effective Operating Area, the further restricted constraints (for output Means) so that Data, with noise, satisfies constraints
4. how close SEO can get to the optimum in the Effective Operating Area

Note the following driving concepts from the Capability Analysis section:

With high Noise (measured by its Sigma) the process Means may have to be more conservative than the constraints in the Optimization Plan. This may be required so that the data generated (with noise) does not violate the constraints by more than MID. This represents a loss in the possible values of the outputs, and therefore of the Performance Index.

As explained in that Section, there is no loss when 3•SIGMA < MID; that is, when the maximum likely prediction error is smaller than what management considers a "Minimum Important Difference" in the output.

"Baseline" is the standard adjustments[26] that the customer is using before applying ULTRAMAX. "Baseline" also includes the mean outputs obtained with the process at those adjustments. If each crew has its own "Baseline", then the customer will have to choose one, from which SEO improvements will be evaluated.

The Baseline analysis consists just of collecting data with regular production runs made with adjustments at Baseline, and to perform a Capability Analysis with the data. In this case the Capability Analysis also provides a Baseline Performance to evaluate SEO improvements (but evaluation is more complex when there are uncontrolled inputs).

Collecting data for the Baseline Capability Analysis is also useful to:

- refine the new procedures for managing the process with SEO
- train plant personnel; especially if done after integration to test and build confidence in the integration modules

⇒ Collecting Data

Most frequently historical data is insufficient to provide confidence in the results, so new regular production data is collected[27].

1. Collect data from the production process running as usual, presumably at the Baseline adjustments. This is not an absolute requirement, however if the adjustments change more data will have to be collected.

   If at all possible, also keep the uncontrolled inputs constant.

   When all inputs remain constant the runs are called "Replicates", and then the manifestation of noise is simply the variation of the outputs.
2. Collect data for all inputs. Run data with any input values missing is useless to predict process outputs. Occasionally missing outputs is OK.
3. Data is collected in at least 10 different days, and spread over different times of the day[28].

[26] Often part of Standard Operating Procedures (SOP).

[27] However, a preliminary analysis could be done with historical data.

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In order to pass the statistical tests the final analysis you will need at least 40[29] more runs than the number of changing inputs. If ULTRAMAX integration is already installed, then simply collect all the data (that meet the other requirements) for two weeks.

4. Start making preliminary Capability Analyses with as little as 10 to 15 runs, as it is easy to detect actual problems with little data.
5. Collect data the same way as planned for doing SEO cycles. This is done very easily AFTER installing integration (the primary reason to do Integration before the Baseline Capability & Consistency Analysis). Some issues to remember:
   - Do not collect data from runs under unusual circumstances, such as at start-up, during or right after a cleaning function, at the beginning of a new shift (with a new crew), etc.
   - If a continuous process, wait till transients (due to readjustments) largely pass, to collect data only for steady-state operations[30]. This does not apply to batch processes.
   - If using ULTRAMAX Stand-Alone, calculate output averages as planned for SEO

   The data is stored by the ULTRAMAX software (Integrated or Stand Alone). If the data is placed first in a spreadsheet, then use the format "General" so as to avoid any commas or $ signs).
6. If the process is changed in any significant way to solve a problem, then data collection has to be started all over again (the old data is no longer valid).

⇒ Creating the Capability Report and Analysis

For how to create the CCA report, understand it and analyzing it go to Process / SEO Conditions & Measures, Capability and Consistency Analysis (CCA) > Capability Analysis Report, page 168.

Note that passing the CCA test with the up-front Baseline Analysis does not assure that future Capability Tests will be passed with the future values of the uncontrolled inputs (known or unknown).

---

[28] The spread is to pick up various sources of variations (such as air temperature and humidity, raw material temperature, operating crews, batches of materials, etc.). Even then, not all likely conditions will be seen in these ten days, and thus with more experience the Noise may actually go up.

[29] The number of runs required is consistent with ty", with sufficient data to create locally accurate, goal-oriented models, a quadrati.

[30] So that the outputs are a consequence of the inputs, and not of the input values before the last adjustments. The rules are different when using the Transient Optimization technology.

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Part 2: OPERATING PROCEDURES

Production Operations with SEO Optimization

An important aspect of optimizing performance operations with existing production assets – while making any one product -- involves adjusting inputs to the process. These adjustments, together with externally determined uncontrolled inputs such as the quality of raw materials and environmental and equipment conditions, are the most important determinants of regular operations performance and value generated for the firm.

Making better adjustments that generate consequent improvements in performance is the most obvious benefits of using ULTRAMAX.

ULTRAMAX uses Sequential Empirical Optimization (SEO) technology to adjust the process during production operations and to help with associated issues – such as detecting upsets in the process. This is done by implementing the Sequential Optimization Cycles described below. We hope that the reader will find these procedures eminently doable and obviously necessary for true, practical optimization.

Recall the SEO should be preceded by a Capability Analysis (page 44)

Adjustments are often made on actual or virtual "knobs" in a control panel, or in a computer screen in the process control room. Adjustments are:

- the setting of setpoints for process control systems
- the manual settings of something physical, such as the position of a damper.
- the setting of biases if there is control logic to determine setpoints – and/or the setting of other control logic parameters

Making "adjustments" is part of "supervisory control". "Tweaking" is an aspect of adjustments which sometimes is overdone and induces extra noise (lower consistency).

The basic idea behind SEO is that if you make adjustments quantitatively, can measure uncontrolled inputs, and can measure the important consequences, then mathematical technology can "learn" the relationship between the inputs and the consequences; and be able to predict output values for new sets of input values. Then a search engine probes the models for all outputs simultaneously to find new adjustments that will lead to predicted (and actual) performance improvements.

This empirical technology works when process operations are fairly repetitive for the same value of the adjustments and uncontrolled inputs (note that this does not imply that the inputs need to remain constant). Thus, SEO is works better when ignoring data obtained during upsets or extraordinary conditions, such as during breakdowns.

The aim is to maximize the process performance in terms of metrics defined by management / engineering, as necessary compensating for the changing values of the uncontrolled inputs (if any), and readily adapting to changes in economic and business requirements. SEO and its Sequential Optimization Cycles are illustrated in Fig. 7.

When a process is run by edict only at the SOP adjustments, then it does not compensate for conditions (uncontrolled inputs, economic conditions). ULTRAMAX is also superior in engineering studies to optimize such SOPs.

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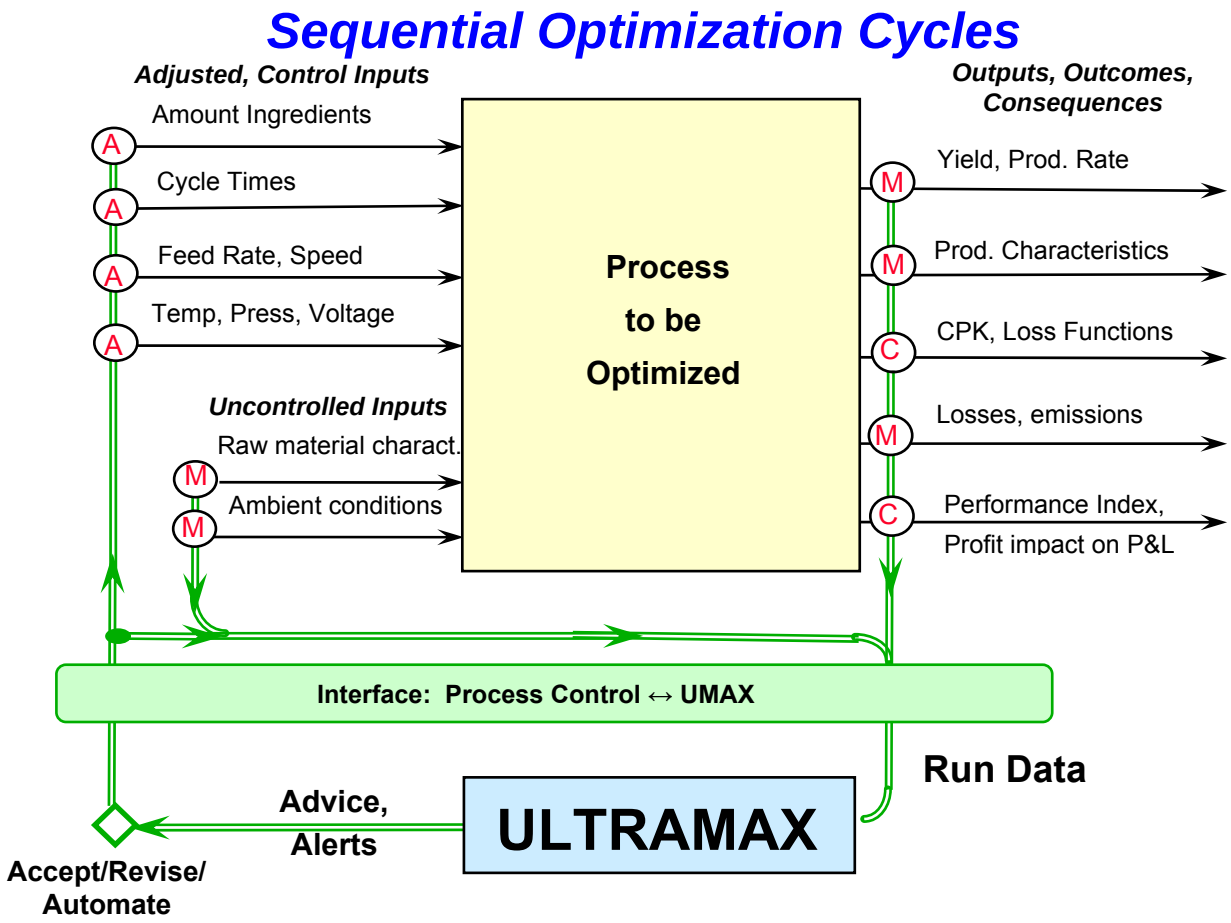


Fig. 7. SEO Cycles: the Decision Input/Output Diagram and flow of Data, Advice and Readjustments of the Process

A = adjust; **M** = measure physical value or calculated outside ULTRAMAX;

C = calculated by equations provided to ULTRAMAX

Implementing the SEO Cycles

The "Sequential Optimization Cycle" -- or "Sequential Optimization" for short -- illustrated in Fig. 7 is basically the following, with details below:

- With **the process**: (above the interface): *make adjustments (the Advice from the previous cycle), measure uncontrolled inputs, run production / manufacturing operations, collect consequent operating run data.*
- With **the software**: (below the interface): *enter and store new run data, update models (with past and new data) and generate new Advice for adjustments (by searching the updated models).*

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In more detail:

1. **Adjust** the process **inputs.** Avoid the possibility of important hysteresis.
2. **Run the process until completing the collection of the consequent output data for a period of time -- or samples are sent to the lab -- to calculate averages or other relevant summaries (e.g., standard deviations). Towards the end, collect uncontrolled input data (if any) as a predictor for the next run (to determine the corresponding optimum). If some data takes too long to obtain, such as from laboratory tests, then proceed with "staggered" cycles (see Timeliness page 195).**
3. **For processes with "memory", such as a continuous process, data collection is done after waiting for the transients due to the new adjustments to fade away, until near the consequent steady-state or stability31.**
4. **Enter data into the computer. This is done manually, or automatically if the installation is "integrated". If integrated, ULTRAMAX has the tools to calculate summaries such as averages, which constitute the actual SEO run data.**
5. **The SEO software will (see The SEO Technology, page 138 for more details):**
   (a) **store** the run data;
   (b) use stored data to create **prediction models** (for each output as a function of all the inputs);
   (c) **search** all the prediction models simultaneously for the best next adjustments, which is called an "**Advice**".
6. **Display Advice for the next adjustments; and Alerts about the state of the process and of optimization**
7. **(If not closed loop optimization) approve or modify the advice.**

   Start next cycle, repeat from 1.

   This **cycle** is done usually every 20 minutes to a couple of hours in continuous processes, or every batch in a batch process (the **Run Time**).

   Production Operations benefits from continuing application of SEO cycles by:
   - Getting periodically the evaluation of performance: Performance Index plus the satisfaction of Constraints
   - Maintaining the compensation for changes in the (included) uncontrolled inputs (feed forward)
   - Detection of upsets and undesirable events
   - Maintaining slow adaptation to changes in the process due to unknown uncontrolled inputs (feed-back)
   - Demanding the "optimization attitude" on the part of plant personnel.

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[31] Unless using the "Transient" version of ULTRAMAX, still in Beta test. The response time is no shorter than approximately the half-time of the transients created by the changing uncontrolled inputs.

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Making Triads

An **important variation** is to run periodic "**triads**" when trying to evaluate improvements due to better adjustments when performance is also affected significantly by changes in the uncontrolled inputs. It also illustrated the effects of changing uncontrolled inputs (known or not).

See On Calculating Gains, page 73 for principles and calculations.

Procedures:

a. Frequency: When actively documenting improvements make a triad approximately every ten runs. Recall that optimal gains are likely to depend on the combination of uncontrolled inputs.

b. The Triad analysis assumes that uncontrolled inputs are very similar. Do not make a triad of cycles with drastically different values of uncontrolled inputs. If there are component inputs (scenarios) make sure that each triad uses the same scenario.

c. To make each triad of cycles, at two near-equal intervals:

Cycle (A): Run with adjustments at optimum advice (see method below). The Advice from the Cycle will be ignored.

Cycle (B): Run with adjustments at Baseline

Cycle (C): Run with adjustments at (new) optimum advice (see method below)

Record in the Event Log which run numbers were each of these three, for future analysis.

Getting optimum advice for triads:

Get the optimum advice as if starting a new series of SEO cycles[32] as follows, done *before* executing the "Cycle" or the "Advice" option in the software…

- (If Integration) press in the OMS button "Restart SEO" (in Advisory mode)
- (If Stand Alone) set Optimization Plan > Parameters… > 'Start new sequence'

  If "Cycle" or "Advice" asks 'Get an Advice?' answer "yes". Success is when the Advice in the Advice Report is headed by Adv/StSeq (= Advice for Start Sequence).

d. Do not make a triad at the beginning of a series of SEO runs (such as after a startup), to assure process stability. Note that making the triad adjustments may involve making changes in adjustments larger than MID.

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[32] What happens then is that the Advice will be:

- in a region of lower prediction error – namely, where we have more experience [restricted Area of Confidence, PAR(10)=0.5]
- NOT limited by the maximum allowed change in adjustments from the last run in the data file, but be free just to get better results [ignore PAR(59)]
- the optimum estimate (within the reduced Area of Confidence); not be burdened with exploration.

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Understanding what is going on and acting upon it

There are four main important issues to be aware of:

A. If a new run has unusual or aberrant data, then:

- If a special cause is discovered that is fixed (prevent future occurrence), then the abnormal data is deleted (which is always done MANUALLY).

  This practice has the well known and established positive impact on better business as in Quality Control – plus in addition reducing noise and thus enabling empirical optimization to converge closer to the ideal optimum (explained in Technical Performance, page 154).

  A rare alternative is that the special cause delivers better results, and then it is adopted permanently, and this starts a new round of optimization (forgetting old data that is no longer representative).

- If the cause is not discovered or discovered and not fixed, the apparently bad data is left, thus doing optimization taking such behavior into account (and thus not being as close to the optimum if that source of noise is removed).

B. Satisfaction of Constraints.

C. Whether SEO extrapolating towards better adjustments – requiring more attention on the part of plant personnel -- or interpolating among past experience.

D. Occasionally confirm that you are really in control of the adjusted (control) inputs; i.e., that adjustments are not changed by some other mechanism or personnel. This assures “independence” of the adjustments.

The easiest way to we aware of issues is to pay attention to the ALERTS described below, except for #4 which the software does not test. This check has to be done by personal observation.

Introduction to Indicators and metrics of Operating Performance and SOE

Let us review certain concepts and indicators necessary to assess conditions:

1. **Data Validity / Reliability:** if the data is very different than recent past then perhaps something is amiss and should be checked. Done one variable at the time for extreme simplicity of concept.
2. **Standard Distance Factor:** A multivariable metric of how far data is from the data base to create the prediction models, which determines the lack of knowledge due to extrapolation.
3. **Standard Error:** How far (in sigmas) actual data is in comparison with the values expected by the prediction models. The Overall Standard Error is a multivariable single metric representing all output Standard Errors.
4. **Output Mean:** the mean value of the output for constant inputs. Actual data will vary around the mean by the sigma of the distribution: the noise plus the lack of knowledge when extrapolating

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5. **Slack:** the distance from a value to a constraint. If negative, it is beyond the constraint. For outputs, if actual data is beyond constraints by more than MID (defined by the end-user), it is considered a definite violation.
6. **Protection:** How far the output mean is from a constraint + MID in number of sigmas of the distribution. This is the protection against the actual data violating the constraint by more than the MID.
7. **Active Output:** An output with active constraints or the Performance Metric; those outputs that one needs to be able to predict well.
8. **Projected Optimum Estimate (POE):** The optimum estimate with protection against constraint violation (assuming no extrapolation). The SDF of the POE indicates whether the optimum is seen outside the existing data, or within (extrapolating or interpolating)
9. **When the SDF of the POE is small, especially when the Average SDF of the POE (in the last eight runs) is small, then there is data around the POE and so the optimum is known well – namely, the process has converged around the (an) optimum.**
10. **Signal:** A metric of how well the model explains variations in outputs based on variations in inputs – the rest is unexplained, or noise. The Signal reflects the quality of the models, how well they represent the underlying process behaviors rather than fitting the noise33. The Signal begins to be truly representative when there is sufficient data to have local models.
    - **Critical Signal:** The minimum signal among active outputs
    - **The Average Critical Signal:** the average in the last eight runs

[33] The Signal is the traditional R-Square in regression analysis, except that adjusted by the degrees of freedom.

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The concepts and metrics are described in detail in SEO Concepts and Metrics, page 159. Here is a copy of the summary table.

Concept / Condition	Measures / Metrics	Where
Data Validity / Reliability, conformance with past data distribution, each variable separately	**Deviation**	Advice Report, Deviation column
Location of inputs, how far inputs are from the data used to generate prediction models, related to ability to predict accurately	**Standard Distance Factor** (**SDF**) **Uncontrolled SDF**	Advice Report, Multivariable Summary Plot
Outputs in Statistical Control	**Standard Error** **Overall Standard Error**	Advice Report, Actual Column, below each output value. Multivariable Summary. Plot
Satisfaction of Constraints	**Slack,** **Min Relative Slack** **Protection**	Advice Report Constraint row for each variable Multivariable Summary Colors in plots
Closeness to Relative Optimum	SDF of the Projected optimum Estimate: **SDF-POE** Average SDF-POE	Advice Report > Multivariable Summary > SDF row; Projected Optimum Estimate (POE) column.
Models suitable for optimization	Prediction Error (**Noise** estimate), **Signal**, **Critical Signal**, **Average Critical Signal**	Signal Report

Description of Alerts

Alerts are a series of simple statements when certain conditions are detected that merit operator attention. This table below becomes an essential tool of production with optimization.

The ULTRAMAX® ALERTS and Guidelines table below displays the thresholds that trigger ALARMs, Warnings and Notes with their corresponding short statements. When none of these conditions apply the display is “No Alerts”.

The thresholds can be modified to suit customer preferences, as are the recommended actions for the operators (done only in this table, not in the software).

Certain levels of Alerts – the ALARMS -- will turn Closed-Loop optimization off, and leave operations management in Advisory mode, for the operator to engage directly in supervisory control.

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When alerts are displayed, the operator can get more specific diagnostics in the full reports generated by ULTRAMAX with data for all variables, indicators (e.g., slacks, standard errors, SDF, noise levels, signals) and full Status Interpretation Statements. These reports are available in the OUI by pressing the button "**ULTRAMAX GUI**" (meaning the Stand-Alone GUI). The most important such report is the Advice Report.

Following is a description of the Alerts and the recommended action for the operators. This table can be tailored by the end-user to suit specific processes.

The following table has several columns:

No	The number or index for each ALERT condition.
Condition	A brief description of the condition it reports.
Metric	A description of the corresponding Metric used
Threshold	The threshold value that determines the severity of the condition. Those values with a PAR(*) below can be changed by the end-user.
ALERT Level	Classification of the condition **ALARM** something abnormal definitely to be aware of; requires attention and likely correction (red) **Warn** something unusual or risky is detected, requires attention (orange) **Note** something is close to borderline; just be aware (yellow)
Alert Text and Trigger	The Alert Text displayed for each condition
Possible Action Steps	Recommended action for the operator. This description appears only in the table of Alerts below, and thus the description can be tailored by the end-user.

Notes:

e.g., A(3) means ALERT(3)

Models are "up to date" if they were built **after** all the stored run data was entered (or after some invalid data was deleted).

Note: It is possible to enter data (e.g., with Enter and Import options in the regular, stand-alone ULTRAMAX) but not update the models. You can bring models up to date by doing Learn or engaging in a Cycle (even with no new data). You should not evaluate new data, evaluate optimality, or get Advices or What-Ifs with obsolete models.

If models are not up to date, then you can no longer create alerts #1 and #2 on the data already entered; however, after the models are up to date, you can review the Current Standard Errors (Residuals) (report or plots) to see if any data seems to be an outlier (over 3-sigma) deviations or to observe special patterns.

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ULTRAMAX® ALERTS and Guidelines

No	Condition	Metric	Threshold	ALERT Level	ALERT Text and Trigger	Possible Action Steps (see Advice Report for detailed analysis)
New Run Data, Actuals (if there are multiple new run data, the metric used is the worst among all runs)						
1	Run Data input location (in relation to past experience)	**Standard Distance Factor (SDF)**	**>1.2** PAR(106)	**ALARM**	**New Actual: inputs were extremely unusual** **SDF > 1.2**	**If surprised, verify that process and data are OK.** **One possible reason is due to unusual uncontrolled inputs (see "Uncontrolled SDF").**
			>0.8 PAR(105)	Note	New Actual: inputs were in new area SDF > 0.8	This is normal when extrapolating towards the optimum, just be aware of it. If surprised, verify that process and data are OK.
2	Run Data Uncontrolled input location	**Uncontrolled SDF**	**>1.2** PAR(106)	**ALARM**	**New Actual: uncontrolled inputs extremely unusual** **Uncontrolled SDF > 1.2**	**If surprised, verify that process and data are OK.**
			>0.6 PAR(104)	Note	New Actual: uncontrolled inputs unusual Uncontrolled SDF > 0.6	Same as above
3	Run Data outputs in statistical control (values as expected, no surprises)	**Overall Standard Error**	**>3.5** PAR(107)	**if A(1) < 0.6** PAR(104) **ALARM**	**New Actual: outputs had unexpected values** **Overall Standard Error > 3.5**	**Standard SPC reaction to an out-of control condition: search for a special cause. If found, prevent for the future and delete data from ULTRAMAX (or correct errors)**
				if A(1) > 0.6 PAR(104) Warn	New Actual: outputs had unexpected values Overall Standard Error > 3.5	Raise your awareness about a possible special cause. This might happen, though, as you extrapolate into the new region towards the optimum.

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No	Condition	Metric	Threshold	ALERT Level	ALERT Text and Trigger	Possible Action Steps (see Advice Report for detailed analysis)
4	Run Data Inputs satisfaction of constraints	**Min slack / MID** (low and hi constraint among all inputs)	**< 0** PAR(111)	**ALARM**	**New Actual: inputs definite constraint violation** **slack < 0**	**This decision made by operating personnel was a violation of standard operating procedures. If this was an exception because of unusual circumstances, such as to compensate for a malfunction somewhere, then delete the data in the ULTRAMAX data base, as the data is not "typical". If this is perceived as "normal", then revise the Optimization Plan.**
			< 0	Warn	New Actual: inputs marginal constraint violation	Appears only if PAR(111)<0
			< 0.5	Note	None, but a '-' appears next to the value in the Advice report	
5	Run Data Outputs satisfaction of constraints	**If slack<0 Min slack / MID** (low and hi constraint among all inputs)	**< -1**	**ALARM**	**New Actual: output(s) definitely violated constraint(s).** **violation > MID**	**The product or process is not acceptable according to establish criteria. Proceed according to company procedures. If product is acceptable, then consider revising the Optimization Plan.**
			< 0	Note	New Actual: output(s) marginally violated constraint(s). slack < 0	If this happens too often (not measured by the software) you are susceptible to violating some output constraint.
		If slack>0 Sigmas protection	< 3 PAR(60)	Note	New Actual: output(s) near constraint(s) Within 3 sigma protection	

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No	Condition	Metric	Threshold	ALERT Level	ALERT Text and Trigger	Possible Action Steps (see Advice Report for detailed analysis)
Models and Projected Optimum Estimate (POE)						
6	Uncontrolled inputs location for models	**Uncontrolled SDF**	**> 1.2** PAR(106)	**ALARM**	**Predicted uncontrolled inputs extremely unusual. Uncontrolled SDF > 1.2**	**Pay careful attention about working in highly unexperienced conditions**
			> 0.6 PAR(104)	Warn	Predicted uncontrolled inputs unusual. Uncontrolled SDF > 0.6	Be alert that unusual conditions are present
7	Models up to date	**true (1) or false (0)**	= 0	Warn	Models: not up to date. Some ALERTS not displayed.	Either wait until the next optimization cycle (which will update the models), or: if using the Optimization Management Screen (OMS), get an Advice; if using Stand-alone GUI, do a Learn.
8	Present models suitable for fast optimization	**Critical Signal**	< 65% PAR(35)	Note	Models: weak for fast optimization Active signal < 65%	If frequent, a probable (but not definite) indicator of not being able to generate (or maintain) SEO improvements. Most meaningful with a lot of data.
9	Recently models suitable for fast optimization	**Average Critical Signal**	< 65% PAR(35)	Warn	Models: recently weak for quick optimization Average active signal < 65%	A probable indicator (but not definite) of not being able to generate (or maintain) SEO improvements. Most meaningful with a lot of data.
10	Operating around relative optimum	**SDF of POE**	> 0.8 PAR(105)	Note	Optimality: Projected Optimum Est. (POE) in unexperienced area SDF of POE > 0.9	There may be operating improvements in an unexperienced area (combination of adjustments) for the current values of the uncontrolled inputs.
11	Regularly operating around relative optimum	**Average SDF of POE**	> 0.6 PAR(104)	Note	Optimality: POE still not regularly around optimal region. Avg. SDF POE > 0.6	You are not close to the optimum for the current values of the uncontrolled inputs.
12	Optimum estimate inputs satisfaction of constraints	**Min Slack / MID**	**< 0**	**ALARM**	**Optimality: inputs violate constraint(s)** **slack < 0**	**This is an unexpected outcome from ULTRAMAX. Something important is amiss. Contact ULTRAMAX expert.**
			< 0.5 PAR(112)	Note	Optimality: inputs near constraint(s) Input slack < 0.5*MID	Be informed that optimal adjustments are likely to be close to an input constraint; you may be close to the edge. This is normal.

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No	Condition	Metric	Threshold	ALERT Level	ALERT Text and Trigger	Possible Action Steps (see Advice Report for detailed analysis)
13	Optimum estimate Outputs satisfaction of constraints	**Sigmas protection**	**< 0**	If A(11) < 0.4 PAR(103) **ALARM**	**Optimality: Unable to assure that all output means satisfy constraints** **Protection < 0 &** **Avg. SDF of POE < 0.4**	**Need to reconsider the viability of the process or the validity of the Optimization Plan.** **See Maintenance, Discovery and Improvements, page 65.**
				(else) Warn	Optimality: Trying to assure that all output means satisfy constraints Protection < 0 & Avg. SDF of POE > 0.4	Operations are still not acceptable according to establish criteria. Proceed according to company procedures. If product is acceptable, then perhaps the Optimization Plan should be revised.
			< 3 PAR(114)	If A(11) < 0.4 PAR(103) Warn	Optimality: Unable to assure that all output data will be within MID-beyond-constraints Protection < 3 sigmas & Avg. SDF of POE < 0.4	Same as a above
				(else) Note	Optimality: Trying to assure that all output data will be within MID-beyond-constraints Protection < 3 sigmas & Avg. SDF of POE > 0.4	Check the possibility that operations or the product is not acceptable according to current criteria.
			< 4 PAR(115)	Note	Optimality: output(s) near constraint(s) Protection < 4 sigmas	This is normal, just be aware of it, you may be close to the edge

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No	Condition	Metric	Threshold	ALERT Level	ALERT Text and Trigger	Possible Action Steps (see Advice Report for detailed analysis)
New Data / What-If Validity or Reliability (if there are multiple, the metric used is the worst among all of them)						
14	Data Validity or Reliability	**Sigma Deviation** from individual history. Larger is worse	**Dev > 6.0 & SDF < 0.6** PAR(104)	**ALARM**	**New Data / What-If: Some value is extremely unusual. See "Deviation" column. Max Deviation > 6.0 sigmas**	**Analysis done for each variable separately as compared with past values. New Data: Some data is very unusual. If it reflects normal operations that should be optimized, then continue; else either correct or delete as non-representative. What If: Either the inputs are unusual, or the expected values are unusual (and unreliable).**
			Dev > 4.0	Warn	New Data / What-If: Some value is unusual. See "Deviation" column. Max Deviation > 3.0 sigmas	Proceed with caution, there is no precedence operating with these values of run data.
Advice / What-If (possible new Adjustments) (if there are multiple, the metric used is the worst among all of them)						
15	Advice / What-If. Within experienced area	**SDF**	**> 1.2** PAR(106)	**ALARM**	**Advice / What-If: inputs extremely unusual, extrapolating SDF > 1.2**	**Proceed with extreme caution, there is no precedence operating with this combination of inputs.**
			> 0.6 PAR(104)	Warn	Advice / What-If: inputs in new area, extrapolating SDF > 0.6	Proceed with caution, there is no precedence working with this combination of inputs.
16	Advice / What-If. Inputs satisfaction of constraints	**Min Slack / MID** (relative slack)	**< 0** PAR(111)	**ALARM**	**Advice / What-If: input(s) definite constraint violation Violation >0**	**This is an unexpected outcome from ULTRAMAX. Something important is amiss. Contact ULTRAMAX expert.**
			< 0	Warn	Advice/What-If: input(s) marginal constraint violation Violation > 0	This is an unexpected outcome from ULTRAMAX. Something important is amiss. Contact ULTRAMAX expert. (Appears if PAR(111)<0).

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No	Condition	Metric	Threshold	ALERT Level	ALERT Text and Trigger	Possible Action Steps (see Advice Report for detailed analysis)
			< 0.5 PAR(112)	Note	Advice / What-If: inputs near constraint(s) Slack < 0.5 MID	This is normal, just be aware of it, you are close to the edge
17	Advice / What-If. Outputs satisfaction of constraints	**Min Sigma Protection**	**< 0**	**ALARM**	**Advice / What-If: output mean(s) constraint violation** **Protection < 0 sigmas**	**ULTRAMAX has not learned yet how to satisfy constraints. It could be that the process is not capable. Proceed with approval of management.**
			< 2 PAR(113)	Warn	Advice / What-If: output run data, significant risk will violate MID-beyond-constraint Protection < 2 sigmas	It is normal for some processes for ULTRAMAX to reduce protection in order to optimize faster. Proceed with care, perhaps with approval of management.
			< 3 PAR(114)	Warn	Advice / What-If: output run data, some risk will violate MID-beyond-constraint Protection < 3 sigmas	It is normal for some processes for ULTRAMAX to reduce protection in order to optimize faster. Proceed with care.
			< 4 PAR(115)	Note	Advice / What-If: output means near constraint(s) but low risk Protection < 4 sigmas	This is normal, just be aware of it, you are close to the edge
Miscellaneous						
19	Access to user supplied calculations	=0 CALC not accessed =1 CALC accessed or not needed	= 0	ALARM	Not accessing CALC equations.	Need to create or use the proper ULTRAMAX application environment. Contact ULTRAMAX expert. Note: This error does not appear when the incorrect CALC is used

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Expected Performance

As a very rough general rule, if the Advices are followed closely

- ULTRAMAX's SEO delivers **steady Improvements** after as many cycles as **four times the number of control inputs** (for each region of similar values of the uncontrolled inputs). This is called the "rule-of-four".
- ULTRAMAX's SEO approaches the **Relative Optimum** in as many cycles **ten times the number of control inputs** (for each region of similar values of the uncontrolled inputs[34]). This is called the "rule-of-ten".
- ULTRAMAX will get as close to the (local) relative optimum as about the noise level (applies to constraints and the Performance Metric). See page 154 for more details.

Both of the factors above depend on many different circumstances, so they are roughly representative only, not accurate.

Variations in SEO Cycles depending on Integration

ULTRAMAX can be used in various ways, as described below. Each affects somewhat how the SEO cycles are implemented; and the differences from the generic description above explained in the following sections:

- Stand-Alone – with manual data exchange with the process, or
- Integrated -- with automatic data exchange with the computer control system of the process. This reduces the chances of human errors, and reduces routine work for operators. The main user interface is the Optimization Management Screen (OUI) shown in page 42.

  "Full integration" means data exchange to and from the DCS and ULTRAMAX. "One-way integration" means data exchange only from DCS to ULTRAMAX (as it may be all that the DCS can support).

  An integrated installation can be used in these three modes:

  1. Manual supervisory control mode, where plant personnel types the desired adjustments and implements by pressing a button
  2. Advisory mode, where the Advice is displayed for plant personnel to approve or modify, and implements by pressing a button

     This is the only mode available mode for one-way integration, where the adjustments are implemented manually.
  3. Closed-loop mode, where the Advices are implemented automatically. Closed-loop optimization is turned on by plant personnel when they determine that the Advices offered by UMAX are consistently reliable (when there are no Alarms with the new data). Closed-loop will shift automatically to Advisory if it detects an undesirable upset, such a variables values wandering beyond acceptable ranges or an Alert which is an Alarm.

[34] These two factors depends significantly on how many inputs there are, how far the Baseline Adjustments are from the Optimum, and whether key calculated variables are calculated inside ULTRAMAX or externally to ULTRAMAX (much faster if calculated internally).

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Even a Closed-loop installation gets started with Manual mode to test and build confidence that the data exchange is as expected; and then goes through a few weeks of Advisory mode optimization until the end-user has built confidence that he/she does not need to make modifications to the Advise generated by SEO.

Stand Alone

In Stand-Alone the only user interface is the Stand-Alone GUI.

The Alerts and Advice is displayed in the **Advice Report**.

- With **the process**:
  - ⇒ Adjust the process manually on the console or physically in the equipment.
  - ⇒ Collect data, probably in a spread sheet to calculate summaries (averages, standard deviations, etc.)
- With **the software**:
  - ⇒ Click the Cycle icon: enter data (automatically update models), produce the Advice report.
  - ⇒ Look at the Alerts and process them as in Understanding what is going on and acting upon it, page 51.
  - ⇒ If accepting the Advice, print the report for the record, and as a means to inform the appropriate operators.
  - ⇒ If modifying the Advice, Do a What If with the desired adjustments. Print the report.

Full Integration in Manual Supervisory Control Mode

(SEO is not engaged, and no Alerts or Advice exist or are displayed)

- ⇒ In the “Adjustment” column plant personnel **types desired adjustments**.
- ⇒ Press the “**Adjust Process**” button to start the next run.
- ⇒ Press “**Start**” to just start collecting data. The “**Status**” section indicates when data collection is completed. This results in a process run and collection of data coming eventually into the “Current” and “Average” column.

Repeat the cycle above as desired.

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Full Integration in Advisory Mode

The Alerts are displayed in a section on top, and the Advice in the column “Advice / Predictions”.

⇒ Look at the Alerts and process them as in Understanding what is going on and acting upon it, page 51. To understand specifics about when failures may be occurring, press “Show Advice Report” to look at the details of the situation.

⇒ The Advice is accepted by pressing the “**Take Advice**” button, and the values move to the “Adjustment” column.

⇒ Optional: In the “Adjustment” column plant personnel **may type modifications** as desired. When completed, press the “**What If?**” button to get the corresponding Alerts and Predictions.

⇒ Press the “**Adjust Process**” button to start the next run

This results in a process run and collection of data coming into the “Current” and “Average” columns. The “**Status**” area indicates when data collection is completed.

⇒ Press the button “**Get Advice**” for the SEO module to create and display the Alerts and the Advice, to repeat the cycle.

Repeat the cycle above as desired.

When the end-user arrives at a state of confidence that the Advices are regularly suitable and do not require modifications, the operator can press the ”**Closed Loop**” button so that Advices are implemented automatically without waiting for review and acceptance.

Full Integration in Closed-Loop Mode

All the steps above take place automatically (without modifications of the Advice).

The SEO Alerts are displayed in a section on top, and the Advice in the column “Advice / Predictions”.

⇒ Look at the Alerts and process them as in ULTRAMAX® ALERTS and Guidelines page 55 below. To understand specifics about when failures may be occurring, press “Show Advice Report” to look at the details of the situation.

If an Alarm is detected (from the UED or the Alerts), Closed-loop mode reverts automatically into Advisory mode.

At any time the operator can press the ”Off” button to revert to Advisory re-adjustment optimization.

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One-way Integration, only Advisory Mode

The Alerts are displayed in a section on top, and the Advice in the column “Advice / Predictions”.

⇒ Look at the Alerts and process them as in ULTRAMAX® ALERTS and Guidelines page 55 below. To understand specifics about when failures may be occurring, press “Show Advice Report” to look at the details of the situation.

⇒ The Advice is accepted by pressing the “Take Advice” button, and the values move to the “Adjustment” column.

⇒ Optional: In the “Adjustment” column plant personnel **may type modifications** as desired. When completed, press the “**What If?**” button to get the corresponding Alerts and Predictions.

⇒ Press the “Print Adjustments” button, and (as in Stand Alone) make the adjustments manually.

When it is perceived that the process has passed the transient state, press the “**Start**” button to start data collection, displayed in the “Current” and “Average” columns. The “**Status**” section indicates when data collection is completed.

⇒ Press the “Get Advice” button for the SEO module to create and display the Alerts and the Advice, to repeat the cycle.

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Maintenance, Discovery and Improvements

In order to maintain production operations with optimization and to reduce barriers, Engineers / Managers engage in several activities, some reactive, some auditing, some proactive, such as:

- Make sure that new run data is valid and representative, which is the same as recognizing something going awry with the process – which is beginning of an engagement in Quality Control (Activity A, page 51).
- Evaluate whether the process is capable to satisfy some basic business requirements (the Capability Analysis). The first evaluation is done up-front with Baseline or regular production data (page 44); which is very valuable on its own right, even if the customer is not interested in optimization.
- Evaluate progress in getting improvements.
- Understand better the process and the environment, leading to Discovery.
- Search for opportunities for further process improvements by considering changes to production, including the physical components of the process, the procedures, and the objectives and goals.
- Maintain the optimization attitude and culture.
- Maintain the OP and integration configuration as the process is changed or re-engineered.

The Macro-Cycle of Improvements

The results from these families of activities, some described in more detail below, may result in

- discovering barriers and refined awareness of ideas and issues
- implementing economically justified improvements to the process and/or to the Optimization Plan.

These discoveries are greatly enhanced by the quantitative awareness of plant personnel when doing true quantitative optimization.

After the improvements a new round of production with SEO takes place. If the old run data is no longer be representative or complete for the new operation:

⇒ ignore the old data by setting PAR(1)=new data run number

⇒ start the new round of SEO with the best known adjustments. Start in “Advisory” mode until you re-establish confidence that the advices do not need modifications, and then set on Closed Loop optimization

which leads to new appraisals and the subsequent improvements.

Thus the **Macro-Cycle of Improvements** that production with optimization enables is, in summary:

a. optimize;

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b. analyze barriers (similar to constraint management);

c. reduce barriers by implementing economically justified changes – and return to "optimize".

This is the "big picture" of continual improvements with optimization.

The possibilities for improvements are limited only by the knowledge and imagination of plant personnel, and the drive of higher management.

Including the Macro Cycle **ULTRAMAX Optimization**[35] or **Advanced Process Management (APC)** is a journey of profitability increases without an end point.

Maintenance

Assure that:

- The OP does not conflict with other stated objectives and requirements, or with how people's performance is judged. There needs to be "**unity of objectives**".

  The interest and attention in Advanced Process Management (APM), and their quantitative awareness of the process and business, can be maintained by management in (at least) a few ways:

  ⇒ Reinforce that optimal operations is important for the business. Maintain the concept alive; be conversant with the ULTRAMAX concepts. Reinforce that better performance for the organization is as defined in the Optimization Plan[36].

  ⇒ Be attuned to the performance appraisals and any barriers encountered.

  ⇒ Provide the appropriate recognition when personnel are implementing ULTRAMAX well and creatively.

  ⇒ Keep up supporting process improvements measured in economic terms

  A particular high level of operations optimization maturity is arrived at when:

  ➢ Management requires that a process being commissioned already include Operation Optimization implemented and tested, as a natural extension of the Process Control and Supervisory Control Systems.

  ➢ Management requires that product/process be at optimum operations before allowing further capital expenditures beyond general maintenance and for new products.

- The Constraint values and the calculations of the Performance Index are kept updated with the dynamics of requirements, economics and priorities.

  In particular, the coefficients in the calculations of output variables, such as the cost of materials and revenues (when calculating profits) needs to be kept updated. It is very easy to maintain these values up-to-date by assigning them

---

[35] Where "true" optimization happens within each round of SEO, and the rest is just a sequence of improvements.

[36] If by any chance the evaluation of performance in the Optimization Plan is becoming irrelevant, then the Optimization Plan is due for updating.

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as Global Factors (see example in the table in the definition of Business Objectives in page 9, and the definition of GLOBALS and the equation used in the OP report), which can be changed easily from the Stand-Alone GUI. Upper management participates in maintaining them up to date.

Note in particular that **updating the Global Factors enables nimble, continual adaptation to the dynamic realities of business.**

- More drastically that the above adapting to gradual changes, is to recognize drastic changes in needs.

  This is a specific actual example of what was possible:

  > A photographic film company was testing a new manufacturing process to produce film. The initial Objectives were to minimize cost for a desired film speed while subject to the constraint of a maximum graininess specification. Within a few days after beginning the ULTRAMAX application, competition introduced a faster film at a low price. The company's response was to continue running ULTRAMAX with revised Objectives. Using the same physical data, the company reset the Ojectives in a few minutes to maximize film speed while satisfying the constraints of maximum competitive cost and maximum graininess.
  >
  > After satisfactorily meeting this objective with SEO, the organization re-focused on the original Objectives to refine the operating conditions for the standard product. Later they found the minimum cost for a specific faster film. All these subsequent analyses were performed with all the collected data and with minimal effort.

- If any changes are made to the process or to the variables in the optimization plan, assure that the OP document, the OP in the application file and the variables in the Configuration of integration in the UED are all updated and consistent with each other.

Appraisals / Discovery

Maintenance of the process, Discovery of problems or opportunities and consequent Improvements is driven mostly by the discipline of regular Appraisals.

There are three major levels of appraisal activities in process management and optimization, each with its own frequency and breath of outlook:

1. Cycle by Cycle: **Recognition of Problems** as they arise as part of the regular SEO cycles was already explained in Understanding what is going on and acting upon it.

   Implied in this activity is a relatively quick response to ALERTs, especially to facilitate discovering the sources of any upsets while they can be noticed or even remembered, such as recognizing an unusual behavior in the process or that a special maintenance event took place.

   The discovery of problems may trigger the need for deeper analysis such as #2 below.

2. ~ Weekly or as called for by #1 above: Perform a Quick Review to take a broader view looking at a few days or weeks of past data.

   See Section Quick Reviews below.

   This review is often done for personal awareness. Higher management may also

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require a brief weekly report from such appraisals to be assured of the good health of optimization.

The discovery of problems may trigger the need for deeper analysis such as #3 below.

3. ~ Quarterly or as called for from #2 above: More in-depth analysis resulting in filling a **Worksheet** and a **Progress Report,** perhaps every one or two quarters.

   See Section In-Depth Progress (Gain) Evaluation below.

As consequence of any of the above appraisals there will be occasional problems to be solved or other improvements to be implemented, as explained above in The Macro-Cycle of Improvements.

Quick Reviews

These are meant to be "quick" reviews, all done within a half-hour or so, if no problems are discovered. The only record generated may be a folder with the plots and software reports generated -- unless higher management requires a weekly summary as an audit of ULTRAMAX's health.

⇒ To do the following, copy the application file to another directory so that the original one can continue to be used during operations.

These are the four regular tasks, which could be modified if there is some specific information.

1. *Plot Key Indicators (vs. run number)*

⇒ Select 2D Graph icon > Basic button. This displays three key performance indicators vs. run number:

- Performance Index (actual data)
- SDF of the Projected Optimum Estimate (SDF-POE) (how close to optimum)
- The Overall Standard Error (to detect outlier data)

Plot of Performance Index

The first thing to notice is the satisfaction of constraints. If constraints are satisfied then values of the Performance Index are relevant. Else, note the frequency of important violations, hopefully getting smaller to indicate making Constraint gains or improvements.

Constraint Problems

The section Plotting Symbols, page 77 below explains in detail the meaning of the run data symbols. The symbol for each run data is the same regardless of the variable plotted, and represents the worse constraint violation among all variables.

Important constraint problems appear as a red diamond. Note that it is normal to have a large sequence of orange diamonds, especially when there are many outputs with active constraints.

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If there are constraint problems, then Perform Capability Analysis page 72 with the appropriate set of run data to see which variables have the problems.

The Capability Analysis, in turn, may lead to further appraisals. Here are two examples:

Example 1:

The Capability Analysis indicates that most of the constraint violations are in the inputs. Since ULTRAMAX never gives advice that violates the input constraints, this is a decision by plant personnel. There are two alternatives:

- Either the decisions were correct, in which case the OP needs revision.
- Or, they were a bad decision, which would be a symptom of lack of coordination between important parties, possibly including "lack of unity of purpose", and certainly "lack of unity in optimization scope". Managmenet has a role to play here.

Example 2:

The Capability Analysis indicates that an output has an average value inside its Optimization Plan constraints but outside its Effective Operating Area (see Fig. 25). You also see that the noise is high in comparison to the management defined MID.

- Plot the "current model std. error" or "current model error" for that output to observe any patterns of the error (residuals) that define the noise.
  - If the errors (residuals) are evenly distributed, then we have an "inherent" process characteristic that to improve is likely to require re-engineering. This will show few outliers.
  - If there are outliers concentrated in some few runs then we might be able to find an assignable (special) cause at that time. It would have been preferable to detect such as situation when the problem arose (if the Advice reports did detect the problem).

Then, if economical, you proceed with resolving the physical problem so as to prevent it in the future, namely, a regular Quality Control (or Quality Assurance) activity.

If you do not have a practical solution, then the MID or the constraint expectations from management are not realistic, and they need to be expanded to be consistent with reality.

Warning: In either of these two examples, if the situation is not remedied then you leave a conflict and contradiction in production that will disturb management and operators, and operators will not be able to take seriously the specifications laid on them. This is trouble with long-term negative consequences, a situation that understood and explained well by

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Dr. Deming. Also, this situation will be a sure contributor to the demise of serious optimization.

Performance Index Performance

The Performance Index becomes relevant only when there are no constraint problems.

If there are no important uncontrolled inputs (known or unknown), then the plot of the Performance Index is sufficient to understand the level of performance, and of the improvements as a function of time.

If there **are** important uncontrolled inputs (known or unknown), then the plot of the Performance Index is not sufficient evidence, as it reflects both the improvements from better adjustments (what is achieved by using ULTRAMAX) and the effect of the changing uncontrolled inputs (and the latter effects could easily be larger than the gains in performance).

To truly understand the improvements in performance due to better adjustments it takes special approach, as explained in Making Triads, page 50, with calculations in page 75.

Plot of SDF of the Projected Optimum Estimate (SDF-POE)

The Standard Distance Factor (SDF) of the Projected Optimum Estimate (POE) is **SDF-POE**.

Convergence to an optimum is a characterized by SDF-POE < 0.4 regularly.

Reminder: The SDF of a set of inputs is a multivariable measure to indicate the position of the inputs in relation to the Area of Confidence (AOC). The AOC is the area that contains the data used to build the prediction models, namely, the area when the model prediction errors are smallest. SDF = PAR(10) indicates the surface of the AOC. Smaller values of SDF indicate being inside of the AOC and having better prediction accuracy. Generally, the smallest possible value of SDF is around 0.2 to 0.3.

The larger the SDF-POE the more SEO is extrapolating; the smaller the more it is interpolating.

When there is a long sequence of SDF-POE < 0.4 this is an indication that ULTRAMAX regularly sees the Optimum Estimate inside the AOC and converging, and repetitive interpolations is an indication of reliability of the assessment. The assessment is true whether or not there are important uncontrolled inputs.

However note that you may fail the test above even though you already know how to optimize for some combinations of uncontrolled inputs, but not for all in that set of runs. This situation could be displayed more clearly if the software had the ability to select data only for similar values of uncontrolled inputs (this feature exists only for one kind of uncontrolled inputs, the component inputs that define the scenarios).

Plot of the Current Overall Standard Error (OSE)

The Overall Standard Error (OSE) for a run data is a single multivariable measure for the individual standard errors for all the outputs. It is defined in

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units of sigmas, where 3.5 sigmas is a practical the threshold into unusual values -- that is outliers, where some special cause is strongly suspected.

To find the individual variables with the outlier data one has to look at the Analysis > Reports > Current Prediction Errors, or plot the errors for all the outputs. Note that there may be a multivariate outlier with any no single output having an outlier; then it is a matter of an unusual combination of the errors.

There is an OSE also for the actual vs. predicted values of new run data in the Advice Report. The Alerts in the Advice Report issue an ALARM when the OSE > 3.5 (when interpolating). This is the ideal time to detect an outlier, as the actual variable(s) with problems can be seen by the individual output standard errors and also by the Deviation column (deviation from the recent historical distribution, an aspect of Data Validation).

As desirable, once evidence is obtained about unusual behavior, plant personnel will search (Discovery) for the cause an engage in Quality Assurance corrections so that future runs would be more reliable and consistent.

2. *Plot All Variables vs. Run #*

⇒ Select 2D Graph icon > "All variables" button > OK.

In addition to seeing quickly a picture of what has happened, this is a good exercise to discover unexpected patterns in the data, such as the variable having only two or three values when more diversity of was expected, extreme high or low values, etc.

⇒ Repeat, but before select Run Average = 10 or so, which will smooth out a lot of the randomness (noise) as display average behavior and trends more clearly.

Depending on your knowledge of first-principles, you might also wish to plot one variable vs. another, to detect expected, or unexpected, relationships.

3. *Review Historical Reports and Event Log*

The ULTRAMAX software stores all the reports it generates in the text file appname.hist.

⇒ Open the plain text file appname.hist with a text editor such as Notepad (or even Word).

⇒ Look at all reports since the last review

⇒ Search for any ALARMS, and make sure that the operating team proceeded (reacted) as directed by management as in the ALERTs table (see page 55). This is checked against the team's Event Log (the log itself is not currently supported by the UMAX software).

If you think that you are getting ALARMS too often when it is not called for, recall that you can tune up with parameters PAR for the thresholds that trigger ALARMS, Warnings and Notes. This is explained in detail in the Description of Alerts.

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More rarely it is possible to see the Usage Errors (if any) in the plain text file appname.err .

4. *Perform Capability Analysis*

The analysis is used to:

- continue validating that the process is capable of satisfying constraints and to evaluate and approve that the Performance Index losses due to noise are tolerable
- find out exactly where constraint violation problems may exists, to aid in problem resolution

The report is created as follows:

⇒ Select any desired filters to select type of run data

⇒ With the Stand-Alone GUI: Analysis > Reports > Consistency >select range of data to analyze

See the Section Capability & Consistency Analysis™ (CCA) below for further details.

This new analysis in UMAX 7.0 has proven to be extremely valuable. Used in old applications did identify problems which were known to exist but not which or where.

In-Depth Progress (Gain) Evaluation

The ULTRAMAX Method offers the following procedure for in-depth appraisal of process performance and improvements: start with the Worksheet, and based on that data, write the Progress Report, as explained below:

Worksheet

Create a worksheet for collecting relevant data and plots, status statements, and analysis of data beyond the software. This is done by following a template called “Sample Evaluation Worksheet.doc”. It describes step by step what to do (referring to the descriptions in this chapter) and it displays various sample status statements.

Technically, the worksheet alone is sufficient to document relevant information.

Progress Report

More formally, write the Progress Report, aimed at management. There is a template “Sample Progress Report.doc” with data, plots, and conclusions from the Worksheet.

Eventually there will be a collection of Progress Reports for each round of SEO.

Note: The Performance Index for any particular process is simply a mechanism to resolve multiple objectives and is not a measure or metric in the mind of the many people. Thus, it is better to emphasize the gains (and losses) of the Key Performance Indicators (KPI) which are familiar with the

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audience, such a throughput, quality, scrap, rework, unit costs, emissions, etc.

Assessing Performance and Improvements

Assessing Performance (in quantitative terms) is the most basic feedback loop for operators, plant personnel and upper management.

Assessing Improvements (Gains) over Baseline operations is done to evaluate the benefits obtained with optimization.

These assessments are a frequent component of the Progress Report.

There are two basic complications to keep in mind:

- The most important requirement for the firm is to satisfy constraints; and when satisfied, next is to improve the Performance Index.
- When there are important uncontrolled inputs, performance is a function of the Adjustments and of the Uncontrolled inputs.

Following is a discussion in more detail about these issues.

On Constraint Gains and Performance Index Gains

1. **Constraint Gains**: If constraints are not regularly satisfied, then improvements is to satisfy constraints more frequently – regardless of the value of the Performance Index.

   It happens often that one of the reasons people use ULTRAMAX is that with adjustments at Baseline constraints are not regularly satisfied, and ULTRAMAX is used to learn how to run a production / manufacturing process to satisfy constraints.

2. **Performance Index Gains**: After constraints are satisfied regularly, improvement is to improve the Performance Index – while maintaining constraint satisfaction.

Both can be easily seen by plotting the Performance Index.

On Calculating Gains

A. When **there are no important changes in uncontrolled inputs** (physical conditions), then just plotting the Performance Index displays both the frequency of constraint satisfaction and the value of the Performance Index.

   To make a statistical comparison between two sets of runs apply the calculations in page 209. It is possible that one of the groups have only one run (such as the first, which is usually a Baseline run).

B. When **there may be important changes in uncontrolled inputs** (physical conditions) -- known or unknown -- then we need to account for the changes in performance due to the changing conditions. This is best done with “Triad” sets of runs.

   A **Triad** is a set of three consecutive runs at equal intervals where:

   - the middle run is at Baseline adjustments

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- the uncontrolled inputs change slowly and gradually, as happens most often (such as the caloric content of coal in a boiler). If there are component inputs, the three runs need to have the same scenario

To be representative, the Triad should not be run shortly after startup or a changeover.

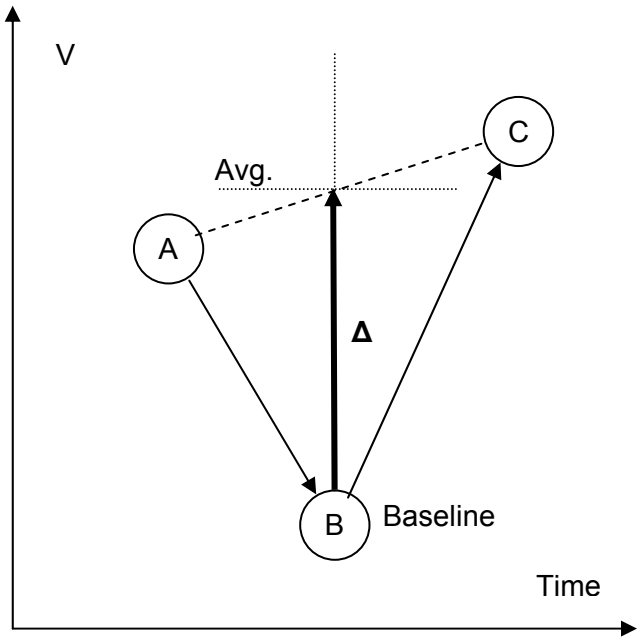
The procedure to make triads was explained in 50, above.

In more detail:

1st run (A) at optimal adjustments, get outcomes V_A

2nd run (B) at Baseline adjustments, get outcomes V_B

3rd run (C) at (new) optimal adjustments; get outcomes V_C



The improvement vs. Baseline (gain in performance index if constraints are satisfied) is the difference

$$\Delta = \frac{V_A + V_C}{2} - V_B$$

Note that the average of new performance -- the average of A and B -- nullify changes in V_A and V_B due to linear changes[37] in the uncontrolled inputs, making it comparable to the conditions in the center Baseline run. The effects that remain are due to the non-linear changes in conditions. The non-linear effects are reduced by making A and C as close to B as possible.

The variations due to unknown reasons in this calculation are: noise plus the non-linear changes in conditions.

[37] Change as a function of time.

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The unknown variation problem is reduced by making several triad tests. Then the improvements are the average difference together with its statistical significance (see calculations below).

By plotting the Baseline V_B (as a function of time) we see the effects of the changing conditions (known or unknown) on the outputs (for constant adjustments at Baseline).

If we could plot Δ (ULTRAMAX cannot) from successive triads for the same family of uncontrolled inputs, then one would see an asymptotic convergence to its optimum as SEO optimizes.

If we could plot the Δ of all successive triads we would get a combination of:

- the improvements for all conditions (families of uncontrolled inputs) as experience is accumulated for each
- converging to different optima as conditions change -- because the Baseline is closer to the optimum for some conditions than for others[38]

Thus, one should see an overall trend towards improvements in Δ until the optimum oscillates only due to changes in conditions.

If the effects of changing uncontrolled inputs are equivalent or larger than the improvements due to optimization, any "performance test" done over a short period will not be representative because we are unlikely to see sufficient variation in conditions. Such a test should be done over a sufficiently long period of time to cover all frequent condition families.

Calculations for triads

For each Triad, in a spreadsheet

- Enter V_A , V_B and V_C
- Calculate $\Delta = \frac{V_A + V_C}{2} - V_B$
- Average Δ and Standard Deviation of Δ
- $\text{Standard Deviation of Average } \Delta = \frac{Std.Dev.(\Delta)}{\sqrt{n}}$

$$Significance = \frac{\text{Average } \Delta}{\text{Standard Deviation of Average } \Delta}$$

We are positive that the average Δ gain is not due to unknown variations when ***significance > 3***, when we have over 3-sigma confidence that it is different than zero.

[38] Another reason for a change in the optimum gain is that the noise is likely to be different for different families and thus ULTRAMAX will approach the different ideal optima by different amounts.

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Various resources of useful data

1. Historical Data Report: Average and standard deviation of the last PAR(68) runs (moving average and standard deviation). PAR(68) also provide moving averages for plots.
2. Capability / Consistency Analysis: For a series of runs (with the option of selecting a filter), calculate:
    - The extent to which the runs have satisfied constraint requirements
    - For each variable:
        - The extent to which the data has satisfied constraint requirements
        - Mean, average, noise, loss in operating region
3. Signal Report:
    - Noise, Ratio of MID-to-Noise (ideally > 3), Spread, Signal, Avg. Signal
    - Which outputs are active in the Optimum Estimate
    - Whether interpolating or still extrapolating (SDF_POE)
4. Historical Achievement Report:

    This is a lengthy internal calculation but with inconclusive results:

    The Historical Achievement Analysis calculates how close each historical run was to the optimum estimate (know today based on all the data available) for the uncontrolled inputs in that run. The difference in Performance Index between Actual and Optimal Estimate is called the "potential gains" or "latent gains", the smaller the closer each actual data to its optimum.

    In the long run one would expect the potential gain to be reduced to near-optimal, which is about the noise level of the Performance Index. There are various factors that affect the potential gains for each sequential run, and thus reduction is not uniform even when making progress approaching the optima:

    - As totally unexperienced combinations of uncontrolled inputs are encountered for the first time we are likely to be further away from their optima, and potential gains could be larger.
    - For adjustments at Baseline (such as at the beginning of a round of optimization, or during triads) the potential gains will depend on the uncontrolled input values (Baseline adjustments are closer to the optimal for some uncontrolled inputs).
    - The noise, which affects how close we can be to the optimum, is likely to be different for different combination of values of the uncontrolled inputs.

    The analysis is an indicator only until the optima (for various combinations of uncontrolled inputs) has been achieved and known.

    The "Potential Gains" can be plotted.

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Sensitivity Analyses

1. After Optimization is achieved, **understand why now the process delivers higher performance** based on your knowledge of first principles, which will lead to better understanding on the key issues to manage the process, and to design even better processes in the future.
2. To further understand the process and effects on the business, perform **Sensitivity** analyses to changes in the **Optimization Plan**: e.g.
   - **Changes in Constraints**, to see what extra POE performance may be achieved from, e.g., enlarging a pump, or arriving at a different set of specifications with the customer.
   - **Changes in Global Factors** which represent costs, revenues, targets and tolerances, i.e., customer's expectations, to see if alternates can be mutually more profitable.

Plotting Symbols

Symbols to plot data, SDF, signal, etc.

Each point has a shape and color that represent the ***worst*** variable achievement in terms of how close the actual run value (regardless of what indicator is being plotted) is to constraints and to the Projected Optimum Estimate[39].

Remember that past data is evaluated with the **current** Optimization Plan.

The criterion for indicating points in the plot is described in the table below, where output "Sigma Protection" is defined as:

$$\text{PROTECTION} = \frac{\text{SLACK} + \text{MID}}{\text{SIGMA}}$$

namely, if the actual data were the output mean, how many sigmas from violating the constraint by exactly MID.

The 3 sigma limit is really PAR(60)=3 by default.

[39] It is a collapse or projection of all dimensions into one.

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GUI		CUI		
white or black	O	+	Window of Operations: **satisfies** constraints as "green" below, and the Objective Function value is **closer** than MID to the projected optimum estimate	**Within AOC** **SDF ≤ 1.0**
green	●	*	All inputs **satisfy** constraints by more than MID; and All outputs have **Protection > 3 sigmas**	
yellow	●	-	Some input **satisfies** constraints by less than MID; and/or Some output has **Protection < 3 sigmas**	
orange	♦	=	Some input or output **violates** constraints by less than MID	
red	♦	X	Some input or output **violates** constraint by more than MID	
blue	●	o	All variables **satisfy** constraints plus MID	**Outside AOC**
violet	●	v	Some variable **violates** constraints plus MID	

Symbols to plot errors and standard errors

green	●	*	Standard Error < 2 sigmas.	
yellow	●	-	Standard Error between 2 and 3 sigmas	
orange	●	=	Standard Error between 3 and 3.5 sigmas	
red	♦	X	Standard Error > 3.5 sigmas	

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Run Data Type

Note that the storage of run data keeps a record of the type of run when integrated, all but the Baseline from the Stand-Alone GUI. It can be seen in the Historical Data part of the GUI.

The following are also preceded by **ACT/** or **ADV/** to indicate Actual or Advice.

	Meaning	Description
POE	Projected Optimum Estimate	(close to) Optimum Estimate without including the lack of knowledge due to extrapolating (i.e., "projected")
AOE	Advice and Optimum Estimate	(close to) Advice which is an Optimum Estimate including the lack of knowledge due to extrapolating. The Safety may be less than 3-sigma when it is determined necessary to converge toward the optimum.
TLP	Travel Limited Path	(close to) Advice moving towards the AOE but limited by "travel" limitations (e.g., each control input cannot be changed by more than its MID).
Expl	Explore	(close to) Advice that includes perturbations in order to maintain information in the data for prediction quality and coverage of future models.
BasLn	Baseline	Run with adjustments at Baseline. This data type is determined by the operator when the data is entered in the Integration environment, it is not determined by the software.
MISC	Miscellaneous	A run made by a different criterion than any of the above, such as decided by plant personnel

Note that the data storage, in addition to the actual run data, also stores the updated POE and the Advice.

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Part 3: EXTENSIONS

Parts 1 and 2 described the standard procedures in the application of the ULTRAMAX Method. Part 3 describes special approaches suitable for particular applications or situations.

ULTRAMAX in Product/Process Development and Design

More Productivity in Design organizations is to produce faster and better designs at lower costs.

A **design** consists of two input elements: **a structure** and **its quantitative parameters**. Thus, in an I beam the "I" is a structure (a square tube would be another structure), and the dimensions and characteristics of the steel are the quantitative elements. The structure is typically a "categorical" factor, while the quantitative elements are continuous or nearly-so. The same concepts would apply to the design of an engine block, a TV tower, a fan, a hull, a detergent, a new pie, a pigment, etc.

Designs are evaluated through tests, some physical, some through simulation when there is good understanding of how nature operates. Tests are done both to validate the expectations of the designers, and to learn more objectively what the consequences from alternate designs are. The designer makes tests both to refine the structure, and given a structure, to refine its parameters.

The traditional discipline to test designs is Design of Experiments (DOE). For instance, Factorial DOE can be used to test several designs, especially, several structures.

When the designer gets to the stage of having converged to the desired structure, then there is a series of tests to determine the best parameters. When "best" is defined in quantitative, measurable terms, this is called (serious) optimization.

It is possible to optimize parameters with DOE. However, when the tests need to be run sequentially, e.g., they are run on a piece of equipment that can run one (or a few) tests at a time, DOE approach requires many more tests (more expensive and time consuming) that optimization-oriented technology such as Sequential Empirical Optimization (SEO). The main reason SEO delivers benefits in fewer tests is that each test is designed using the data already collected from previous tests; as compared to the practice of DOE of waiting until a set of tests are completed before analyzing the data to extract the useful information.

In addition, optimization solutions such as ULTRAMAX® also enable running a sequence of simulations automatically, thus relieving design engineer's time and attention. For instance, the sequence can be run overnight or during a weekend. More details about simulations in a section below.

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The ULTRAMAX Method offers several benefits when started earlier in the life of a product, in Product/Process Development rather than on the production floor.

1. The first family of benefits is conceptual, inducing much earlier a more complete awareness of the whole criteria for product/process design and production. This results in laying the groundwork for both
    - ... optimizing a new production unit before it is commissioned. This leads to a smoother and faster start-up.
    - ... commissioning a new product/process which includes tested conditions which set the stage for continuous improvements and quick adaptation to changing conditions and objectives. This leads to improvements in profitability performance from the very beginning.
2. The second family of benefits deals with the effectiveness of using SEO optimization in laboratory, pilot plant and full-scale tests, and in the analysis of the data so obtained. The benefits are:
    - Since it is so simple to address multiple objectives with ULTRAMAX, it encourages the researchers to think up front about **all** aspects of the product/process which are potentially important, such as product performance, materials' cost, producibility and throughput, robustness, safety/reliability, and various aspects of customer satisfaction as measured through market research tests.
    - When the tests have to be run sequentially, e.g., when using a single piece of equipment (i.e., with only one set of adjustments at a time), ULTRAMAX saves significant amount of development time. In this environment ULTRAMAX converges to the optimum in a fraction of the tests required by alternate methods. This effectiveness results in finding out sooner whether or not the current conceptual elements are adequate for the business needs, and thus:
        - a product goes to market sooner; or
        - you become aware sooner that some aspect of the product concept or structure needs revision, and avoid the delay of trying to tune product concepts which are incapable of meeting needs
    - ULTRAMAX offers a practical solution to take models from one level of research (e.g., pilot plant) and use them to guide the optimization at the next level (e.g., full scale operations), without depending on the models being complete. This streamlines further the tuning up and optimizing of processes. (See, Prior Models, page 95.)

ULTRAMAX with simulators

ULTRAMAX offers a time-saving method to achieve better design optimization when designs are evaluated through simulators. In this environment the design engineer...

1. Creates (codifies) a conceptual design in the simulator. This design has categorical *conceptual structure* elements such as shapes, materials, type of process; and quantitative *design parameters* such as dimensions, ingredient mix, and if it includes a process, running conditions such as temperature, speeds, pressures, etc. The function of the simulator is to predict design suitability.

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2. Defines design objectives based on the results predicted by the simulator. The Goals are represented by a *measure to be maximized or minimized*. This may be a simple simulated result, or it may include calculations to balance various objectives related to technical and economical issues. *Constraints* can be used on design parameters and/or on simulation result variables, to represent performance, capacity, safety and specifications.
3. Selects some *design parameters* to be optimized (while the others remain constant). In slightly more sophisticated applications, families of design parameters may be set with rules, where the optimization parameters may include coefficients and thresholds of the rules.
4. Starts the integrated, automatic computer parameter optimization cycles whereby:
   - ULTRAMAX indicates combinations of *design parameters* to be use in the simulation, and
   - the simulator provides the consequent simulated *output values*

   which is automated when there is data exchange between the two softwares. The cycle is repeated as long as desirable, or until the practical optimum has been achieved. Unless the simulator already includes an equivalently capable optimizer, this not only saves significant amount of time of the design engineer as compared to doing what-if simulations by hand.
5. Evaluates the logical quantitative consequences of the *categorical structural elements* and the *design parameters*. This information is used to determine that either satisfaction has been achieved, or to improve the conceptual/ structural elements of the design, and repeat the parameter optimization cycles.

The benefit of using ULTRAMAX for design functions is to reduce time by:

- Allowing the engineer to arrive at satisfactory results sooner with earlier conceptual structures, which may be impossible to optimize with time-consuming simulation trials or other less efficient technology/methods.
- Realizing sooner that an optimal realization of a conceptual structure is insufficient to satisfy results. This will lead sooner to revise the conceptual structure rather than spending time trying to tune incapable conceptual structures.

Examples

1. **Aircraft Engine Structural Part**

   The goals were to design the minimum weight structural ring that would not fail due to fatigue. Some of the design parameters were the flexible dimensions which did not have to match other important parts. Alternate design parameters were evaluated with a finite element analysis program which calculates the maximum stress for a set of force and thermal loads, and the volume of the part. With ULTRAMAX the designer achieved a reduction in weight of 5% beyond what had been possible earlier, while maintaining the maximum stress below an upper safety limit, i.e., assuring fatigue life expectancy.

2. **Electric Motor**

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The goals were to design an electric motor with certain torque vs. RPM characteristics suitable to aircraft operating at various altitudes. Some of the design parameters were the wire gage, the number of turns, and the capacitance. Alternate design parameters were evaluated by a computer program which calculated the motor performance characteristics and the manufacturing cost. With ULTRAMAX, the designer was able to find a design that fully satisfied requirements and at a competitive cost in **one day** rather than the normal five days.

3. Electronic Circuit

The goals were to design a high-frequency Field Effect Transistor (FET) circuit to provide large and uniform gain in a particular frequency band, with low input and output reflection coefficients, and insensitive to the normal variations induced by the production environment (i.e., robustness). Some of the design parameters were thickness and distances between conductors in the FET chip circuit. Alternate design parameters were evaluated by a commercially available simulation program, which included limited optimization abilities. With ULTRAMAX the designer was able to obtain results at least 10% better than with the program's optimization, an improvement deemed significant.

ULTRAMAX was set up closed-looped with the design evaluation program, and the cycles of exploring alternate design parameters were done overnight, without user intervention. In the morning, upon seeing the logical optimum consequences of the design concept, the design engineer changed some fundamental structural elements repeated the optimization, and in three days achieved full satisfaction.

4. Plastic Injection Molding

The goals were to design a part and the mold so as to produce parts with tolerable internal stresses, namely, dimensional stability. Most of the design parameters were dimensions and process control inputs such as injection temperature, pressure, and hold and cooling times. Alternate design parameters were evaluated by a commercially available simulation program. The evaluation included cooling rates and remaining stresses in the part. The results were 15% better than a Taguchi DOE, and the final results were 25% better.

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Noise Sources and Reduction

This provides guidelines as to what are likely sources of noise and solution methods.

Noise is defined as the variations that occur in output run data (including summaries and calculated variables) for constant values of the inputs in the Optimization Plan, namely, caused by unknown or unmeasured causes.

Recall that output variation due to known changes in uncontrolled inputs from run to run[40] is **not** noise; they are understood by ULTRAMAX, and properly managed.

The Noise is measured as the standard deviation of the unexplained variations in the outputs.

Noise is illustrated in the first 10 runs in Fig. 2. Improvement Profile for the Example, page 14., obtained for the Baseline Capability Analysis[41].

Noise reduces the effectiveness of any empirical solution because it: (1) tends to obscure or hide the true underlying behavior of the process, and (2) requires more safety to avoid constraint violations. In particular it is a fundamental barrier to achieving further process improvements.

The noise is too high when:

- The Noise is larger than 1/3rd the MID. This problem is sometimes resolved by collecting more data and giving ULTRAMAX the opportunity to have better fitting model, including locally accurate models.
- The *Signal* is smaller than 50%. It represents a ratio of the unexplained variance in outputs to the total variance of the outputs[42]. There is some correlation between Signal and improvements with SEO, but not sufficiently high to pay much attention to the Signal.

The typical sources of noise are:

- **unknown variations in materials** (e.g., purity, dimensions, hardness, viscosity, color, moisture, temperature, cleanliness, amount of rework)
- **unknown variations in the environment** (e.g., ambient temperature, humidity)
- **variations in the process itself** (e.g., imperfection of first level control devices, vibrations, material buildup, heat buildup, inconsistent implementation of procedures)
- **errors in measurement** (e.g., poor or uncalibrated instruments, inconsistent measurement procedures)
- **using invalid, unrepresentative data** (e.g., because the data is old and the process has changed, because it is in the transient due to readjusting control inputs, or using outliers due to special causes outside the optimization procedures)
- **errors in entering data** into ULTRAMAX software

---

[40] More precisely, changes in the average values of the uncontrolled inputs from run to run. Changes of the values of the uncontrolled inputs within a run are a source of noise.

[41] The analysis really requires much more data.

[42] It is the R-Square adjusted by the degrees of freedom

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- **not deleting data which was obtained under abnormal conditions** not represented by known uncontrolled inputs

Except when measuring noise with real replicates (constant inputs, in which case noise is just the variation of the outputs), then a contribution to the residuals (which determine the "noise") is the "lack of fit" of the models used in the technology. This is particularly noticeable when forcing descriptive models (using all data equally, like in regular regression analysis and neural networks) rather than the usual "local models" as in the ULTRAMAX technology.

Noise reduction with ULTRAMAX

The most obvious way is the detection of upsets. Since the inputs change from run to run, the common cause deviations are those from the values predicted by the models. This is called "Regression or Generalized SPC".

All these approaches are implemented by modifying the Optimization Plan (OP).

- Noise from inputs not in the OP are resolved by including those uncontrolled inputs – or something highly correlated to it – in the OP, and measuring its value during operation. This is a very common occurrence.
- Common cause or inherent variability[43] can be reduced by: (1) measuring it – usually a standard deviation or CPK; and (2) including in the objectives its reduction (such as through adding the proper performance loss function or placing constraints on it).
- Reorganize the process control strategies by identifying adjusted inputs which could be determined better after knowing the value of intermediate variables (see Intermediate Variables as Inputs, page 87).

Noise reduction with standard methods

There are several disciplines whose main objective is to increase consistency or reduce variability by changing the process or procedures. Among the: All Quality Control, Dr. Deming, Dr. Juran, Dr. Taguchi, Six-Sigma, and then, fundamental good engineering.

One aspect in common with several of these solutions is to do within-Run SPC analysis. Note that it must be self-contained within each run. There cannot be references to raw data from previous runs, since that data could be significantly different because the inputs were different. This SPC is not supported by ULTRAMAX.

[43] when it depends on the values of the inputs, as it usually does

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Intermediate Variables as Inputs

Introduction

Les us regain clarity on a couple of basic issues:

1. It has been established that for an effective OP the control inputs need to be independent of each other. "Independence" means that in adjusting one input there is not any mechanism that will automatically change some other input. Namely, all control inputs can be adjusted independently of the others (regardless of the effect it may have on performance).

   If this were not so, then it would be impossible to implement sets of adjustments, which is the main product from the ULTRAMAX software.

2. Another perspective is that we need models that remain valid when the user intervenes in the process being optimized. "Intervention" means to add a new mechanism into the process, such as readjusting inputs to get improvements in performance.

   Empirical models fitted to data reflect the repetitive mechanisms that created the data[44]. The empirical models remain useful as long as the mechanisms remain essentially the same. Thus, for just an observer, any combination of inputs and outputs will work fine. The quality of the model is reflected by the noise in the predictions.

   However, Process Optimization implies being more than an observer, but intervening on the process. Thus, for the prediction models to be useful, they need to remain valid even when the user intervenes, when there is a new mechanism being applied.

These models are called "cause-and-effect". One way -- probably the only one -- to assure that such models are "cause-and-effect" is when we there are no mechanisms tying the control inputs. Then the only mechanisms left are the process itself after the inputs.

By comparison, prediction models when there are mechanisms tying the inputs are sometimes called "correlation" models, which are valid only for observers, such as for predicting unknown variable values.

When we can assure that there are no mechanisms interrelating the control inputs then we have both desired attributes:

- The inputs are independent.
- The models are "cause-and-effect", and thus valid for optimization.

Note that the above indicates that uncontrolled inputs should be independent of the control inputs; however, the uncontrolled inputs can be interdependent among themselves since there only role is to predict for an observer.

[44] Without necessarily knowing what the mechanisms are.

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Note that recycle variables[45] typically are not independent of control input, especially of their values some time ago. When there are no alternatives we treat them as uncontrolled inputs, but it is better to define the process boundaries so that they are internal variables.

Let us consider a process with inputs and outputs such as in Fig. 8, where A and C are sets of variables in the OP. "A" includes all the inputs (Control and Uncontrolled). "a" is the unknown source of noise. This matches the standard Decision I/O Diagram such as in Fig. 1 page 11.

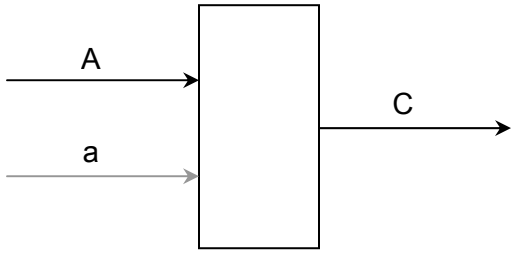


Fig. 8. Standard view of a System

In particular we shall be splitting the process into sub-systems so that some of the intermediate variables appear as Control or Uncontrolled inputs in the sub-system Optimization Plans. There are two potential advantages to using Intermediate variables as inputs:

- **Noise isolation/compensation**: avoid the effects of noise sources which affect only one intermediate variable by compensating for the noise up to this point.
- **Fewer inputs**: avoid the complication of the many ways the same value on an intermediate variable can be controlled. Further, some intermediate variables may have fundamental properties which are worth exploiting and understanding, e.g., a ratio whose optimal value ought to remain constant under a great variety of situations.
- **More effective use of the run data**: Take advantage of known values of intermediate variables for the implementation of the best adjustments made afterwards.

These benefits can be obtained with sufficient prior knowledge and care, as explained in three Sections, where each can be read independently of the others:

1. Noise Isolation and Fewer Inputs,
2. Serial Cascading Control,
3. Selection of Intermediate variables to control.

[45] Recycle flows from process by-products.

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1. Noise Isolation & Fewer Inputs

The main objective in what follows is to isolate the effects of the variables into subsystems which are easier to manage or ignore.

Let us imagine that we can conceptually (not necessarily physically) split the system into blocks such as in Fig. 9. The A set of inputs is separated into subsets {A1,a1} and {A2,a2} and corresponding intermediate variables {B1,b1} and {B2,b2} which **independently** carry all the effects from A to C.

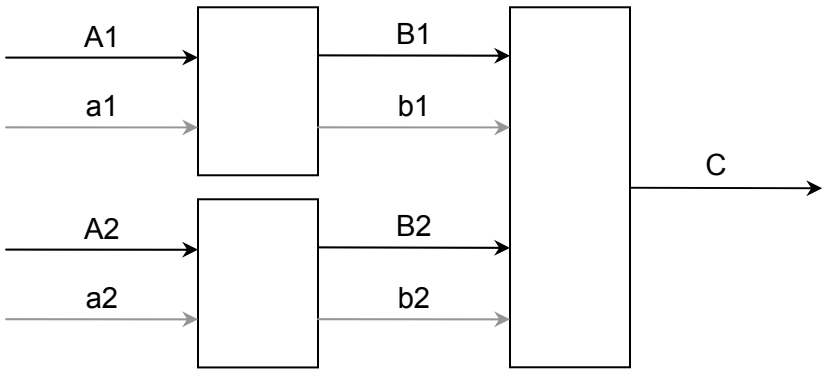


Fig. 9. Independent sets of Intermediate variables

Let us assume further that:

- Unknown effects b1 are minimal (this, and independence between 1 and 2 are critical assumptions)
- The a1 noise source has some permanence. For instance, it would make sense to change certain control inputs in A1 so as to compensate for the unknown a1 so that B1 achieves a desired target value (e.g., change a valve position in A1 so that a flow in a flow meter in B1 has a desired value as the upstream pressure changes with time). *Further, this compensation can be done rather quickly, in a fraction of the Run Time*.
- The B1 could be conceptual variables, such as energy delivered, which as such would be calculated intermediate.

B1 with only one variable

For the time being let B1 be a single variable.

In a situation like in Fig. 9, an approach to manage the process could be as follows: let B1 become:

- a target value to be reached by adjusting A1 (making B1 a controlled variable for the A $\rightarrow$ B subsystem; and
- an ULTRAMAX Control input for the rest of the system as in the Decision Input/Output Diagram in Fig. 10.

Note that in this section the intermediate variables used as an input are of type "Control",

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while in Serial Cascade in the next section the intermediate variables used as inputs are "Uncontrolled".

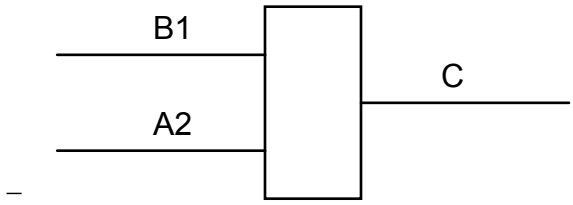


Fig. 10. Decision Input/Output Diagram with Intermediate variable B1 as Control Input

In doing this we achieve two main benefits:

1. We exchange ***all*** the variables in A1 for ***one*** variable B1.
2. We remove most of the concerns due to (any) noise a1.

To achieve this we need to:

- Identify the appropriate B1. Often it is a recognized critical variable for running the process (e.g., a critical temperature). However, sometimes there is room for conceptual breakthroughs, as in the following example:

  A1 consists of *Temperature* and *Time*, and we decide, based on knowledge of first principles, that *Energy* = Δ *Temp* * *Time* is the real driver of *Yield (Lb._out / Lb._in)*. Then, B1 will be *Energy*, each value obtainable by many combinations of *Temperature* and *Time.* For further ideas see Section below 3. Selection of Intermediate variables to control.

- If a1 is important and the compensation adjustments of A1 are done by hand, make sure that the procedures are consistent and executed at the correct frequency. If the compensation adjustments are done with first-level controller, then the setpoint B1 would be a typical primitive variable and the operation of the A$\rightarrow$B subsystem would be taken just as an internal mechanism of the system.

Recall that this makes sense when we know that A1 does not have effects on B2. A problem could be, for example, that either:

- *Temperature* has a side affect on the clarity of the product, which is a quality objective in C to be managed.
- *Time* has an important side effect on *Production rate* (Lb./hr.), which is an objective in C to be managed to meet demand.

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B1 with multiple variables

These principles apply just as well when there are more than one Intermediate variable like B1. Let us assume that there are four single, intermediate, independent variables B1 to B4 as in Fig. 11 where A1 to A4 are each sets of variables.

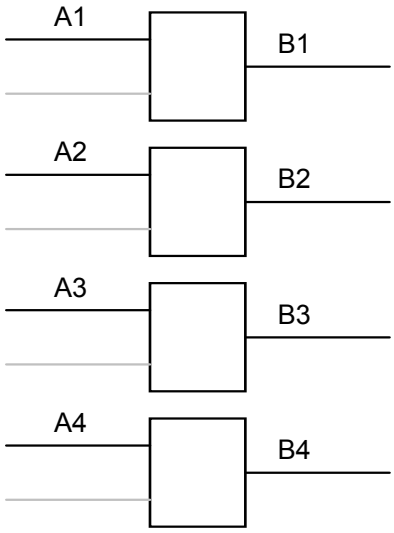


Fig. 11. Several Independent Bs.

Here there are no cross-effects between individual As (or their noise) and the other Bs. Then each A is adjusted so that its corresponding B achieves the desired value, and readjusted independently as frequently as necessary (e.g., to compensate for its noise source, remember, assuming quick feedback).

A variation of this that is still manageable is the **parallel cascade** as in Fig. 12.

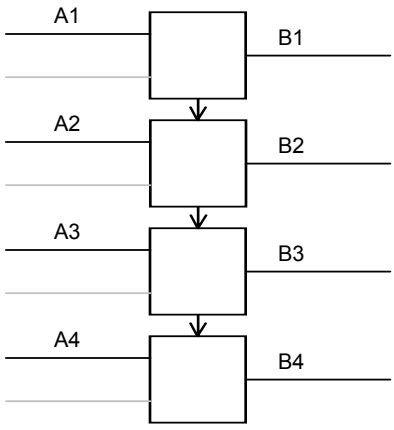


Fig. 12. Parallel Cascading effects

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Here the strategy is to Adjust A1 first; then, in order, A2, A3 and A4, so as to achieve the desired values of the corresponding Bs; and repeat the whole sequence as often as necessary. In this case if a2 is important, changing and unknown, this is a source of noise affecting only variables B2, B3 and B4, and thus adjustments A2, A3 and A4.

Note: "Parallel" cascading is different from "serial" (as described in the following section) in that in parallel cascading the effects of adjusting A are relatively quick and without analytical support; while in the "serial", a full operation needs to be completed before the effects become evident, and it uses variations of the Decision Input/Output Diagram to give Advice for the control inputs.

2. Serial Cascading Optimization

Let us consider three consecutive production steps, each step with control inputs requiring adjustments decisions such as in Fig. 13. Here D and E would be Intermediate variables when looking at everything as one process as in Fig. 14. Imagine these examples: three consecutive units in continuous processes, or three consecutive stages in a batch process. The dotted (gray) arrows are unknown effects which may manifest themselves in subsequent production steps.

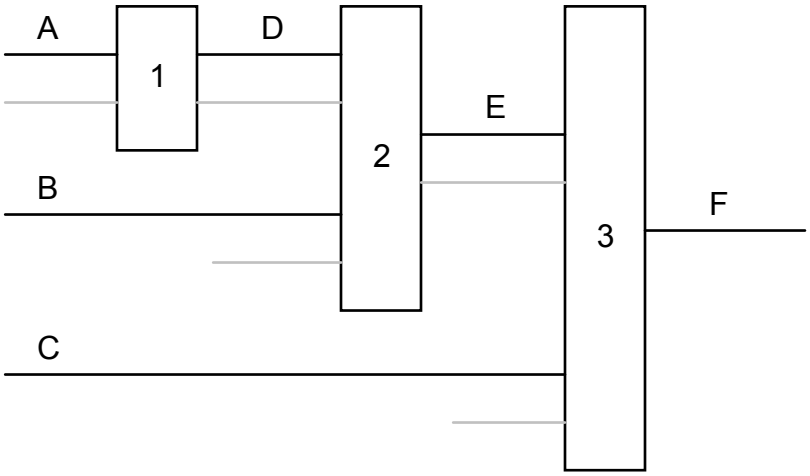


Fig. 13. Three consecutive operations

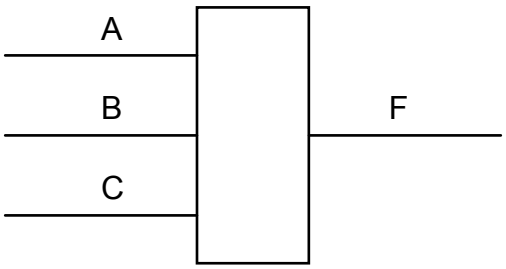


Fig. 14. Simplest ULTRAMAX Decision Input/Output Diagram

The standard simpler solution is to represent the application to ULTRAMAX as in Fig. 14, ignoring (any) knowledge of D, E. Thus, with this simple solution we give up an opportunity, e.g., to compensate for the A noise effects that manifest in D (and E). A particular advantage of Serial Cascading Optimization is to make a final adjustment

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decision on the value of B after we observe the value of D; and similarly with C after observing the values of E.

If we wish to take advantage of the values of D and E then we would use three simultaneous Optimization Plans (OP#1, OP#2 and OP#3, each with its own application file), as in Fig. 15, all with the same identical data. Variables D and E are here uncontrolled inputs.

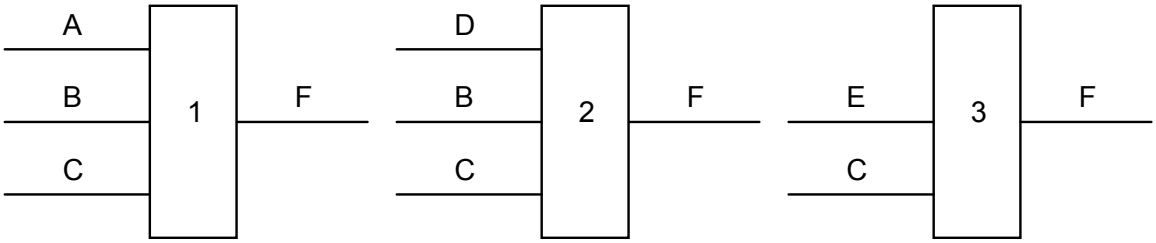


Fig. 15. Serial Cascading Optimization with three Optimization Plans

Then, the procedure (most obvious for a batch operation) would be:

1. Define new adjustments A (and B and C) with OP#1. Run step 1 with A adjustments to get the output value of D.
2. Define new adjustments B (and C) with OP#2, now with D as uncontrolled inputs. Run step 2 with B adjustments to get the output value of E.
3. Define new adjustments C with OP#3, now with E (and perhaps D) as uncontrolled inputs. Run step 3 with C adjustments to get F.

All applications share the same data and variables. What is different is the role the variables play for each **Input/Output Diagram**, as in this chart:

	OP#1	OP#2	OP#3
A	control	unused	unused
B	control	control	unused
C	control	control	control
D	unused or output	uncontrolled	Unused (or uncont.)
E	unused or output	unused or output	uncontrolled
F	ouput	ouput	ouput

Note that the hidden effects "d" would be a source of noise for F (and E); and to the extent that “d” is produced by A OP#1 may pick up this relationship. This is not like Fig. 10 where we are separating the A1→B1 sub-system in Fig. 9 from the ULTRAMAX solution.

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It is possible to simplify OP#1 by not including all of the variables in B, as follows:

- If some Bs are used mostly to compensate for the A noise effects on the values of D, then it is not valuable to include them in OP#1, as the A noise values (by definition) are not known, and therefore B cannot be predetermined in OP#1.
- However, any Bs which have a significant impact on E (and F) outside of compensating for the A noise have to be included in OP#1.

Note: The difference between the two situations above could be determined if a questionable B variable is included in OP#1 and OP#2, and seeing its detected effects (when the weights of the data in the models are the same[46]). If not significant in OP#1 but yes in OP#2, this is a sign of the first condition; while if significant in OP#1 ***and*** in OP#2 this is a sign of the second condition.

Similarly it is possible to simplify the analysis by not including some of the Cs in OP#1 and OP#2.

3. Selection of Intermediate variables to control

Dr. Taguchi teaches that desirable design inputs should not interact, and that the people designing the experiments ought to know enough about engineering to identify "fundamental" variables which may have independent effects. This is not easy to do, even when having sufficient knowledge. However, the principle is applicable for identifying practical intermediate variables which may be only conceptual (calculated). A previous example was used to illustrate this concept for the intermediate variable *Energy* as a function of *Temperature* and *Time*.

Prof. Emanuel Sachs of MIT, Department of Mechanical Engineering, proposed that one approach to identify fundamental variables may be through dimensional analysis. One example is the Reynolds number, a dimensionless number which determines whether flow in a pipe is laminar or turbulent. The Reynolds Number may have important basic effects such as resistance to flow, heat transfer, and possible rate of chemical reactions. Another approach he recommends is to find fundamental dimensionless variables with the use of the Buckingham Pi theorem (Bridgman, 1931). Ms. Michele Storm master's thesis (April 1989) at MIT, with Prof. Sachs, gathers evidence to support this view. A paper helicopter design concept was first sequentially optimized based on primitive inputs, namely, the actual dimensions. Then it was optimized based on fundamental dimensionless variables developed with the Buckingham Pi theorem which turned four primitive inputs into two fundamental intermediates. The later approach made ULTRAMAX's SEO converge to the optimum in fewer cycles than the former.

These approaches are suited for SEO because they help converge faster due to less noise and fewer variables. A secondary benefit is that they also make the team acquire more incisive insights to guide future process/product development.

[46] The only way today to assure having the same weights is to have descriptive models, with PAR(45)=2

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Prior Models

Introduction

A Prior Model (PM) is a model represented by computer code to predict various outputs as a function of various inputs.

Introducing Prior Models (PM) in the Optimization Plan provides prior knowledge which enables the SEO to generate a more valuable, higher performance sequence of runs early on in the SEO Cycles. This feature requires using the DLL calculations.

A PM does not have to be too accurate nor include all the variables to be useful.

In most cases just a linear model would be sufficient. The most important role is to provide a rough idea of the gradient for output variables. Bias errors (a constant displacement in the output) are irrelevant as the models will be calibrated after the first run data is entered[47].

A simple PM is also desirable for quicker computations. A PM will be probed hundreds of times when generating an Advice.

Of course, if a PM is distinctly incorrect, this will slow down getting performance improvements, as ULTRAMAX will go in the wrong direction and get inferior performance before it can unlearn the knowledge represented by such a PM.

The models may be obtained from:

- The same process. This is likely to happen when:
  - there is a major change in the process, and thus the old data is no longer as valid (but the models would be useful because the change is not that great).
  - there is a change in the Optimization Plan that includes with new inputs for which there is no historical data.
- Other similar processes
- Pilot Plants (scale up)
- Laboratory tests (scale up)

Prior Models can be created by:

- A previous application of ULTRAMAX
- Regression analysis with empirical data
- Fitting data from a Factorial DOE
- Representation of first principles

Note that introducing prior models requires tripling the number of output variables.

[47] By "Calibration" we mean that the prediction value will be the Prior Model plus the Correction Model, which is a function of the difference between the Actual Data and the Prior Model. For instance, right after the first data set is entered the estimate for the inputs in the first data set will be exactly the value of the past data itself.

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Procedures

The only variables that can use or accept Prior Models (PM) are Dependent variables (types 5,6,7,8) that are not Calculated (C)[48].

Each dependent variable with a PM requires (see example below):

- three variables in the Decision Input/Output Diagram of the Optimization Plan:
  1. The normal variable with the **Actual data**, of entry mode 'H' or 'A'. Expected values in What-If and Synthesis equal to the Prior Model (provided by the user) plus the Correction Model (internal model based on stored data).
  2. A corresponding **PM variable**, of mode 'C', whose values are defined like regular calculated variables. It holds the values of the Prior Models as a function of the run inputs, and the expected values in What-if and Synthesis are the Prior Models themselves.
  3. A corresponding **Correction variable**, of mode 'D' (Delta). It automatically holds the value of the Correction = Actual - PM. Expected values for the Correction are the internal regression model based on this data. The correction models are used to calculate the expected "Actual" value as described in No. 1 above.
- in the subroutine CALC in the DLL form of user-supplied calculations, enter this code in this order for each variable with a PM:
  1. Type the equations for the PM variables just as for regular calculated (C) variables. Each has to be an equation of Inputs (control and uncontrolled, of types 1 and 2), and may be an equation of other PM variables, but not of any other kind of variables. These rules are such so that the PM values can be calculable even without any data.
  2. Then define the interrelationship between the three corresponding variables by calling "Include the Prior Model" as follows (arguments in this order: actual, prior model, correction):

     ```
     CALL IPM(VALUE(11), VALUE(16), VALUE(19))
     ```

  3. Then, enter any equations for regular calculated variable, using the "actual" value for those variables with PM.

  Prior Models, as any other user-given equations, can be provided to CALC through function or subroutine calls.

[48] Otherwise the Prior Models are irrelevant, and further, the calculations would overwrite the PMs

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What happens

The basic relationship which should hold true in the data and in the results estimates is

ACTUAL = PRIOR MODEL + CORRECTION.

The first run, as usual, is made with adjustments at Baseline, the best adjustments known to the user.

The advice for the second run will be based on the Prior Model plus a correction equal to the difference between the Actual values of the first run data and the Prior Model at the inputs of that run data. That is, the first run data serves as calibration. Also, the Optimum Estimate with the calibration run data indicates where the optimum of the Prior Model is (within the Area of Confidence, the ellipsoid within MID around the first data point).

Thereafter, the Advices are based on predictions equal to the Prior Model plus the internal Correction regression models.

If the PMs are only linear or quadratic, then after there is enough data so that the correction models are full quadratics, ULTRAMAX's awareness of the process behavior is exactly the same as if the PMs were no longer there. You know this is true when the Shape and Coefficient models indicate "full quadratic" or "local quadratic models".

The example displayed below has as a prior model which is the linear portion of a previous ULTRAMAX application. Here the coefficients were obtained from the Coefficients Report with the Reference point at the new Baseline or starting adjustments.

The subroutine IPM(Actual, Prior Model, Correction) in the CALC subroutine does the following:

- When **entering data** the IPM takes Actual and Prior Model as inputs, and it calculates

  Correction = Actual - Prior Model

  The Correction Data will be used internally by ULTRAMAX to create Correction Models.

- When **estimating results** (such as in What-if and Advice), CALC comes in with the value of the Correction Model, and the IPM takes Prior Model and Correction Model as inputs, and calculates Estimated results:

  Estimation of Actual = Prior Model + Correction Model

The Shape and Coefficient reports always display the internal ULTRAMAX model created with the data regardless of any user-supplied models (prior or otherwise).

Miscellaneous

If PAR(30)=1 (ignore CALC equations when estimating outputs, use only internal models) then the "Actual" and "PM" variables will be estimated only by the internal regression models in ULTRAMAX. While data will maintain the basic relationship described above, this will not be exactly true with the internal ULTRAMAX models.

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PROCESS OPTIMIZATION PLAN

23:09 28 FEB 2005

APPLICATION: C:\Devel...t\chempm.umax Simulation of chemical process w/pr

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VAR #	NAME	UNITS	DEPENDENCY	RO LE	MO DE	MID	CONSTRAINTS LO	CONSTRAINTS HI
1	CRUDE FEED	K GAL/HR		1	H	0.5	25.	--
2	REACTANT	% VOL		1	H	0.5	--	--
3	TEMPERATURE	DEG.F.		1	H	5.	--	190.
11	VISCOSITY	K CPS		5	A	--	--	--
12	SOLIDS	% WT		5	A	--	--	--
13	PRESSURE	PSI		5	A	5.	--	310.
14	TPL	UOL		6	C	0.1	--	--
15	AVG TPL	UOL		7	A	--	--	--
16	VISCOSITY pr	K CPS		5	C	--	--	--
17	SOLIDS prior	% WT		5	C	--	--	--
18	PRESSURE pri	PSI		5	C	--	--	--
19	VISCOSITY co	K CPS		5	D	--	--	--
20	SOLIDS corr	% WT		5	D	--	--	--
21	PRESSURE cor	PSI		5	D	--	--	--

MINimizing objective function, variable # 14, "TPL" (Type 6)

List of non-default PARAMETERS (if any). See Blue Book, page 206.
Parameters control Operations, Data entry, or Reports.
Parameters with no description should be reset to default.
SELECTION FILTERS for: Plots, Consistency, Averages
All runs regardless of constraint satisfacton & run type

List of GLOBAL FACTORS

#	Value	Units	Name
GLOBAL(1)	= 850.	K CPS	VISCtarget
GLOBAL(2)	= 50.	K CPS	VISCtol
GLOBAL(3)	= 45.	% WT	SOLtarget
GLOBAL(4)	= 2.5	% WT	SOLtol
GLOBAL(5)	= 26.	KGal/Ba	CFref
GLOBAL(6)	= -10.	KGal/Ba	CFtol

There are no GUI calculations entered.

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```
CALC.DLL
c ------------------------------------------------------------------------
      IF (APPLIS('chempm') ) THEN
          ! Calculate Prior Models

        IF (      VALUE(1) /= -777       ! for dealing with missing data
     :      .AND. VALUE(2) /= -777
     :      .AND. VALUE(3) /= -777
     :                              ) THEN
          DX1 = VALUE(1) - 25               ! crude feed
          DX2 = VALUE(2) - 14               ! reactant
          DX3 = VALUE(3) - 170              ! temperature

          VALUE(16) =   920.19
     :                - 16.71 * DX1
     :                + 32.13 * DX2
     :                - 2.320 * DX3
          VALUE(17) =   45.232
     :                + .7217 * DX1
     :                - 1.618 * DX2
     :                - .1312 * DX3
          VALUE(18) =   308.84
     :                + 6.387 * DX1
     :                + 9.414 * DX2
     :                + .8640 * DX3

            ! Include Prior Models (actual, prior model, correction)
          CALL IPM(VALUE(11), VALUE(16), VALUE(19))         ! viscosity
          CALL IPM(VALUE(12), VALUE(17), VALUE(20))         ! solids
          CALL IPM(VALUE(13), VALUE(18), VALUE(21))         ! pressure
        ENDIF

        IF (      VALUE(11) /= -777       ! for dealing with missing data
     :      .AND. VALUE(12) /= -777
     :      .AND. VALUE(1) /= -777
     :                                ) THEN
          VALUE(14) =   PLS(VALUE(11), GLOBAL(1), GLOBAL(2), 'P')
     :                + PLS(VALUE(12), GLOBAL(3), GLOBAL(4), 'P')
     :                + PLS(VALUE(1), GLOBAL(5), GLOBAL(6), 'L')
        ELSE
          VALUE(14) = -777
        ENDIF
c ------------------------------------------------------------------------
```

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Part 4: PRINCIPLES and BACKGROUND

Goal Setting

Introduction

Goal Setting is to balance the performance for all individual objectives such as Key Performance Indicators (KPI) into a way of determining overall process performance.

Goal Setting for production operations performance is portrayed by the ***Performance Index***[49] (or *Objective Function,* the output to be maximized or minimized) and ***Constraints***[50] on inputs and outputs within which production is acceptable and safe.

Recall from the "**Measurability**" key success factor that this excludes processes where achievement is assured only by making adjustments following fixed predetermined rules – namely, where production procedures preclude any flexibility. In the latter case, use ULTRAMAX only for optimization studies to optimize the fixed procedures.

The *Performance Index and Constraints* are *real-time performance metrics* which are updated as often as the summary measurements are taken, such as every half-hour or every batch. These metrics are ideal to evaluate overall production operations performance (even if not doing optimization).

Having Performance Index and Constraint metrics helps plant personnel resolve frequent conflicting demands placed on them, so that they can concentrate in improving overall performance in a more consistent and productive way.

In this context, and as implemented in the software, improving[51] **the Performance Index or Objective Function is done only after minding the satisfaction of constraints.**

> *The "performance index" is an overall metric of performance (e.g., economic impact) which is valid to compare operating states when constraints are satisfied. Else the satisfaction of constraints (e.g., safety) dominates the evaluation of performance.*
>
> *Thus, there is "performance improvement" moving from a state of constraint violation to a state of constraint satisfaction regardless of the performance index value; and no increases in performance index is a "performance improvement" if moving into a state of constraint violation.*

In the setup procedures of the ULTRAMAX Method, goal setting is done as described in Steps to Formulate the Optimization Plan, page 28, where the ULTRAMAX implementation team...

1. ... Defines the *general business objectives* to be achieved (step 1).

---

[49] What in economic theory would be called "utility", plus the constraints, of course.

[50] Functionally, constraints are interpreted as satisfied or not, regardless of by how much they are satisfied or violated. If this does not represent reality, then include the gradual effect in the objective function.

[51] Note that the phrase 'optimizing the objective function' includes the holistic concept of satisfying constraints.

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2. ... Selects various *individual objectives* which as a whole represent the business objectives and includes all important company and customer needs (step 4). These objectives are typically related to:
   - Production rate, yield, efficiency;
   - Quality (average characteristics and consistency);
   - Costs, losses, revenues;
   - Safety, reliability (protect process/product);
   - Compliance with regulations (such as emissions).

   They surely include the Key Performance Indicators (KPI).
3. ... For each individual objective the team selects one or more *variables* (inputs or outputs) to be measured/set (some being summaries of multiple raw data such as percent defective, averages and standard deviations) or calculated based on other variables (step 4).
4. ... Integrates the individual objectives into a statement of the *Goals*, the balance of the individual achievements, into a quantitative description of what is better performance (step 6, page 33).

This section develops the background and various approaches to do the last three of the above steps to do Goal Setting. The focus is on principles and ways to do Goal Setting, with an implication as to which variables need to be used.

Describing the Business Objectives and the Goals for process management can become a key role management, turning it into a tool to properly represent the interests of the whole enterprise, including the voice of the customer as perceived by marketing. Modification to the goals becomes the key mechanism by which top management assures that production quickly realigns with evolving company needs, and aspect of Agile production and manufacturing.

Ideally, in most applications of ULTRAMAX the goals reflect directly the impact on the P&L statement of the company, both short- ***and*** long-term. The long-term effects such as quality levels, reasonable safety, and compliance with regulations often include objectives which “are more important than short-term economics”. It is very beneficial to realize that even if economics are not described explicitly the criteria imbedded in the Goals are indeed intimately related to the welfare of the enterprise.

There are three major approaches to goal setting (all explained in detail in this chapter):

1. The Performance Index or Objective Function is an economic calculation (e.g., variable profit $/time in a capacity constrained environment; and variable unit cost $/unit in a sales constrained environment), plus constraints for other considerations which can be represented by “hard limits”, such as capacities, safety/reliability limits, regulatory requirements, agreements with other parties (political effects), and even constraints on economic evaluations.
2. The Performance Index or Objective Function is a quasi-economic calculation such as:
   - indicators of satisfaction, e.g., linearly weighted variables (page 111)
   - addition of Performance Loss Functions, page 112)

   plus constraints for other considerations such as quality, safety, political and regulatory requirements.

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3. The Performance Index or Objective Function is a measured output, plus constraints as necessary on other variables to represent other objectives (Approach 3: Performance Index is a Single Output, Plus Constraints, page 132).

Combinations of these approaches are also possible.

On the nature of CONSTRAINTS:

Satisfying constraints represent aspects of performance that are more important than improving the value of the Performance Index or Objective Function. Thus, the constraints must represent true requirements, not just "vaguely desirable" aspects of performance.

Constraints are limits you want to impose, from which ULTRAMAX will try to stay away from. It is possible to place constraints on the Performance Index.

- Constraints on **control inputs** or **decisions** are interpreted as the threshold into *undesirable* or *impossible* values (such as a negative amount of an ingredient). Obeying these constraints is the highest priority of ULTRAMAX's Advice.

  Constraints on control inputs are absolute, since it is a decision. If a decision is not implemented exactly (nothing is), that is a variation associated with the process itself.

- Constraints on **outputs** or **consequences** (dependent variables, including "ruled inputs") are interpreted as the threshold into *undesirable* values only.

  As a consequence, ***Do not include constraints on outputs to rule out things that will happen anyhow*****,** such as trying to force Standard Deviation ≥ 0, Yield $\leq 100\%$, Percent defective ≥ 0, Purity $\leq 100\%$, etc. These limits are, in fact, highly desirable. If these limits were entered as constraints then ULTRAMAX will try to stay away from them, which would be detrimental to higher production performance.

  There are two criteria for satisfying output constraints:

  a. The output process Means (for given inputs) need to satisfy the Constraints in the Optimization Plan (OP).

  b. The actual data produced by operations (Mean + randomness or noise) needs to be within MID beyond constraints most of the time. The risk of going beyond – of a definite constraint violation – is defined by a protection of so-many-sigmas, such as 3_sigmas. If MID < 3_sigmas then we may have to sacrifice potential performance by having to keep the Mean more conservative than the Constraint (see Protection and Safety Buffer in Fig. 25 page 142

  From a **functional** point of view, most often crossing the values of a constraint has a smooth effect on performance. This expected behavior is consistent with the second of The Four Critical Success Factors: Graduality (page 18).

  When the effect of crossing the constraint is drastic, such as creating the failure of a piece of equipment, then the constraint is made conservative, a typical engineering solution. From a **cultural** or **contractual** point of view crossing over

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the constraint may also have drastic effects, such as violating a quality specification or legal regulations.

The user can bypass this logic by setting MID=0 to represent total intolerance in violating the constraint by all runs – which necessitates that the mean process value(s) be away from the constraint(s).

Note that constraints are applied by SEO to Run Data, most often **summaries** such the average -- not to the individual raw data used to calculate the summary. In order to control the incidence of **individual raw data** violating the constraints, such as meeting quality specifications in discrete production, you should include another summary variable, the *standard deviation*, which describes the variation of the raw data around the average. Then you can use both summaries as follows:

- Have a PL (page 112) for the average (with its target and tolerance) and a PL for the standard deviation (with target of zero and most frequently with the same tolerance), using the Parabolic Form.
- Create a new calculated variable, the CP or CPK (page 133) for this output variable, and place an adequate lower constraint on it.
- Let the summary be *percent defective* (non-conforming), and either to include it into the Performance Loss functions, or place upper constraints on it.

For the calculation of *standard deviation* or *percent defective* note the required Minimum Sample Sizes, page 195.

On the nature of GOOD QUALITY:

- The traditional and legalistic concept "***good quality means to be within specifications***" is best implemented through ***constraints***. This implies that anything within specifications -- i.e., within constraints -- is deemed to be equally good or acceptable (to the company and/or to the customer).

  When the constraints apply to each raw datum rather than to its average, it is implemented by placing constraints to calculated measures such CPKs. The CPK (defined in page 133) relates to the probability of raw data violating specifications and also directly implies that anything within specifications is equally good [52].

- The advanced or "world quality" concept "***good quality means to be consistently close to targets***" -- in short "less variation from targets" as a representation of customer satisfaction -- is implemented through the ***performance loss functions*** (or if at all possible, through direct economic analyses). Performance Loss Functions (PL, defined in page 112) measure the variation of the data from a target or reference in quasi-economic terms.

[52] In other words, the basic reason that CPK is larger the closer to the target is to increase the probability that the individual values are within specifications, not to make the deviation from target smaller.

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Note that the same process could be run under two conditions with the same CPK (with two different combinations of mean and standard deviation) but with very different PLs (page 133).

The rest of this section is divided into three main parts:

- General considerations in goal setting
- The three major approaches to goal setting
- Miscellaneous topics

General considerations in Goal Setting

The procedures for defining objectives and goals to represent the organization's success may be influenced by some common errors:

- Paying too much attention to relatively unimportant factors
- Failing to account for some important factors
- Failing to keep the goals up to date

The following are some principles and conceptual examples which will help you streamline your approach to make things simpler and representative. Two key concepts are:

- Focus only on values that vary
- Focus on all important issues

and, of course, keep the description of goals up to date.

Focusing only on Effects that Vary

Optimization only concerns changes in performance generated by the changes in the control inputs (decisions) in the Optimization Plan, where usually the ultimate desirable effect is on the P&L statement. Aspects of performance which will not be affected are irrelevant. In the application of the ULTRAMAX Method, only *variable* [53] parameters need to be included.

This means that adding an item which is a constant to the objective function and/or to a variable with constraints (and adding the same constant to the constraint) does not affect either the optimal decision nor, of course, its effects on the P&L statement (whether known or not). Also, including outputs that do not change significantly (and obey constraints by more than the noise) will not affect the optimal decision.

For example:

1. Let the goals include reducing variable unit cost ($/part). The components of the unit costs which may remain constant (e.g., painting, packaging, certain materials) may be ignored.
2. Let the goals include increasing profitability rate (K$/hr.). If the annual values of overhead, sales, and advertising costs are not affected by adjustments to the control

[53] That is, variable as it relates to the scope of the decisions in the current Optimization Plan. Do not confuse with "variable costs" in accounting, which typically deals with "variable" is it relates to scheduling decisions.

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inputs; they can be ignored. Including these variable, however, will not distort the optimization, but make the analysis more complicated and probably add undesirable noise if they are measured.

3. When the goals include reducing unit costs ($/part) for a fixed amount of production, lower profit decisions might result if unit cost includes the distribution of fixed overhead based on equipment utilization time.

   For instance, production rate might be increased to achieve a reduction of constant overhead costs distributed according to production time (and thus passing the burden to other products) while incurring a (smaller) out-of-pocket increase in the unit consumption of energy. However, the net effects on the P&L statement are greater costs due to increase in energy consumption, and of course no difference in the constant overhead component (also revenues remain the same because of the fixed amount of total production).

On the other hand, if the "constant" output item has high measurement noise, this becomes a barrier to optimization.

Focusing on ALL Important Effects

This section is the essence of the ideal *holistic* or *comprehensive* approach addressed in pages 23 and 101. The essence of the idea is to reflect all impacts on the company...

- ... in your departments, in other departments, and in the market
- ... this period and in future periods (short-term and long-term effects).

Although many important effects are difficult to estimate, estimating them is better than ignoring them. This section shows two examples where simply requiring better management control in the definition of objectives can result in significantly better performance.

Example 1: Acceptable Quality Levels.

Imagine making a product that requires expensive rework at some threshold level of imperfection.

This threshold may be based on functional considerations as in Fig. 16. "Rework" example. Let the cost of alternate decisions be as follows:

- The ***cost of accepting*** a production lot without rework is proportional to the level of imperfection (such as percent defective), that reflects the negative effect on customers (both internal and external to the organization), including if external, future effects on market share[54].
- The ***cost of rework*** is as follows:
  1. Due to the extra work and disturbance to normal product flow, rework incurs a fixed cost regardless of the quality level.

[54] This is meant to reflect the effects which are implicitly taken into account, not that the effects need to be evaluated explicitly. In fact, usually what the organization understands well (through years of experience and the culture instilled by management) is the quality threshold which we are defining, from which we can infer the magnitude of the effects which would make it logical.

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2. In addition to a fixed cost, the cost varies with the level of product quality (imperfection, e.g. percent defective or nonconforming product) because:

   — the actual cost of rework depends of the level of imperfection (greater the more imperfection), and

   — the rework only removes part of the imperfection, leaving behind a constant fraction of the original quality.

The logical decision as to whether to ***accept*** the product lot as is or to ***rework*** it is based on ***minimizing cost***. Thus when the level of imperfection is to the left of the crossover it is better to accept, and when to the right to rework. The crossover defines the threshold of "Unacceptable Quality Limit" (UQL).

The UQL is determined by experience, not by an economic analysis. However, the following logic applies anyhow.

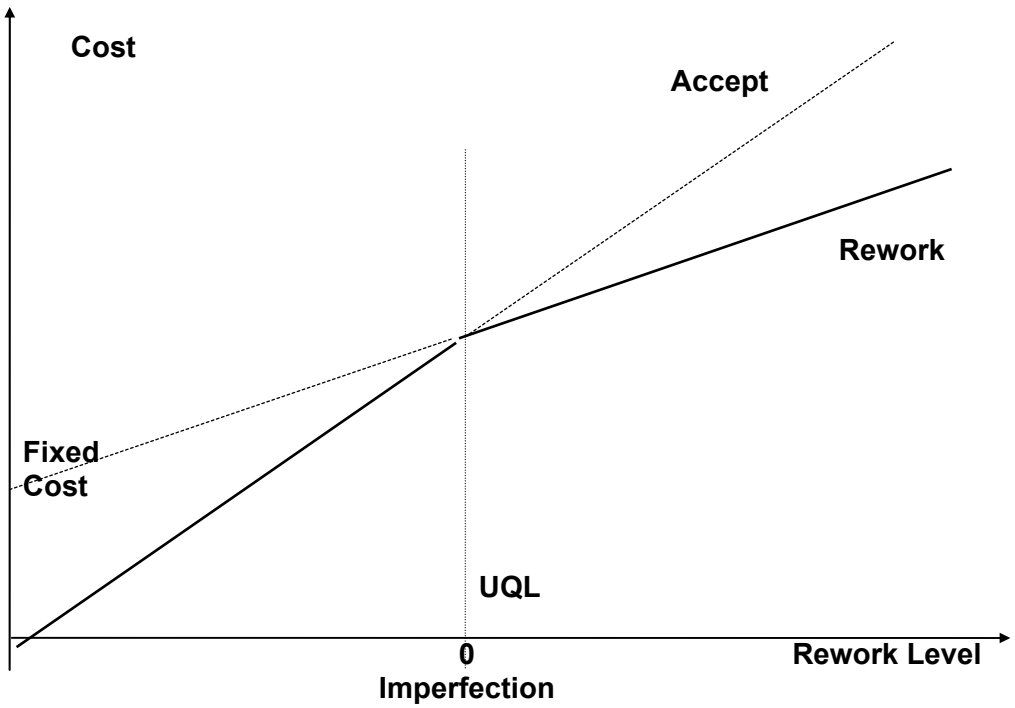


Fig. 16. "Rework" example

It turns out that the cost of rework is charged to the producing department. However, the costs of imperfections delivered to the customer are incurred elsewhere (e.g., warranty, loosing future sales), and which hurt the total company profitability whether they are accounted for or not (if not, they are "intangible").

Now, let us imagine that the manager of this operation treats the threshold UQL as an upper limit of "acceptable" quality in the sense that any quality level below UQL is equally good (that is, his/her objectives are **not** to reduce the level of imperfection as much a possible (taking other things into account). At the same time, this manger spends significant efforts reducing other lesser costs which do appear in his accounts. This results in a quality close to UQL and savings in

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books only, while missing opportunities for higher firm profitability.

This attitude is provincial and ignores the needs of the organization as a whole, and probably reflects how the manager's performance is being evaluated. This is not a "holistic" attitude.

Management could discourage this behavior by including in the evaluation of the managers performance the quality achieved. Supporting this in the accounting operating statements should include an estimated costs of imperfect quality; and equivalently, this is reflected in the Performance Index of an ULTRAMAX Optimization Plan.

Example 2: Amount of Active Ingredient.

In this example more active ingredients in the product provides better product performance, better consumer satisfaction, and increased future demand. It also increases production costs.

Marketing calculated the profitability depending on the amount of ingredient, as shown in the next figure, and determined the ideal target as the maximum of the profitability curve.

Centered at the ideal target, Engineering determines the variability it can control as illustrated by the old, large bell shaped distribution. This determined the practical upper and lower specification limits (USL, LSL), that defines how much active ingredient is "acceptable" for daily production.

Eventually these limits were misinterpreted as "anything between the limits is equally good" in terms of value to the firm (as it is still implied, e.g., in Six Sigma). Production management, guided largely by cost reduction concerns, installed advanced process control devices to reduce variability. Then they took advantage of the greater consistency of ingredients, as represented by the narrow bell shape distribution, and centered the average level of ingredients as low as possible – thus saving costs -- and still be "99+% confident" of meeting the lower specification limit.

Production managers proclaimed this capital outlay and manpower allocation as a "savings" project, even though as we can see they ended up operating at a lower profitability level, as indicated in the average New and Old profitability levels in the next figure.

Management could discourage this behavior and waste by:

- specifying quality first by targets, and secondarily by tolerances (rather than just tolerances)
- including in the monthly evaluation of performance the long term "costs" associated with deviations from targets, and not just by the frequency of specification violations (non-conforming product).

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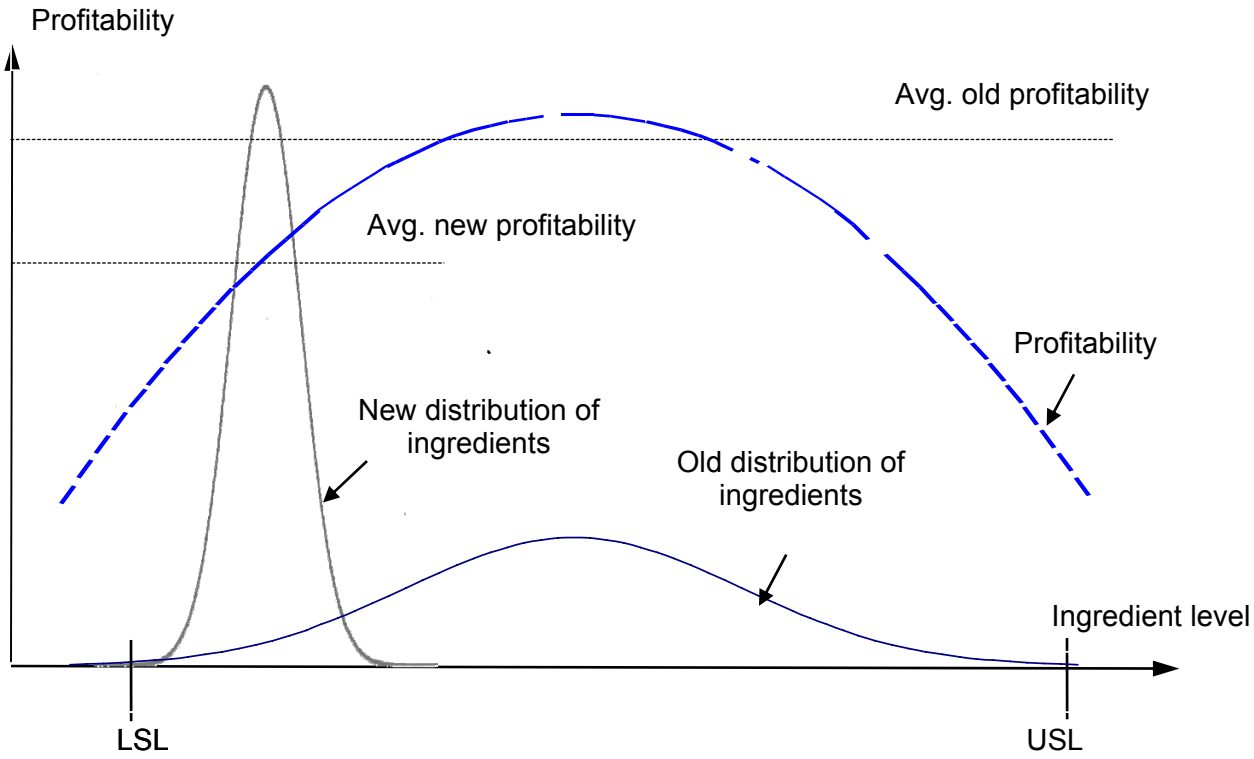


Fig. 17. "Amount of ingredients" example

The previous two examples illustrate the outcomes that occur when companies (and accounting) use the wrong criteria to evaluate performance. When the evaluation is allowed to be very local (provincial), profitability can get hurt.

Note, as well, that some of the intangible effects are often far more important than minor tactical effects that are more readily calculated. The same issue is sometimes described as organizational objectives having both short-term (e.g., tactical) and long-term (e.g., strategic) components.

Approach 1: Full Economic Analysis

In this approach the Performance Index is an all-inclusive measure of variable profitability to maximize, or a measure of variable cost to minimize. Those results for which the profitability impacts are difficult to determine, such as quality considerations, could be resolved with the Performance Loss function or with Constraints.

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Process adjustments affecting total production

When the decisions to be made can affect total production and sales (e.g., in a capacity constrained environment[55]), the **revenue** is affected and a profitability equation is necessary. The most common equation is fundamentally of the form:

$$\text{Var. Profit (\$/t)} = \text{Var. Revenue (\$/t)} - \text{Var. Cost (\$/t)}$$
$$= \sum \text{Prod.Rate(P/t)} * \text{Revenue (\$/P)} - \sum \text{Consumption(U/t)} * \text{Unit Cost (\$/U)} - \text{REF}$$

where:

- *t* stands for time, e.g., per hour of production making this product
- *P* is the unit for products produced (e.g., cases, pounds, gallons)
- *U* is the unit for usage of resources consumed (e.g., materials, energy, machine time)
- the $\sum$ for revenue stands for the addition of several co (or by)-products; and the $\sum$ for costs stands for consumption of several items.
- the *Revenue ($/P)* is the net variable revenue at the output end of the process being optimized. The revenue may[56]:
  - be the net revenue further downstream minus the variable costs until that point, such as packaging and transportation.
  - be a function of quality levels such as purity. Significantly different quality levels may even produce outright different products, such a premium, standard or seconds. Also, the revenue may reflect some premium/penalty depending of the consistency of results (standard deviations).
  - be negative for certain by-products which are undesirable, such as having to dispose certain materials;
- there are a variety of ways to calculate *Consumption (U/t)*, such as directly through rate measurement gauges, or through:

  *Prod. Rate (P/t) * Usage (U/P)*

  If all cost items are in this form and there is only one product, increasing production rate *Prod. Rate (P/t)* will increase profits as long as

  Revenue ($/P) > ∑Usage (U/P) * Unit Cost ($/U).[57]

  (i.e., when marginal[58] profit is positive, or marginal revenue > marginal costs).

---

[55] That is, the company can sell all it makes and more. The decision of *what* product to make in the allocated time is made outside ULTRAMAX, it is part of production planning and scheduling.

[56] In even more sophisticated analysis the unit revenue could also be a function of the production rate to reflect the demand curve, to support the extra sales.

[57] This justifies the frequent direct use of Production Rate in the Performance Loss Functions in the next section.

[58] Here "marginal" means associated with the next unit of production.

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"Yield" expresses in units *lbs.out/lbs.in* is an example of the inverse of *Usage (U/P)*.

Note that *labor* is a **constant** in *$/t* and thus does not have to be included (unless the readjustments call for more or less manpower... e.g., more hand packing if increasing throughput).

- *REF* stands for some reference value of variable profit such as "budget", or such that the *Var. Profit* = 0 at the starting Baseline (so as to display directly the improvements achieved when constraints are not violated).

In **a capacity constrained business** there is a distinct possibility that unit costs due to *Usage (U/P)* could actually increase (e.g. increasing drying rate at higher unit energy consumption) as a necessary aspect of increasing production rate, and still increase overall profitability due to extra sales (while the marginal profit is still positive: marginal revenue > marginal costs).

Note in particular in this case how insufficient it could be to represent the objective as reducing Product Unit Cost ($/P) = Usage (U/P) * Unit Cost ($/U) plus some distribution of overhead (on a time basis) because "this rewards higher production rates." Clearly this approach is not sufficiently related to profitability (the P&L statement) because it ignores the revenue part, which will vary and is important.

As an example imagine a process with very low overhead. Imagine further that unit consumption (usage) increases with production rates, e.g., in order to maintain quality. Thus, following the objective to minimize product unit cost ($/P) one may end up barely increasing, or even reducing production rate, and thus giving up the revenues and profits from higher production rates.

Process adjustments not affecting total production

When total production is not affected then revenues remain constant. This may happen either because the control inputs in the optimization plan will not change production rate, or because it is a sales constrained business environment. Then the economic objective function is typically of the form

Product Unit Costs ($/P) = $\sum$ *Usage (U/P) * Unit Cost ($/U)*

= *Variable Cost ($/t) / Prod. Rate (P/t)*

Remember the topics discussed in Focusing only on Effects that Vary, page 105. Be mindful than even though *Labor (U/P)* is usually included by Accounting, it may in fact be a constant in the short-term.

Simple Economic Indicators

Sometimes organizations have a simple and straight-forward understanding of the main impacts of their business.

In this example of food/beverage manufacturing, the objective is optimizing a process to find the combination of yield and flavor that contributes the most economic value. Each one percent increase in yield saves about $0.8M/year, and each 0.5 increase in flavor increases marginal profit due to increased sales by about $1.2M/yr. Assuming the past yield is 62% and the past flavor is 5.4, the Performance Index or objective function to be maximized is calculated as follows:

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$$\text{Variable Profit (\$M/yr)} = 0.8 * (\text{Yield} - 62) + 1.2 * \frac{(\text{Flavor} - 5.4)}{0.5}$$

Notice that this company, in rewarding these Goals, is aware that benefits do not need to be determined with much accuracy to recognize important improvements.

Approach 2: Quasi-Economic Analysis

Economic analysis is not always possible or complete. An alternative to resolve the balance between multiple objectives is to evaluate the effects of individual objectives in additive, standardized units of impact on the company.

Performance Loss Functions

Performance Loss (PL) Functions address the balance of multiple objectives by measuring outcomes from an agreed upon "zero point" and in compatible units which are additive. The ***zero*** stands for either:

- Ideal, perfection, nominal or **target** value (e.g., standard deviation = 0, dimension = 3.456", pH = 4.8). Deviations from a target in either direction represent a loss, as compared to making perfect product (as defined by the formula card, blue prints and marketing).
- A **reference** value from which deviations indicate a loss or a gain depending on the direction (e.g., production rate, purity, costs).

The ***scale*** or unit of the PL is such that losses are standardized to have the value of 1 when the value of the characteristic is at the maximum tolerable deviation from target, called the ***tolerance*** (but which people should think of as "***intolerance***", the threshold of beginning intolerable deviation from target). ULTRAMAX refers to this as the "Unit of Loss", abbreviated UOL.

The definition of the value of TOL is based on the cumulative learning and culture that exists in the business, the product of experience, problems and their resolution. It is often arrived as a consensus among the optimization team, eventually with heave influence by upper management. As optimization is achieved based on the assigned values, experience usually leads to refining its value to best represent corporate needs.

In a rational world, all the variables with PL functions would be such that PL=1 indicate similar losses in the P&L statement. This property is called 'equivalency'. Then if we can determine the economic value of one UOL, then multiply the PL functions by this value to give an approximation of the economic impact. This is readily done when one of the variables evaluated through the PL is, e.g., *cost*.

The next several sections will discuss different types of loss functions and show how they are represented in the software. At the end there is a Table of Performance Loss Functions (page 124) to use a handy reference. In practice people use most of the time just linear and quadratic loss functions.

Quadratic (parabola) Loss Functions.

For some objectives, the characteristic being evaluated is such that deviations from target in either direction generate losses for the business. The

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specifications (maximum tolerable deviation from the target) are the values at which the losses are so high that the product is no longer desirable as is. For instance, if the dimensions of the sides of a box are too small, the fit may be too loose, however, if the sides are too large, the box may be difficult to assemble.

The PL functions estimating these losses are particularly suitable to represent current advanced concepts in quality management that better quality is less deviation from target – as compared to reducing the risk of violating quality specifications.

To understand these loss functions, let us play a mental game. Assume that the "true" losses are estimated for each deviation from target for a product characteristic, and include all future effects of this deviation from target. Some of these losses are due to the:

- cost of rework or extra costs downstream of further processing or assembly,
- effects on other parts and overall process performance,
- appearance issues that may influence product purchase,
- longer-term effects on product life, customer satisfaction, warranties, law suits and future market share.

Taguchi would call these "societal costs" due to insufficient service.

Fig. 18 represents what the losses might look like if we could accurately represent them.

This curve could display some strange behavior. For instance, at some deviation from target the true loss may not grow as fast because of corrective actions such as rework (as in Fig. 16, page 107), or even scrap. At no point should the loss be larger than that for scrap as long as we make the correct operating decisions.

While this complex cost function is likely to be extremely difficult to estimate, a parabola may be an adequate way to represent its main characteristics. A parabola exhibits the basic behavior of minimal loss when the deviations are near zero, and increasing losses at an increasing rate the greater the distance from target as shown in Fig. 19.

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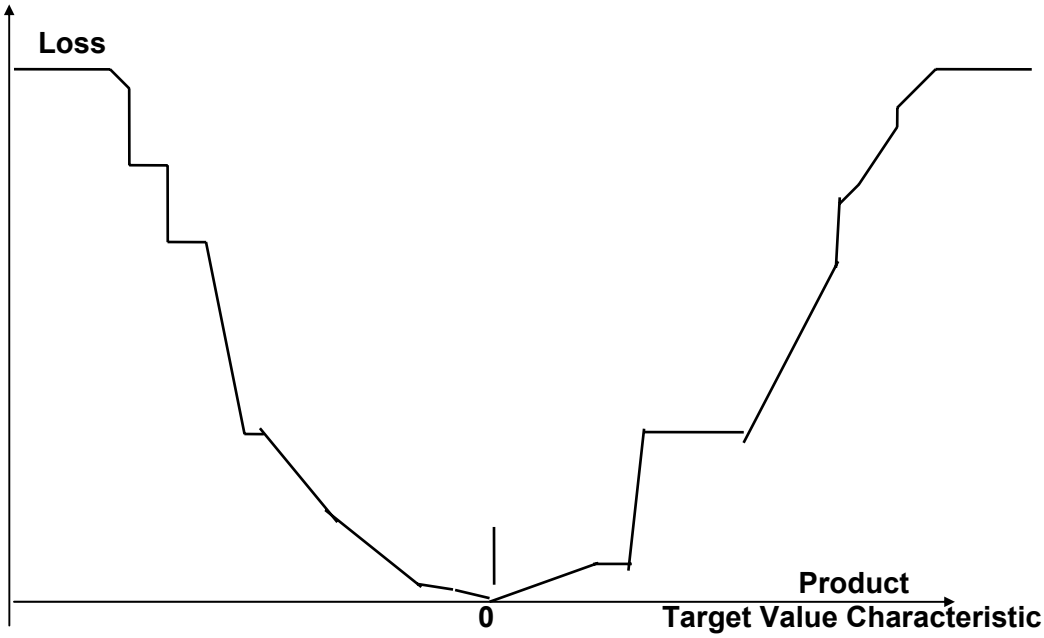


Fig. 18. True Loss

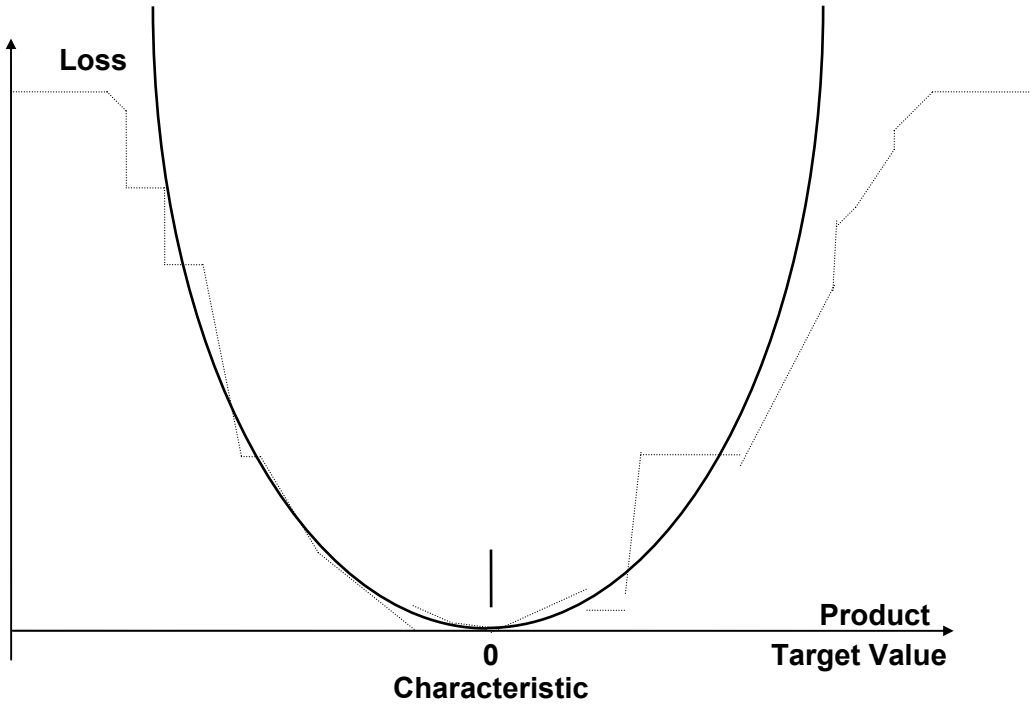


Fig. 19. Approximated Loss

The rest of this section will help determine the equations for quadratic (parabolic) loss functions. The following notation is used in equations

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TGT	Target
REF	Reference (for linear functions)
TOL	Tolerance, maximum tolerable deviation from target or reference, the onset of intolerable deviations for this effect alone
VAL	Value (level) of the product or process characteristic
AVG	Average of several VALs
SD	Standard deviation of several VALs[59]
PL	Performance Loss function, including Avg. PL

The equation for the parabola with the zero and the scale displayed in Fig. 20, is:

$$PL(VAL) = \left(\frac{VAL - TGT}{TOL} \right)^2$$

Note: when VAL = TGT, then PL = 0

when VAL = TGT ± TOL, then PL = 1

Notice that the (the inverse of) Tolerance TOL acts like a **weighing factor**, with similar impact to the coefficients used for "economic indicators".

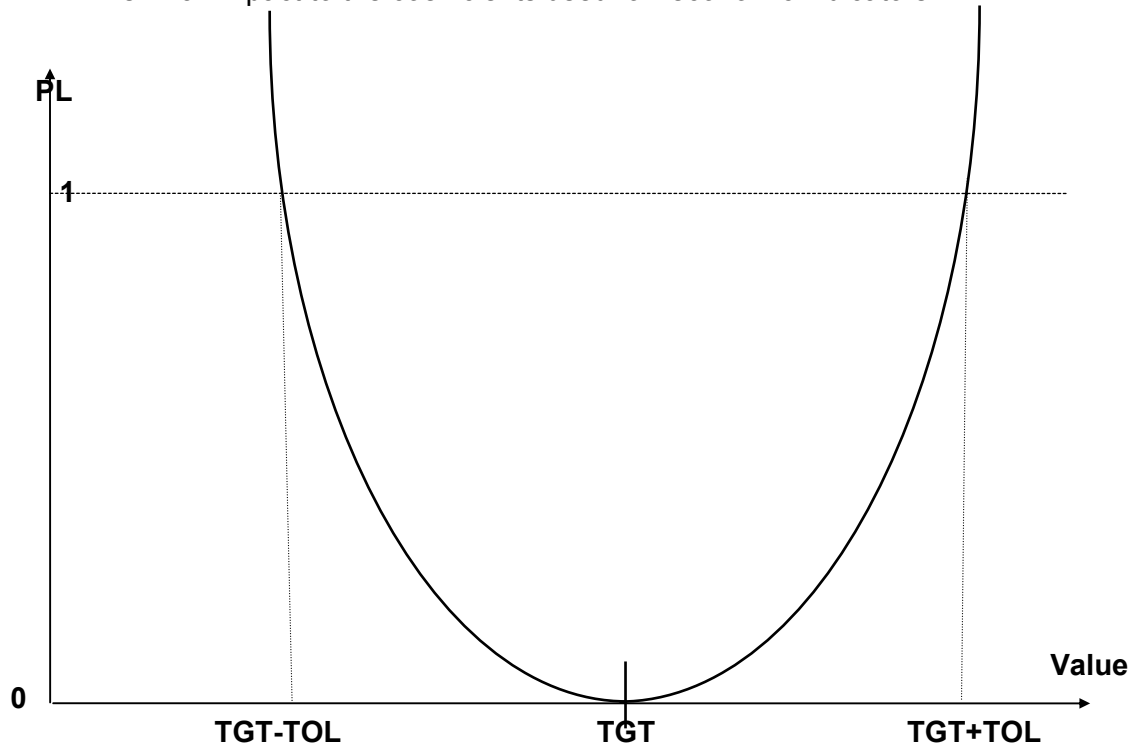


Fig. 20. Quadratic Loss Function

[59] These standard deviations calculated with the N denominator, rather than the (N-1) to estimate the standard deviation of the population.

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Parabolic PL for summary data

An objective related to a value such as *paint thickness* may have several raw measurements (VAL) per run. The PL for this objective is the average PL for each raw measurement. Based on the average (AVG) and standard deviation (SD, a measure of variability) of the raw VALs, the average Performance Loss turns out to be:

$$PL = Avg.PL = \left(\frac{AVG - REF}{TOL}\right)^2 + \left(\frac{SD}{TOL}\right)^2$$

The PL for summary data is thus decomposed into the addition of the PL for poor targeting LPA = PL(AVG); and the PL for the variability from part to part PLS = PL(SD), as follows:

$$PLA = PL(AVG) = \left(\frac{AVG - REF}{TOL}\right)^2$$

$$PLS = PL(SD) = \left(\frac{SD}{TOL}\right)^2$$

PL = PLA + PLS

So, to calculate this PL we need two output variables, the average Paint AVG and the standard deviation Paint SD, and its PL is the addition of the PLs for each of these. This is but one example of how PLs are additive.

A PLA > PLS indicates that the greatest improvements will be obtained by centering the process properly. If PLS > PLA, then the most effective solution is to reduce variability, but this is typically more difficult to do (e.g., by better first level controllers, more accurate measurements).

In some cases the PLA component is treated with a different tolerance than the PLS component. For instance:

- the requirements may be that parts within a batch be consistent, where the requirements for average are relatively lax because normally the customer can make and easy adjustment for the average.
- the requirements are about the average, where the variability from part to part is not critical, such as with density of granules.

In these cases, the tolerances of the critical characteristics will be much smaller than the others, some to the extreme that the components PLA or PLS may be ignored altogether.

Note 2: As a matter of reference, in a properly targeted process the relationship between CP=CPK and Avg.PL=PLS is:

$$PL = \left(\frac{1}{3 * CPK}\right)^2$$

Then, e.g., a CP = CPK = 1.50 has an Avg.PL = PLS = 0.047.

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Linear Loss Functions.

For some objectives, the characteristic being evaluated is such that the-larger-the-better, such as purity and production -- the opposite of this would be material usage. In these cases the loss function is linear, as described next.

For instance, if the output being measured is "cost", it obviously has a linear, economic, loss function, not parabolic. Linear PLs are frequently expressed in relationship to a reference (REF, where the loss is defined as zero[60]). The linear relationship enables one to have negative losses, or gains, compared to the reference value.

The sign of the Tolerance (TOL) indicates immediately in which direction from the reference the loss exists. In the case of cost, the TOL (direction of hurt or more losses) is positive and represents the level of intolerable extra cost (all other things remaining constant).

The linear equation is:

$$PL(VAL) = \frac{VAL - REF}{TOL}$$

Note that changing the reference only changes the PL by a constant, which as described earlier (Focusing only on Effects that Vary, page 105) does not change the decisions. Thus, **in (true) linear functions the tolerance value is really independent of the reference**[61]. The TOL represents an intolerable *change* in the value regardless of the REF. See plots with negative and positive TOL in page 119.

This linear approach applies to variables with economic effects that are proportional to the deviations from the reference within a workable range. Here are various examples:

- the larger the better (TOL<0) (see Fig. 21)
  - strength,
  - purity,
  - profitability,
  - production rate (parts/hour)
- the smaller the better (TOL>0) (see Fig. 22)
  - percent defective (that is, non-conforming),
  - failures per hour,
  - production rate (min/part)

---

[60] For instance, the ideal of zero cost is so unrealistic that it does not make a good reference value. However, zero percent defective and zero standard deviations are also technically unrealistic, but a useful conceptual point of reference (e.g., zero-defect program).

[61] This fact may sound counter-intuitive, e.g., when people are being rewarded in the ways we were trying to avoid in Examples 1 and 2. If it still appears counter-intuitive, then perhaps the PL is really not linear, but some other function which crosses the value of PL=0.

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Quality items are typically referenced to the ideal values, such as zero percent defective, zero standard-deviation.

Note that sometimes the inverse, e.g., hours/part, may be more linearly related to economic impact.

Note in particular that the Tolerance (TOL) acts like a **weighing factor**, the inverse of the coefficients used for Economic Indicators.

Linear PL for summary data

The average PL for a series of raw data is simply the PL function for the average:

$$PL = \frac{AVG - REF}{TOL}$$

The standard deviation SD, if calculated at all, plays no role.

Other variations

As far as the ULTRAMAX software is concerned, any reasonable smooth function can be used.

Variables such as *strength* or *purity*, may be linear in the short term, but once the variables are sufficiently large, further increase is of no value. Some solutions include:

- Use the linear approximation, but do not give credit to a production rate beyond a specified limit (for example, 45 parts/hr, because thereafter increased production rate is of little value). Here TOL<0:

$$PL(VAL) = \frac{\min\{val, 45.) - REF}{TOL}$$

- Examine the inverse, with an ideal value of zero as in production rate. Using the square of the inverse appears as:

$$PL(VAL) = \left(\frac{REF}{VAL}\right)^2$$

This is taken from a signal-to-noise ratio used by Taguchi, but lacks obvious quasi-economic value.

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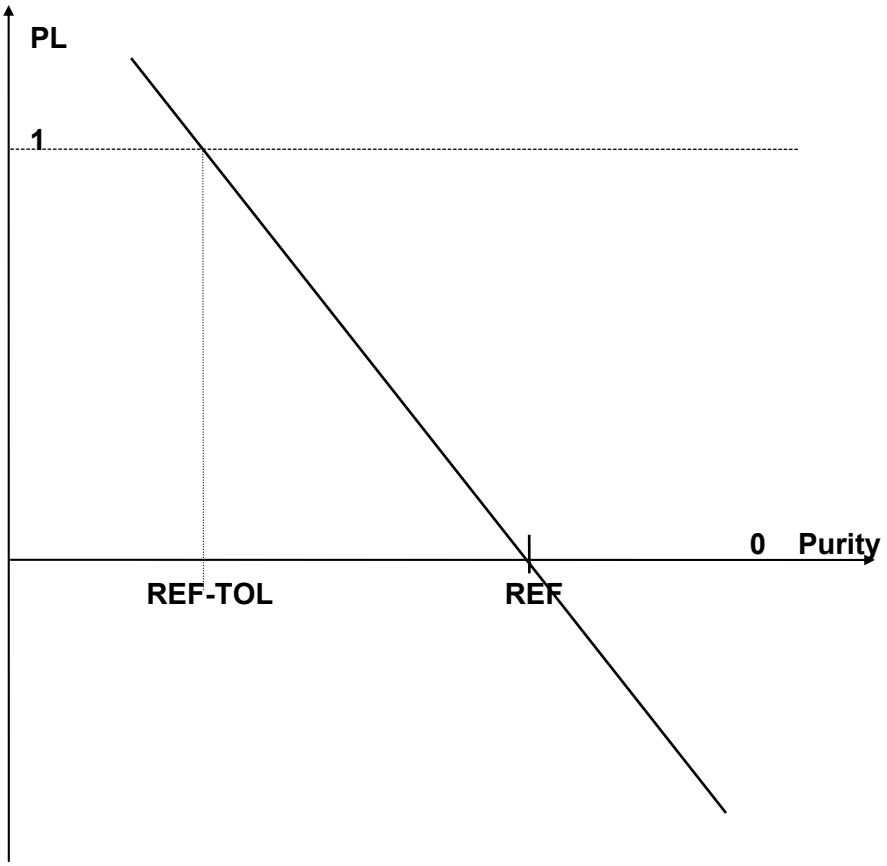


Fig. 21. A Decreasing Linear Loss Function (the-larger-the-better, TOL<0)

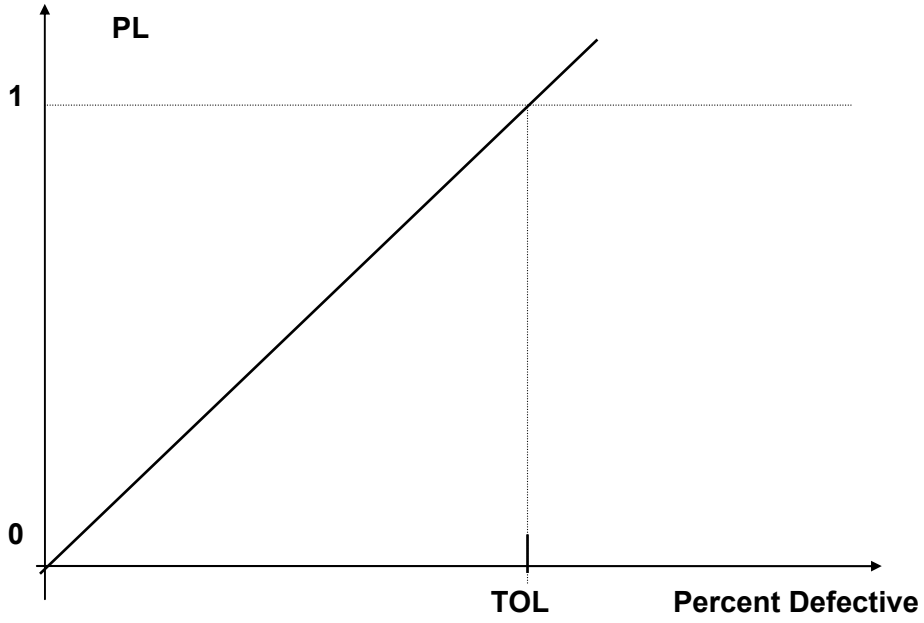


Fig. 22. An Increasing Linear Loss Function (the-lower-the-better, TOL>0)

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The ULTRAMAX Built-In Functions for PLs

The performance loss equations described before, and several combinations which arise, can be entered directly into the equations to define the calculated variables, or entered more easily with the Symmetric and Asymmetric built-in functions described below (DLL calculations only). The two following sections have examples.

Nomenclature	
VAL	Value of the variable for which one wants to determines the loss function, either a: input measured output (single or summary value) calculated variable.
TGT, REF TOL LTOL, HTOL	Target or Reference, and Tolerance deviation from target. They are typically either numbers (with decimal points) or Global Factors. LTOL and HTOL are Low and High TOL for asymmetric LPs
PL	Performance Loss function
'P'	Parabolic form of loss function
'L'	Linear form of loss function
'M'	Mixed: parabolic when positive, linear when negative (this is particularly suitable for reducing estimated standard deviations, where negative estimates imply better values, not worse when applying the square in the quadratic PL function)
'N'	No loss or gain (the equivalent to TOL = ∞).

Performance Loss, Symmetric (same parameter high and low)		
PL = PLS (VAL, TGT, TOL, '**P**')	$\left(\frac{VAL - TGT}{TOL}\right)^2$	parabolic
PL = PLS (VAL, TGT, TOL, '**L**')	$\frac{VAL - TGT}{TOL}$	linear
PL = PLS (VAL, TGT, TOL, '**M**')	for VAL>0 $\left(\frac{VAL - TGT}{TOL}\right)^2$ for VAL<0 $\frac{VAL - TGT}{-TOL}$	mixed
PL = PLS (VAL, TGT, TOL, '**N**')	0	null

Performance Loss, Asymmetric	If VAL < TGT	If VAL > TGT
PL = PLA (VAL, TGT, LTOL, 'P', HTOL, 'L') The form can also be 'N' A low linear loss function requires a negative LTOL, and a high one a positive HTOL.	e.g., if 'P' $\left(\frac{VAL - TGT}{LTOL}\right)^2$	e.g., if 'L' $\frac{VAL - TGT}{HTOL}$

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An Example with Multiple Objectives

Since the PL is fundamentally an economic evaluation, it can be added to the PL generated by other product/process characteristics. The only requirement is that the units be approximately the same. The actual dollar losses incurred by each UOL -- the deviation of one tolerance when all other characteristics remain constant -- must be similar for each loss function. An error of a factor of two appears to be acceptable. This property is called **equivalent tolerances**.

Thus, the variable to be minimized (the Performance Index) becomes the addition of the PLs., the **Total Performance Loss (TPL)**. In addition, constraints may represent other objectives, or they may even represent the fact that the PL reflects performance only once certain limits are satisfied, such as specification limits.

In the following page is an example of an OP with three individual objectives. Each objective has:

- A description of the individual objectives
- A description of the necessary variables, with their units.
- The PL equations for each variable, each including the
  - Target or Reference
  - Tolerance
  - Form of the PL Function (P=Parabolic, L=Linear)
- How the desired PL is entered into the Optimization Plan. There are two options:
  - Code (DLL, “C” for GUI)
  - Built-in functions (DLL only)
- A numerical example for each variable including the actual value of the variable and its corresponding PL.

At the bottom of the table is the Total Performance Loss, the addition of the actual data for the three objectives in this Optimization Plan.

Following the table is:

- A sketch of the Decision Diagram, which is how the team would record the information during the formulation of the Optimization Plan.
- The equations to calculate the TPL as entered into the software.

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AN EXAMPLE WITH MULTIPLE OBJECTIVES

Objective 1. Dimension 'A' should be .8700 inches for all parts, with a ± out-of-tolerance deviation starting at 0.001 inches. This individual objective requires two summary variables, the average (AVG) and the standard deviation (SD).

Objective 2. The surface appearance should meet specifications and have no defectives parts, with an out-of-tolerance deviation starting at 3%.

Objective 3. The process should make as many parts per hour as possible to reduce unit cost. It is now making 45 pts./hr. The deviation from reference at which the loss begins to be intolerable is –30%.

GOAL SETTING						SETUP [62]	NUMERICAL EXAMPLE	
VARIABLES		TGT / REF	TOL	Form	Perf. Loss Equations	Entry of the calculation (below, using built-in DLL form)	ACTUAL VALUE	PL (UOL)
#1	Dimension A: *Avg*. (in) Name: *Dim A AVG*	0.87	0.001	P	$\left(\frac{AVG-0.8700}{0.001}\right)^2$	((*AVG*-.8700) / .001)**2 --- PLS(*AVG*,0.8700,0.0010, 'P')	*AVG* = 0.8697 in	PLA = 0.087
	Dimension A: Std.Dev. (in) Name: *Dim A SD*	0.	0.001	P	$\left(\frac{SD}{0.001}\right)^2$	(*SD* / .001)**2 --- PLS(*SD*, 0., 0.0010, 'P')	*SD* = 0.0002 in	PLS = 0.040
#2	Percent Defective Name: *PercDef*	0	+3	L	$\frac{PD}{3}$	*PD* / 3 --- PLS(*PD*, 0., 3., 'L')	*PD* = 0.37%	PL = 0.123
#3	Production rate (pts./hr) Name: *Productivity*	45	-30%	L	$\frac{PR-45}{-0.30*45}$	(*PR* - 45) / (-0.30*45) --- PLS(*PR*, 45., -0.30*45, 'L')	*PR* = 48 pts./hr	PL = - 0.222
PERFORMANCE INDEX = TPL = Σ PL = 0.028								

[62] The software User's Guide describes how to enter the PL you want in the Stand Alone GUI. Else use a .DLL.

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Note that the Total Performance Loss (TPL) is the addition of the PLs for the three objectives:

Thus, of two separate Runs with different inputs, the one with the lowest TPL is more desirable (particularly if the difference is larger than the MID). Fig. 23 shows the Decision Diagram associated with this application and Fig. 24 shows the equations used to calculate the TPL.

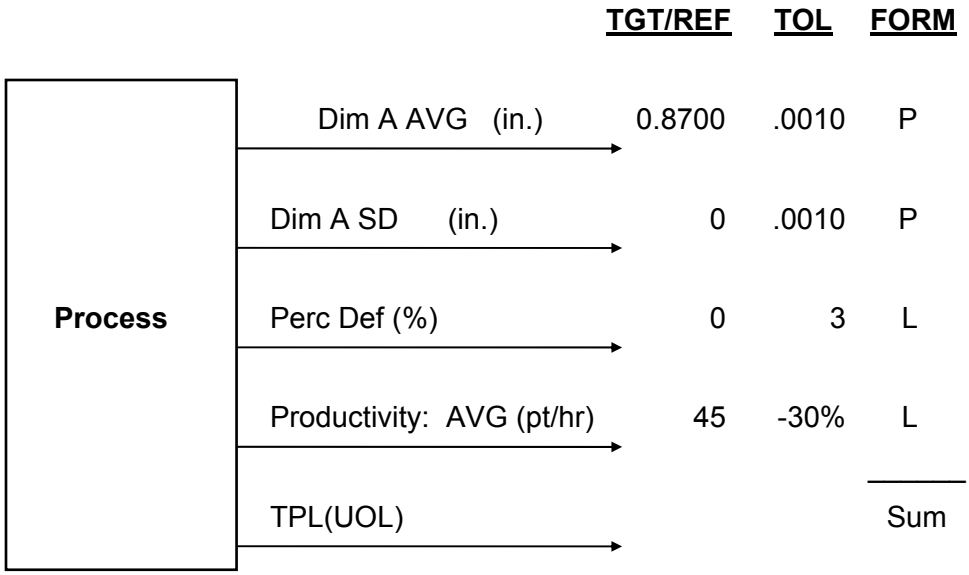


Fig. 23. Dec Diagram Showing PL Information

GUI	
TPL =	((Dim_A_AVG - 0.870)/0.001)**2 + (Dim_A_SD / 0.001)**2 + PercDef / 3 + ((Productivity / 45 - 1.) / (-0.30)
DLL in Fortran[63]	
TPL =	((VALUE(*) - 0.8700)/0.0010)**2 + (VALUE(*) / 0.001)**2 + VALUE(*) / 3 + ((VALUE(*) / 45 - 1.) / (-0.30)
ULTRAMAX Built-in function	
TPL =	PLS (VALUE(*), 0.870, 0.001, P) + PLS (VALUE(*), 0., 0.001, P) + PLS (VALUE(*), 0., 3., L) + PLS (VALUE(*)/45, 1., -0.30, L)

Fig. 24. How PL equations are entered

[63] The value of the variable is represented by an array VALUE, where the (*) stands for the number associated with the variable. In GUI environments the user only needs to click the name of the variable in its place.

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TABLE OF PERFORMANCE LOSS FUNCTIONS

The Loss Function provides a mechanism to balance the achievement of several individual objectives. Each objective gives rise to one or more variables to be measured (inputs or outputs).

The PLs for each variable for all objectives are added to form the **Total Performance Loss** (TPL) to be minimized. In addition, you can impose any constraints you wish (with the possibility that the process is incapable to satisfy them all, in which case the TPL is irrelevant).

The tolerances TOL is the deviation from the target or reference which is the threshold into intolerable results. For the PLs to be additive, the TOLs must be "**equivalent**", i.e., the negative effect to the enterprise in a deviation from the target or reference by the TOL (all other things being equal) and must be about the same for all variables, within a factor or about two.

Each objective should be included only once. Let us imagine that a minimum strength is required, and the resulting thickness that should be sufficiently large to provide this strength be known. It would be misleading (but not a failure) to include both the Performance Losses for strength and for thickness. In general, it is better to include just the ultimate objective (here, strength). On the other hand, you may include a PL of the thickness for objectives other than strength (e.g., cost).

Notes:

For the calculation of summaries remember the rules for minimum sample sizes to detect differences in performance, as described in page 195.

All constants such as targets, references, tolerances, etc. can be **Global Factors**, easily changed at the software user interface level. These and the values of **Constraints** become a main tool by which production management refine the definition of objectives to steer production into better satisfaction of evolving company and customer needs.

It lends clarity to think of maximum "tolerable" deviations, the tolerance (TOL), as the beginning of "**in**tolerable" deviations.

GOAL SETTING					**SET UP**	
Individual Objective	**Variables to measure, type (Avg., Std. Dev.), units**	**TGT / REF**	**TOL onset of intolerable deviations**	**FORM**	**EQUATION PL = Fortran code (all additive)**	**BUILT-IN FUNCTION PL = (all additive)**
Average or single measure is important. The consumption of a surface treatment chemical should be 1.1 gr./part. The tolerance deviation is .07. Part-to-part variation is not a critical issue, in fact it is immeasurable (just as it is for single measures of bulk processes, such as concentration, viscosity, pH, etc.).	Consumption: AVG gr./pt	1.10	0.07	P	((AVG- 1.10)/0.007)**2	PLS (AVG, 1.10, 0.07, 'P')

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GOAL SETTING					SET UP	
Individual Objective	**Variables to measure, type (Avg., Std. Dev.), units**	**TGT / REF**	**TOL onset of intolerable deviations**	**FORM**	**EQUATION PL = Fortran code (all additive)**	**BUILT-IN FUNCTION PL = (all additive)**
Specifications for individual parts Dimension A should be .8700 inches for all parts. The specifications are ± .0010 inches.	Dimension A: AVG in. Dimension A: SD in.	0.87 0.00	0.001 0.001	P p	((AVG - .87)/0.001)**2 (SD/0.001)**2	PLS (AVG, 0.87, 0.001, 'P') PLS (SD, 0., 0.001, 'P')
Specifications for individual parts in % deviation. Dimension B should be TGT inches for all parts, where the value of TGT depends on the model of part being made. The specification are ± 0.01 (1%).	Dimension B: AVG in. Dimension B: SD in.	TGT 0	0.01 0.01	P P	((AVG/TGT -1)/0.01)**2 (SD/0.01)**2	PLS (AVG/TGT, 1.,0.01, 'P') PLS (SD/TGT, 1., 0.01, 'P')
Average deviations from target are undesirable mostly in one direction (asymmetric).[64]						
a. Moisture content should be 5%. It reduces product performance to have more, but it does not subtract or add any value to have less moisture. The intolerable deviation is +2% moisture.	Moisture: AVG %	5	+2	P	IF (AVG. LT. 5) PL= (AVG – 5) / (2))**2 IF (AVG. GT. 5) PL = 0	PLA (AVG, 5., 2., 'P', 0., 'N') (the 0. right TOL is irrelevant because of the 'N')
b. Similarly, but less moisture reduces production rate linearly if processes downstream. The intolerable negative deviation is -4%. Alternatively, the effect could be quadratic (parabola).	Moisture: AVG %	5	–4, +2	P	IF (AVG. LT. 5) PL = (AVG – 5) / (–4) IF (AVG. GT. 5) PL = ((AVG – 5) / 2)**2	PLA (AVG, 5., -4., 'L', 2., 'P') (if it were 'P', there is no difference between -4 and +4, but the sign lends clarity)
Only variation from average is important. Dimension D should be approximately .4 inches. What is important is that it must be the same for all parts, as any constant deviation is regularly adjusted for in the setting of the fixtures. The deviation from average at which the loss begins to be intolerable is .00025.	Dimension D: SD in	0	0.00025	P	(SD/0.00025)**2	PLS (SD, 0., 0.00025, 'P')

[64] Note that sometimes this lack of symmetry can be resolved by working with the logarithm of the result variable, such as measuring noise levels with decibels.

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GOAL SETTING					SET UP	
Individual Objective	**Variables to measure, type (Avg., Std. Dev.), units**	**TGT / REF**	**TOL onset of intolerable deviations**	**F O R M**	**EQUATION PL = Fortran code (all additive)**	**BUILT-IN FUNCTION PL = (all additive)**
Percent defective. The surface appearance should meet specifications, and no parts should be defective. Intolerable percent is 3%.	Percent Defective: PD%	0	3	L	PD/3	PLS (PD, 0., 3., 'L')
Production rate. The process should increase production rate. It is now making 45 pts./hr. Intolerable deviation begins at -30%.						
a. Capacity constrained environment. The most important contribution to P&L profitability is more sales. Use parts/hour.	Production rate: AVG pt/hr	45	-30%	L	(AVG/45 - 1.) / (-0.30)	PLS(AVG/45., 1., -0.30, 'L')
b. Sales constrained environment. The most important contribution to profitability is reduced variable costs. Use minutes/part. The reference is 1.333 min/pt (60/45). The deviation from reference at which the loss begins to be intolerable is .571 min/pt (the inverse of 45 - 30% pts./hr, minus the reference).	Production rate: AVG min/pt	1.333	0.571	L	(AVG - 1.333)/.571	PLS(AVG, 1.333, 0.571, 'L')
Costs. The process energy consumption should be as low as possible. It now costs \$785/hr. The loss which begins to be intolerable is \$120/hr. Note that this objective means that each UOL is worth \$120/hr (after the TOL for all objectives are "equivalent").	Cost of Energy: AVG \$/hr	785	120	L	(AVG - 785)/120	PLS (AVG, 785., 120., 'L')

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GOAL SETTING					SET UP	
Individual Objective	**Variables to measure, type (Avg., Std. Dev.), units**	**TGT / REF**	**TOL onset of intolerable deviations**	**FORM**	**EQUATION PL = Fortran code (all additive)**	**BUILT-IN FUNCTION PL = (all additive)**
Material strength (high but consistent). The strength of the composite material should be as high as possible, but consistent, so that the product has the same feel. The reference strength is 15 kg/mm^2. The deviation from reference strength at which the loss in strength begins to be intolerable is -30%. The individual part deviation from the average at which the loss begins to be intolerable is 10%.						
a. In product development the aim the outcome of increasing the material strength is the reduce the dimension to save material cost. Therefore, the inverse of strength, mm^2/kg, or material requirement, is linearly related to cost. The current reference is .0667 mm^2/kg (the inverse of 15 kg/ mm^2). The tolerance deviation is +.02 mm^2/kg (30% of .0667). The deviation from average at which the loss begins to be intolerable is 10%.	Strength: AVG mm^2/kg Strength: SD mm^2/kg	0.0667 0	+0.02 10%	L P	(AVG - 0.0667)/.02 ((SD/AVG)/.10)**2	PLS (AVG, 0.0667, 0.02, 'L') PLS (SD/AVG, 0., 0.10, 'P')
b. In production operations the dimension of the part is fixed, and the only advantage is to provide more part strength, with longer product life. The value to the customer is not linear with strength, and it reaches a limit where further gains are irrelevant. Unless a more functional analysis of the situation is required, the Taguchi approach, a quadratic of the inverse, may be appropriate. The deviation from ideal inverse strength (0) at which the loss begins to be intolerable is .095 (the inverse of 15 kg/mm^2 minus 30%). The deviation from average at which the loss begins to be intolerable continues to be 10%.	Strength: AVG mm^2/kg Strength: SD mm^2/kg	0 0	0.095 10%	P P	(1/AVG**2) /.095 (SD/AVG)/.10)**2	PLS (1/AVG**2, 0., 0.095, 'P') PLS (SD/AVG, 0., 0.10, 'P')

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GOAL SETTING					SET UP	
Individual Objective	**Variables to measure, type (Avg., Std. Dev.), units**	**TGT / REF**	**TOL onset of intolerable deviations**	**FORM**	**EQUATION PL = Fortran code (all additive)**	**BUILT-IN FUNCTION PL = (all additive)**
Individual Objective	Variables to measure, type (Avg., Std. Dev.), units	TGT / REF	TOL onset of intolerable deviations	FORM	EQUATION PL = code (all additive)	BUILT-IN FUNCTION PL = (all additive)
Average or single measure is important. The consumption of a surface treatment chemical should be 1.1 gr./part. The tolerance deviation is .07. Part-to-part variation is not a critical issue, in fact it is immeasurable (just as it is for single measures of bulk processes, such as concentration, viscosity, pH, etc.).	Consumption: AVG gr./pt	1.10	0.07	P	((AVG- 1.10)/0.007)**2	PLS (AVG, 1.10, 0.07, 'P')
Specifications for individual parts Dimension A should be .8700 inches for all parts. The specifications are ± .0010 inches.	Dimension A: AVG in. Dimension A: SD in.	0.87 0.00	0.001 0.001	P p	((AVG - .87)/0.001)**2 (SD/0.001)**2	PLS (AVG, 0.87, 0.001, 'P') PLS (SD, 0., 0.001, 'P')
Specifications for individual parts in % deviation. Dimension B should be TGT inches for all parts, where the value of TGT depends on the model of part being made. The specification are ± 0.01 (1%).	Dimension B: AVG in. Dimension B: SD in.	TGT 0	0.01 0.01	P P	((AVG/TGT -1)/0.01)**2 (SD/0.01)**2	PLS (AVG/TGT, 1.,0.01, 'P') PLS (SD/TGT, 1., 0.01, 'P')
Average deviations from target are undesirable mostly in one direction (asymmetric).[65]						
a. Moisture content should be 5%. It reduces product performance to have more, but it does not subtract or add any value to have less moisture. The intolerable deviation is +2% moisture.	Moisture: AVG %	5	+2	P	IF (AVG < 5) PL = (AVG – 5) / 2)**2 IF (AVG > 5) PL = 0	PLA (AVG, 5., 2., 'P', 0., 'N') (the 0. right TOL is irrelevant because of the 'N')

[65] Note that sometimes this lack of symmetry can be resolved by working with the logarithm of the result variable, such as measuring noise levels with decibels.

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GOAL SETTING					SET UP	
Individual Objective	**Variables to measure, type (Avg., Std. Dev.), units**	**TGT / REF**	**TOL onset of intolerable deviations**	**FORM**	**EQUATION PL = Fortran code (all additive)**	**BUILT-IN FUNCTION PL = (all additive)**
b. Similarly, but less moisture reduces production rate linearly if processes downstream. The intolerable negative deviation is -4%. Alternatively, the effect could be quadratic (parabola).	Moisture: AVG %	5	–4, +2	P	IF (AVG < 5) PL = (AVG – 5) / (–4) IF (AVG > 5) PL = ((AVG – 5) / 2)**2	PLA (AVG, 5., -4., 'L', 2., 'P') (if it were 'P', there is no difference between -4 and +4, but the sign lends clarity)
Only variation from average is important. Dimension D should be approximately .4 inches. What is important is that it must be the same for all parts, as any constant deviation is regularly adjusted for in the setting of the fixtures. The deviation from average at which the loss begins to be intolerable is .00025.	Dimension D: SD in	0	0.00025	P	(SD/0.00025)**2	PLS (SD, 0., 0.00025, 'P')
Percent defective. The surface appearance should meet specifications, and no parts should be defective. Intolerable percent is 3%.	Percent Defective: PD%	0	3	L	PD/3	PLS (PD, 0., 3., 'L')
Production rate. The process should increase production rate. It is now making 45 pts./hr. Intolerable deviation begins at -30%.						
a. Capacity constrained environment. The most important contribution to P&L profitability is more sales. Use parts/hour.	Production rate: AVG pt/hr	45	-30%	L	(AVG/45 - 1.) / (-0.30)	PLS(AVG/45., 1., -0.30, 'L')

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GOAL SETTING					SET UP	
Individual Objective	**Variables to measure, type (Avg., Std. Dev.), units**	**TGT / REF**	**TOL onset of intolerable deviations**	**FORM**	**EQUATION PL = Fortran code (all additive)**	**BUILT-IN FUNCTION PL = (all additive)**
b. Sales constrained environment. The most important contribution to profitability is reduced variable costs. Use minutes/part. The reference is 1.333 min/pt (60/45). The deviation from reference at which the loss begins to be intolerable is .571 min/pt (the inverse of 45 - 30% pts./hr, minus the reference).	Production rate: AVG min/pt	1.333	0.571	L	(AVG - 1.333)/.571	PLS (AVG, 1.333, 0.571, 'L')
Costs. The process energy consumption should be as low as possible. It now costs $785/hr. The loss which begins to be intolerable is $120/hr. Note that this objective means that each UOL is worth $120/hr (after the TOL for all objectives are "equivalent").	Cost of Energy: AVG $/hr	785	120	L	(AVG - 785)/120	PLS (AVG, 785., 120., 'L')
Material strength (high but consistent). The strength of the composite material should be as high as possible, but consistent, so that the product has the same feel. The reference strength is 15 kg/mm^2. The deviation from reference strength at which the loss in strength begins to be intolerable is -30%. The individual part deviation from the average at which the loss begins to be intolerable is 10%.						
a. In product development the aim the outcome of increasing the material strength is the reduce the dimension to save material cost. Therefore, the inverse of strength, mm^2/kg, or material requirement, is linearly related to cost. The current reference is .0667 mm^2/kg (the inverse of 15 kg/mm^2). The tolerance deviation is +.02 mm^2/kg (30% of .0667). The deviation from average at which the loss begins to be intolerable is 10%.	Strength: AVG mm^2/kg Strength: SD mm^2/kg	0.0667 0	+0.02 10%	L P	(AVG - 0.0667)/.02 ((SD/AVG)/.10)**2	PLS (AVG, 0.0667, 0.02, 'L') PLS (SD/AVG, 0., 0.10, 'P')

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GOAL SETTING					SET UP	
Individual Objective	**Variables to measure, type (Avg., Std. Dev.), units**	**TGT / REF**	**TOL onset of intolerable deviations**	**FORM**	**EQUATION PL = Fortran code (all additive)**	**BUILT-IN FUNCTION PL = (all additive)**
b. In production operations the dimension of the part is fixed, and the only advantage is to provide more part strength, with longer product life. The value to the customer is not linear with strength, and it reaches a limit where further gains are irrelevant. Unless a more functional analysis of the situation is required, the Taguchi approach, a quadratic of the inverse, may be appropriate. The deviation from ideal inverse strength (0) at which the loss begins to be intolerable is .095 (the inverse of 15 kg/mm^2 minus 30%). The deviation from average at which the loss begins to be intolerable continues to be 10%.	Strength: AVG mm^2/kg Strength: SD mm^2/kg	0 0	0.095 10%	P P	(1/AVG**2) /.095 (SD/AVG)/.10)**2	PLS(1/AVG**2, 0., 0.095, 'P') PLS (SD/AVG, 0., 0.10, 'P')

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Approach 3: Performance Index is a Single Output, Plus Constraints

One way to represent Goals with multiple objectives is to optimize a single objective while using Constraints to represent all the other objectives[66].

Some typical Goals that include Constraints are:

- Maximize yield while quality and costs are within acceptable levels (keep main feed rate constant).
- Maximize fuel efficiency (minimize heat rate in steam boilers) while maintaining various emissions within regulatory limitations and intermediate process variables within safety/reliability constraints.
- Maximize production rate while quality and costs are within acceptable levels (vary feed rates).
- Minimize level of impurities while production rate while other quality characteristics and costs are within acceptable levels.
- Minimize the deviation from target calculated based on the average and the standard deviation, while maintaining production above a minimum acceptable level and quality levels within limits.
- Minimize unit costs while maintaining process capabilities and production levels within acceptable levels. This example involves using the "CPK function" to describe Process Capability.

As described in General considerations in Goal Setting above, this approach, while being the simplest, suffers from the following fallacy: everything with "specifications" is frequently not equally satisfactory or of equal value to the customer.

[66] Remember that the higher priority in the software is to satisfy constraints. Only after this is achieved does the software attempt to improve the objective function. It is possible to have constraints in the objective function, so that this is dealt with at the same priority as other constraints.

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Miscellaneous topics

The following sections discuss miscellaneous topics:

- Traditional measure of capability, the CPK
- Variable constraints
- Mixture problems
- Taguchi method.

CPK Measure of Process Capability

CP and CPK measure the relationship between product quality and specification limits for multiple production parts.

This type of measure is particularly representative if any value within the specifications may be assumed to be equally good for the enterprise and for the customer.

The measurements required for calculating CPK and CP are the average (AVG) and the standard deviation (SD) of multiple raw measurements made under the same input conditions.

CP reflects the potential performance if the process is perfectly targeted. This information is useful when adjusting the average is easy and will not affect other results.

$$CP = \frac{Specification_spread}{\mathrm{Pr}\,ocess_spread} = \frac{USL - LSL}{6 * SD} = \frac{TOL}{3 * SD}$$

where:

TGT	=	Target, nominal value
TOL	=	Tolerance deviation from target
USL	=	TGT + TOL: Upper Specification Limit
LSL	=	TGT - TOL: Lower Specification Limit
USL - LSL	=	Specification or Tolerance Range
6 *SD*	=	Process Spread

CPK reflects the current process performance, including the lack of targeting.

$$CPK = \min\left\{\frac{USL - AVG}{3 * SD}, \frac{AVG - LSL}{3 * SD}\right\}$$

Note that a process could run in two different ways with the same CPK with different AVG and SD, but with very different variations from target as calculated with the Performance Loss:

Let:

TGT = 10; TOL = 1

CPK = 2

in two ways:

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AVG1 = 10.3; AVG2 = 10.7

Calculations:

$$CPK = 2 = \frac{10+1-10.3}{3*SD1} = \frac{10+1-10.7}{3*SD2}$$

or

$$SD1 = \frac{.7}{2*3} = 0.117$$

$$SD2 = \frac{.3}{2*3} = 0.058$$

since :

$$PL = \left(\frac{AVG-TGT}{TOL}\right)^2 + \left(\frac{SD}{TOL}\right)^2$$

$$PL1 = \left(\frac{.3}{1}\right)^2 + \left(\frac{.117}{1}\right)^2 = .090 + .014 = .103$$

$$PL2 = \left(\frac{.7}{1}\right)^2 + \left(\frac{.058}{1}\right)^2 = .490 + .003 = .493$$

which, as we can see, is dominated by the location of the average.

Variable Constraints

Variables (inputs and outputs) do not always have **constant** constraints. The constraints could be a function of other variables. The ULTRAMAX software only handles directly constant constraints. Thus, variable constraints conditions have to be represented by new variables with constant constraints.

For instance, let us assume that the situation is that the ratio of two variables $y_i/y_j < 0.35$.

1. We could define a new calculated output $R = y_i/y_j$ with the constant upper constraint 0.35 so that

   $$R = y_i/y_j < 0.35$$

2. Equivalently, thinking of y_i as the variable with a variable constraint,

   $$y_i < 0.35*y_j$$

   introduce a calculated slack output with a lower constraint of 0

   $$0 \le S = 0.35*y_j - y_i$$

.

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The general description of these two approaches to represent variable constraints with constant constraints is:

1. Create a conceptual, calculated output that has constant constraints.

   Introduce a conceptual variable ***v*** which is:

   - a function of other variables, where V stands for the set of other variable values:

     $$v = F(V)$$

   - restricted to constant upper, lower, or both constraint(s).

2. For the desired existing output create a conceptual, calculated slack output with lower constraint of zero.

   Let an existing variable ***y*** be restricted by a lower constraint which is a function C(V) of other variables, where V stands for the set of other variable values. Let the requirement be that

   $$C(V) \le y$$

   Introduce a conceptual variable which is a slack

   $$0 \le S = y - C(V)$$

   which has the constant constraint that the software can handle.

   It could also be that variable ***y*** has an upper constraint which is a different function D(V), which will be handled in the same way with another slack calculated output.

Mixture Applications

Decision diagrams have "mixture" aspects when the Optimization Plan addresses percent of various ingredients (rather than the total amount of each ingredient). The total percent must add to 100.

This approach is typically used in the development of product formulations to excel in certain properties given cost and producibility considerations, such as creating new detergents, chemicals, pharmaceuticals, and alloys.

If there are N ingredients, then there are really N–1 decision (control) inputs and the other ingredient is a dependent variable (a ruled input) being a direct consequence of the values of the other ingredients.

Example: Let Alpha, Beta, Gamma, and Delta be the four ingredients. Beta will be a ruled input (a dependent variable, a consequence of the other inputs).

Name	Type	Entry Mode	Constraints	
			Low	High
Alpha	Control		0	100
Beta	Ruled	Calculated	0	100
Gamma	Control		0	100
Delta	Control		0	100

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The equation defined in the setup of the optimization plan in the software is:

Beta = 100 - Alpha - Gamma - Delta

The Constraints can be other numbers between 0 and 100 if functional, safety or legal restrictions exist.

In a case like this, the detected effects of Alpha in the Shape Report and the coefficients in the coefficient report -- (keeping Gamma and Delta constant -- reflect the effect of increasing Alpha **and** decreasing Beta by the same amount.

Of course, this exercise can be just a part of a larger Decision Diagram, with a variety of other inputs and results, with their own Constraints (if any), and any Performance Index.

The Taguchi Method

The Taguchi Method includes two distinct issues:

1. The engineering concept that a better design is one where future performance of the product is more insensitive to the variation of “noise” factors; and that it is practical to run controlled experiments to design products and processes to learn how to reduce the sensitivity to these factors. This insensitivity of product performance to such noise factors is called *robustness*.

   Given a set of design or process parameters, the Taguchi method recommends experiments to evaluate the sensitivity to these changes. The experiment consists of several Runs with the "noise" factors at particular controlled levels. The evaluation includes, typically, a quadratic loss function summarized into a signal-to-noise value which is to be maximized.

2. How to run several of these experiments in order to identify the most desirable combination of design or operating parameters.

   When the Taguchi experiments are done sequentially and the results can be obtained on a timely basis; ULTRAMAX’s SEO offers the ability to identify the optimal region of the design parameters better, and generally with fewer experiments[67] than with Taguchi’s tables. This is particularly true when *either*:

   - "Optimization" is interpreted as finding the best combination of all possible levels of the design parameters; i.e.,

     ⇒ not restricted to those values of the design parameters selected for the factorial DOE;

     ⇒ within and outside the original exploration region.

   - Multiple objectives exist, which may include cost and safety constraints.

[67] Each design “experiment” consists of several simulations with controlled changes in the “noise factor”. The data for ULTRAMAX are the average and standard deviations summaries (and perhaps maximum and minimum values).

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- Experimental results are obtained through computer simulations. Then, linking the two programs (e.g., through transfer files) would enable automatic optimization.

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The SEO Technology

Introduction

Basically SEO manages a sequence of periodic readjustment[68] to the control inputs of an input/output system with the aim to maximize the cumulative sequence of overall value delivered by the process. SEO continually updates its learning from the stored run data.

Some earlier approaches to solve this problem were EVOP (e.g., Box and Draper, 1969) and SIMPLEX (Spendley, Hext and Himsworth; 1962, Walters, F.H. et al. 1991). As computer power became accessible to production in the late 60's, Dr. Moreno lead efforts within Procter & Gamble to develop and implement new algorithms taking advantage of what is possible with computer programming and advanced statistical modeling in order to vastly overcome the limitations of those earlier methods.

The SEO started in 1982 is the third version started from scratch. Several hundreds of individual processes have been optimized with ULTRAMAX.

Desirable SEO features

These are the properties which are necessary, or highly desirable, for managing process adjustments, and define SEO (from Moreno, 1993, 1994):

- **User Control:** As one aspect of flexibility, users are not required to follow the sequential advices provided by the technology. This is desirable for early acceptance. It is also desirable during the earlier adjustment cycles when very little data has been collected, to enable existing experience to be included in the run data base.

  Sequential cycles can be stopped at any time to allow you to continue taking advantage of the gains achieved so far. Later on, you can continue where you left off, making use of the stored run data.

- **Multiple inputs, multiple outputs:** It deals with multiple inputs (some adjusted, some uncontrolled) affecting multiple outputs (some directly measured or estimated, some calculated as a function of other inputs and outputs with internal equations written by the user). Control inputs include physical variables adjusted manually, setpoints, and control logic parameters (such as gains for first-level controllers). Some important families of calculations are:

  1. mass and energy balance equations to estimate certain results which cannot be measured directly
  2. estimations of final results based on intermediate and input variables (e.g., NOx based on more readily measurable operating variables)
  3. calculations of consistency vs. constraints or specifications, such as CPK and Loss Functions; and
  4. economic equations taking all costs (and potential revenues) into account.

[68] In this book “adjusting” control inputs is used the same as “setting” control inputs, including adjusting or setting things by hand. Unlike feedback mechanism, SEO does not directly recommend a change or correction in an adjustment, but advises specific values.

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- **Multiple objectives:** The evaluation of process performance is determined in quantitative terms defined by the team, where objectives are set by management. Individual objectives might relate to measures such as yields, production rates, economic evaluations (costs and revenues), quality, safety and equipment life, losses, emissions, byproducts, and satisfaction of regulatory constraints. Basically, if performance can be measured or estimated with sufficient accuracy and on a timely basis to learn from it, it can be taken into account.

  Goals are represented by:

  a. a "Performance Index" or "objective function", an output to be maximized or minimized (which can be a directly measured value, or a function of other variables

  b. by necessary constraints on inputs and outputs

- **Processes with noise:** It deals effectively with process outputs that have "noise". Noise is the level of output data variations when inputs remain constant. Like all empirical solutions, the larger the noise the less effective it will be.
- **Can use prior valid data:** The analysis can be primed with valid data from previous runs.
- **Prior Model free:** It does not require that the user provide mathematical / computer models of the behavior of the process. However, if the models are available it can use them to give the technology a jump-start. These models could just be approximations for some outputs as a function of some inputs.
- **Feed-forward adjustments:** It can determine optimal adjustments for known values of uncontrolled inputs (e.g., resource allocation, materials characteristics, ambient conditions, process conditions). This is feed-forward optimization, based on predictive models (without requiring the process to deviate from objectives before corrective action takes place, such as in feedback control).
- **Feed-back adjustments:** It can do feed-back optimization when the process changes slowly due to changes in unknown uncontrolled inputs.
- **Closed-loop implementation:** It is applicable in the spectrum from Stand-Alone (no integration), to integrated but full hands-on human control (an Advisory capacity), to fully automatic on-line closed-loop optimization without requiring human intervention and interpretation (where ULTRAMAX provides reliable direct supervisory control, including efficient alerts).

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Mathematical Formulation of SEO

The Process, Requirements and Optimum

The process behaves as follows (**X, U, Y** and **ε** are vectors):

Y = **F**(**X**,**U**) + **ε**

where:

- **X** is the vector of adjusted inputs (control inputs), the decision that controls operational performance
- **U** is the vector of uncontrolled inputs, with values determined elsewhere (by people or nature)
- **Y** the vector of outputs (after transients due to readjustments), the outcome or consequence from the inputs **X**,**U**. Y_0 is the value of the Performance Index or **Objective Function**
- **F**(**X**,**U**) is the steady-state mean (i.e., after transients due to readjustments) process output vector, or the "response". As we shall see, the form of **F** is unknown, the same as for **ε** below.

  F, as representative of production processes, is relatively "smooth" in the area of interest. **F** could possibly be changing slowly with time (or equivalently, changing because of slow changes in unknown uncontrolled inputs not included in **U**).

- **ε** is an bell-shaped[69] noise vector with mean zero and covariance matrix Σ with the diagonal of variances, whose square root is the standard deviation, sigma or noise vector **N**. It is most likely affected by **X,U.**

  ε is a property of the process AND of the OP, in particular, the inputs included and how **Y** is measured (e.g., averages of more raw data may have lower noise[70]).

  This concept of noise is smaller than in most other quality control analyses because changes in the known uncontrolled inputs do not contribute to noise, while they do in regular methods.

Note that in this model "errors" are assigned only to the outputs. The inputs are presumed to be absolutely correct.

---

[69] With the understanding, contrary to remarks in vogue in the last few years in certain fields, that very few processes produce any data following a Normal distribution. Note that there are almost no physical principles that indicate that process outputs should be normally distributed (except, e.g., the energy emitted by a black object vs. the log of the frequency; the addition of several distributions of similar independent standard deviations – Central Limit Theorem). On the contrary, for instance, a process often places limits – resulting in trimming tails -- which destroys a normal distribution. This has been confirmed by experience, where with sufficient data one almost always could prove that process output distributions are not normal.

[70] Especially if **U** does not change much.

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The requirements and optimal operations are defined as:

Maximize the objective function (Performance Index) by selecting the adjustments X that satisfies the constraint requirements, given the value of U. The constraint requirements are:

a) ***The mean[71] process values almost never violates the upper (UC) and lower (LC) constraints.***

b) ***The actual data for each run almost never violates a constraint by more than an amount "Minimum Important Difference" (MID). The MID allows for a "gray" area for the actual data constraint violation.***

Mathematically:

$\mathbf{Max_{X/U}}$ **{**F_0**(X,U)}** 0 for objective function

s.t.[72] subject to; that is, while satisfying these the constraint requirements

Requirement **(a)**

$X_i \le UC_i$ i=inputs with upper constraint UC

$X_i \ge LC_i$ i=inputs with lower constraint LC

$F_i(X,U) \le UC_i$ i=outputs with upper constraint UC

$F_i(X,U) \ge LC_i$ i=outputs with lower constraint LC

Requirement **(b)**

$Y_i = F_i(X,U) + \varepsilon_i \le UC_i + MID_i$ most of the time

Yi = Fi(X,U) + εi ≥ LCi - MIDi most of the time

The "most of the time" (b) requirement is translated into requiring that the practical worst noise level ε_i to be satisfied be 3-sigmas[73] ($3*N_i$), called the 3-sigma Protection:

$F_i(X,U) \le UC_i - 3*N_i + MID_i$

$F_i(X,U) \ge LC_i + 3*N_i - MID_i$

Similarly, practical optimization is when the achieved F_0 is within MID_0 of the optimal one defined above.

---

[71] In this book "**mean**" is used to express the average for the same values of the inputs; "**average**" is used for the average across time (with possible changes in inputs).

[72] s.t. means "subject to", or, within these constraint limits as a higher priority

[73] We are not using the criterion of the probability of violating constraints in order to avoid having to make assumptions about the distribution of the data (see footnote 69).

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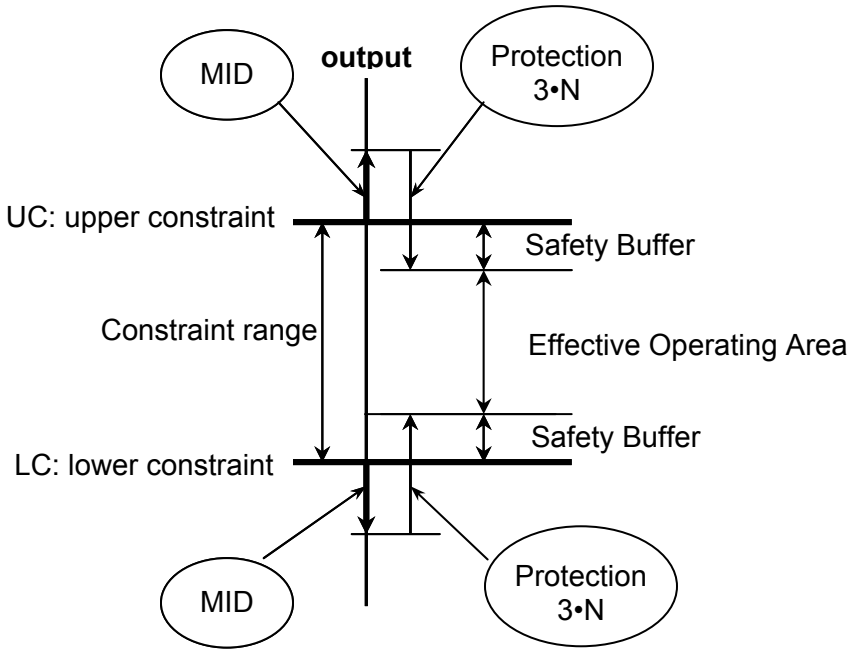


Fig. 25 : The "Effective Operating Area" to satisfy (b) requirement; the area of acceptable output means.

So, bringing both constraint requirements (a) and (b) into one composite set of equations, the optimum we are searching is defined as:

Given the values of U,

$\mathbf{Max}_{\mathbf{X/U}}\ \{F_0(\mathbf{X},\mathbf{U})\}$ 0 for objective function

s.t. $X_i \le UC_i$ i=inputs with upper constraint UC

$X_i \ge LC_i$ i=inputs with lower constraint LC

$F_i(X,U) \le UC_i - \max\{3*N_i - MID_i, 0\}$ i=outputs with upper constraint UC

$F_i(X,U) \ge LC_i + \max\{3*N_i - MID_i, 0\}$ i=outputs with lower constraint LC

which defines the *optimum adjustment* $\mathbf{X}^*$ and the *mean optimum outputs* $\mathbf{Y}^* = \mathbf{F}(\mathbf{X}^*,\mathbf{U})$; both for each value of U.

The practical objective is not quite to find the optimal **X**, but to define **X**s that satisfy all constraints and produces mean outputs no further away from the optimal $F(\mathbf{X}^*,\mathbf{U})$ than MID_0. The set of such **X**s defines the Window of Operations.

If there are upper and lower constraints and the Safety Buffer (defined below) is sufficiently large, there is no Operating Range and the process is totally incapable. In this case optimization makes little sense – the problem should be fixed.

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An equivalent formulation of the output constraints is:

$UC_i - F_i(X,U) \geq \max\{3*N_i - MID_i, 0\}$ i=outputs with upper constraint UC

$F_i(X,U) - LC_i \geq \max\{3*N_i - MID_i, 0\}$ i=outputs with lower constraint LC

This form has a useful interpretation. The left-hand-terms are the mean Slacks (how far from constraints, negative is beyond); the right-hand-term is defined as the Safety Buffer; as follows:

$\text{Mean Slack}_i = UC - F_i(X,U)$ (upper)

$\text{Mean Slack}_i = F_i(X,U) - LC$ (lower)

$\text{Safety Buffer}_i = \max\{3*N_i - MID_i, 0\}$

(recall, Protection = $3*N_i$)

Thus, the optimum output constraint requirements are simply that:

Mean Slack ≥ Safety Buffer

Note that for the active outputs (that is, with active constraints) if the noise is sufficiently small, $N_i < \frac{MID_i}{3}$, then the Safety Buffer is zero, and the only constraint requirements is for the mean process values [requirement (a)]; and reduction of noise does not help except for the Performance Index. If the (mean output) constraints and the MIDs are properly defined from a business point of view, this is the sign of a process under good process control (and quality data).

On the other hand, if MID is zero, the only constraint requirement is for the actual process actual values [requirement (b)]. In this case, any reduction of noise for an active output results in potential improvements of the Objective Function (Performance Index).

Sequential Empirical Optimization (SEO)

The characteristics of a Sequential Empirical Optimization solution to optimize a process are:

- It is Empirical: that is, series of run data {**X**,**U**,**Y**} are known, but some (or all) **F**(**X**,**U**) and **ε** are not. In particular the structure of **F** is also not known except to assume reasonable "smoothness".
- It is Sequential: that is, it continually stores operating (run) data and continually uses the stored run data and extracts from it information and knowledge to create updated sequential advice to adjust the process next. Then a process run produces a new {**X,U,Y**} which in turn is stored in the database and repeat the cycle. Thus SEO creates a series of $\{\mathbf{X,U,Y}\}_t$ stored in a data base. The first $\mathbf{X}_t$ is the adjustments at Baseline, defined by the user.

  Note how the sequential analysis above emulates the process of a mind gaining experience through repeated action, and using the remembered actions, conditions and outcomes to make better

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action decisions in the future, and aspect or learning and skill development.

Sequential analysis has a very valuable advantage (à la [3] Wald): it uses the information in each sequential run immediately in order to refine knowledge and to decide where to run next. By comparison, in other more parallel empirical studies (DOE, Neural Networks) the value of the information is not used until the data is analyzed at the end. This is the most important reason why a properly implemented SEO technology is the fastest empirical optimizer available today[74].

SEO Maximization is meant as a local optimum at the end-path of continuous improvements from the starting adjustments, and ideally jumping over minor local optima. In particular, finding "other mountains" to climb is taken as a responsibility of engineering or R&D, not of daily production.

The performance of a SEO solution -- for the likely process forms of $\mathbf{Y} = \mathbf{F}(\mathbf{X},\mathbf{U}) + \boldsymbol{\varepsilon}$ that will be optimized -- is evaluated, basically, by:

- How quickly the series $\{\mathbf{X},\mathbf{U},\mathbf{Y}\}_t$ converges to the optimum $\{\mathbf{X}^*,\mathbf{U},\mathbf{Y}^*\}$.
- How closely the series $\{\mathbf{X},\mathbf{U},\mathbf{Y}\}_t$ approach to the optimum $\{\mathbf{X}^*,\mathbf{U},\mathbf{Y}^*\}$

In manufacturing, which is a continual value generation situation, the performance of SEO is evaluated by the cumulative process performance while following the advice provided by the technology[75] -- or more specifically, since the implementation of optimization is started[76].

(In engineering and development the performance would be evaluated by the costs and delay incurred in the number of "experiments" and by how close the solution approaches the optimum.)

Depending on the SEO technology used (or on how it is adjusted), achievements in terms of the objective function F_0 may frequently be increased at the sacrifice of increasing the incidence of violating some constraints, and thus the SEO needs to proceed carefully. The sequence of $\mathbf{X}_t$ needs to be very "intelligent": low risk and effective. It must have a correct balance between:

- being conservative so as to avoid making the process violate constraints or sacrifice performance too much,
- being bold, moving towards improved performance,
- creating new data so that future models have improved prediction capabilities,

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[74] The other synergistic reason (bootstrapping) is that since SEO approaches the optimum faster, the database has relatively more data around the optimum, which is the most valuable data to understand the location of the optimum (recall, SEO does not assume knowing the structure of the process transfer function).

[75] The concept is very clear when there are no constraint violations: find the cumulative or average Performance Index. When there are violations, this approach requires placing a penalty for constraint violation, which is something somewhat arbitrary.

[76] This clarification is made to take into account the performance lossses and time that it takes to collect experimental data for model-based alternative approaches (such as DOE and NN); and the time to design and implement the first-principle models and collect data to validate it

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- providing compensation for known values of the uncontrolled inputs (feed-forward optimization), and
- reacting to changing behavior due to unknown uncontrolled inputs (feedback optimization).

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The ULTRAMAX solution for SEO

Fundamentals

In addition to storing run data, the solution includes of two basic functions: Learning (creation of predictive models); and Synthesis (using the models to generate advice for the next run).

1. **Learning, model building**

   Models of generic form **M(X,U)** $\cong$ **F(X,U)** are created for new values of **U** or when new run data is collected in the database, based on the set of historical run data $\{\mathbf{X,U,Y}\}_t$.

   At the same time the technology creates:

   - An Area of Confidence (AOC), which is the region in **X,U** where **M(X,U)** is most accurate – within the region covered by $\{\mathbf{X},\mathbf{U}\}_t$.

     It is calculated by an elaborate pattern application of the Mahalanobis [1936] distance which allows the AOC to be concave or composed of disjointed areas.

   - An estimate $\sum$ of **ε** of and the noise **n**

   The prediction models enable "What-if" estimation or outputs given the inputs (forward analysis).

2. **Synthesis, Advice**

   Synthesis is almost the reverse of What-if. Synthesis is finding the values of the inputs which satisfy some criteria for the estimated outputs. Here it is "which-is-best" for decision-making.

   The solution is the same as for the theoretical definitions of optimum, except that:

   - It is obtained with estimated functions **M** and noise **N**. In particular, the prediction error E_i depends of the level of extrapolation, as typical of regression models.
   - {**X**,**U**} belonging to the AOC, where **M** is most accurate.
   - Adds certain perturbations to that future models be more representative but without sacrificing performance much[77].

[77] This is the leftover of the benefits of orthogonality in DOE.

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The Advice **X** for the next run is produced by:

$\mathbf{Max}_{\mathbf{X/U}}$ $\{M_0(\mathbf{X},\mathbf{U})\}$ 0 for objective function

s.t. $X_i \le UC_i$ i=inputs with upper constraint UC

$X_i \ge LC_i$ i=inputs with lower constraint LC

$F_i(X,U) \le UC_i - \max\{3*E_i - MID_i, 0\}$ i=outputs with upper constraint UC

$F_i(X,U) \ge LC_i + \max\{3*E_i - MID_i, 0\}$ i=outputs with lower constraint LC

$\{\mathbf{X},\mathbf{U}\} \in$ AOC Area of Confidence

plus adding periodic **perturbations**

Including the constraint of being in the Area of Confidence, in addition of assuring acceptable prediction errors, enables dealing with input data which follows certain patterns, such as near-collinearities.

Note: For people accustomed to desiring orthogonal data, let us clarify that a properly formulated Optimization Plan will have independent control inputs, thus any collinearities are not the result of cause-and-effect relationships between the inputs. Note further that in moving the adjustments from the current region to the optimal ones, it is desirable to move along ridges so as achieve maximum cumulative performance. This leads to having near collinear inputs, but only due to an intelligent optimizer getting desired outcomes.

Note these two properties about **M** for SEO which yield significant simplification and effectiveness in calculations:

1. It is not required to understand the effects of each input separately; it is just necessary to be able to predict results in the AOC region. Thus, with the protection afforded by the AOC, confounding of input effects is of little consequence.
2. Trying to make **M** accurate away from the optimum (e.g., by making its generic form too involved or by fitting data in that region) is not only is irrelevant but it distorts the fitted models resulting in lesser accuracy around the optimum.

These properties are less demanding on **M(X,U)** than the usual requirements for DOE, Neural Networks and First Principle models (each for different reasons).

Note this characteristic of SEO: if a sequential advice is relatively poor because of making an incorrect inference, the run is likely to provide data that reveals the issue: (good) sequential empirical decision analyses are self-healing.

Details

Learning (update models)

The ULTRAMAX solution creates prediction models **M** in three ways.

The solutions 1 and 2 are a multivariable quadratic or second-order Taylor approximation of the response surface F for each output as a function of all

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inputs.

1. **Bayesian Statistics**

   Models are crated with Bayesian statistics when there is little data. This compares with the computationally much faster Classical or Fisherian statistical models least-sum-square fits. Also, in general the Bayesian framework is superior for decision making analysis – as compared to scientific true-false assessments best made with classical statistics.

   Bayesian solutions enable us to utilize the information available in as little as two runs[78] to already be able to move sequentially in a direction of likely improvements.

   This thought problem illustrates what is desirable. Imagine:

   - ⇒ A process with two adjustments and one result to be maximized
   - ⇒ The two adjustments are represented by two sides of a room, and the results by height
   - ⇒ Two process runs (with different adjustments and different results), which correspond to two points in space in the room (while explaining this, hold the point at the tips of the thumb and index finger of each hand).

   Now, ask the question: in which direction should we change the adjustments next with the highest likelihood of improving results? Obviously we would move the adjustments from the low performing run to and passing the high performing run.

   - ⇒ The simplest prediction model structure that would recognize the effects of the adjustment inputs is a linear model with three parameters (a constant and a coefficient for each input).
   - ⇒ Classical statistics cannot create a prediction model with the above data (there are two data points to estimate three parameters).
   - ⇒ Bayesian statistics can create such a model; and the result matches what our intuition indicates: move adjustments from the lower height run towards the higher height run, going beyond the higher run adjustments[79].
   - ⇒ The above holds true for any number of inputs! Thus, if we have 20 adjustments, ULTRAMAX's sequential Bayesian models will likely be obtaining performance improvements starting from the third run, while classical statistics cannot even venture a guess until the 22nd run. Neural Networks, having many more coefficients, would be much worse in this respect.

   The application of Bayesian regression starts with linear models with very limited data, and then move on the full quadratic with more data.

   The solution is explained by Hurwitz, 1993. True Bayesian analysis, involving multidimensional integrals, are very time consuming – as compared to classical regression analysis. Quasi-Bayesian regression models are created with this computational short-cut: use dummy "prior data" with "null" knowledge and classical regression analysis. The result is very similar to

[78] With different input values

[79] The MID indicates how much further we can go.

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ridge regression, which was illustrated by Hurwitz to be similar to the Bayesian analysis.

When there is sufficient data to calculate all coefficients with Classical statistics and have enough extra to calculate the noise with classical regression, the models are said to be "**complete**".

Full quadratic models have $\frac{(N+1)(N+2)}{2}$ coefficients, where N is the number of inputs. The minimum number of run data required to get complete models is this value plus the desired number of degrees of freedom, approximately 2*N+. Then PRIOR=0 in the Model and the Coefficients reports, and models are "complete". Then also most indicators of model suitability such as the "noise" and "signal" became more reliable.

2. Goal-oriented, Locally Accurate Models

A breakthrough awareness and principle is to realize that:

For optimizing, we only need to predict well around the optimum (or best predicted running conditions)[80].

- When more data than necessary for complete models becomes available, **weighted** linear regression is used to focus on optimal operations, to include only data which produces the most accurate models around where the best estimated results[81].

  Note that for doing optimization, representing the response surface away from the optimum is actually a detriment since this tends to distort the fit of the models around the optimum (unless the model structure is very representative).

  Locally accurate quadratic models largely eliminate the concern of lack-of-model fit for "smooth" response surfaces.

- The models also weight older data less when it detects deviations in process behavior from past patterns[82].

- The input spread of the series $\{\mathbf{X,U,Y}\}_t$ depends on the noise and the **MID**.

3. Calculated outputs

Calculated outputs are a function of inputs and/or of other predicted outputs.

These models have the following characteristics:

- they provide guidance as to which direction to follow for improvements even with limited data.

- they adapt well to slow dynamic changes in process behavior due to unknown causes

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[80] There are a few more requirements. The important thing is to recognize that predicting equally well everywhere is not very relevant for optimization, ans is the burden of virtually all empirical solutions.

[81] Obviously, this is not regular regression.

[82] This happens when the process is deigned "dynamic"; see PAR(40).

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- they tend not to be badly fooled into fitting noise rather than the underlying process behavior (due to the Bayesian analysis with little data, and due to heuristics to select data when there is excess data to generate locally accurate models).
- since they have relatively few coefficients (e.g., as compared with Neural Networks), they are capable to extrapolate further going from the starting set of adjustments to the optimal ones.
- deal effectively with smooth response surfaces with structural non-linearities and smooth non-constant interactions among the inputs (e.g., curved ridges).

Synthesis (generate advice)

The SEO resolution of the mathematical formulation above is done with a non-linear programming algorithm which can:

(a) solve conditional optimization

(b) has reasonable abilities with mixed-integer solutions (but not when approaching few categorical values such as zero-one programming)

Synthesis is created by this prioritization:

1st : inputs obey their constraints.

2nd: when giving Advice only (not for the Optimum Estimate), give Advice where each input does not change by more than its MID[83]

3rd : {**X**,**U**} $\in$ AOC, to assure relative certainty of estimates

4th : Multivariable limit on the "travel" of the re-adjustments: Travel[84] $\leq$ PAR(41)

5th : outputs obey their constraints

6th : optimize the Performance Index or objective function (often plus a perturbation function)

Synthesis could be seen as the coding of "volition", or a goal oriented attitude in the part of the technology, to attempt desired better results not experienced before. A basic reason this is possible is because of the implicit assumption of the reasonableness of interpolation and extrapolation in systems with gradual variables. With the current technology this is much more difficult to achieve with categorical (extremely discrete) decision problems.

Other specifics about the technology used were published in Moreno & Yunker 1992, Hurwitz 1993, Moreno 1986 & 1988.

Reference to Artificial Intelligence

All together, Learning and Synthesis, like most algorithms, are a series of heuristics or rules which are based on the accumulated experience of experts

[83] There is also a multivariable limit on the "travel" of the re-adjustments: Travel $\leq$ PAR(41).

[84] A Mahanobis distance.

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and the creators of the technology. As such, the technology could be seen as an "Expert System" to learn how to adjust a process to optimize its performance.

The ULTRAMAX User's Guide (Moreno, 1984-87), in Chapter 14, makes the distinction that ULTRAMAX has a domain of a relatively small number of variables with infinite values, while traditional Expert Systems tend to deal with very many elements, each of which is characterized by a few discrete (categorical) values. Further, it suggests that "Artificial Intelligence" should have the characteristics of Cognition and Volition, where *Cognition* is defined as "Learning about how a system is and could be", and *Volition* as "Creating or selecting options based on an evaluation of consequences". Reflex behavior, such as a standard PID control unit for production environments, has neither cognition nor volition, and therefore it is not Artificial Intelligence.

Some further characteristics

- **Types of variables:**

  It is restricted to using control inputs and outputs which behave gradually, in order to be able to use the modeling methods described above. "Gradual" means that the variables are potentially continuous, but can be discretized in small steps. It does not handle categorical factors except for the component inputs.

- **Time frame:**

  ULTRAMAX runs "real-time" with a time frame for process adjustments in the order of every several minutes to hours, or batch time. While software CPU is a consideration for short cycles, the most frequent determining factors of the time between adjustments (Run Time) are:

  - in continuous processes: how long it takes to get to steady-state due to readjustments of the control inputs, so that the output data truly reflect the current inputs, rather than effects from previous ones (unless using the “Transient Version” of SEO)
  - in batch processes: the length of a batch processing
  - during the first cycles while converging the optimum , how long it takes after the completion of a run to measure the values of **Y** – e.g. from a laboratory example**.**
  - how frequently **U** changes

  By comparison, the time frame for (first-level) "process control" is in the order of a second or fractions. In process control this is what most people understand as "real-time".

- **Quick adaptation to alternate goals:**

  The evaluation of process performance can be made through calculated variables (most frequently the Performance Index is calculated), which can be a function of Global Factors. The Global Factors represent targets, tolerances, unit costs, etc., and they can easily be changed from a central location in the software.

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In particular, calculated variables are recalculated with new factors to portray old (valid) physical data with today's objectives and evaluation factors, and the sequence of advices will move quickly to the new optimal conditions.

This also means that the Optimization Plan does not have to be worked through in total detail. In fact, it is common for the awareness of true optimization objectives to evolve significantly during the course of SEO.

- **Noise:**

  Changes in ***known*** uncontrolled inputs **are not** a source of "noise", as in most other statistical approaches. For ULTRAMAX their effects are understood, and managed. While uncontrolled inputs are "special causes" for SPC, ULTRAMAX provides the means to cope with them. The ***unknown*** uncontrolled inputs, and changes in the known uncontrolled inputs within a run are noise factors.

- **Optimum Estimates:**

  There are three “flavors” of Optimum Estimates made by the software:

  1. The **Potential** or **Projected Optimum Estimate (POE)** (within the AOC), where the expected prediction errors are calculated on the presumption that there is plenty of data around it as if it were interpolation, as when there is actual data around this optimum.
  2. The **Advice & Optimum Estimate (AOE)** has sufficient protection to be an advice to actually run the process. Here the expected prediction errors take into account the increase in uncertainty due to extrapolation into regions with less or no data[85]. When AOE is close to POE, then ULTRAMAX has converged to the optimum – which happens when the SDF of Opt.Est. ≤ 0.4 regularly.

- **Advices:**

  3. The **Travel-Limited Path (TLP)** are intermediate advices moving from the current adjustments towards the AOE, but which cannot move directly to the AOE because of constraints on of maximum changes in adjustments[86].
  4. **Advice w/ Exploration**, the product of the perturbation.

Performance Characteristics

Position Claim

The main claim made by Ultramax Corporation is:

Starting from the current operations, ULTRAMAX’s Sequential Optimization provides higher cumulative process performance -- cumulative contribution to profits or any metric defined by the

[85] For the AOE, when extrapolating, the protection sometimes needs to be less than 3-sigma in order to enable faster progress moving towards an optimum close to output constraints.

[86] Defined by the user through PAR(59) and the MID.

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end-user -- than any other empirical technology (and perhaps than ANY technology[87]) **today.**

This is equivalent to taking fewer runs (readjustments) to approach the optimum, and to get closer to the optimum.

This property is valid when starting away from a (local) optimum[88]; and applies:

(a) Almost definitely when there is only one local (and thus global) optimal operating performance.

(b) Almost always when the process is dynamic (slow changes in unknown uncontrolled inputs), as alternate technologies today tend to be much more static[89].

(c) Most probably for a year or longer, when there are multiple local optima and ULTRAMAX converges to an inferior local optimum.

ULTRAMAX may converge to a local optimum because it starts from current operating conditions and follows a maximum-gain[90] path towards the top of a "mountain of performance", which could be a local optimum.

By comparison, Neural Networks (NN) [like Design of Experiments (DOE)] require up-front many experimental runs with the production process itself, aiming at covering a large area of potential running conditions. Note that during these experiments the cumulative performance suffers significantly (constraint violations, unit costs, throughput, quality, emissions, equipment life, etc.) – which constitutes a significant hidden cost of the implementation.

However, in the case of NN, the model could represent various "mountains of performance" within the explored input area, and the attached optimization logic would be able to select the best such mountain. This best mountain could be better than the one that ULTRAMAX converges to. In such case, in the course of time ***and*** as long as the process is not changed or improved, the NN cumulative performance will eventually exceed that obtained with ULTRAMAX. If the process changes then the NN models will have to be recreated with a new set of experimental runs, incurring again the associated operating costs; while ULTRAMAX, with a lag, automatically keeps up with it.

Note the contrast: SEO is both an interpolation and a sequential extrapolation tool, while Neural Networks and Design of Experiments are basically interpolation tools. Note also that no interpolation technology can assure identifying the global optimum, as it might exists outside the area chosen up front for gathering the

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87 The alternate to "empirical" is "first-principle based" models, which take a very long time to make and validate to the same degree of accuracy – while the process continues working at the current conditions. One case where first-principle models can do better is when required measurements are not available on-line.

88 If starting at an optimum, the best performance is obtained by doing nothing different, which is a strategy which does not allow getting improvements when starting away from the optimum.

89 Still, ULTRAMAX can only handle slow changes in unknown uncontrolled inputs.

90 Considerable more effective than gradient-ascent, especially when there are interactions between the inputs.

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experimental data, and will remain so until the experimental area is expanded to where the global optimum is. On the other hand, *if there is only one local optimum, ULTRAMAX will converge towards it wherever it is*[91]; and *will maximize the cumulative performance*.

Technical Performance

The performance capability of ULTRAMAX is, in general and simplified terms, as obtained through replicated experiments optimizing a variety of simulated processes:

Regularly, ULTRAMAX's SEO average performance will get relatively quickly as close to the optimum as the order of magnitude of the noise level.

The losses vs. ideal Performance Index in the Effective Operating Area (in turn reduced by the noise in the outputs) increase with the number of inputs of the application; and is relatively better the more active outputs are calculated[92]. Further conditions are described below.

Recall that for ULTRAMAX, ***noise*** DOES NOT INCLUDE the effects of changes in uncontrolled inputs for which there is data from run to run, while noise does with many quality control solutions.

This assumes not running into a region of "insensitivity" (defined below).

As a practical consequence of the above:

- Let us assume that at the starting operating conditions constraints are regularly satisfied, namely, gains are only on the Performance index or objective function. To get improvement the noise of the objective function N_0 needs to be small (like 1/3rd) in comparison to the potential improvements in going to the optimum. If the potential improvements are lower than the noise then we will only be able to prove that we are running at near-optimal conditions for the current noise level.
- It is possible to get better optima by reducing the noise of the objective function and active constraints, which is practical to do until 3*N = 3*sigmas ≤ MID.

  The condition

  3*N = 3*sigmas ≤ MID:

  makes sense because such level of control on operations would regularly satisfy management daily concerns from production about forecast and safety (as represented by MID values management defined)[93]. This is a lesser requirement than Six-Sigma, with the consequence that where Six-Sigma is

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91 More narrowly, ULTRAMAX will always converge so some optimum, while NN and DOE only to one within the area of data coverage.

92 To the extreme that if all outputs are calculated then no data is necessary – the reality is fully represented by calculations.

93 This has been an underlying message in Quality Control, such as Dr. Deming, Dr. Juran, Six-Sigma, etc.

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already implemented the data will have even less noise and this will make it very easy to optimize well.

Note that the variables used, MIDs, objectives and constraints may change with time as the business conditions and awareness of issues change (easily done with an sequential analysis solution).

ULTRAMAX is a tool for active on-going process management, not to be placed in the background to run-by-itself and be virtually ignored, otherwise the optimization plan will sooner or later become obsolete; and further, the end-user will miss the advantages detecting problems (out of statistical control condition), and will miss developing a better understanding of the process.

Why better?

ULTRAMAX's SEO is faster for two basic reasons:

1. Most importantly, because of the Sequential Analysis; the fact that the information in new data is used right away to generate knowledge, rather than leave the information unused in model-centric solutions until the data is analyzed at the end.
2. Bayesian statistics, that enable moving into possible improvements starting with the third run

ULTRAMAX's SEO is likely to get closer to the ideal optimum because it uses locally accurate models.

Other Specifics

The run averages deviate from the optimum $F_0(\mathbf{X}^*,\mathbf{U})$ due to:

1. Imperfect knowledge because of the existing noise N in the data, and lack of model fit. In static processes these effects decrease as more data becomes available[94].
2. Failing the (B) condition below if gradients get reduced too much in the path converging to the optimum.
3. The need to maintain a spread of the input data so that the continually updated empirical models maintain the knowledge about F(X,U) around the optimum, and also to pick up slow dynamic changes from unknown causes[95]. The amount of spread can be managed.

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[94] The amount of sequential data converging to the optimum has two positive effects: (1) increases precision by the traditional $1/\sqrt{n}$ factor, and (2) enables the local model to be accurate in the smaller region where the lack of model fit (or variations in the noise level) is not important in comparison to the existing noise.

[95] The mechanism to pick up slow changes of unknown origin is to treat the process as dynamic (PAR(49)=1), which enables the data weighing to discount data because of its age. ULTRAMAX will gradually realize that using later data creates better fits, and this provides for gradual and slow feedback adjustment of the new modes of operations (not unlike the concept of running average). Conceptually, continuously changing unknown uncontrolled inputs could change at such frequencies that ULTRAMAX might enter a situation of resonance; but this has never been seen.

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The above performance characteristics apply when:

A. All the ULTRAMAX Parameters are set at default, or with some refinements suggested by Ultramax Corporation.

B. Do not run into an insensitivity condition.

   Common sense (and theory) indicates that convergence to an optimum will not happen if the noise is too large in comparison to the gradient[96] times the willingness to change the inputs[97].

   Limits to changes in the control inputs from one set of adjustments to the next is determined by their MID and constraint ranges on inputs[98]; and if small enough it will hung up Sequential Optimization. A metric that reflects this situation is a small Signal – but amazingly, ULTRAMAX has optimized with small signals also.

Bibliography

Adelman, A., and W.F. Stevens (1972). "Process Improvement by the 'Complex' Method," AIChE Journal, Vol. 18, No. 1, p. 20.

Bhateja, C.P. and C.W. Moreno, (1989) "Practical Optimization of Production Grinding Systems", SME Modern Grinding Technology Clinic, October 10,12, 1989; Detroit, Michigan

Box, M.J. (1965). "A New Method of Constrained Optimization and a Comparison With Other Methods," Computer Journal, Vol. 6, p. 42.

Box, G.E.P., and N.R. Draper (1987). Empirical Model Building and Response Surfaces, John Wiley & Sons, New York, New York.

Box, G.E.P., and N.R. Draper (1969). Evolutionary Operations, John Wiley & Sons, New York, New York.

Catasús-Serviá, J.J., Smoak. R.A., Squires, R.T., Baldwin, A.L., and Sorge, J.N. (1993) "A test for an Optimization Method applied to Controlling NOx Emissions in a PC-Fired Boiler", Joint ISA, POWID/EPRI Controls and Instrumentation Conference, June 7-9, 1993, Phoenix, Arizona.

Draper, N., and H. Smith (1966). Applied Regression Analysis, John Wiley & Sons, New York, New York.

Dharmarajan, N.N., and Patterson, P.D. “Boiler Tuning with SPO: Critical First Step in NOx Compliance Strategy of Central & South West Corporation.” Paper presented at AWMA Conference and Expo, Nashville, TN, June 1996.

Hurwitz, A.M. (1993) "Sequential Process. Optimization with a Commercial Package", Arizona State University; 6th National Symposium: Statistics and Design in Automated Manufacturing, Phoenix, AZ, February 1993.

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[96] More explicitly, this is the “effective” gradient, the change in output for the likely changes in inputs, not the infinitesimal gradient of calculus. It could be said that we are concerned with “difference analysis” rather than “differential analysis”, as it befits making sequential adjustments with significant changes in values. Thus, in addition to the requirement of the slope of a ridge, we are concerned with the curvature of a ridge.

[97] The same reason why Screening Experiments sometimes fail.

[98] See PAR(59) in page 205.

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Krueger, M.S.; Stopek, J.J.; Yunker, S.B.; and Patterson, P.D. “Illinois Power’s Operator Advisory System to Reduce Heat Rate and Control NOx Based on Sequential Optimization.” Paper presented at Power-Gen Americas ‘95, Anaheim, CA, December 1995.

Lang, R and Jarc, C.A. (1998) “Using Closed-loop Dynamic Optimization to Improve Boiler Efficiency at Chemopretrol’s Litvinov Plant”, Int’l Joint Power Generation Conference, Baltimore, Maryland, Aug 24, 1998.

McVay, M. and Patterson, P.D. (1998) “Illinois Power’s On-Line Dynamic Optimization of Cyclone Boilers for Efficiency and Emissions Improvement”, Int’l Joint Power Generation Conference, Baltimore, Maryland, Aug 24, 1998.

Mahalanobis P.C. (1936). “On the generalized distance in statistics”, Proceedings of Natural Sciences, India 2:49–55.

Moreno, C.W. & Yunker, S.B. (1992) "ULTRAMAX: Continuous Process Improvement through Sequential Optimization", EPRI.

Moreno, C.W. (1979). "A Performance Approach to Attribute Sampling and Multiple Action Decisions", AIIE Transactions, September, pp. 183-97.

Moreno, C.W. (1984-1987) *"ULTRAMAX User's Guide"*, Ultramax Corporation, Cincinnati, Ohio, USA.

Moreno, C.W. (1988). "Statistical Process Optimization," P-Q Systems Annual Conference, Dayton, Ohio, 1987, pp. 1-14.

Moreno, C.W. (1986). "Self-Learning Optimizing Control Software", Instrument Society of America, Robotics and Expert Systems Conference (Robexs'86), Houston, Texas, June 1986, pp. 371-377.

Moreno, C.W. (1993) "How Modern 'Smarter Not Harder' Technologies can Simultaneously Maximize the Combination of Pollution Reduction and Business Success", Conference on Environmental Commerce, CONEC'93, Chattanooga, TN, October 17-20, 1993.

Moreno, C.W. (1993) "Improving Process Performance: Which is Better, SPO or DOE?", Conference on Industrial Statistics and Quality Improvements, Oakland University, Rochester, MI, August 20-22, 1993.

Moreno, C.W. (1994) “Gaining Control of Plastic Forming Machines with new Technology for On-line Adjustments”, Structural Plastics Conference, April 10-13, 1994, Washington, DC.

Moreno, C.W. (1999) “Improvements Through Process Adjustments”, American Statistical Association Quality and Productivity Research Conference, May 19-21 1999, Schenectady, New York, USA.

Moreno, C.W. (2001) “Dynamic Optimization Of Production Operations”, IIE Annual Conference 2001, Dallas, Texas, USA, May 20-23

Moreno, C.W. (2001) “Comparison of two well known Methods for Optimizing Power Plant Operations”, 44th Annual ISA Power Industry Div. Conference, Orlando, Florida, USA July 7-13, 2001

Moreno, C.W. (2006) “Software: A gentler step-up”, InTech, ISA (Instrumentation, Systems and Automation) Society, April 2006, pp. 45. www.isa.org/InTechTemplate.cfm?Section=Article_Index1&template=/ContentManagement/ContentDisplay.cfm&ContentID=53257

Nachtsheim, C.J. (1987). "Tools for Computer-Aided Design of Experiments", *Journal of Quality Technology*, Vol. 19, No. 3, July 1987, pp. 132-160.

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Press, S.J. (1972). Applied Multivariate Analysis, Holt, Rinehart & Winston, Chicago, Illinois.

Redlinger, S.E. (1988). "The ULTRAMAX Method and Practical Applications", P-Q-Systems Annual Conference, Dayton, Ohio, June 15-17, 1988, pp. 2-24.

Redlinger, S.E. (1990). "Application of the ULTRAMAX Program in Decision Making and Optimization", R's Company Innovation Solutions Publication, Cincinnati, Ohio.

Redlinger, S.E. and C.W. Moreno (1993), "Applications of Sequential Optimization in Injection and Blow Molding Operations", *3rd. International, Conference on Advances in Polymer Processing,* Cincinnati, Ohio, March 29-31, 1993

Sachs, E.M., R. Guo, S. Ha, and A. Hu, (1990), "On-line process optimization and control using [run-by-run] sequential design of experiments", *1990 Symposium on VLSI Technology Rec.*, Honolulu, HI, June.

Sachs, E.M., R. Guo, S. Ha, and A. Hu, (1991), "Process Control System for VLSI Fabrication", *IEEE Transactions on Semiconductor Manufacturing*, Vol. 4, No. 2, May 1991 Abstract

Spendley, W., G.R. Hext, F.R. Himsworth (1962). "Sequential Applications of SIMPLEX Designs in Optimization and EVOP," Technometrics, Vol. 4, pp. 441-61.

Wald, A. (1947). Sequential Analysis, John Wiley & Sons, New York, New York.

Walters, F.H. et al. (1991). Sequential SIMPLEX Optimization, CRS Press, Inc., Boca Raton, Florida.

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Process / SEO Conditions

This section covers the metrics to detect issues associated with the progress of SEO and the status of process operations. There are two foci:

1. What can be achieved by the analysis of the stream of data coming in through SEO cycles.
2. The Capability / Consistency Analysis with a collection of valid operating data.

SEO Concepts and Metrics

Below is a series of concepts and their metrics, first a summary table and then a detailed explanation. Extreme values trigger **ALERT**s as explained in the If a new run has unusual or aberrant data.

Keep in mind ULTRAMAX® ALERTS and Guidelines, page 55.

Concept / Condition	Measures / Metrics	Where
Data Validity / Reliability, conformance with past data distribution, each variable separately	**Deviation**	Advice Report, Deviation column
Location of inputs, how far inputs are from the data used to generate prediction models, related to ability to predict accurately	**Standard Distance Factor** (**SDF**) **Uncontrolled SDF**	Advice Report, Multivariable Summary Plot
Outputs in Statistical Control	**Standard Error** **Overall Standard Error**	Advice Report, Actual Column, below each output value. Multivariable Summary. Plot
Satisfaction of Constraints	**Slack,** **Min Relative Slack** **Protection**	Advice Report Constraint row for each variable Multivariable Summary Colors in plots
Closeness to Relative Optimum	SDF of the Projected optimum Estimate: **SDF-POE** Average SDF-POE	Advice Report > Multivariable Summary > SDF row; Projected Optimum Estimate (POE) column.
Models suitable for optimization	Prediction Error (**Noise** estimate), **Signal**, **Critical Signal**, **Average Critical Signal**	Signal Report

Conformance of data to past experience or patterns of behavior

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There are three concepts and metrics as explained below.

Data Validity / Reliability; separately for each variable, how close it is to past values

Metric:

***Deviation*:** is the distance of a new run value to the average of the reference data in units of the standard deviation (sigma) of the distribution of the data. Deviation is inversely related to the **Validity** or **Reliability** of the data.

It is displayed in the "Deviation" column of the Advice Report when entering new run data (and Cycle) and doing a What-If.

Flags:

A deviation exceeding about 3.5 sigmas is a likely candidate of representing some abnormality in behavior or in its sensor. Values larger than 3.5 sigmas are marked with a ' **=** ', and larger than 6 sigmas with a ' ***** '; and in ALERT 14. The maximum Deviation (among all variables) is displayed in the "Multivariate Summary" heading in the Advice Report.

Use:

It is used to find out whether there was a change in the process behavior (including sensors), and to identify the likely variables.

When only one variable is way out, something associated with it may be the matter; when several variables are out, then a more fundamental cause seems to be in effect, such as a change in known or unknown inputs.

Location of inputs, prediction ability (Standard Distance Factor SDF), Area of Confidence (AOC)

"Inputs" in the Optimization Plan (OP) include the control inputs being adjusted and the uncontrolled inputs being measured. Based

Note: they do not include the "ruled inputs", which since they are affected by the other inputs they are logically "outcomes" (dependent or output variables).

Metrics:

***Standard Distance Factor (SDF)**:* The location (distance) of the inputs in relation to the run data used to build the latest prediction models is given by the multivariable (among inputs only) Standard Distance Factor (SDF).

The region of input values where the SDF is sufficiently small to create the most reasonable predictions is called the Area **of Confidence (AOC),** where the SDF< PAR(10)=1 by default. The AOC could be concave and may even consist of disjoint sub-regions. See definition of PAR(10) in page 204. When there is a lot of data, it is difficult to have an SDF < 0.2 .

The SDF is based on the Mahalanobis (Mahanobis) distance (built on the inverse covariance matrix of the reference data[99]); similar in nature to the distance to the average of the run data in units of its standard deviation. The

[99] The weighted data used to create locally-accurate models

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loss in precision when predicting in extrapolation depends directly on the SDF.

Uncontrolled SDF: Equivalently to the SDF (for ALL the inputs), there is an Uncontrolled SDF (SDFU) only for the set of uncontrolled inputs. Unlike the regular SDF, the SDFU is calculated using all the run data that exists in the data base[100].

SDF and SDFU are only multivariable metrics, which appear in the Multivariable Summary in the Advice Report.

Flags:

A SDF>0.6=PAR(104) is the threshold into "extrapolation", with prediction abilities sufficiently poor not to bother evaluating an ALARM respecting whether the process is in Statistical Control.

No flags except for the ALERTs.

Use:

The SDF is used to control the amount of extrapolation in SEO's Advice, to avoid large prediction errors. When the optimum is perceived to be outside the AOC, the Advice will be at its edge (with some exceptions explained below). Thus, the movement of operation adjustments from Baseline to the near optimal ones involves a sequence of readjustments, each with a combination of re-adjustments with relatively large but limited SDFs.

The SDF of the Advice or of the POE could be larger that PAR(10), i.e., outside of the Area of Confidence, when:

- there are such unusual uncontrolled inputs which will force any combination of inputs (control and uncontrolled) to be unfamiliar.
- constraints on inputs have been changed so that all past data violates these new constraints (because ULTRAMAX has as the highest priority the satisfying of input constraint, see Priorities in page 150).

Thus, ULTRAMAX will not propose an Advice outside the AOC in order to meet output constraints. Certainty about prediction of outputs is considered a higher priority (more valuable) than taking unknown chances to meet constraints.

Note: It could be that the SDF of a "What-If" input be large but the inputs still be in a region of past experience. This may happen when ULTRAMAX creates locally accurate models around the optimum, and the What-If inputs are far from this area.

See usage of SDF of the Optimum Estimate in the section "Close to Relative Optimum" below to determine if the process has already converged to the Relative Optimum.

Outputs in Statistical Control, Standard Errors

Metric:

[100] Or more precisely, all runs after PAR(1).

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Standard Error: For each output variable the metric is Standard Error, the difference between the value and the model expected value (for the values of all the inputs, adjusted and uncontrolled) in units of the sigma of the expected prediction error.

$$\text{standard error} = \frac{\text{actual prediction error}}{1\,\sigma\ \text{expected prediction error}}$$

Recall that the expected prediction error is a function of the noise and of the input values, more explicitly, of their SDF, as explained above.

While it applies to each output separately, it is a multidimensional metric because it depends on the values of all the inputs. But there is a "more" multivariable metric for all output standard errors, called the **Overall Standard Error**.

The Standard Errors is displayed in the column of Actual Data, just below the values of the outputs.

Note that "Errors" are assigned only to the outputs. The inputs (decisions or uncontrolled input measures) are presumed to be absolutely correct.

Flags:

If the Overall Standard Error is very large (usually, larger than 3.5 sigmas) this is the detection of a possible upset or aberrant behavior. There are two kinds of situations:

If the Overall Standard Error is very large and the inputs are in a familiar (experienced) area (the Standard Distance Factor SDF defined below is small) – then a special cause is detected and shown as an ALARM.

If the outputs have undesired values due to strange values of the inputs (adjusted or uncontrolled) – that is, extrapolating and the SDF is large -- then only a Warning appears. If the reason is due to unusual uncontrolled inputs, then the Uncontrolled SDF (also) is large.

Standard Errors larger than 3.5 sigmas are marked with a '**=**', and larger than 6 sigmas with a '*****'.

The Overall Standard Error triggers its own ALERTs.

Use:

It is traditional (a la Shewhart) to consider a standard error larger than 3 sigmas (in absolute value) as a practical indication that something unusual may have happened, that chances are that there is an assignable or special cause rather than a random event.

Theoretical note: this assessment does not require that the errors be distributed "normally", as it is so commonly believed today. Regardless of the distribution, it is rare to have by random causes alone generate an error larger than three sigmas (e.g., Chebyshev's Inequality indicates that, in the worst, case the probability of random data lying beyond 3σ for any bell-shaped symmetrical distribution has to be less than 5%). The assumption of "normality" is necessary only to numerically ascertain the actual risks (which is not done, or necessary, within ULTRAMAX's SEO technology).

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Experience indicates that seldom does a process behave with "normally" distributed errors: with sufficient data, almost always it is possible to demonstrate a deviation from normality. Also, there are plenty of theoretical reasons why a process should not behave with normally distributed noise (e.g., the effect of any process step which chops off distribution tails), and few theoretical reasons why it should be normal. The best known reason why noise should be normally distributed is when the Central Limit Theorem applies, that applies to the sum of many independent random variables from similar distributions, a condition which seldom exists.

By the way, it is always possible to take a variable with any distribution and transform it so that the distribution of the transformed variable is "normally distributed", but since this distribution is seldom known and even less seldom applied, this fact is irrelevant.

One of the possible "special causes" created by doing SEO is to extrapolate into relatively unknown regions. Then the patterns of the process behavior may be different than the patterns perceived in the region where there is data.

Satisfaction of Constraints (Slacks and Protection):

Recall (page 103) that the constraint requirements are that the Mean process outputs (as predicted) satisfy the OP constraints, and that the data generated by process operations satisfy MID beyond the constraints.

Metrics:

There are two metrics related to constraints: Slack and Protection:

The calculation of Slack and Protection of past data is done with the current OP and the noise as calculated with the latest models created.

Slack: the amount of room left before meeting the OP constraint.

For a lower constraint: SLACK = Value - Upper constraint

For an upper constraint: SLACK = Upper constraint - Value

The Slack is negative when going beyond the constraint.

In the Advice report the Slacks are displayed for each variable in the row where the constraints are defined

Relative Slack: SLACK / MID. It applies only to inputs, to see how close to the Constraints they are in relative terms.

The minimum input Relative Slack is displayed in the heading "Multivariate Summary".

Protection: "Protection" or "Sigma Protection" of data and estimates is how many sigmas the output value is away from MID beyond the constraint:

$$\text{PROTECTION} = \frac{\text{SLACK} + \text{MID}}{\text{SIGMA}}$$

"Sigma" here is:

- for actual data, the noise

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- for estimates, the standard deviation of the expected prediction error.

The minimum # sigma protection is displayed in the “Multivariate Summary”

Flags:

Input Slacks<0 have a ‘ ***** ‘ to their right.

Output actual data slacks:

Beyond the constraint (SLACK < 0) have a ‘ **=** ‘and generate an ALARM

MID beyond the constraint (SLACK < -MID) have a ‘ ***** ‘

Output estimates slacks:

With Protection < 0 have a ‘ ***** ‘ and generate an ALARM

With Protection < 3 have a ‘ **=** ‘ and generate an Warning (for more details see Alerts Table)

Use:

For inputs:

SLACK >=0 Advices always satisfy this requirement.

For outputs:

SLACK >=0 and

Protection > 3 sigmas. This is slowly relaxed to that SEO can move faster towards to optimum.

According to the priorities in generating the Advice (page 150), the advice may not satisfy output constraints when:

- Output constraints are not usually satisfied in the past data.
- Output constraints are not being satisfied in the region of the current values of control input constraints.

An output is considered **"active"** when:

- it is the output variable being maximized or minimized (the "Performance Index or objective function”), or
- or when the protection < PAR(115) = 4 by default

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Close to Relative Optimum (SDF of Projected Optimum Estimate, SDF_POE)

Metrics:

SDF of Projected Optimum Estimate (SDF_POE): The SDF-POE will be large (close to PAR(10)=1) when extrapolating, when the optimum is estimated to be away from the region where data exists.

By the definition of POE it will not be larger than PAR(10), the outer limit of the AOC (with few exceptions as described above in page 159). The SDF of the Opt. Est. may be close to PAR(10) (see definition in page 204)...

- when moving from the starting adjustments extrapolating towards the new near-optimal ones and there is no experience in that area, what is called "mountain climbing".
- after the optimum has been approached, because of:
  - a new unusual combination of uncontrolled inputs
  - dynamic changes in the behavior of the process
  - changes in costs or objectives
  - there are inputs which do not have major effects and the numerical analysis may appear to indicate an optimum far away. Therefore the perceived effects of those inputs are not stable and the location of the optimum as it relates to those inputs moves around[101].

The Standard Distance of the Projected Optimum Estimate (POE) will normally be less than 0.40 when the process runs have settled around the optimum. This number increases the larger the number of inputs.

Average SDF_POE: Since the SDF_POE keeps on changing, the **Average SDF_POE** of the latest cycles is calculated (and displayed in the Status Statements), which if less than 0.40 is taken as a good indicator that the process has converged to a (local) optimum.

[101] The stability of estimated input POE can be plotted by setting the plot option "optimum estimates"

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Models suitable for optimization (Expected Error, Signal, Critical Signal, Average Critical Signal)

ULTRAMAX's SEO fits the data for each output variable to create models as a function of the inputs (whether the variables are calculated or not). Useful models for SEO represent the underlying behavior of the process -- as compared to fitting the noise.

Metrics:

Noise: Each output has a noise estimate to represent the variations which cannot be explained with the prediction models (the standard deviation of the residuals from regression, adjusted by the degrees of freedom).

***Signal*:** The relation between noise to the total variation of the data in each output gives a relative measure of how much of the total variation can be explained. This metric is the Signal (the R-square adjusted by the degrees of freedom).

Common sense indicates that is the Signal is high, then SEO gets possible improvements well. However there have been many instances of getting good and quick improvements with low Signal. The signal becomes reliable after having “complete” models (page 149), and by then SEO is usually around the optimum already.

The measure of suitability for all models is called the **Critical Signal**, which is the minimum signal among all outputs which are "active".

***Average Critical Signal*:** A better indicator, to avoid being misled by occasional poor Critical Signals, is the **Average Critical Signal** in the last few runs.

The Signal, Critical Signal and Average Critical Signal are displayed in the “Signal Report”.

Flags:

There are no flags for Signal, as it is not a strongly determining factor.

Use:

When selecting data for creating a Locally-Accurate model, a criterion is to have Signal > 65%, if possible.

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Capability & Consistency Analysis™ (CCA)

Introduction

The process capabilities must be consistent with the minimum performance requirements. Else the process needs improvements or the requirements need relaxing before optimizing. This can be tested up front.

The main tasks performed with the Capability & Consistency Analysis are to:

- Determine performance of the actual run data being analyzed
- Detect whether some barriers exist which need to be fixed now; i.e., that require change(s) to the Optimization Plan and/or to the Process, that not any optimization can resolve
- Evaluate barriers to maximum performance due to lack of consistency (noise)

The CCA is not necessary to do production with ULTRAMAX optimization. If there are problems they will eventually be discovered as you do production; however it will save time and keep issues clear if the problems are solved before starting / continuing doing production operations with SEO.

The up-front Baseline CCA (page 44) also performs a secondary and useful function: to get practice collecting operating data the way that will be necessary for the Sequential Empirical Optimization (SEO) cycles: if Stand-Alone especially how to calculate outputs which are averages of several individual measurements; if Integrated, how to manage the process from the Optimization Management Screen, and to develop confidence that the data exchange and calculation of averages are correct.

Capability is the competence of the process to satisfy the company's requirements; e.g., satisfy constraints and get as good a Performance Index as possible.

As used here "capability" does not include "how well the process could operate with better adjustments", which is resolved through applying SEO.

Consistency -- the opposite of noise -- greatly affects Capability. Other than when the output means already do not satisfy constraints, noise is a determining barrier for **data** to satisfy constraints and for the Performance Index to be better.

The Capability & Consistency Analysis (CCA) evaluates the compatibility of current process operations with business needs / objectives; and this makes the analysis extremely valuable in its own right to plant managers.

Specifically, the CCA indicates which specific variables have any historical or capability problems. In particular it analyzes:

1. How well the selected actual run data has satisfied constraints
2. The capability of future runs (with same mean outputs) to satisfy constraints

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3. Losses in mean output Effective Operating Area in order for the data to satisfy constraints

4. For outputs that have upper and lower constraints, whether there is evidence of lack of capability (negative Effective Operating Area); and if capable, whether there is sufficient room to apply SEO

5. How close ULTRAMAX® can get to the optimum Performance Index -- within the Effective Operating Area -- due to the Performance Index noise level (sigma of prediction error)

6. When the runs are with adjustments at Baseline, the outputs give a reference to evaluate future SEO performance improvements

The Capability Analysis is useful in two principal ways:

A. Is a diagnostic tool to find out exactly which variables may have some constraint satisfaction problems and the magnitude of the problem.

B. Is part of the big picture of Advanced Process Management (APM), the customer will keep on making the Capability Analyses periodically (with the regular or SEO production data) to be assured about the continuing suitability of the process for the business needs.

Basics

The conceptual "Model"

The basic representation to understand output values (actual data) is:

actual = mean + noise

The only value known is the Actual; the Mean and Noise are estimated (inputs are presumed to be known exactly, which is true of Decision Inputs).

The "mean" process output is explained by the values of the inputs through the "prediction model". The prediction model is created by fitting (regression) the run data in the database, or including calculations if provided by the end-user.

The "noise" is a random event, whose specific value for one run is not explained by the values of the (known) inputs. However, the noise spread, level or magnitude is known by the standard deviation or sigma of its distribution. The "noise level" is often just called "sigma" to match the nomenclature of Statistical Quality Control and the Six-Sigma approaches.

Capability Analysis Report

Procedures

There are two aspects of creating a CCA report: (1) calculating the **noise** for the analysis, and then (2) creating the report with the **analysis**.

1. The **noise** is calculated with the LEARN option – by selecting the Learn icon in the tool bar. While most often the latest noise calculations are used, there are circumstances where various options should be set before doing LEARN:

   - If doing a Baseline CCA or with a limited number of run data:

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For this purpose the calculation of noise is not as reliable if using Bayesian statistics. For this we also need to use linear regression. Thus, set

PAR(76) = 0, to not use Bayesian regression

PAR(85) = 1, to create linear models[102]

Also, the calculation of noise needs to be relevant for all run data, thus we need to avoid creating locally accurate models. Thus set:

PAR(45) = 2, to not create local models

An interesting benefit of setting these parameters is to use standard statistical methods which would let other statistical experts duplicate the results obtained.

- It is possible also to select a sequential set of data to LEARN from, by assigning these values to these parameters:

  PAR(1) = B, to use only data with run number B or larger

  PAR(2) = E, to use only run numbers up to number E; **or**

  = -N, to use only the last N run data

  Remember to set these parameters back to default before continuing with SEO.

- When there are Component inputs, and therefore Scenarios, LEARN can only be done with data for only one selected scenario at a time.

2. The CCA Report is **calculated** in the Stand-Alone GUI, with the option Analysis > Reports > Consistency.

   - This analysis takes place with whatever data is already selected with Filters for data selection, page 208 for PAR(125) through PAR(139) – most often using all data [PAR(125)=1]. For instance an analysis could be made for only Baseline run data, and separately for SEO run data, where the comparison will give insights about improvements made. It is possible, also, to exclude runs that have some constraint violation, and/or – if there are scenarios -- to use only the runs in the Scenario used for LEARN.
   - The first step in the Consistency option is for the GUI to ask the user to define a set of run data to analyze (regardless of the set used to calculate noise in LEARN).

*NOTE: If this is an up-front Baseline CCA – ideally replicates -- then SEO will be started in a different application file **without the Baseline prior data**.*

[102] The "noise" is estimated by the standard deviation of the residuals adjusted by the degrees of freedom.

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The Report Contents

The statistical calculations and tests performed are described below in the section Capability, Appendix: Tests in the Analysis.

The Capability Analysis Report is organized as follows:

Run Data Selection and basis for the Noise estimates

A. STATUS: Description of the main findings, including recommendations for Next Steps. It includes a reference to when the end-user needs to accept the Constraint and Performance Index losses due to noise – or fix the situation.

B. OBJECTIVE FUNCTION (Performance Index): Past Performance by types of runs (e.g., Baseline, SEO, optimum estimates, etc.), Future Barriers due to noise, Future Convergence (how close an SEO can get to the optimum within the Effective Operating Area).

C. CONSTRAINT SUMMARY: Number of failures observed by type of variables and by runs

D. OUTPUT CONSTRAINT SATISFACTION & EFFECTIVE OPERATING AREA (for Means). It includes a section on which noise level and/or MID values could resolve output range failures (for outputs with both upper and lower constraints)

E. CONSTRAINT SATISFACTION TESTS: Displays the number of sigma satisfaction for the tests: D (actual data), P (predicted distribution) and R (Ranges) tests for all the variables. It also includes a section of the SHORTAGES for those variables with failures.

F. INDIVIDUAL VARIABLE STATISTICS: This includes values such as averages (among the data selected), standard deviations, noise and Losses. The averages for Baseline data are the reference for doing Performance Analysis.

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Analysis, Decisions

From the tests above we:

A. Understand whether the selected runs satisfy constraints

B. Understand whether future similar runs (at the same output average) will satisfy constraints

C. Understand whether conditions 1 and 2 in the table below are satisfied or it is required to fix before doing (or continuing with) SEO

D. Get diagnostic data for conditions 3 and 4 for the end-user to accept or fix.

	Incapability Condition	Purpose for fixing the problem
1	The Effective Operating Area is negative: the process is incapable of satisfying constraints. This situation becomes evident to the analysis only for outputs with both upper and lower constraints. It is inferred for all other outputs after optimizing performance.	It is mandatory to resolve this process incompatibility with business objectives and needs… for the health of the business, of management, and of operators (along the lines of Deming's principles and basic Statistical Quality Control).
2	The Effective Operating Area has a size less than 3 sigmas. This situation becomes evident to the analysis only for outputs with both upper and lower constraints.	3 sigmas are needed so that SEO has sufficient "room" to explore and learn without undue sacrifice of data constraint violations[103].
3	The Effective Operating Area is judged to be too restrictive to allow the Performance Index to achieve desirable values	To have the opportunity to generate higher profitability.
4	The Performance Index's noise is too large in comparison to the MID. On the average, SEO gets the Performance Index as close to the optimum (in the Effective Operating Areas) as the order of magnitude of its the noise.	To have the opportunity to generate higher profitability.

There are three possible outcomes from the Capability Analysis:

1. **Some of the four incapability conditions** (in the table) **are discovered**.

   The process needs to be fixed (for the reasons explained in the table), which may involve:

   A. Refining the Optimization Plan. This solution could include:

      (a) reducing the noise by including important input(s) that explain the changing values of the outputs;

[103] E.g., to avoid the constraint violations incurred in collecting data for Neural Network or Design-of-Experiments solutions.

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(b) increasing MIDs which are unrealistically too small in comparison to the noise;

(c) relaxing some constraints which were set unrealistically too tight

(d) reduce noise by calculating averages with more raw measurements

B. Improving the Process. This solution could include:

(a) reducing the sources of noise (e.g., bringing raw materials to a desired temperature, re-tune control logic, clean a sensor, make something more rigid)

(b) expanding the constraints (e.g., installing a higher capacity pump)

Note that the ULTRAMAX concept of "process" includes the physical process (one or more production units), control logic, sensors and the procedures used.

C. Improving the Conditions (uncontrolled inputs). This solution could include: improving the availability of cooling water; getting materials / fuels from a higher quality source.

If there are physical improvements and the analysis is done for an up-front Baseline Capability Analysis, then the test has to be repeated, until obtaining the outcomes 2 or 3 below.

2. The four tests above are passed, but run data does not satisfy all constraint requirements.

Then SEO is the correct tool to try to solve the problem. Process improvement will be to resolve constraint violations (if possible) and then improve the Performance Index.

If after reaching the optimum all constraints cannot be satisfied, then the process is incapable, and like condition #1 in the table, this needs to be fixed.

3. The four tests in the table are passed, and the run data satisfies all constraint requirements[104] -- the process has proven to be capable (for the current values of the uncontrolled inputs).

Then the process and operators have a clean bill-of-health, and SEO performance improvements will be just to improve the Performance Index (while maintaining constraint satisfaction).

A potential problem is that new conditions (known or unknown uncontrolled inputs) may appear in the future which may generate difficulties satisfying constraints. The Advice Report would detect when such unusual operating conditions take place, to be reviewed and prevented in the future (an aspect of continual Quality Control).

[104] That is, output data average is inside the Effective Operating Area. In other words, current operators know how to satisfy constraints.

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Capability, Appendix: Tests in the Analysis

These are tests on past data behavior, and tests on forecasts based on the "distribution" of the noise.

Constraint Satisfaction Requirements

(a) **Means:**

Output Means[105] satisfy the constraints in the Optimization Plan.

(b) **Data:**

Control Input (adjusted, decision inputs) Data[106] satisfies constraints, as they are the decisions made.

Output Data satisfies MID beyond constraints (almost always). An end-user only interested in data considerations will set MID=0.

Below is a table of all the tests.

(D) refers to data, (P) to prediction through distributions, and (R) to prediction through distributions related to variables with constraint ranges (variables with upper and lower constraints).

Nest, the (a) (Means) and (b) (Data) refer to the corresponding constraint satisfaction requirements above.

Note that the Loss tests LC and LS are passed by acceptance by the end-user based on their knowledge and experience, not exclusively on analysis of the data.

Failing tests D and P could be resolved with SEO, failing the others are just process lack of capability problems – that is, they need to be resolved outside ULTRAMAX.

[105] We use "mean" to indicate the average output for constant input values; and "average" for the average values across several runs with possibly different input values.

[106] There are no constraint requirements for uncontrolled inputs. Note that control input values are the decision made, and thus have no uncertainty or noise.

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#		Past Data (counters)	Future runs (distribution) Future runs with output means at data output averages	Capability (distribution) Property of the process, independent of adjustments
	Inputs	Db		
	Output means Output data	Da Db	Pa Pb	
1	Effective Operating Range exists for basic process capability to meet business needs			R > 0
2	Effective Operating Range sufficiently large for effective SEO			R > 3
3	Losses to define Effective Operating Range			LC
4	Performance Index Loss			LS

All distribution tests have this structure in common:

- A metric defined with:
  - A numerator that is a measure of interest, the difference between two values
  - A denominator that is a measure of distribution spread of the numerator. It has the same units as the numerator. It is often the noise "sigma".
- The ratio is a dimensionless number, which is a "number of standard deviations" or "number of sigmas".

  The logic of the test is to determine the minimum number of standard deviations that gives us certainty that future actual values will seldom have undesirable values. It is often 3 sigmas for the standard 3 sigma confidence[107].

  The tests are not black-and-white: failing but being close to passing the tests means to have a slightly higher likelihood of violating the criteria, which could still be acceptable for business. The Capability Report displays both the number of sigmas for all tests for all variables, and the shortcoming vs. minimum desired only for those variables with shortcomings.

[107] As in Shewhart, 3 sigma limits are used even though it is known that the actual distribution of the data is virtually never normally distributed (contrary to the assumption in many simplistic analyses).

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Test for Past Actual Data (Fraction)

The metric is fraction of runs satisfying the requirements as defined below…

For each ***input***:

Test **Db**) 100% of runs satisfy constraints [108]

For each ***output***:

Test **Da**) 50% or more runs satisfy constraints

Test **Db**) 100% are within MID beyond constraint[109]

The Capability Analysis report is an excellent tool to identify the variables with historical data constraint problems (from only one run to a set of runs). Recall, also, that each Advice report indicates when each new actual run data violates constraints, and in which variables.

Test for Future Runs (Distribution, # of sigmas)

This test is made for an imaginary run made with such adjustments that the mean outputs are the average values among the selected runs for the Capability Analysis. This is perfectly representative for true replicate data; and an idealization when the inputs are not true replicates[110].

(a) Basic: Output Means satisfy the constraints in the Optimization Plan.

There are two levels to this test:

- Basic constraint satisfaction

  $$\textbf{Mean Slack} > \textbf{0}$$

  Note that if the **Mean Slack < 0** either the process or the way it is being managed simply does not satisfy the (a) basic constraint requirement. This is NOT a Consistency problem.

- **3 sigma confidence that mean satisfies constraints**

  The standard deviation of the estimate of the mean and the mean slack is sigma/$\sqrt{n}$, where "n" is the number of degrees of freedom in the calculation of noise sigma.

  Test **Pa** = $\dfrac{\textbf{Mean Slack}}{\textbf{sigma/}\sqrt{\textbf{n}}} > 3$

  This is to have at least a "3 sigma protection" for the mean.

  Satisfying this test satisfies the previous one.

---

[108] Or, if relaxed, within PAR(111)*MID beyond constraints

[109] Truly, for large number of run data, we could expect some outputs to exceed MID beyond constraints, but we avoid this extra complication in the analysis.

[110] Note that the noise would be less than the standard deviation of the data (because the noise is the part of output variability not explained by the changing inputs), unless the selected runs are true replicates.

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(b) Data Consistency: Output Data satisfies MID beyond constraints almost always.

Test **Pb** = $\frac{\text{Mean Slack} + \text{MID}}{\text{sigma}} > 3$

This is to have at least a "3 sigma protection" [111], as illustrated in Fig. 25, page 142.

Note that when MID < 3 sigma, then we need the Safety Buffer or a minimum mean slack to satisfy the (Pb) data Consistency constraint requirement.

A Safety Buffer > 0 is a loss or reduction in the tolerable output mean operating area (from the constraints in the Optimization Pan) to comply with data constraint requirements, and a potential missed opportunity to have better Performance Index values (if the constraint is "active").

Loss = max {0, 3*sigma-MID}

Upper effective constraint = UC – Loss (if any)

Lower effective constraint = LC + Loss (if any)

Failing either of the P tests is resolvable with SEO unless the process is just basically incapable, which except for the constraint ranges below, will be known only after approaching optimal operations.

Outputs with both upper and lower constraints:

The following analysis detects some process incapabilities dependent only on the consistency of the process (and regardless of the mean output in the selected data).

When an output has both upper and lower constraints the Effective 'Operating Area' is an Operating Range

Effective Operation Range = Constraint Range – 2*Loss

There are two levels of requirements for the Effective Operating Range.

1. The minimum Consistency requirement is that

   $\text{Operating Range} > 0$

   If Operating Range < 0 the range does not exist and the process is incapable. This is NOT an operator ability problem, and therefore not solvable with SEO. Operations with this process make both management and operators frustrated, and business objectives will not be satisfied, along the lines of Deming's principles. The situation needs to be resolved for the health of the business and all plant personnel.

2. For SEO to be able to create prediction models fitting diverse output values, the rule-of-thumb is that the effective operating range needs to be at least 3 sigmas [112]

[111] Note that "*having sufficient consistency or sufficient process control*" to satisfy constraints is different than being in "*statistical control*". Being in statistical control is neither sufficient (the noise level could still be too large) nor necessary (there may be noticeable special causes but their noise effects may be small in comparison to the MID).

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Test **R** = $\frac{\text{Operating Range}}{\text{sigma}} > 3$

which includes the basic requirement of R > 0

Failing this test indicates that the situation needs to be resolved in order to apply SEO.

If P fails, passing R is not sufficient to assure capability, as the process may not be capable of operating within the Effective Operating Area for all outputs simultaneously.

However, passing all the tests above is not sufficient for effective optimization. There may still be too much a sacrifice in Performance Index values due to noise:

1. by having to force output means into the narrower "Effective Operating Range"

   Note that this is a process limitation; it is not affected by ULTRAMAX being an empirical solution.

2. by how close SEO can get the Performance Index -- on the average -- to the optimum within the Effective Operating Range"

   Note that this ULTRAMAX's solution limitation; being affected by the uncertainties in data due to noise.

Test for Acceptable Constraint Losses

As explained above:

Upper effective constraint = UC - Loss

Lower effective constraint = LC + Loss

The Capability Report displays the Lower and Upper Effective Constraints when there is a loss, and the values of the Losses.

Test **LC**: The test is passed when the end-user determines that the Effective Constraints are acceptable based on their knowledge and experience.

The solution of increasing MID to 3*sigma will enlarge the Effective Operating Area to the full allowed area for means in the Optimization Plan.

If the end-user continues with SEO it is assumed that he/she passed the test LC.

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[112] This is not a confidence issue. If not optimizing, Six-Sigma calls for at least 1 sigma.

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Test for Acceptable Performance Index Losses

The technical performance of ULTRAMAX's SEO is that the average Performance Index will (frequently) quickly get as close to its ideal optimum (without noise) as about three sigmas of the noise[113]. We want this loss to be unimportant, namely, less than MID.

$$\text{Test } \mathbf{LS} = \frac{\text{MID}}{\text{scorecard sigma}} > 3$$

The solution of increasing MID to 3*sigma will NOT result in improvements of the Performance Index values. Reducing the noise (sigma) will.

If this test fails and the end-user continues with SEO it is assumed that he/she accepts the discrepancy and the lower expected gains in performance.

[113] Or better. Sometimes for conceptual simplicity, we mention that the Performance Index on the average will get as close to the optimum as the noise level (one sigma). Things are not that precise.

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APPENDICES

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GLOSSARY

This is a compendium of concepts and semantics used in ULTRAMAX, including some relevant comments to better understand their implications.

Active Constraint : See Constraint.

Active Data : See Data

Active Output : See Variables.

Actual Data : See Data.

Adjust : To set control inputs such as: the position of levers or valves; the values of setpoints or biases; the values of control logic parameters and thresholds of rules (e.g., constants in PID first level controllers). Also called **to tune** the process. It is an aspect of Supervisory Control.

Adjusted input : See Variables, Control Inputs.

Adjustment : The act and result from adjusting or setting process inputs. Same as “setting”; the action and the result.

Advice : The suggested adjustments of the Adjusted (Control) inputs provided by the software. The software also displays the predicted values of outputs, and a variety of indicators of conditions such as slacks, protection, prediction errors, and SDF. In ULTRAMAX it is not required that plant personnel follow the Advice.

Advice: **Advice and Optimum Estimate (AOE):** Advice which is also the optimum estimate (with current knowledge).

Advice: **Travel Limited Path (TLP):** Advice aiming at the Advice and Optimum Estimate but limited by maximum allowed change in the adjustments (typically to maintain process stability).

Aggressiveness : The willingness to extrapolate Adjustments into regions of unexperienced values. The position of an Advice is limited to be within the Area of Confidence (with few exceptions). The AOC is defined by the maximum allowed SDF, the value of Parameter PAR(10).

Alerts : The software interprets the status of the process and of optimization in series of brief ALERT texts. They are classified as "Notes", "Warnings" and "ALARMS".

Application : The process or system whose performance is being optimized with ULTRAMAX. Associated with each there is: an Optimization Plan document; an application file “appname.umax” with the configurations of the optimization plan, data, models, etc; and a historical file “appname.hist” with the recent transactions made with the software.

Application File : A file created by the ULTRAMAX software containing the Optimization Plan, run data, models, and indicators for each application. It has the typical name " appname.umax” (before 2004 it could have been “appname.um6”), where "appname" stands for a unique application name.

Area of Confidence (AOC) : The AOC is the region of combination of inputs for which the models best predict the results given the current data and the (possibly) local models built with it. It is defined by those inputs with a SDF $\leq$ PAR(10). The AOC can be a concave region, and even a disjoint region. When the AOC is a locally accurate region, its location can change a lot after “climbing the mountain of performance” and with changes in the uncontrolled inputs.

Average Standard Distance Factor of the Projected Optimum Estimate: The Average SDF-POE of the last eight cycles, but not including the first as many cycles as the number of inputs. This number is highly related to being around the (local) relative optimum.

Balanced Scorecard (BSC): A method of evaluating the operating performance of an organization unit created by Drs. Robert S. Kaplan, David P. Norton. It is characterized by measuring and tracking **all** important indicators, called Key Performance Indicators (KPI).

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With ULTRAMAX we concentrate on Production Operations. Its solution to measure performance is consistent with the concept of BSC in that plant managers can evaluate all KPIs in their physical units (plus it adds statistical significance).

In addition to BSC ULTRAMAX offers tools to evaluate the individual KPIs in common, additive units (which can be actual economic calculations or the quasi-economic Performance Loss). The Performance Index is a single metric that is the addition of the individual KPI evaluations. This is very useful even when not doing BSC or ULTRAMAX optimization.

Baseline: The reference values of the variables for evaluating improvements. They are determined by the process adjustments at past experience and preference, such as standard operating procedures or where daily production was run before SEO, and the corresponding mean outputs (which depend on the values of the uncontrolled inputs).

Baseline : **Baseline Adjustments :** Adjustments at the Baseline values.

Baseline : **Baseline Run :** A process run with adjustments at Baseline.

Business Objectives : See Objectives.

Calculated Variable : See Variables

Character User Interface (CUI) : Menu-driven user interfaces in plain text characters used for process development (bypassing the need for the GUI) – and used in the past for certain types of unusual integrations.

Closed-loop : Integration where both: Run Data is passed automatically from the DCS or SCADA to ULTRAMAX; and where Advices are implemented automatically unless undesirable upsets such as ALARMS are detected.

Consistency : The ability of the process to generate the same outputs data for the same values of the inputs. Also, **Reproducibility**. The variations that remain – and are not explained by the known inputs – is the **noise**

Constraint : Limits on inputs or outputs to represent objectives which are typically not included in the Performance Index or Objective Function (page 103), and whose satisfaction is more important than improvements in the Performance Index. Functionally, constraints are interpreted as satisfied or not, regardless of by how much they are violated. If this is not representative of reality, then include the gradual effect in the Performance Index or objective function.

Constraint: **Active Constraint :** A constraint which if relaxed would yield improvements in the Performance Index or Objective Function. Applied especially at Optimum conditions.

Control : There are various usages of this term:

Control : **Process Control System (**First Level Control)**:** It includes control schemes such as **Regulatory Control** to whereby manipulated variables are changed so that a controlled variables are maintained closed to setpoint values. A typical example is a PID controller (e.g., manipulate *steam* flow to maintain *temperature* as close to its setpoint or target value as perturbations affect the process). The PCS may also include **Control Logic**, to adjust setpoints depending on process conditions – a function at some level similar to ULTRAMAX's, but unlikely to be near-optimal.

Control : **Supervisory Control :** Decisions and actions by plant personnel to manage production operations; including the adjustment of setpoints and biases, and the management of start-up, shutdowns and maintenance. ULTRAMAX automates the adjustment decisions being driven by optimization objectives defined by management.

In relation to First Level Controls it is called "Second Level Control". In some circles Third Level Control relates to production planning and scheduling.

Control Inputs : See Variables

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Controlled Inputs : Old name given to Control or Adjusted inputs. It was used to distinguish between the adjustable inputs that were adjusted (controlled) through SEO from those which were not. This terminology created conflicts with control engineers who call "controlled" the output variable from regulatory control, not the decision inputs such as setpoints or P, I, D parameters. In this context, the controlled variables are here consequences, outcomes or outputs. See Variables.

Critical Signal : The minimum Signal among all Active outputs.

CUI : See Character User Interface.

Curvature : Quadratic effects of inputs on outputs (which are an addition to the linear effects). They approximates many effects which are non-linear -- while the other non-linear effects due to lack-of-model-fit appear as noise.

Cycle : Short name for the SEO Cycle: Adjust, Collect run data, Update models, Generate Advice. In the software alone “Cycle” stands for the software portion (enter data, update models, generate advice).
Do not confuse with the various **cycle times** of the process itself, as set with control parameters. These are typical control inputs for SEO.

Data :

Data: **Active Data :** Data actually used in creating the local regression models, with positive weights. This has nothing to do with Active Output.

Data: **Actual Data :** Data obtained from process runs by measurements, summaries or calculations.

Data: **Historical Data :** The collection of Actual Data stored in the application file. It may include Prior Data.

Data: **Missing Data :** Missing values in a Run Data. The software supports occasional missing output values. On import/export they are represented by the value -777. Run data with missing inputs will be stored and ignored until the missing data is entered.

Data: **Prior Data :** Data available from Runs made before SEO begins. Prior data is useful if it is Valid.

Data: **Raw Data :** Raw Data are individual measurements associated with a process run, frequently collected every few seconds.

Data: **Summary Data :** is more representative cycle data provided to ULTRAMAX such as averages, standard deviations and percent defective, to reduce the amount of noise.
In steady-state optimization – the most frequent – data for summaries is collected only after the process achieves a certain level of stability after the transients due to readjustment fade away.
Synchronization with data obtained late such as waiting for laboratory tests is resolved by hand.
Summaries are calculated in the ULTRAMAX Event Driver (UED) discussed in page 41, or outside of the software if done Stand-Alone.

Data: **Run Data :** Set of operations data representing one run, where uncontrolled inputs and outputs are ofern averages or other summaries. Also, **run data set.**

Data: **Valid Data :** Data which is representative of how the process operates (runs) now and in the future in what is called “normal” operations being optimized with SEO. Old data may no longer be valid because of changes to the process.

DCS : Distributed Control System, the computer system that controls the process, functionally equivalent to SCADA

DDE : A data communications protocol: Dynamic Data Exchange

Decision Input/Output Diagram: A graphical presentation of the system/process to be optimized. It may include information to define the input and output variables, as illustrated in Fig. 6, page 31.

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They are different from Engineering I/O diagrams in that in the former intermediate variables are placed as outputs or consequences, since they are outcomes of the inputs; while in the Engineering diagrams they are often placed as inputs, since they affect the outputs. This is why there tends to be fewer inputs in a Decision I/O diagram.
The Decision Input/Output Diagram also reproduces the information of an Ishikawa diagram in the sense that the inputs are the cause of the outputs, but the relationship here is not logical, but functional [Y=F(X)].

Descriptive Model : See Model.

Discovery : The process of becoming aware of: 1) malfunctions in the process, 2) finding new insights about the process, or 3) understanding fundamental changes to improve the process. Discovery is triggered by the discipline induced by the ULTRAMAX Method and the checks it calls for. It is a powerful aspect of the ULTRAMAX Method for continuous improvements, perhaps even more valuable than the tangible improvements through optimization.

Dynamic : See Process.

Economic Evaluation : A Performance Index that represents the direct calculation of economic outcomes, ideally, the present value of short and long-term (variable) impacts on the P&L statement.
Experience indicates that not much accuracy is required to uncover significant process improvements, even if the estimated gains are off by some percentage. It greatly enhances the success of ULTRAMAX when top management and accounting personnel participate in the creation of the calculations and attest to the representativeness of the economic calculations.

Empirical : Based on observation or experience. It is the opposite of approaches based on formulating the behavior of the systems through first principle relationships. Empirical methods are limited by the ability to observe reality, while the first-principle approaches are limited by overlooking important real-life effects.

EMT (Extra Measurement Time) : See Time.

Error : The difference between the actual results and the results predicted by the model.

Error: **Estimated or Expected Error:** The standard deviation of the Prediction Error for an output. It is a function of the Noise estimate and of the SDF of the inputs.

Error: **Overall Standard Error:** A single multivariable metric of the Standard Errors for all outputs, taking into account the covariance of the prediction errors. If too large, then some combination of results is statistically different from the expected results and the historical prediction error distribution.

Error: **Prediction Error:** The Actual value minus the Predicted value of an output.

Error: **Standard Error:** For an actual value of an output, the ratio of the Prediction Error divided by the Expected Error (1 sigma), i.e., in units of sigma or standard deviation. The true technical name is "Standardized Error".

Estimate : Value predicted by the models, same as Predicted or Expected.

Event Log : A log kept by production personnel of all events that may have a bearing with an SEO implementation, such as: what was done when Alarms appeared such an out-of-control event was detected; when a breakdown happened; which runs are a Triad, important maintenance procedures that may render old data invalid, etc.

Expected Error : See Error.

Expected Value : Value predicted by the models, same as Predicted or Estimate.

Export : To place the historical run data in the application file in column and row spread-sheet format.

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Extra Measurement Time (EMT) : See Time.

Fixed Input : See Variables

Game Plan : Name given to the Optimization Plan until 1999.

Global Factor : Factors (constants, parameters) used in formulas to define calculated variables. Global Factors typically represent economic factors, targets, and/or tolerances.
Once ULTRAMAX is properly adopted by production managers, the Global Factors and Constraints become the main mechanism by which top management guides the performance of the process.

Goals : The way overall performance is evaluated; the balancing or resolving the tradeoffs among several individual objectives, namely, the Performance Index or Objective Function and the Constraints.

Gradient : A measure of the direction and magnitude of the impact of small changes in process inputs on the output; linear effects, devoid of interactions.

Graphical User Interface (GUI) : User Interface characterized by point-and-click with the mouse on drop menus and buttons, multiple windows, etc. which tends to be intuitive, visually appealing, and with more option alternatives at any one time.
It was frequently used for the **Stand-Alone GUI**. The other GUI is the one supporting the ULTRAMAX Event Driver.

GUI : See Graphical User Interface.

Historical Achievement Analysis: Improvement analysis when the optimum is not constant because of changing Uncontrolled Inputs.
The technology first calculates the Projected Optimum Estimate for each past run (conditional on the value of the Uncontrolled inputs), and then computes the difference between the actual results and the optimum estimate, i.e., the "lost opportunity" which should be minimized.
This analysis is valid and relevant AFTER the optimum is regularly approached.

Hysteresis: The effect that, when making an adjustment going up in value, the consequence on (some) outputs is different than making the same adjustment going down. ULTRAMAX does not have the capability to take hysteresis into account. Consistency in results for the same numerical adjustments is important for data analysis.

Implementation : The project consisting of: interfacing with upper management to lay out expectations, initial training of plant personnel, guiding them in the planning and engaging in production / manufacturing operations with optimization to the point that they are fully trained to continue on their own; and later review of their practices, the process and gains achieved present to upper management.

Import : To get (typically a lot of) data in row and column format (spread-sheet format) to be added to and/or substitute current data. See "Export".
This term is not used for automatic data entry through "Integration".

Important Difference : A difference in value of a variable that in itself or through other related variables creates an effect which is deemed significant to management, that management would pay attention to.
It is neither the measurement accuracy nor the setting accuracy, which are only related to differences which are noticeable or manageable, not important. Also, this concept has nothing to do with "statistically significant difference", which relates to differences which are truly detected. See Minimum Important Difference.

Improvement Profile : Plots of run data obtained with SEO vs. Run Number, to see improvement trends.

Independent inputs : See Variables.

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Input/Output Diagram : See Decision I/O Diagram.

Inputs : See Variables.

Integration : Automatic data exchange between the ULTRAMAX software and DCS or SCADA systems.

Key Performance Indicators (KPI) : The main variables by which production performance is evaluated (usually outputs but may include adjusted – decision -- inputs). For optimization the KPIs have constraints and/or are used to calculate the Performance Index (the metric to be maximized or minimized).

Learn : The action of creating prediction models based on stored historical data. It is an icon in the Stand-Alone GUI.

Local Model : See Model.

Localization Parameters : Parameters that indicate the localization of the models.

Loss Function : See Performance Loss Function.

Measured : See Variables.

MID : See Minimum Important Difference.

Minimum Important Difference (MID) :

See Important Difference (thus, it would be desirable in a "well running" process that the maximum measurement imprecision be lower than the MID).

- For the Performance Index the MID indicates the minimum change in value which is noticeable to management.
- For outputs with constraints, the MID indicates the violation of constraints that is important, that needs to be avoided.
- For inputs, the MID is two things: (a) the maximum change in a re-adjustment; (b) the minimum change which is either important by itself, or which is likely to create important differences in at least one important output.

Missing Data : See Data.

Model : A mathematical relationship to predict the values of an output as a function of the values of all the inputs. Models suitable for decision making (as compared to only for observer prediction) need to be cause-and-effect, i.e., remain valid when the decision maker intervenes on the process.

Model : **Calculated Model :** Same as Calculated output (see Variables). It could be based on first principles or economics, on statistical analysis, or be a hybrid.

Model : **Descriptive Model :** A regression model created with all historical data counted equally, as done in regular (global) regression. As all empirical models, they are accurate in the region of data. They are less accurate than local models in the region of interest, but tend to have larger Signals.

Model : **Local Model (Locally Accurate Model):** A regression model which is accurate only in the region of interest, where they are more accurate than Descriptive Models.
With sufficient data and for relatively "smooth" responses, proper local quadratic models mostly resolve the perennial problem of lack-of-model-fit.
Local models are created by doing weighted regression with higher weights in the historical data which is in the region of interest (e.g., with the current values of the uncontrolled inputs), and other relevant past experience. This is the Active Data.
The concept of Local Model includes the possibility of Descriptive Model as an extreme.

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Model : **Prior Model :** A model which represent the knowledge available at the beginning of a new round of SEO. This models do not have to be accurate (a linear approximation is already very useful), nor does it have to include all the inputs and output. It is a practical way to transfer knowledge from one round of SEO to the next when the historical data is no longer valid or sufficient, and therefore it is deleted or ignored.

Model : **Regression :** A model created by fitting a particular mathematical structure to data, typically to minimize some measure of error between output data and overall model estimate. ULTRAMAX uses quadratic models and subsets. Neural Models are another example.

Monitored Output : See Variables

Noise : Unexplained variation in output Actual Data, easily seen if all inputs remain constant. Note that for ULTRAMAX ***noise*** is lower than in regular statistical analysis, as the variations created by known uncontrolled inputs is not included (as they are managed with ULTRAMAX).
Noise is created by machine vibrations, the (inevitable) imperfections of the process control system, unknown material impurities, measurement inaccuracies, changes in undefined uncontrolled inputs, wear and tear, accumulation of materials, etc. Some Noise effects are random (the common causes, which change rather independently and from a predictable distribution from run to run), while others could be much larger or be shifts and drifts (special causes, from unknown uncontrolled inputs).
Note that when the output measure is a Standard Deviation, noise is the variation in the Standard Deviation. The notion of unexplained variability in Raw Data within a run – or lack of repeatability -- is "noise" also, but this meaning is not used in SEO

Non-linear Optimization : See Optimization

Objective Function : See Variables

Objectives :

Objectives : **Business Objectives :** Also **"General Business Objectives"**. A short statement of the overall business objectives to be satisfied in the current round of SEO, in terms which top management can readily understand and relate to. Typically it is not very technical in nature. Specific objectives in measurable terms are called the "Individual Objectives".

Objectives : **Individual Objectives :** An individual aspect of performance which (all other things being equal) would be desirable to improve or to keep within constraints, such as "higher production rate", "lower unit cost", "closer to targets", "lower emissions ", "lower process wear", "lower consumption of energy", "maintain process parameters within safety/reliability limits", etc.
The team describes individual objectives which are related to the Business Objectives in quantitative and measurable terms. This, in turn, dictates which are the Inputs and/or Outputs that should be measured or calculated. An individual objective may require more than one variable to represent/evaluate its performance -- e.g., average and standard deviation of a variable that is desired to be close to target.
See "Goals" for the balance of all individual objectives for the evaluation of overall performance, which in turns determines the Performance Index or Objective Function, the variable to be maximized or minimized subject to the Constraints representing other individual objectives.

Operating Scenario : See Scenario for short.

Operating Variable : About component inputs, See Variables

Operation: The running or functioning of a production process while making a product. For ULTRAMAX we are interested in the aspects of operations that are affected by how the process is controlled, how the operator operates it, the results from managing the operation, where the most important decision is how to adjust (set) the control inputs.

OPC : A data communication protocol: Object linking and embedding – OLE -- for Process Control

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Optimization : The quantitative determination of the most effective or functional state possible by affecting the actionable decisions within a domain or scope of interest. This requires mathematical technology. In our case the domain is the area of production / manufacturing operations management.

As done with ULTRAMAX's SEO, the process of improving process/product performance by adjusting the control or adjusted inputs to obtain the best practical performance in quantitative, concrete terms.

Mathematically, optimization is the maximization or minimization of the Performance Index (Objective Function) while constraints are satisfied.

Other things leading to improvements are not called (serious) Optimization. There are a variety of tasks leading to improved performance, such as gaining understanding of the critical aspects of the process, resolving issues, refining the Optimization Plan to better represent business objectives and include the truly most relevant variables, discovering erroneous assumptions and opportunities for improvements, removing bottlenecks by improving the process itself, etc. These are all improvements and benefits achieved through applying the ULTRAMAX Method.

Optimization: **Feed forward Optimization:** Optimization knowing values of the Uncontrolled inputs before the Run.

Optimization: **Feedback Optimization:** Optimization without knowing values of changing Uncontrolled inputs before the Run. The ULTRAMAX technology at this time can deal only with slow changes due to unknown (or unmeasured) causes.

Optimization: **Non-linear Optimization:** Algorithm for optimizing mathematical (or computer) models. The ULTRAMAX software includes a powerful non-linear optimizing engine that is used both for creating local models and to generate advice.
When all outputs in an ULTRAMAX Optimization Plan are calculated values (eventually a function of the inputs), then ULTRAMAX performs non-linear optimization without using any data.

Optimization Cycle : The sequential cycles within SEO which starts with an existing Advice to adjust the process, and includes:
1) Adjust the control inputs, make a process run, and acquire the raw data to calculate averages or summary run data.
2) Enter the run data into the ULTRAMAXsoftware.
3) Learn: update the models with all of the historical data.
4) Synthesis: generate Advice(s) for the next cycle using the updated models.

Optimization Plan : The objectives and the scope of SEO; the Optimization Strategy. This is a conceptual model that includes the definition of business objectives, individual objectives, variables, goals, Baseline, minimum important differences, constraints, calculations, Prior Models, etc. to define the process/product to be optimized, the evaluation of performance (the Goals), and what variables will be adjusted and which not in this round of Sequential Optimization. The ULTRAMAX Method calls for the Optimization Plan to change with time to adapt quickly to changing needs and levels of awareness.

Optimum : A state of running the process such that any change in adjustments does not yield any (mean or average) improvements. The concept of "optimization" is conditional on the given values of the uncontrolled inputs (if any) and includes the satisfaction of constraints (if any).

Optimum: **Advice and Optimum Estimate (AOE):** Estimate of the equivalent to Safe Optimum but where – to merit being an Advice -- the protection is based on the sigma of prediction errors, including the lack of knowledge due to extrapolating.

Optimum: **Local Optimum :** Equivalent to Optimum, but restricted to <u>small</u> changes in adjustments. It may be the global optimum – and if not, one needs to travel through a valley of lower performance before reaching the global optimum.

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Optimum: **Optimal Region :** The region of Control or Adjusted inputs values beyond which important losses in performance take place. See Window of Operations.

Optimum: **Practical Optimum :** The situation in SEO where one is at the Relative Optimum, and MID>3*noise for all Active outputs. That is, further improvements through noise reduction are not important, and in the case of Process/Product Development or Design Optimization we are done with SEO.

Optimum: **Projected Optimum Estimate (POE):** The predicted Safe Optimum.

Optimum: **Ideal Optimum :** Potential optimum without noise.

Optimum: **Relative Optimum :** The situation in SEO where the process is regularly running around the safe optimum for the current Optimization Plan and Noise levels.

Optimum: **Safe Optimum** : Optimum within Active Operating Area – i.e., having sufficient protection against output data violating constraints beyond the MID. Same as Ideal Optimum if all constraints have MID>3*noise.

Output : See Variables.

Overall Standard Error : See Error

PAR : The name of the computer variable with the values of the ULTRAMAX Parameters.

Parameter : Values which control the operations of control logic, including gains and thresholds. The ULTRAMAX technology also has its own parameters whose values characterize various operations and displays.
The word “parameter” is also used as “process parameter”, what we call “variables”.

Performance: Production operations (overall) performance is evaluated first by seeing that all constraints are satisfied, and then by the Performance Index.

Performance Index: More explicitly, the **Production Operations Performance Index** (also, occasionally, Performance Metric or Measure of Performance) is a single metric to provide plant personnel an overall, real-time notion of how a well a production operation is running -- as long as constraints are satisfied. The Performance Index is a single, holistic, composite metric that is the addition of the evaluations of the individual Key Performance Indicators (KPI) expressing upper management’s values and priorities – as long as constraints are satisfied. Depending on whether it is a “profit” or a “cost” or “loss” metric, the higher or the lower the better the balance of all KPIs levels is for the firm.

The **Performance Metrics** are the Performance Index, the Constraints and their MIDs.

The great benefit of a holistic Index is to know the total benefits to the company of a change; and to avoid the possibility of thinking that the change is an improvement when it also incurs a greater cumulative loss elsewhere, now or in the future. Such a Performance Index is practical when there are simple ways to define it, as explained in Goal Setting, page 101.

When the Index is in economic or quasi-economic terms it represents – even if the user is not aware of it -- the present value of the marginal cash flow impact on the P&L statement – more than accounting systems do. KPIs represent individual simpler objectives such as lower unit cost, higher throughput (higher sales if capacity constrained), better quality (higher future demand and sales), lower emissions, reduced losses, etc.

Certain aspects of performance do not lead themselves to be evaluated in economic or other gradual terms, e.g. safety (e.g., max boiler pressure or max grind-wheel RPM) and legal requirements (e.g., max NOx); and thus they are represented much more simply as constraints (categorical yes/no evaluation). Thus, the Performance Index is only meaningful when the constraints are satisfied.

The Performance Index also serves as the mathematical **Objective Function** when applying true (technical, quantitative) optimization to production operations. The constraints act simply as optimization constraints.

Previous names since 1982 for the Performance Index have been: Measure of Performance, Goal, Objective Function and Scorecard.

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Performance Loss Function : A simplified quasi-economic analysis of process performance used instead of an outright Economic Evaluation of a run (page 112).
It is based on calculating a standardized "cost" in "**units of loss**" which is zero (0.0) when the variable value is at a desired **target** or some reference value; and it is one (1.0) when the undesirable deviation from the **target** or **reference** reaches the "**tolerance**" value, which is the threshold into intolerable results.
The values of "tolerance" for various variables are adjusted so that they are "**consistent**" in the sense that the negative impact for being at the Tolerance deviation is about the same (within about a factor or about two).
This makes the individual Performance Losses (standardized costs) for all practical purposes additive, in order to generate a **Total Performance Loss** (**TPL**) calculated Performance Index or objective function to be minimized.

Predicted Metrics : The Performance Index, the Constraints and their MIDs.

Predicted Value : Value predicted by the models, same as Estimate or Expected value.

Prediction Error : See Error.

Prior Data : See Data.

Prior Model : See Model

Product Family : A set of products going through one process that the learning done in one product is approximately applicable to the other products. All products in one family will be managed with **one** Optimization Plan, with different products defined by "operating" inputs.

Profits : Increases in "profits" are the same thing as (net) increases in the "P&L statement". Many people use indicators (such as production rate) instead of calculating the impact on the bottom line, but the principle is the same. In some non-economic applications of ULTRAMAX, "profits" is philosphically speaking what the user wants the most of or the least of.
More especifically, often the impact of production has long term impacts (such as the benefits from better quality and customer satisfaction – which lead to more demand and more future business). In such case the concept of "profits" is the present value of the impact flow on the P&L statement, even if this concept is not explicitly employed but the user.

Process : A part of a unit operation, a unit operation, or a sequence of unit operations which affect each other. The "process" includes the procedures used to run it, such as bringing raw materials to room temperature, ways to adjust flows, rules and logic for adjusting setpoints, way to handle product, ways to take measurements, etc. Called "System" for a more general concept to apply SEO to non production processes.

Process: **Dynamic :** A process is called dynamic when the behavior changes due to unknown causes such as wear-and-tear, accumulation of materials, depletion of catalysts, etc. ULTRAMAX can track a dynamic process which changes slowly through feed-back optimization. The order of the data is critical for the software to deal with hidden dynamic effects.

Process: **Static :** A process is called Static when the Dynamic effects can be neglected. When a process is known to be static, then the Learn function can be improved by setting PAR(49) = 0. You also do this also when the order in the data has been lost, and thus dynamic effects cannot be detected.

Production Operations Performance Index : See Performance Index

Projected Optimum Estimate : See Optimum

Raw Data : See Data.

Relative Optimum : See Optimum.

Relative Slack : See Slack.

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Report : The information provided back by the software when the user executes an option or requests such report. Most reports can be scrolled on the computer screen and/or printed.

Response Variable : See Variable, Dependent

Role : See Variable Definition

Round of SEO : The set of SEO cycles with essentially the same Process/Product and Optimization Plan. A new round of SEO starts after a major change in the OP or the process itself.

Ruled Input : See Variables

Run : The operation of the process during the Run Time, with constant control or adjusted inputs.

Run Data : See Data and Run Type below.

Run Number : A sequential number provided by the software to each new run data. Deleting data does not change the run numbers allocated to the other run data.

Run Time : See Time.

Run Type : Type of run data depending on the SEO advice. The types and brief descriptor are:

Undef	***Undefined***
Misc	**Miscellaneous**, not close to following SEO advice
BasLn	**Baseline**
Explr	**Advice with Exploration**
TLP	Advice & **Travel-Limited Path** (trying to go to AOE)
StSeq	**Start up new sequence** of SEO runs (as requested by user)
AOE	**Advice** & **Optimum Estimate**
POE	**Projected Optimum Estimate**

Run Data, Run Data set: A set of actual values for all variables for one Run of the process. This includes any required time synchronization for data obtained from different sources at different times (done outside ULTRAMAX proper). If there were 6 inputs and 8 outputs a run data set would include the values of 14 variables.

Safety : For the value of an output, the number of prediction sigmas away from MID beyond constraints. In general we want SAFETY > 3 SIGMAS

Scenario : Short for “operating scenario”. In processes that have components that may or may not be running during operations, a scenario is a set of operating components (e.g., coal mills in a boiler).

SCADA System : Supervisory Control and Data Acquisition System, functionally equivalent to DCS, on which the first level Process Control System is configured.

Sequential Empirical Optimization (SEO) : The technology which enables one to sequentially (or when needed) determine the values for the process Adjustments (setting of control or adjusted inputs) based on data collected from running (operating) the process. The objective is for the process performance to converge safely, quickly and closely to the optimal running conditions.

Sensitivity Analysis : The study of:

1) the effects of changes in inputs on the values of outputs, e.g., to determine how accurately the control inputs need to be set and the uncontrolled inputs need to be measured, or to see whether the effects match what is expected from knowledge of first principles.

2) the effects of changes in the Optimization Plan such as Constraints, Performance Index and Global factors on the behavior of the optimum, e.g., to evaluate the benefits from moving or removing constraints through re-engineering projects, re-negotiating specifications with a customer, evaluating whether improvements according to a new evaluation can be achieved, etc.

Series of SEO Cycles : Series of cycles performed one right after the other. A new series is started, e.g., after process startup.

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Sigma : The symbol σ given the true Standard Deviation; and sometimes used loosely for the estimated Standard Deviation.

Signal : The extent to which the models explain the underlying behavior of the output as a function of the inputs, what is desirable to easily achieve and prove optimality. It is a measure of the variability in the outputs that is explained by the variability in the inputs: 100% means all is explained, and 0% means nothing is explained, all is noise. It is related to the R-square: it is the R-square adjusted by the degrees of freedom.
Getting a Signal means to get a Critical Signal > 65%, the level generally associated with achieving quick improvements or proving optimality. The objective of creating local models is not to maximize the Signal, but to minimize the Prediction Error while keeping the Signal above a threshold value. See Critical Signal and Average Critical Signal.

Situation : The aspect of production which may change due to changes in materials, environment, requirements, unit costs, etc.

Slack : Difference between the value of a variable and its Constraint, such that a positive value means to satisfy the constraint, and negative to violate. For an actual output data, a violation is deemed important when it exceeds the MID, i.e., when SLACK < -MID.

Slack : **Relative Slack :** The ratio of Slack/MID. It used for inputs and for outputs when Slack is negative.

Stability : See Steady State.

Staggered SEO : SEO when the data is not available at the end of the Run. Thus, Data is entered when available, and Advices obtained when necessary. The first few runs without run data are planned by getting multiple Advices.

Standard Distance : A dimensionless measure of the proximity of one set of inputs to the inputs of the data used to create the models in units of Sigma (Standard Deviation of the distribution of inputs data values). It is used in the calculation of the Standard Distance Factor (SDF).

Standard Distance Factor (SDF) : The ratio of the Standard Distance to the maximum Standard Distance that defines the limit of the Area of Confidence. It is a dimensionless measure of the proximity of one set of inputs to the inputs of the data used to create the models. It is related to the calculation of Expected Error. The Area of Confidence is defined as the set of input values with SDF < PAR(10) = Aggressiveness.

SDF of the Predicted Optimum Estimate (SDF-POE) : It indicates the location of the optimum estimate as it relates to the Area of Confidence. If it is at the edge of the AOC this indicates that there is more learning to be done to achieve near-optimal running conditions. See Average SDF-POE.

Standard Error : See Error.

Static : See Process.

Steady State : The state of the process when its results are the effects of the current values of the adjustments, rather than previous adjustments. Also see Time: Steady-State.

Summary : See Data.

Synthesis : The process in the SEO technology where all models are integrated to predict all results for given values of the inputs, and where a search is made for new values of the inputs which give the Advice which will ultimately result in effective SEO

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System : The subject of interest, for which the best possible performance is to be achieved. It is a well defined set of elements which as a whole produce varying results as inputs change. The most popular example is a Production Process. Others are product formulations, product/process through computer simulations, a laboratory procedure, a business plan, mechanisms such as a racing car and a sailboat, a cooking recipe, a medical treatment, etc.

Target : Desired value for an output. Used to define loss functions and CPK.

Time :

Time: **Extra Measurement Time (EMT) :** The time from the completion of a Run (the process adjustments for the next run) until data is obtained. If larger than 0 (zero), then you are implementing Staggered SEO.

Time: **Run Time :** The time that elapses from the adjustments at the start of a Run with new inputs, to the adjustments at the start of the next run.
Do not confuse with the "run time" as the time it takes to make a production run for production scheduling, even though sometimes they are the same. They are different when being able to make a new set of adjustments, i.e., starting a new run before the old one is finished, such as when the process consists of several unit operations working in sequence.

Time: **Transient Time :** The time it takes for the system to approach stability after adjustments are made. No performance data should be collected during this time.

Tolerance : In the calculation of the Performance Loss Function, the maximum deviation from the target or reference which is tolerable, the onset of intolerable results. Tolerance limit is more clearly thought of as ***in*tolerance.** With an explicit sign it means that the intolerance is arrived at in that direction.

Transient : Generally speaking, A distinguishable pattern of process behavior due to changes in process inputs, whether known or not.
One concern for SEO is the transient process behavior created by new adjustments to the inputs (until reaching **steady state**). One way to manage transients is to include as inputs in the input/output Diagram previous inputs or other process behavior characteristics which explain and predict the transients, but this makes the analysis more involved.

Travel : A standardized measure of the movement of input values from the input values in the last run data, given in units of SDF.

Triad : In SEO, a sequence of closely spaced runs to detect the effect of current uncontrolled input values (whether known or not) and its comparison with the progress made with SEO runs; the sequence three runs with adjustments at: Optimal, Baseline, Optimal.

Tuning : See Adjust.

Type : See Run type.

ULTRAMAX : A method and technology/software to optimize the performance of a process, emphasizing the tasks of:
1. achieving with the team unity of purpose and of scope with the endorsement of top management
2. adjusting the control inputs/parameters to the process;
3. detecting malfunctions;
3. understanding possibilities and bottlenecks;
4. extracting information on how to resolve barriers to further improvements.

ULTRAMAX does not include but complements tasks such as: first level process control system, maintenance, scheduling, and process improvements through Six-Sigma, etc.. ULTRAMAX includes the most effective SEO technology today.

Uncontrolled SDF (USDF) : A dimensionless measure of the proximity of one set of uncontrolled inputs to the uncontrolled inputs of all the historical data in the data base.

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Uncontrolled Inputs : See Variables.

Unused Variable : See Variables.

User Interface : The part of the software that represents the link between the user and the computational program: The Stand-Alone GUI and the Optimization Management Screen GUI in integration..

Valid Data : See Data.

Variable Definition : A series of 10 fields used by the software to define a variable in the Optimization Plan (see page 197). In addition the variable is defined by a variable number.

Variables : Inputs and outputs to the system or process being optimized; the process/product parameters.

Variables: **Active Output :** An output whose value needs to be predicted well. For those, ideally the Prediction Error < MID/3. The most obvious ones are the Performance Index and the outputs with Active Constraints. Also, but unknown to the software (at this time), may be output variables which are used in the calculations of calculated active outputs. This has nothing to do with Active Run Data.

Variables: **Adjusted input :** An adjustable variable that plant personnel is adjusting to improve performance. Identically the same thing as the “specific” Control input or Decision input. Perhaps its use is increasing.

Variables: **Automatic :** Variables (entry mode) whose values are entered automatically by Integration (data communications) with other systems, such as DCS or SCADA .

Variables: **Calculated input :** An input whose value is a function of Unused variables (only), entered by the user as equations in subroutine CALCI. They are used, e.g., to define as control inputs some fundamental variable such as energy (an intermediate conceptual variable), rather than two control inputs such as time and temperature (only DLL calculations).

Variables: **Calculated output :** An output whose value is an equation entered by the user as function of other input and/or output variables.

Variables: **Control Input :**
Generally, a process input parameter that the user can adjust largely independently from the setting of other control inputs. Also, **Adjustable input, Decision input** or **Manipulated variable.**
Specifically, the **adjusted** inputs in the SEO cycles, as defined in the Optimization Plan. Other adjustable inputs are fixed or ruled, or are ignored because really they have no appreciable effect on the outputs of this Optimization Plan. Control inputs are constant within a run.
They are also called **Decision Variables** in operations research and decision sciences. A control input that is difficult to change could be defined as "control" when willing to adjust it, and as "uncontrolled" with a fixed value when not.

Variables: **Dependent variables :** The set of outputs and ruled inputs. Also called **Response** variables in some fields.

Variables: **Fixed Input :** An adjustable input which is kept constant. It does not have to be included in the OP, but the operator must recognize that it has to remain constant.

Variables: **Important Output :** An output which could be Active: the Performance Index, outputs with Constraints, and outputs used in Calculations.

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Variables: **Independent inputs :** Control or adjustable inputs are “independent” of each other when, in adjusting any inputs, there is not any mechanism that would automatically change another input. The set of control inputs needs to be independent of each other in order to create empirical models useful for optimization. The uncontrolled inputs need to be independent of the control inputs, while they can be interdependent among themselves.

Variables: **Input :** For the Process, adjustable or uncontrolled inputs that affect the outputs. For the Optimization Plan, the adjustments (decision inputs) and variables which predict the outputs (uncontrolled inputs).

Variables: **Intermediate variable :** A variable which is a function of some control input and which affects the final outcomes or outputs of interest. If included in the Optimization Plan, they are placed as outputs, frequently only Monitored. Usually they are placed as inputs in Engineering I/O diagrams.

Variables: **Measured, Manual** : a variable entered into the GUI by hand, through the key board.

Variables: **Objective Function :** See Performance Index.

Variables: **Operating :** A component input which is operating (as compared to not). Has values 0 or 1.

Variables: **Output :** A variable whose value is an outcome or consequence of the values of the inputs.

Variables: **Performance Index** : see Performance Index

Variables: **Ruled Input :** An adjustable (a potential control input) that is consistently being adjusted by a specific rule which depends on the values of other inputs, outputs, or even measurements which are not entered into the software.
It is a dependent variable that behaves logically like an output. The rule may be a calculation in CALC (not CALCI).

Variables: **Uncontrolled Input or influence:** A process input that is not easily adjustable by the user and is not affected by the adjustments. It is useful when its variations create important variations on the outputs.

Variables: **Unused variable:** A variable that is entered, stored, plotted and displayed in the Historical Report, but is not used in any analysis. It is simple to change to other types of variables to study the implications.

What-if Analysis : It determine the models' predicted outputs from user-selected input values.

Window of Operations : That region of control or adjusted inputs that assures constraint satisfaction and is less than one MID away from the Safe Optimum value of the Performance Index.

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ULTRAMAX DATA

Timeliness

As explained in the Critical Success Factors, the advantages of SEO require that run data be available on a timely basis when extrapolating to different areas of adjustments, the "mountain climbing".

When Extra Measurement Time[114] EMT > 0, SEO cycles are run under the **staggered** procedures, getting Advice when necessary, and entering data when available. The basic rule is that the EMT should be less that 10 times the Run Time, ideally less that 5 times.

Once the region of optimality is reached and there is a sizable database, the timeliness requirement is highly reduced, where it might be acceptable to enter new run data only once a week.

If it is not practical to satisfy this rule, then it is more practical collect data for model-centric approaches such as Design of Experiments (DOE) or Neural Networks. If the problem is ever resolved, than one continue with SEO.

Minimum Sample Sizes

For certain summaries to be minimally accurate:

- **To estimate Fraction Defective (or non-conforming rate according to some criterion) the sample size should be sufficiently large that at least 20 defectives are observed.**

  Basis: For a Binomial distribution with small fraction defective "P", the standard deviation of the estimated or sample "p" is $\sigma_p = \sqrt{\frac{P(1-P)}{n}} \approx \sqrt{\frac{P}{n}}$. Now, we want a sample such that the one tail z•sigma is within the desired one-way accuracy of the estimate P; or

  $$z \times \sigma_p \leq f \times P$$

  Where: "f" is the accuracy required (fraction of P)

  "z" is the confidence required (in number of sigmas).

  Let **d = nP** be the expected number of defectives found. Substituting we get

  $$\frac{z}{f} \leq \sqrt{d} \text{ or}$$

  $$d \geq \left(\frac{z}{f}\right)^2$$

  So, if z=1.3 sigmas (roughly, 10% tails or 80% total confidence) and f=0.25 accuracy, then d = 27 defectives. Note that these are, truly, modest requirements.

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[114] Extra Measurement Time (EMT) is the time it takes to get the measurements after the adjustments for the next run

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Thus, as you decrease P by improving the process adjustments with ULTRAMAX, the sample size increases… until for very small P the sample size may become so large as to be impractical, i.e., it is not effective to measure quality by finding defectives. Alternates:

- Define more stringent non-confirming requirements that would increase the maximum acceptable P -- which may have negative political consequences.
- Rethink the concept, ideally translating the evaluation into a continuous variable. E.g., in a tensile test, instead of evaluating whether a sample breaks or not at a particular stress, measuring the stress at which it breaks.

- **To estimate a Standard Deviation (sigma) the sample size should be larger than 40.**

  Basis: For a Normally distributed variable it takes about 50 independent[115] runs to estimate a standard deviation with a 25% accuracy with 80% confidence (10% tails); as can be observed from a χ^2/df distribution (chi-square over degrees of freedom; estimated sigma / actual sigma). Note that these are, truly, modest requirements.

  Also note that within ULTRAMAX we do not make the assumption that the distribution is normally distributed, just bell-shaped.

  Since estimating sigmas is so uncertain, when estimating a 3•sigma protection (e.g., to protect against violating a constraint), the 3 does not have to be determined very accurately. For the same confidence as 3•sigma in a Normal distribution, the number of sigmas would be different for different distributions. The wisdom of taking 3 as a simple number regardless of the (bell-shaped) distribution was determined by Shewhart over 60 years ago.

[115] The condition of "independent" runs, i.e., non-serially-correlated deviations from the average, is very important, and it affects the procedures given for the Consistency Analysis.

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DEFINITION OF VARIABLES

These are the process variables or parameters used in the Decision Input/Output Diagrams to represents the managements of the process, the key conceptual model of the Optimization Plan.

Usually, any variables in the Optimization Plan which are intermediate process variables -- i.e., variables which are affected by inputs but affect outputs -- are also classified as Outputs (outcomes, results, consequences)[116]. With some advanced conceptual models otherwise "intermediate" variables can be control or uncontrolled inputs, as described in "Intermediate Variables as Inputs", page 87.

Field Descriptors

- **VAR**iable Number: up to 3-digit number[117]. The Stand-alone GUI assigns values automatically and sequentially as variables are defined.
- Variable **NAME**: a descriptive name of no more than 12 characters. The GUI assigns an underscore to any blanks within a name.
- **UNITS**: units of measurement for the variable, limited to 8 characters. The GUI assigns an underscore to any blanks within a unit name. The UNITS of measurement should be such that the values are between 0.0001 and 10000 in absolute value most of the time.

  Entering the “units name” is a good indication that there are ways to measure the variable.

  NOTE: If you have calculations, it is better to write the Units in physical format rather than abbreviation. E.g., to indicate "parts-per-hour", pt/hr is better than PPH, so as to be able to make cost and dimensional analyses easier.
- **DEPENDENCY** Field: The name of the component input upon which the operation of this variable depends. This variable has value when the component input has the value of one (and if the component input is zero this variable will have “missing” value; it is not operating).

[116] By comparison, in engineering input/output diagrams, intermediate variables are frequently listed as inputs to try too how they affect final outputs.

[117] In general the algorithms will run slightly faster with smaller variable numbers – since this requires smaller arrays.

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- **ROLE**: In reports it appears as the Number, in the Stand-Alone GUI is appears as the Name; and in the Optimization Plan document it can appear as a letter identification (or as the number).

Roles				Description
Group	**Name(s)** Alternatives in parenthesis	**Letter**	**Number**	
Unused	**Unused (Ignored)**	I	0	Not used in ULTRAMAX analysis, but to keep data for future analysis.
Inputs	**Control (Adjusted) (Decision)**	C A D	1	A process setting that is adjusted by the operator –a setpoint or bias in regulatory control, or manually. Control inputs must be independent of each other. At least one is necessary.
	Uncontrolled (External)	U	2	An input that affects outputs. Ideally is estimated before the run. It should not affected directly or indirectly by current control inputs (they are independent of the control inputs), however they can be interdependent among themselves (the reason is that they are used as predictors only, not as decision variables). Uncontrolled inputs can be adjustable or decisions variables made externally – and before – the optimization logic of ULTRAMAX.
	Component			Entered as Uncontrolled Inputs. Can only have the values of 0 (not operating) or 1 (operating).
	Transient Indicator	T	3	A summary of past behavior (any variable; inputs, intermediates, outputs) during a past time span that affects the value of the current or future outputs. Calculated outside the ULTRAMAX Core, such as by the UED (see Integration).
	Ruled	R	8	Adjustable input but their value is determined by user-defined rules (control logic). The rule may be empirical or calculated. Since its value is a consequence of other control and uncontrolled inputs, it behaves like an output. The parameters of the rule may be control inputs, thus improving the rule itself.

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Roles				Description
Group	**Name(s)** Alternatives in parenthesis	**Letter**	**Number**	
Outputs (outcomes) (consequences)	**Monitored**	M	4	Outputs not affecting objectives, to know how they behave (set to this type rather than as "important" only to save computer calculation time, which with the fast computers today it is of little benefit). Being discontinued. Do not use for variables used in the calculations of other variables.
	Important	O	5	Output that affects the evaluation of objectives: Performance Index, constrained outputs, or used in calculations.
	Performance Index **(Objective Function)**	PI OF	6	The output variable to **maximize** or **minimize.** Only one. Most of the times it is a calculated output.

- Entry **MODE**: how data is entered into the software

  H **Measured**: values collected and entered manually (by Hand) through keyboard (it could be Summary data calculated outside of ULTRAMAX).

  C **Calculated**: value calculated as a function of other variables (inputs and/or outputs) by user-supplied calculations. The calculations or equations can be entered directly into the Stand-Alone GUI, or accessed through DLL (created in coordination with Ultramax Corporation).

  A **Automatic**: value entered through Integration or for Transient Indicators.

- **Baseline:** The reference variable values to determine improvements and to start a round of SEO. It includes the adjustments at Baseline values and the corresponding mean output values. For the Optimization Plan Baseline values are necessary only for the Adjustable or Control Inputs (Adjusted and Fixed) – the output values are produced by the process and this will be repeated a few times, especially to see their dependence on the values of the uncontrolled inputs. Baseline values are defined in the Optimization Plan document but are not entered into the software.

- Minimum Important Difference (**MID**):

  All control inputs, all outputs with constraints, and the Performance Index require an MID.

  In few words, the MIDs are:

  - for the Performance Index, the minimum difference in this performance metric considered important by management
  - for outputs with constraints, the minimum violation of constraints (in the OP) which are

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considered important by management

- for control inputs, the maximum tolerable change (the minimum change that is undesirable from one run to the next) – also, the minimum change that is necessary to change outputs by their MID. Modified by PAR(59).

MIDs represent sensitivities to what is important for the business; not the capabilities of the process, sensors or control systems. For the sake of getting performance improvements with SEO, the MIDs do not require to be defined very accurately.

The MID is the most unusual value that the end-user has to define. For those interested the meaning of the MID is described in more detail at the end of this chapter in page 200.

- **CONSTRAINTS, LOw and/or HIgh**:

  These are values that should not be exceeded to satisfy objectives. Neither low nor high constraints are necessary. See "On the nature of CONSTRAINTS:" page 103.

  Recall not including constraints on outputs to represent things that will happen anyhow, such as Yield $\leq$ 100%, Percent defective $\geq$ 0, etc. Entering these constraints will tell ULTRAMAX to avoid approaching such "undesirable" values, which is not what we want.

Summary of requirements

NUMBER	Automatically assigned
NAME	Always required
UNITS	Always required
ROLE	Important=5 required for all outputs used in calculations of important outputs
ENTRY MODE	Always required
BASELINE	Conceptually meaningful in the Optimization Plan for Control Inputs only (not entered in the software)
MID	Required for: Control Inputs, Performance Index (Objective Function) and Outputs with constraints
CONSTRAINTS	Optional, ignored in uncontrolled inputs if using both upper and lower constraints then: 2*MID < Hi Const - Low Const

Details about the MID

The MID for the Objective Function (Performance Index) is the minimum difference in value that managers consider important from a business point of view; that if they knew about would either concern or truly satisfy them.

The MID for an output with constraints is the maximum violation of constraints by actual run data that is tolerable. Higher violations are considered "important"; lesser violations are in the "gray" area.

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The two above indicate that ideally, for prediction reliability from a business point of view, the sigma of the noise should be such that 3-sigma is less than the output MID[118] (similar to the concepts of Deming, Total Quality and Six-Sigma).

Making MID smaller-than-real for outputs with constraints could require operations to pull the process mean away from the constraint in order to provide a 3-sigma protection against the actual run values having important constraint violations (> MID). This reduction in the Effective Operating Range for "active" constraints results in lower possible Performance Index values.

Note: The customer might prefer to treat the constraints in the Optimization Plan not as a limit for the mean process value, but as a limit for the run data. This can be represented by setting MID to zero; however this does away with the "gray" area concept, and with the evaluation of noise vs. MID.

Making MID larger-than-real, especially for the Performance Index, generates larger spread of data around the optimum, which results in lower average Performance Index.

In net, the output MIDs do not have to be accurate but they should be realistic.

MID for control inputs plays two roles:

1. MID defines that maximum change in adjustment from one run to the next that plan personnel consider prudent (e.g., to maintain process stability and control). It is modified by the factor PAR(59).

   Note: the ULTRAMAX Advices are not forced to provide an MID change in adjustments; ULTRAMAX will be doing a balancing act among all simultaneous input changes so as to control uncertainty risks to satisfy constraints and to improve the Performance Index.

2. The minimum change that leads to early recognition of the effect of the input on the outputs. It is related to the outputs noise levels and MIDs.

   Note: Unlike output MIDs, control input MIDs are not related to any "tolerable input constraint violations". There is no need for a gray area since the control inputs are decisions variables and they have no noise.

This "effect detection" requirement can be described formally as follows:

Definition: there is a "**key**" relationship between an input and an output when an MID change in the input (x) -- keeping all other inputs constant -- creates a mean output (y) change ≥ max{3*noise$_y$, MID] [119].

Until the noise is known, assume that the noise satisfies the ideal 3*noise<MID. Then a "key" relationship exists when *an MID change in the input creates a mean output change ≥ MID* [120].

Then we explain the requirements as: For quick and effective SEO, the control input MIDs should be large enough that, for the Performance Index or outputs with constraints (which are likely to be active):

---

[118] To provide a 3 sigma protection without placing tighter requirements to the process means.

[119] Conceptually, $gradient = \frac{\partial y}{\partial x} > \frac{\max\{3 * noise_y, MID_y\}}{MID_x}$ or $MID_x > \frac{\max\{3 * noise_y, MID_y\}}{gradient}$

[120] This was the only criterion in the earlier versions of the ULTRAMAX Method.

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- each output has at least one key control input, and
- each control input has at least one key output.

Then, SEO can figure out relatively quickly the effects of the all inputs on all outputs. Otherwise it will still optimize, but take more runs (require more data, and move less) unless it runs into the condition of insensitivity (page 156).

Note: In practice it has been sufficient to determine sufficient MID based on the perception of operating personnel when formulating the Optimization Plan and applicable to the region of starting adjustments[121].

A *simple exercise* to help assure satisfying the above requirement is to complete the following table, placing a ✓ for each input-output pair that in the opinion of operating personnel fulfill the "key relationship":

	Output variable numbers								
Control input variable numbers			✓			✓			
		✓					✓		
		✓			✓				
	✓					✓		✓	✓
		✓					✓		

Then the requirements in that all columns and all rows should have a ✓. From this chart we can see that the 4th output does not have a key relationship with any input; and thus some input MIDs need to be increased until some input will have a key relationship with that output.

If it turns out that increasing the MID of an input does not create a desired change in any output, then that input is a good candidate to be considered "fixed" for this scope of optimization, even to the point of not being included in the Optimization Plan.

When the chart is completed, include it in the Optimization Plan document, for future reference.

In addition, the control input MID/50 defines the minimum increment (grid, rounding) in the values considered for the advice[122].

Note: this is not a statement about adjustment changes. The data increment restriction is done so as not to burden the operating personnel with unnecessarily "precise" advice.

[121] There is no need to run any kind of "experiment" to calculate the gradients.

[122] Really PAR(61)*MID with default PAR(61) = 0.02

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ULTRAMAX PARAMETERS

ULTRAMAX has Parameters by which you can control its Operations, Alerts, Data Entry and Reports. The following Parameters are the most useful for general users, organized by functionality.

Parameters are identified by a number, like PAR(10), and by a key descriptor shown in **bold** below. The **bold** value displayed is its default value.

You should change parameters from default values only for good reasons. Parameters that are not at their default values are shown and described in the middle of the Optimization Plan Report. Parameters displayed with a value but with no description should be reset to default value, unless indicated by the Ultramax Corporation.

Resetting any one parameter to default is done by entering the '--' for 'undefined' value. There are options to reset (almost) all parameters to default.

Affecting both LEARNING and OPTIMIZATION

PAR(1) = **0** **Low Limit**: Lowest run number (of data) to include in the analyses, data with lower run numbers are ignored.

Used to ignore old data when the process has changed. Ignored data can be plotted, and reintroduced simply by changing PAR(1).

PAR(20) = **10** **Thoroughness**: Maximum CPU time (min) per cycle, and other limitations.

Used to reduce the amount of CPU compute time. However, the software will take the minimum needed to produce acceptable results. If the time limit is active faster computers will produce better results. The limitations apply to the thoroughness of the mathematical algorithms in Learn and Synthesis.

PAR(20) = **70** **Calculations**:

0 → not using any

1 → calculations in Stand-Alone GUI

2 → calculations in **ucalcdll.dll**, being **unocalc.dll** when no calculations

Affecting LEARNING (Model Building)

PAR(35) = **0.65** **Minimum Signal**: Minimum desired Signal of the active outputs.

This is done to maintain the representativeness of local models (that otherwise could be lost due to many runs around the optimum). In general, getting a larger signal happens by including more data in the local models. Using more data for local models may decrease the prediction accuracy, and will take longer to identify a change in process behavior through feedback optimization. This requirement can be relaxed without major impact on SEO performance.

PAR(45) **Model Localization**:

= **0** Create LOCAL MODELS with redefined Localization Coefficients. This demands significantly more compute time than the other options. See Thoroughness, PAR(20).

= 1 Create LOCAL MODELS with fixed (latest) Localization Coefficients: B(2), B(3) and B(4).

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ULTRAMAX will use the latest localization coefficients calculated when creating models with PAR(45)=0.

= 2 Create DESCRIPTIVE MODELS (global models, all data counts equally), rather than local. This is the same as in regular statistical regression and neural networks. The Localization Coefficients are ignored and not changed.

Typically, Descriptive Models are less accurate than Local Models because they fit data away from optimal running conditions, and this may distort the quadratic models used. Thus, the determination of optimum estimates and advice is not as effective.

PAR(49) **Type of Process**:

= **1** DYNAMIC: A process whose behavior may change with time for unknown reasons, such as due to changes in unknown uncontrolled inputs. The algorithms consider gradually ignoring old data when it does not contribute to the ability to predict.

When a process changes its pattern of behavior the local models weight older data less. Large unexplained upsets[123] will create less suitable advices until there is sufficient data after the upset that the prediction models are more reliable.

= 0 STATIC: A process whose behavior remains constant with time, or for which the order of the data has been lost. Old data is not ignored just because of age.

The age (sequence) of data is not considered in defining the weights for creating local models. This is necessary when not doing sequential optimization, e.g., when analyzing experimental data and the order of experimentation has been lost. In such case see also PAR(41) (Optimization, Maximum Adjustment Travel) to remove limitations when generating an Advice.

Affecting OPTIMIZATION (Synthesis)

PAR(10) **Aggressiveness**: Maximum SDF [124] which defines the limit or size of the Area of Confidence, which in turn determines the maximum movement of an Optimum Estimate or Advice. (See also PAR(59))

The smaller PAR(10) the less the risk from bad extrapolation – however you will see a slower improvement rate. Large values will speed up improvements in rather "flat" response surfaces, but slow progress when there are curvatures, especially if the curvatures are irregular (non-quadratic).

= **1.0** Default value, rather conservative. Values up to 1.3 are also very acceptable

= 0.5 Very conservative value for specially demanding low-risk operations. SEO movement towards optimal running conditions will be slower or nil.

= 1.5 Bold for production. Prediction ability at the edges of the AOC (large SDF) will be poor. Care should be taken following the Advices.

= 2.0 Very bold, only for laboratory or simulation experimentation. Prediction accuracy severely limited at the edge of the enlarged Area of Confidence.

PAR(25) = **3** **Concentration**: How frequently the Advice tries to be also an Optimum Estimate. It can be a fractional number.

This applies only when there is one run per cycle -- if more, the first Advice is always the AOE. If trying to be an AOE and other things limit is, then the run will be a TLP (travel limited path).

---

[123] Detected through the Standard Errors of new run data.

[124] The Standard Distance Factor (SDF) is a property on the combination of inputs values only.

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PAR(30) **Use Calculations** or internal Regression Models:

= **0** Equations entered by the user (calculations) override internal regression models.

ULTRAMAX uses the user-supplied equations (calculations) both to calculate calculated outputs in new data, and in estimating results such as for What-If, Synthesis and, plotting models.

= 1 Regression models override equations when estimating results; supplied calculations used only on new data entry.

This is useful when internal model estimates show improvements, but the calculation would defeat it. One example is when the internal models predict a negative standard deviation – indicating potential improvements but impossible results -- but using the calculations (involving a square power) will give large positive values, that does not represent reality and the values are deemed undesirable. This syndrome is avoided when using the built-in loss functions.

PAR(41) = 1 Multivariable Maximum Adjustment Travel: Limits a new Advice to a maximum Travel (in SDF units) from the last data point equal to PAR(11)*PAR(10). This limitation does not apply when generating a Projected Optimum Estimate or when there is a large shift in Uncontrolled inputs.

In addition are the travel limits imposed on each control input separately by PAR(59)*MID.

= 0 Removes this limitation. This is necessary when not doing SEO. For instance, when optimizing with models obtained through standard parallel Design of Experiments you should not limit the maximum amount of travel (distance of optimum estimate from last data point). See also PAR(49) (Learning, Type of Process).

PAR(59) = 1 Individual **Maximum Adjustment Change**: Number of MIDs to define maximum change in inputs for an Advice. This limitation does not apply when:

- generating a Projected Optimum Estimate (POE)
- there is a change in component inputs
- there is a large change in (other) uncontrolled inputs
- start of new series of SEO cycles [PAR(80)=1]

PAR(60) = **3.0** **Safety**: Desired protection in Optimum Estimates and Advices so that future actual data will not violate a constraint by more than MID, measured in units of prediction sigmas. Number of estimation sigmas away from MID beyond constraint.

The smaller PAR(60) the closer the advice prediction can get to the constraint. A smaller Safety will allow SEO to search more aggressively, but also will increase the risk of constraint violation, as already done automatically within the SEO algorithm.

PAR(61) = **0.02** **Minimum Adjustment Increment** from which select input values in units of MID, to avoid unnecessary apparent precision in the reports.

The minimum adjustment increment (not the same thing as delta adjustments) is the smallest of:

- PAR(61)*MID
- 1/5th the range between upper and lower constraints (to have room the change values)
- one-sigma the spread of the control input data used to build models.

An optimum estimate or advice is not the “rounded” values of a continuous answer,

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but actually the discrete optimum among allowed values.

= 0 Inactive, use continuous values. The report writing logic will also limit the number of decimals displayed.

PAR(80) **Re-start sequence of SEO runs** (not be bound by last run in database).

= **0** Inactive. Also inactive if PAR(25)=1, always optimum estimate.

= 1 Do re-start. Avoid travel-limited advice (see PAR(59)) from the last stored run data and limit Advice to SDF≤0.5 for more accuracy. PAR(80) will reset automatically back to default after new data is entered, to continue normal SEO. This is useful, e.g., after a startup of the process or of applying SEO, and after doing occasional Baseline runs (with adjustments at Baseline, to test improvements for current uncontrolled inputs, as in the triads).

Affecting Reports and Plots

PAR(47) **No. of Columns** in Advice Report.

= **0** Automatic, displaying only the data relevant to the latest option selected by the user.

= N N latest (≤50), to see some information that may have been missed in previous cycles.

PAR(64) = **2** **Prediction Range** In Model Plots: abs(PAR(64)) is the plus/minus number of sigmas for the confidence curves for the models.

> 0 Prediction interval for the output for one run, reflecting the process noise and ability to predict depending on SDF.

< 0 Confidence interval for expected (mean) value of the model. These boundaries can be misleading for two reasons: 1) it presumes that the models structure being fit is very good, which may not be so; and 2) the input can be confounded with others that may have the actual effects.

PAR(68) **Moving averages & Standard Deviations**:

Plots display moving average for the last PAR(68) runs, where missing data is ignored. The Historical Data report display running average and standard deviation.

= **0** Inactive, display last run data (same as =1).

= N length for moving averages and Std. Deviations.

PAR(93) **Display Actual vs. Model Report** in Shape and in Coefficients reports

= **0** Does not display (done to avoid extremely long reports when there is a lot of data – prediction errors can be found in the Error Reports and can also be plotted).

= 1 Display them.

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Affecting ALERTS (Thresholds for different messages)

This set determines the level of the ALERTS numbers after the description, as indicated in the ULTRAMAX® ALERTS and Guidelines, page 55.

Inputs: Location of inputs in the Area of Confidence (AOC) measured by the Standard Distance Factor SDF. SDF=PAR(10) is the edge of the AOC, the maximum accepted prediction error, the furthest location of Advice (when possible).

PAR(103) = **0.4** Upper SDF of the **High Confidence** prediction area.

Upper SDF of Projected Optimum Estimates indicating a **Relative Optimum**.

13

PAR(104) = **0.6** Upper SDF of the **Good Confidence** area.

Upper SDF of "prediction by **interpolation**"; beyond is extrapolation

2, 3, 6, 11, 15

PAR(105) = **0.8** Upper SDF of the **Fair Confidence** area ('Note' alert for new Actual Data)

1, 10

PAR(106) = **1.2** SDF threshold into **Major Extrapolation**

1, 2, 6, 15

Outputs: Location of Actual outputs measured in Overall Standard Error (multivariate sigma deviation from expected values). The basic 'standard error' applies to individual outputs.

PAR(107) = **3.5** Overall Standard Error threshold into area called process **out of control**

3

This following set is about each variable separately:

Inputs constraints: Measured by Relative Slack = Slack / MID = fraction of MID

PAR(111) = **0** Upper input constraint violation (in fraction of MID) considered "**marginal**" violation. Beyond is the "**definite**" constraint violation (recall, with violation Slack is negative, and so is the relative slack).

4, 16

PAR(112) = **0.5** Upper input slack (in fraction of MID) considered "**close to constraints**".

12, 16

Output constraints: Measured in '**#** sigma protection", where sigma is the expected prediction error.

Note: the objective for outputs is that (most of the time) process means be limited by the constraints; and actual data be limited by MID-beyond-constraints

PAR(113) = **2.0** Upper output sigma protection of "**significant risk**" (ALARM) of violating constraint

17

PAR(114) = **3.0** Upper output sigma protection of "**some risk**" (Warn) of violating constraint

13, 17

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PAR(115) = **4.0** Upper output sigma protection to be "**close to constraints**" (Note) with minimal risk 13, 17

Filters for data selection

Data filter selection apply to Plots & Capability/Consistency Analysis (not to Learn)

PAR(125) = **0.** **Use** selection **filter** below

= 1. Select ALL RUNS; override selections below

PAR(126) = **1.** Select all runs even if **Constraint Violation**

= 0. Select only runs that satisfy constraints

PAR(127) = **1.** Select runs from all **Scenarios** (defined by component inputs operating)

= 0. Select only runs in scenario selected in last LEARN or CYCLE.

PAR(130) = **0** Select runs as in **Run Type** PAR(131) through PAR(139)

= 1 Select runs from all Run Types; override selections below

In run types the selection is made based on the largest of this set of values.

If all = 0, **all** run types selected

If some = 1, **only** those run types are selected

If some = -1, those are **excluded**, the rest selected

The text in italics is the descriptor as displayed in the historical data and Advice report.

Run types

PAR(131) = **0** *Misc* **Miscellaneous** runs, not close to following SEO advice

PAR(132) = **0** *BasLn* **Baseline** Run

PAR(135) = **0** *Expl* **Advice** with **Exploration**

PAR(136) = **0** *TLP* Advice along **Travel-Limited Path** (trying to go to AOE)

PAR(137) = **0** *StSeq* **Start up new sequence** of SEO runs (as requested by user)

PAR(138) = **0** *AOE* **Advice** & **Optimum Estimate**

PAR(139) = **0** *POE* **Projected Optimum Estimate**

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CALCULATIONS OF STATISTICAL TESTS

Comparing Performance between two sets of runs

This comparison is made first for the incidence of Constraint Violations, and then for the values of each variables separately.

Let there be two groups 1st (reference) and 2nd (which we wish to compare to the reference). Any one group could have only one run, such as a Baseline run.

For each variable and each group we have:

n (number of runs); m (average value; sd (standard deviation)

$\text{Std. Dev. m} = \frac{sd}{\sqrt{n}}$, that determines the confidence of knowledge about the true 'm'

Let Δ be the difference between the means

$$\text{Avg.}\,\Delta = \text{m}_2 - \text{m}_1$$

$$\text{Std. Dev. Avg.}\,\Delta = \sqrt{\left(\text{Std.Dev m}_1^2 + \text{Std.Dev m}_2^2\right)}$$

Then the significance[125] of the difference is

$$significance = \frac{\text{Avg. }\Delta}{\text{Std. Dev. of Avg.}\,\Delta}$$

We are practically positive that the average Δ change is not due to unknown variations when ***significance > 3 sigma***, when we have over 3-sigma confidence that it is different than zero.

Since the average reference values at the Baseline runs is a constant, the plot of an output and of its Δ follow the same pattern, both approaching asymptotically the optimum.

[125] Really, the "standardized difference"

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TRAINING SELF-TEST QUESTIONS

These questions are for readers to test their understanding of ULTRAMAX. Many were questions asked by students and practitioners after training.

QUESTIONS	ANSWERS
1. How many run data do you need to have before the ULTRAMAX software can give advice on where to adjust the control inputs next?	One, the run entered with first adjustments selected by the end-user.
2. What do the Slacks tell you?	There is a Slack for each constraint. The Slack indicates how close a variable value is to the constraints. A negative value means violations of constraints. Page 163
3. What does the Standard Distance Factor (SDF) tell you?	The SDF is a measure of distance of a set of the inputs (for data, advice, what-if, etc.) to the inputs of the data used to create the prediction models. It is a factor in determining prediction accuracy. The Area of Confidence is determined by the most accurate combination of inputs, where the SDF is the smallest. Page 159.
4. What does the Expected Error tell you?	The expected error represents the level of noise, plus the uncertainty which increases with the SDF – with more extrapolation.
5. What does the historical Standard Error tell you?	The historical standard errors, which appears in the "actual" column of the Advice report, depict how far the actual outputs are from the expected value determined by the model (before the actual data was entered). It is in units of sigma. Page 161. When the ABS(Standard Error) > 3.5 sigmas this is an indication that something unusual has happened.
6. Does a large Standard Error indicate that necessarily something different (a special cause) happened to the Process being optimized?	No. It could be that new data has inputs in a region with a new pattern of behavior which cannot be extrapolated from the old data.
7. What does the SDF of the Projected Optimum Estimate tell you?	The SDF of the POE indicates whether the optimum estimate (as determined by the models) lies within or outside of the area of confidence. If outside, then the optimum estimate is for the best process performance at the edge of the area of confidence, where the estimates are relatively more accurate. Page 166.
8. What is the guideline to determine if the process is at a Relative Optimum?	The average SDF of the POE for several cycles is less than 0.4. Page 166.
9. When data becomes available sequentially, what is the fundamental reason that sequential statistical analyses yield desired results faster than parallel analyses?	Sequential analysis yields better results sooner simply because they use the data when it becomes available to decide what to do next, rather than collecting data for later analysis at the end of data collection.
10. What are some of the key aspects of SEO that enable you to make runs during regular production with low risk of making off-quality product?	Good tools for Operations performance metrics Model based Advice within area of confidence Addresses multiple objectives The end user is not forced to follow the advice
11. How much is "Discovery" an important aspect of arriving at performance improvements?	"Discovery" is one of the critical complements to just doing SEO optimization, to detect events and understand barriers and solution. Page 65

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QUESTIONS	ANSWERS
12. Should you enter into ULTRAMAX data which even though valid, was from a run with very poor performance?	Yes. ULTRAMAX will learn from this data that this is a region to avoid. What is important is that the data be valid.
13. What are the priorities programmed into the software to give Advice?	control inputs obey their constraints. when giving Advice only (not for the Optimum Estimate), give Advice where each input does not change by more than its MID*PAR(59) {**X**,**U**} ∈ AOC, to assure relative certainty of estimates Multivariable limit on the "travel" of the re-adjustments: Travel[126] ≤ PAR(41) outputs obey their constraints optimize the Performance Index or Objective Function (often plus a perturbation function) ıge 150
14. How would you tell that an ULTRAMAX advice does not meet constraints because it wants to make sure that the estimates are reasonably reliable?	When the Advice violates a constraint and the SDF of the advice has a value close to PAR(10).
15. Why doesn't ULTRAMAX give the Optimum Estimate as the Advice all the time?	To avoid the risk of making a series of runs in a sub-space (e.g., in a straight run), and thus not maintaining knowledge of the behavior of the process away from this region. This creates the zigzag effect.
16. Under which conditions could the Advice knowingly fail to meet constraints?	unusual combination of uncontrolled inputs, or drastic change in the constraints of the control inputs. in the stored data output constraints are not satisfied (and ULTRAMAX rather give advice with predictability in the area of confidence rather than uncertain advice which obeys constraints). ge 150
17. How do you tell if a Process cannot satisfy the constraints in the Optimization Plan?	You get to the relative optimum and the optimum estimate does nor satisfy all constraints (some slack is negative). The "optimum" will be some point in the unfeasible region where some metric of constraint violation is minimized.
18. In the sequence of SEO cycles the advice is changing inputs too little. How do you make ULTRAMAX bolder?	Increase the "aggressiveness" parameter, PAR(10), so as to increase the size of the area of confidence. Page 204
19. Does changing PAR(10) change the value of the SDF?	No. The SDF is a metric; while PAR(10) is the criterion for defining the edge of the area of confidence. Page 159.
20. Does lack of detected effects by an input mean necessarily that changes in the values of that variable are not important? How so?	No. It could be that the input was not changed sufficiently (for the existing noise) to be able to detect its effect. A very important input would have all detected effects equal to zero if it has not changed at all.
21. What are the advantages of using the Performance Loss Functions instead of Constraints?	The PL functions provide a practical method of representing the quality concept that "closer to the target" is better; while constraints implement the quality concept that "everything within specifications" is equally satisfactory to the customer. Pages 112.
22. Does it make sense to have a Performance Loss function on an input?	Yes. Any process variable can be used to evaluate its performance. See example in page 16.
23. What condition should the Tolerances satisfy so that several Performance Losses are additive?	The tolerances should be such that the cost associated with a deviation from their target/reference, for each one separately, be about the same (within a factor of two or three). This requirement is that the tolerances to be "equivalent". Page 121.

[126] A Mahanobis distance.

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QUESTIONS	ANSWERS
24. Why are "percent non-conformance", "production rate" and "costs" better represented by linear performance loss functions rather than parabolic?	Because they do not have "targets" such that high or low deviations incur a penalty. "Cost" has obviously a linear effect in the P&L statement. Percent defective is typically such that half the defect level, half the pain; i.e., linear. Page 117.
25. Could the Performance Loss be something other than a straight line or a parabola?	Yes. It can be any Piece-wise gradual equation provided by the user.
26. How critical is the assumption of "normally distributed noise" on the performance of ULTRAMAX, and on the interpretation of reports? What statistical risks does ULTRAMAX assume in its criterion to satisfy output Constraints?	ULTRAMAX does nor require any distribution assumptions except that it be roughly bell-shape. Distributions are necessary to make risk statements, which are never made within ULTRAMAX. The protection it provided is numbers of sigmas. Page 163.
27. Why is it recommended that initially an Optimization Plan should be simple as practical, rather than trying to solve all problems at the beginning?	This will most likely lead to sub-optimization (by keeping other control inputs constant rather than optimizing them). However optimization is much more than the numerical analysis. It involves learning about the objectives, the process behavior, how to best measure results, and discovery, and changes to the process and the optimization plan. These benefits can be achieved as well in a simpler environment. Continual improvements is a never-ending journey. In the course of time, in subsequent rounds of SEO other inputs will be introduced, and some current ones dropped.
28. Why is it recommended that the Optimization Plan Report list all known inputs, including those Control inputs which were decided to keep fixed, and Uncontrolled inputs which were decided to ignore?	To document that they have not been ignored. The control inputs shall be kept fixed (constant) need to be understood by everybody involved. This is an important aspect of the management of process adjustments to avoid noise sources. ge 32.
29. If you add a new input to the Optimization Plan, what should you do to the historical data in the software?	Process runs with missing input data are ignored by the analysis. Thus, to be able to learn from the data that already exists one needs the historical values of the new input. Else, just start a new round of SEO.
30. Why is it recommended that the Data Log Sheet have data for all possible variables?	Because if any of these variables were to be included in a future evolution of the optimization plan, then we want to have their historical values so as to be able to learn from them.
31. If we were to reduce noise by improving components of the process (rather than by including a new variable which explains the variability), what should we do with the historical data?	Ignore the historical data, as it is no longer valid (representative of how the process behaves now and in the future). Adjust PAR(1).
32. Did you know that it is undesirable to enter replicates (runs with same inputs) as separate runs?	Yes, the ULTRAMAX technology does not readily manage replicates and the special handling that is necessary when there is limited information.
33. How does the ULTRAMAX software calculate summaries (e.g., Avg. and Std. Dev. of raw data)?	ULTRAMAX calculates summaries only when integrated, in the interface communications module.
34. What are the advantages of writing the Optimization Plan Report?	To document the optimization plan. To achieve unity of purpose and scope within the team. To be in agreement with top management. To enable people to change priorities and discover improved approaches. To create a comprehensive training tool of process management. ge 35.

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QUESTIONS	ANSWERS
35. Sometimes management reacts negatively to major performance improvements of current production equipment which is obtained simply with better process adjustments and discipline. The reaction may be annoyance because these benefits not obtained in the past are perceived as incompetence of the people. How can this be avoided or defused?	Management needs a paradigm shift. Management need to recognize that even well run processes have significant untapped process capabilities which can be realized practically only with solutions such as ULTRAMAX. Abstractly, this is seldom news to management. What is frequently surprising -- and disconcerting -- is to find the level of realized improvements. This goes against the repeated experience (paradigm) of getting improvements only through capital expenditures.
36. How would you tell the ULTRAMAX SEO software that your objective is just to maximize an input?	Create a Performance Index (output) which is calculated to be equal to the input, and to be maximized.
37. Once we learn how to run a process to achieve very low percent non-conformance, is percent non-conformance a good measure of quality achievement in order to maintain optimization?	No. In order to be able to detect differences in percent defective of about 25% the sample size must be sufficiently large to detect over 20 defectives. This requires a very large sample size when the percent is very low. (page 195) A better solution, when possible, is to work with a "variable" (rather than "attribute") measurement which is related to the onset of the defective happening, such as strength, pressure, thickness, etc.
38. Someone comes to you saying: "my application has six inputs, I made twenty runs following the SEO advices, and I am not getting any improvements -- ULTRAMAX does not work for my application." How would you help this person?	Note that the rule-of-four is still not satisfied in this round of SEO Page 61. Note that ULTRAMAX is much more than the SEO technology. It is a Method that helps users define objectives and give coherence to managing the adjustment of processes to deliver the highest value to the enterprise. It also helps understand where the barriers are to achieve improvements (e.g. too much noise).
39. What is the use of Global factors? What happens with the old calculated values when you change the values of the global factors?	The global factors are a handy way to change parameters in the optimization plan to represent new business situations, such as cost of materials, customer specifications, etc. Old physical data is still valid. The evaluation of old data with the new criterion is done automatically for all the past data. Pages 27, 33, 66
40. How do you determine whether you achieved performance improvements when there are important uncontrolled inputs which have changed and also affect performance (i.e., the optimum values of the control inputs depend on the uncontrolled input values)?	When the gains in the Performance Index are much greater than the effects of changing uncontrolled inputs, then just observe the values of the Performance Index. When the above is not true, then much more detailed analysis is necessary such as the triads: periodically make a run with adjustments at Baseline, to compare SEO performance just before and after with similar values of the uncontrolled inputs. Also, the evaluation has to be made over a long period of time to have a chance to evaluate improvements under varied sets of uncontrolled inputs. Page 50
41. Since with ULTRAMAX we typically get to a relative optimum in about as many runs as 10 times the number of (important) inputs, and Factorial DOE and Taguchi often have designs with fewer runs than 3 times the number of inputs, how come we say that ULTRAMAX converges to the optimum in fewer runs or tests than other DOE technology?	Our claim is based on process with continuous variables and when defining "optimization" as the best performance, whether we know where it is (to run a DOE around it) or not. The reason is that to get to know the location of the optimum with the same accuracy as determined by ULTRAMAX, typically one has to perform several sequential DOEs, first to locate the optimal region which is typically away from current operating practices, and then to find the right area to cover. The other great benefit of using ULTRAMAX is to do this while engaged in production, at much lower cost than making experimental runs.
42. Under which conditions it is better to start a round of SEO with data obtained from Parallel DOE (Design of Experiments)?	DOE has significant advantages the process is not “sequential”, such as in agriculture and certain parallel laboratory studies, and making poor performance runs is not costly. This is an issue related to the timeliness of the availability of the data. SEO is most effective when the extra measurement time EMT<5*run_time. Page 195.

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QUESTIONS	ANSWERS
43. What benefits can you derive from using ULTRAMAX to analyze data already obtained through (parallel) DOE?	Easily optimize control inputs addressing multiple objectives. Automatically integrating the models for overall process performance optimization. Introduction of calculated variables. Automatically creating models with curvatures (when there is sufficient data). Automatically create models which are more accurate around the optimum (when there is plenty of data).
44. How do you detect outliers?	On Stand-Alone data entry, there is a data editor detecting data individually outside normal ranges. The advice report, in the "actual" columns, displays the standard errors, indicating deviations from expected values. The error report and plots display the prediction error
45. What are the benefits of using Prior Models?	Approximately correct prior models enable SEO to start moving in the direction of improvements much faster, and thus produce higher cumulative performance. Prior models are particularly valuable to harness knowledge obtained in lab, pilot plant and other similar processes; including the same process with older data that is not longer complete or valid. Page 95.
46. In SPC (Statistical Process Control), improvements in quality take place when special causes are identified and eliminated (or their effects blocked) for future runs, thus steadily reducing sources of variability. What is the equivalent task in ULTRAMAX? To what extent can you get improvements if you do not perform this task?	Upsets or special causes (that create deviations larger than the noise) are identified by responding to the Alerts, and the specifics by observing the individual standard errors. Page 51 If the upset data is not deleted and the special causes are not prevented for the future, the variations they generate become part of noise, and the larger noise will prevent average performance to be closer to the ideal optimum. Page 154.
47. What is the apparent conflict between ULTRAMAX and SPC? How can both be applied at the same time?	Superficially, the conflict is that ULTRAMAX keeps on changing the Control inputs, while standard SPC wants to keep them constant. ULTRAMAX compensates for the changes of inputs by engaging in Regression SPC. Regular SPC can be applied within each run, when the control inputs remain constant, while ULTRAMAX has run-to-run SPC built in (the standard errors).
48. Historically, what aspect of the ULTRAMAX solution has led to the greatest improvements of the understanding of what are the important (first-principle) mechanisms which affect results?	Simply observing optimal running conditions. To an experienced engineer, this by itself lends clarity as to the relative effects of various mechanisms.
49. How frequently could there be missing inputs?, missing outputs?	The ULTRAMAX software cannot handle missing inputs (in which case the whole run data set is ignored). However it can handle occasional missing outputs.
50. What are the benefits from understanding the shape of each response surface, and the effects of each of the inputs?	It is an aspect of "listening to the process -- or its data". To confirm what you know, or be challenged by the data.
51. Why should you set PAR(41)=0, PAR(49)=0 and PAR(59)=0 when generating an Optimum estimate based on prior data from a parallel DOE?	ULTRAMAX normally treats data as if it were sequential. Normally models are built taking into account the possible existence of slow dynamic changes in the process due to unknown time-dependent factors. Selecting Optimization Plan > Parameters > Static (PAR(49)=0) treats the process as static, removing this consideration, treating all data as equally representative of the present (DOE is likely to have the tests randomized). Page 203. Normally advice is given not too far away from the latest run. PAR(41)=0 and PAR(59)=0 removes this requirement. Page 204.

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QUESTIONS	ANSWERS
52. How do you determine when ULTRAMAX detects that two or more inputs interact (i.e., the effect of one variable depends on the values of the other variable)?	Select Optimization Plan > Parameters > Detailed. The shape report for the dependent variable displays significant curvature effects (>.50) for more than one input for some "individual input contribution".
53. Describe ways in which an application of ULTRAMAX's SEO can fail to provide gains.	Failing any of the four Critical Success Factors. Page 18. Starting at the optimum. Not knowing what is important for the organization or how to measure it.
54. What could happen if a presumed Control input is not really so, that is, it is dependent by other variables (aggravated when the "other" variable is unknown)?	Basically, the Advice cannot be followed, as something will change the supposedly control input.
55. Does collinear input data mean that the variables are not independent?	No. Interdependent variables will give near-collinear data. However independent inputs can be collinear because this is what is necessary in production to get good results in the case of sharp ridges.
56. It is known that if two (or more) inputs are collinear, then it is difficult (if possible at all) to detect the separate effects of the inputs. What does this do to the ability of the regression models to predict results?	Nothing, as long as we recognize that in practice models fitted to data (such as regression and neural networks) are likely to be valid only in the area where data exists.
57. In what ways are "interactive" and "interdependent" in inputs the same thing?	They are different things: "interactive" deals with their effect on something else; and "interdependent" deals with one affecting the other.
58. If the MIDs were sufficiently large, can we be assured that ULTRAMAX would arrive at a Global (rather than a Local) Optimum?	It is possible that large MIDs would make inputs "jump" over local optima that are not very distinctive. But this is not sufficient to assure reaching a global optimum.
59. Why do orthogonal experiments tend to produce so much off-quality product during the experiment?	Because it is typically in the combination of things that bad -- and superior -- results are obtained. Bad results tend to occur when combining extremes of different inputs (that is why they are avoided in ULTRAMAX).
60. Since ULTRAMAX fits only a quadratic (second order) function, how can ULTRAMAX's SEO optimize when there are important higher order effects?	Because of the condition of "graduality", with sufficient data to create locally accurate, goal-oriented models, a quadratic fit is sufficient to determine the optimum with sufficient accuracy.

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See Glossary (page 180), for definition of terms with special meaning for the ULTRAMAX solution.

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